Volume II: Modern Applications

The Quantum Theory of Fields

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Abstract: Weinberg's QFT, typeset in nicer notation. Typesetting done entirely by coding agents, namely GPT 5.6. Figures done by Fable 5. Enjoy!

Contents

Preface to Volume II

Notation

15 Non-Abelian Gauge Theories

The quantum field theories that have proved successful in describing the real world are all non-Abelian gauge theories, theories based on principles of gauge invariance more general than the simple $U(1)$ gauge invariance of quantum electrodynamics. These theories share with electrodynamics the attractive feature, outlined at the end of Section 8.1, that the existence and some of the properties of the gauge fields follow from a principle of invariance under local gauge transformations. In electrodynamics, fields $\psi_n(x)$ of charge e_n undergo the gauge transformation $\psi_n(x) \longrightarrow \exp\{ie_n\Lambda(x)\}\psi_n(x)$ with arbitrary $\Lambda(x)$. Since $\partial_\mu\psi_n(x)$ does not transform like $\psi_n(x)$, we must introduce a field $A_\mu(x)$ with the gauge transformation property $A_\mu(x) \longrightarrow A_\mu(x) + \partial_\mu\Lambda(x)$, and use it to construct a gauge-covariant derivative $\partial_\mu\psi_n(x) - ie_nA_\mu(x)\psi_n(x)$, which transforms just like $\psi_n(x)$ and can therefore be used with $\psi_n(x)$ to construct a gauge-invariant Lagrangian. In a similar way, the existence and some of the properties of the gravitational field $g_{\mu\nu}(x)$ in general relativity follow from a symmetry principle, under general coordinate transformations.[1] Given these distinguished precedents, it was natural that local gauge invariance should be extended to invariance under local non-Abelian gauge transformations.

In the original 1954 work of Yang and Mills [1], the non-Abelian gauge group was taken to be the $SU(2)$ group of isotopic spin rotations, and the vector fields analogous to the photon field were interpreted as the fields of strongly-interacting vector mesons of isotopic spin unity. This proposal immediately encountered the obstacle that these vector mesons would have to have zero mass, like photons, and it seemed that any such particles would already have been detected. Another problem was that, like all strong-interaction theories at that time, there was nothing that could be done with it; it seemed that the large coupling constant of the theory would preclude any use of perturbation theory.

Gauge theories were soon generalized to arbitrary non-Abelian gauge groups [2], and their quantization continued to be studied mathematically, notably by Feynman [3], Faddeev and Popov [4], and De Witt [5], in part as a warming up exercise for the harder problem of quantizing general relativity. They showed that the naive Feynman rules obtained by simply inspecting the Lagrangian need to be supplemented by additional 'ghost' loops. However, the physical relevance of these theories did not begin to be understood until the late 1960s. It eventually turned out that all of the observed interactions of elementary particles are generated by vector fields associated with local gauge symmetries; the corresponding spin 1 particles are either very heavy, as a result of a spontaneous breakdown of the gauge symmetry, or are 'trapped', as a result of the rise of the coupling constant at long distances. These matters will be the subjects respectively of Chapters 21 and 18. In this chapter we shall explore the formulation of the non-Abelian gauge theories, and the derivation of their Feynman rules.

[1] Of course, both local gauge invariance and general covariance can be realized in a trivial way, by taking $A_\mu(x)$ and $g_{\mu\nu}(x)$ to be non-dynamical c-number functions that simply characterize a choice of phase or coordinate system, respectively. These symmetries become physically significant when we treat $A_\mu(x)$ and $g_{\mu\nu}(x)$ as dynamical fields, over which we integrate in calculating S-matrix elements.

15.1 Gauge Invariance

We assume that the Lagrangian of our theory is invariant under a set of infinitesimal transformations on the matter fields $\psi_\ell(x)$

$$\delta\psi_\ell(x) = i\epsilon^\alpha(x)(t_\alpha)_\ell{}^m\psi_m(x), \tag{15.1.1}$$

with some set of independent constant matrices[2] t_α, and with real infinitesimal parameters $\epsilon^\alpha(x)$ which (as for gauge transformations in electrodynamics) are allowed to depend on position in spacetime. We assume that these symmetry transformations are the infinitesimal part of a Lie group; as shown in Section 2.2, this requires that the t_α obey commutation relations

$$[t_\alpha, t_\beta] = iC^\gamma{}_{\alpha\beta}t_\gamma, \tag{15.1.2}$$

where $C^\gamma{}_{\alpha\beta}$ are a set of real constants, known as the structure constants of the group. The antisymmetry of the commutator immediately tells us that the structure constants are similarly antisymmetric:

$$C^\gamma{}_{\alpha\beta} = -C^\gamma{}_{\beta\alpha}. \tag{15.1.3}$$

Also, from the Jacobi identity

$$0 = [[t_\alpha, t_\beta], t_\gamma] + [[t_\gamma, t_\alpha], t_\beta] + [[t_\beta, t_\gamma], t_\alpha] \tag{15.1.4}$$

we see that the Cs satisfy the further constraint

$$0 = C^\delta{}_{\alpha\beta}C^\epsilon{}_{\delta\gamma} + C^\delta{}_{\gamma\alpha}C^\epsilon{}_{\delta\beta} + C^\delta{}_{\beta\gamma}C^\epsilon{}_{\delta\alpha}. \tag{15.1.5}$$

Any set of constants $C^\gamma{}_{\alpha\beta}$ that satisfy Eqs. (15.1.3) and (15.1.5) define at least one set of matrices t^A_α:

$$(t^A_\alpha)^\beta{}_\gamma = -iC^\beta{}_{\gamma\alpha}, \tag{15.1.6}$$

that satisfy the commutation relations (15.1.2) with structure constants $C^\gamma{}_{\alpha\beta}$:

$$[t^A_\alpha, t^A_\beta] = iC^\gamma{}_{\alpha\beta}t^A_\gamma. \tag{15.1.7}$$

This is known as the 'adjoint' (or 'regular') representation of the Lie algebra with structure constants $C^\gamma{}_{\alpha\beta}$.

For example, in the original Yang–Mills theory, the matter fields were the doublet consisting of proton and neutron fields ψ_p and ψ_n:

$$\psi = \begin{pmatrix}\psi_p \\ \psi_n\end{pmatrix}$$

[2]In this book we shall generally label symmetry generators with letters α, β, etc. from the beginning of the Greek alphabet, in order to keep these labels distinct from the indices μ, ν, etc. from the middle of the Greek alphabet that are used to label spacetime coordinates. Later, in dealing with broken symmetries, we will often use letters a, b, etc. from the beginning of the Latin alphabet to label generators of spontaneously broken symmetries, and letters i, j, etc. from the middle of the Latin alphabet to label generators of unbroken symmetries.

and the t_α with $\alpha = 1, 2, 3$ were the isospin matrices

$$t_1 = \frac{1}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad t_2 = \frac{1}{2}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad t_3 = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

These satisfy the commutation relations (15.1.2) with

$$C^\gamma{}_{\alpha\beta} = \epsilon_{\gamma\alpha\beta},$$

where as usual $\epsilon_{\gamma\alpha\beta}$ is $+1$ or -1 if γ, α, β is an even or an odd permutation of 1, 2, 3, respectively, and vanishes otherwise. We recognize this as the same as the Lie algebra (2.4.18) of the three-dimensional rotation group; the matrices t_α here furnish what we recognize as the spin 1/2 representation of this Lie algebra. The matrices (15.1.6) of the adjoint representation are here (in a basis with rows and columns labelled 1,2,3):

$$t_1^A = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}, \qquad t_2^A = \begin{pmatrix} 0 & 0 & i \\ 0 & 0 & 0 \\ -i & 0 & 0 \end{pmatrix}, \qquad t_3^A = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

This is the spin 1 representation of the Lie algebra of the rotation group.

Now consider what is needed to make the Lagrangian invariant under the transformations (15.1.1). If there were no derivatives acting on the fields, the task would be easy—any function of the matter fields that was invariant under the transformation (15.1.1) with ϵ^α constant would also be invariant with ϵ^α arbitrary real functions of the spacetime coordinates. This is not the case if the Lagrangian involves derivatives of the fields (as it must), because with position-dependent functions $\epsilon^\alpha(x)$, the derivatives of the matter fields do not transform like the fields themselves. Differentiating Eq. (15.1.1) gives

$$\delta\big(\partial_\mu \psi_\ell(x)\big) = i\epsilon^\alpha(x)(t_\alpha)_\ell{}^m\big(\partial_\mu\psi_m(x)\big) + i\big(\partial_\mu\epsilon^\alpha(x)\big)(t_\alpha)_\ell{}^m\psi_m(x). \tag{15.1.8}$$

To make the Lagrangian invariant, we need a field $A^\alpha{}_\mu$, whose transformation rule involves a term $\partial_\mu\epsilon^\alpha$, which can be used to cancel the second term in Eq. (15.1.8). Since this field carries an α-index, we would expect it also to undergo a matrix transformation like Eq. (15.1.1), but with t_α replaced with the adjoint representation matrices (15.1.6). Let us therefore tentatively take the transformation relation of these new 'gauge' fields as

$$\delta A^\beta{}_\mu = \partial_\mu\epsilon^\beta + i\epsilon^\alpha(t_\alpha^A)^\beta{}_\gamma A^\gamma{}_\mu$$

or, using Eq. (15.1.6),

$$\delta A^\beta{}_\mu = \partial_\mu\epsilon^\beta + C^\beta{}_{\gamma\alpha}\epsilon^\alpha A^\gamma{}_\mu. \tag{15.1.9}$$

This allows us to construct a 'covariant derivative':[3]

$$(D_\mu\psi(x))_\ell = \partial_\mu\psi_\ell(x) - iA^\beta{}_\mu(x)(t_\beta)_\ell{}^m\psi_m(x). \tag{15.1.10}$$

[3]As discussed in the next section, in writing Eq. (15.1.10) we are tacitly supposing that any coupling-constant factors like the electric charge are included in the t_β, and hence also in the structure constants.

As planned, the term $\partial_\mu \epsilon^\beta$ in the transformation of $A^\beta{}_\mu$ in the second term of Eq. (15.1.10) cancels the term proportional to $\partial_\mu \epsilon^\beta$ in the transformation of the first term, leaving us with

$$\delta(D_\mu \psi)_\ell = i\epsilon^\alpha (t_\alpha)_\ell{}^m \partial_\mu \psi_m - iC^\beta{}_{\gamma\alpha}\epsilon^\alpha A^\gamma{}_\mu (t_\beta)_\ell{}^m \psi_m \\ + A^\gamma{}_\mu (t_\gamma)_\ell{}^m i\epsilon^\alpha (t_\alpha)_m{}^n \psi_n$$

or, using Eq. (15.1.2),

$$\delta(D_\mu \psi)_\ell = i\epsilon^\alpha (t_\alpha)_\ell{}^m (D_\mu \psi)_m, \tag{15.1.11}$$

so that $D_\mu \psi$ transforms just like ψ itself.

We also need to worry about derivatives of the gauge field. In order to eliminate the term $\partial_\nu \partial_\mu \epsilon^\beta$ in the transformation of $\partial_\nu A^\beta{}_\mu$, we antisymmetrize with respect to μ and ν, just as in electrodynamics. However, we still have terms in the transformation of $\partial_\nu A^\beta{}_\mu - \partial_\mu A^\beta{}_\nu$ proportional to first derivatives of $\epsilon(x)$, arising from the second term in Eq. (15.1.9). The easiest way to construct a 'covariant curl', $F^\gamma{}_{\nu\mu}$, in whose transformation rule all such derivatives of $\epsilon(x)$ cancel, is to consider the commutator of two covariant derivatives acting on a matter field ψ:

$$\big([D_\nu, D_\mu]\psi\big)_\ell = -i(t_\gamma)_\ell{}^m F^\gamma{}_{\nu\mu}\psi_m, \tag{15.1.12}$$

where

$$F^\gamma{}_{\nu\mu} \equiv \partial_\nu A^\gamma{}_\mu - \partial_\mu A^\gamma{}_\nu + C^\gamma{}_{\alpha\beta} A^\alpha{}_\nu A^\beta{}_\mu. \tag{15.1.13}$$

Eq. (15.1.12) makes it obvious that $F^\gamma{}_{\nu\mu}$ must transform just like a matter field that happens to belong to the adjoint representation:

$$\delta F^\beta{}_{\nu\mu} = i\epsilon^\alpha (t^A_\alpha)^\beta{}_\gamma F^\gamma{}_{\nu\mu} = \epsilon^\alpha C^\beta{}_{\gamma\alpha} F^\gamma{}_{\nu\mu}. \tag{15.1.14}$$

The reader may check by direct calculation (using the relation (15.1.5)) that the quantity $F^\alpha{}_{\nu\mu}$ defined in (15.1.13) actually has the simple transformation rule (15.1.14).

For some purposes, it is useful to know that these infinitesimal gauge transformations can be upgraded to finite transformations. A group element can be parameterized by a set of real functions $\Lambda^\alpha(x)$ so that it acts on a general matter field $\psi_\ell(x)$ through the matrix transformation

$$\psi_\ell(x) \longrightarrow \psi_{\ell\Lambda}(x) = \big[\exp(it_\alpha \Lambda^\alpha(x))\big]_\ell{}^m \psi_m(x). \tag{15.1.15}$$

We want the covariant derivative to transform in the same way:

$$(\partial_\mu - it_\alpha A^\alpha{}_{\mu\Lambda})\psi_\Lambda = \exp(it_\alpha \Lambda^\alpha)(\partial_\mu - it_\alpha A^\alpha{}_\mu)\psi, \tag{15.1.16}$$

so we must impose on $A^\alpha{}_\mu$ the transformation rule $A^\alpha{}_\mu \longrightarrow A^\alpha{}_{\mu\Lambda}$, with

$$\partial_\mu \exp(it_\beta \Lambda^\beta) - it_\beta \exp(it_\alpha \Lambda^\alpha) A^\beta{}_{\mu\Lambda} = -i \exp(it_\alpha \Lambda^\alpha) t_\beta A^\beta{}_\mu$$

or in other words

$$t_\alpha A^\alpha{}_{\mu\Lambda} = \exp(it_\beta \Lambda^\beta) t_\alpha A^\alpha{}_\mu \exp(-it_\beta \Lambda^\beta) \\ - i\big[\partial_\mu \exp(it_\beta \Lambda^\beta)\big] \exp(-it_\beta \Lambda^\beta). \tag{15.1.17}$$

Eqs. (15.1.15) and (15.1.17) reduce to the previous transformation rules (15.1.1) and (15.1.9) in the limit where $\Lambda^\alpha(x)$ is an infinitesimal $\epsilon^\alpha(x)$.

From Eq. (15.1.17), we can see that by a suitable choice of $\Lambda^\beta(x)$, it is always possible to make $A^\alpha{}_{\mu\Lambda}(x)$ vanish at any one point, say $x = z$. (Simply take $\Lambda^\alpha(z)$ to vanish, and $\partial\Lambda^\alpha(x)/\partial x^\mu = -A^\alpha{}_\mu(x)$ at $x = z$.) Also, it is always possible to choose $\Lambda^\beta(x)$ so that any one spacetime component of $A^\alpha{}_{\mu\Lambda}(x)$ vanishes for all α everywhere in at least a finite domain around any given point. For instance, to make $A^\alpha{}_{3\Lambda}(x)$ vanish, we must solve the set of ordinary first-order differential equations for the parameters $\Lambda^\beta(x)$:

$$\partial_3 \exp(it_\beta\Lambda^\beta) = -i\exp(it_\beta\Lambda^\beta)t_\alpha A^\alpha{}_3, \tag{15.1.18}$$

which always have a solution in at least a finite domain around any ordinary point.

However, in general it is not possible to choose $\Lambda^\alpha(x)$ to make all four components $A^\alpha{}_{\mu\Lambda}(x)$ vanish in a finite region. For this purpose, we would have to be able to satisfy the partial differential equations

$$\partial_\mu \exp(it_\beta\Lambda^\beta) = -i\exp(it_\beta\Lambda^\beta)t_\alpha A^\alpha{}_\mu, \tag{15.1.19}$$

which cannot be solved unless certain integrability conditions are satisfied. In particular, if $A^\alpha{}_{\mu\Lambda}(x)$ vanishes everywhere then so does $F^\alpha{}_{\mu\nu\Lambda}(x)$, but since the field strength transforms homogeneously, $F^\alpha{}_{\mu\nu\Lambda}(x)$ can vanish only if $F^\alpha{}_{\mu\nu}(x)$ does. A gauge field $A^\alpha{}_\mu(x)$ is called a 'pure gauge' field if there exists a gauge transformation which makes it vanish everywhere. It is not difficult to show that the condition that $F^\alpha{}_{\mu\nu}$ should vanish everywhere is not only necessary but also sufficient for $A^\alpha{}_\mu(x)$ to be expressible in any simply connected region as a pure gauge field [6].

* * *

There is a deep analogy between the construction here of objects that transform simply under gauge transformations and the construction in general relativity of objects that transform covariantly under general coordinate transformations. Just as we use the gauge field to construct covariant derivatives $D_\mu\psi_\ell$ of matter fields with the same gauge transformation properties as the matter fields themselves, so we use the affine connection $\Gamma^\mu{}_{\nu\lambda}(x)$ to construct covariant derivatives of tensors $T^{\rho\cdots}{}_{\kappa\lambda\cdots}$:

$$T^{\rho\cdots}{}_{\kappa\cdots;\nu} \equiv \partial_\nu T^{\rho\cdots}{}_{\kappa\cdots} + \Gamma^\rho{}_{\nu\lambda}T^{\lambda\cdots}{}_{\kappa\cdots} + \cdots - \Gamma^\mu{}_{\nu\kappa}T^{\rho\cdots}{}_{\mu\cdots} - \cdots\ ,$$

which are themselves tensors. Also, from the derivatives of the gauge field we constructed a field strength $F^\alpha{}_{\mu\nu}$ with the gauge transformation property of a matter field belonging to the adjoint representation of the gauge group; correspondingly, from the derivatives of the affine connection we may construct a quantity:

$$R^\lambda{}_{\mu\nu\kappa} = \frac{\partial\Gamma^\lambda{}_{\mu\nu}}{\partial x^\kappa} - \frac{\partial\Gamma^\lambda{}_{\mu\kappa}}{\partial x^\nu} + \Gamma^\eta{}_{\mu\nu}\Gamma^\lambda{}_{\kappa\eta} - \Gamma^\eta{}_{\mu\kappa}\Gamma^\lambda{}_{\nu\eta},$$

which transforms as a tensor, the Riemann–Christoffel curvature tensor. The commutator of two gauge-covariant derivatives D_μ and D_ν may be expressed in terms of the field-strength tensor $F^\alpha{}_{\mu\nu}$; similarly, the commutator of two covariant derivatives with respect to x^ν and x^κ may be expressed in terms of the curvature:

$$T^{\lambda\cdots}{}_{\mu\cdots;\nu;\kappa} - T^{\lambda\cdots}{}_{\mu\cdots;\kappa;\nu} = R^\lambda{}_{\sigma\nu\kappa} T^{\sigma\cdots}{}_{\mu\cdots} + \cdots - R^\sigma{}_{\mu\nu\kappa} T^{\lambda\cdots}{}_{\sigma\cdots} - \cdots .$$

The necessary and sufficient condition for the existence of a gauge in which the gauge field vanishes in a finite simply connected region is the vanishing of the field-strength tensor, and the necessary and sufficient condition for the existence of a coordinate system in which the affine connection vanishes in a finite simply connected region is the vanishing of the Riemann–Christoffel curvature tensor. The analogy breaks down in one important respect: in general relativity the affine connection is itself constructed from first derivatives of the metric tensor, while in gauge theories the gauge fields are not expressed in terms of any more fundamental fields.

15.2 Gauge Theory Lagrangians and Simple Lie Groups

The transformation rules of the gauge-field tensor $F^\alpha{}_{\mu\nu}$ and the matter fields ψ and their gauge-covariant derivatives do not involve the derivatives of the transformation parameters $\epsilon^\alpha(x)$, so if the Lagrangian is constructed solely from these ingredients, and if it is invariant under global transformations with ϵ^α constant, then it is invariant under gauge transformations with general position-dependent $\epsilon^\alpha(x)$. We therefore assume that the Lagrangian satisfies these conditions: that is,

$$\mathcal{L} = \mathcal{L}\big(\psi, D_\mu\psi, D_\nu D_\mu\psi, \ldots, F^\alpha{}_{\mu\nu}, D_\rho F^\alpha{}_{\mu\nu}, \ldots\big) \tag{15.2.1}$$

with the invariance condition:

$$\begin{aligned} &\frac{\partial\mathcal{L}}{\partial\psi_\ell} i(t_\alpha)_\ell{}^m \psi_m + \frac{\partial\mathcal{L}}{\partial(D_\mu\psi_\ell)} i(t_\alpha)_\ell{}^m (D_\mu\psi_m) \\ &\quad + \frac{\partial\mathcal{L}}{\partial(D_\nu D_\mu\psi_\ell)} i(t_\alpha)_\ell{}^m (D_\nu D_\mu\psi_m) + \cdots + \frac{\partial\mathcal{L}}{\partial F^\beta{}_{\mu\nu}} C^\beta{}_{\gamma\alpha} F^\gamma{}_{\nu\mu} \\ &\quad + \frac{\partial\mathcal{L}}{\partial D_\rho F^\beta{}_{\mu\nu}} C^\beta{}_{\gamma\alpha} D_\rho F^\gamma{}_{\nu\mu} + \cdots = 0. \end{aligned} \tag{15.2.2}$$

On the other hand, the Lagrangian may not depend on the gauge field itself, except insofar as it appears in $F^\alpha{}_{\mu\nu}$ and in gauge-covariant derivatives D_μ. In particular, a mass term $-\frac{1}{2}m^2_{\alpha\beta}A^\alpha{}_\mu A^{\beta\mu}$ is ruled out.

We shall concentrate now on the terms in the Lagrangian that depend only on $F^\alpha{}_{\mu\nu}$. Just as in electrodynamics, for any massless particle of unit spin the Lagrangian must contain a free-particle term quadratic in $\partial_\mu A^\alpha{}_\nu - \partial_\nu A^\alpha{}_\mu$, and gauge invariance then dictates that this free-particle term should appear as part of a term quadratic in the field-strength tensor $F^\alpha{}_{\mu\nu}$. Lorentz invariance and parity conservation dictate its form as

$$\mathcal{L}_A = -\frac{1}{2} g_{\alpha\beta} F^\alpha{}_{\mu\nu} F^{\beta\mu\nu} \tag{15.2.3}$$

with a constant matrix $g_{\alpha\beta}$. If we do not assume parity (or CP or T) conservation, then we may also include in the Lagrangian a term

$$\mathcal{L}'_A = -\frac{1}{2}\theta_{\alpha\beta}\epsilon^{\mu\nu\rho\sigma} F^\alpha{}_{\mu\nu} F^\beta{}_{\rho\sigma}$$

with another constant matrix $\theta_{\alpha\beta}$. This term is actually a derivative, and therefore does not affect the field equations or the Feynman rules. Such a term would, however, have non-perturbative quantum mechanical effects, to be discussed in Section 23.6.

Before going on to consider the properties of the matrix $g_{\alpha\beta}$, it is worth drawing attention to the fact that it is not possible to introduce a kinematic term for the gauge field $A^\alpha{}_\mu(x)$ without also including interactions, the terms in Eq. (15.2.3) arising from the quadratic part of the field strength $F^\alpha{}_{\mu\nu}$ defined by Eq. (15.1.13). This is one more respect in which non-Abelian gauge theories resemble general relativity, where the kinematic part of the Lagrangian for the gravitational field is contained in the Einstein–Hilbert Lagrangian density $-\sqrt{g}R/8\pi G$, which also contains self-interactions of the field. The reasons in both cases are similar: the gravitational field interacts with itself because it interacts with anything that carries energy and momentum, and the gauge field interacts with itself because it interacts with anything that transforms according to a non-trivial representation (in this case the adjoint representation) of the gauge group. This is in contrast to the case of electrodynamics, where the photon does not carry electric charge, the quantum number with which it interacts, and it is consequently possible to introduce a kinematic term $-\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$ for the electromagnetic field that does not entail interactions.

The numerical matrix $g_{\alpha\beta}$ may be taken symmetric, and must be taken real to give a real Lagrangian. In order for this term to satisfy the gauge-invariance requirement (15.2.2), we must have for all δ:

$$g_{\alpha\beta}F^\alpha{}_{\mu\nu}C^\beta{}_{\gamma\delta}F^{\gamma\mu\nu} = 0.$$

In order for this to be true without having to impose any functional relations among the Fs, the matrix $g_{\alpha\beta}$ must satisfy the condition:

$$g_{\alpha\beta}C^\beta{}_{\gamma\delta} = -g_{\gamma\beta}C^\beta{}_{\alpha\delta}. \tag{15.2.4}$$

There is one more important condition on the matrix $g_{\alpha\beta}$. Just as in quantum electrodynamics, the rules of canonical quantization and the positivity properties of the quantum mechanical scalar product require that the matrix $g_{\alpha\beta}$ in the Lagrangian (15.2.3) must be positive-definite. (That is, $g_{\alpha\beta}u^\alpha u^\beta$ is positive for all real u, and vanishes for some real u only if $u^\alpha = 0$ for all α.) This is analogous to the requirement that in the kinematic Lagrangian $-\frac{1}{2}Z\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}m^2\phi^2$ for a real scalar field ϕ, the constant Z must be positive-definite.

These requirements on the matrix $g_{\alpha\beta}$ have far reaching implications. They form one of a set of three equivalent conditions:

a: There exists a real symmetric positive-definite matrix $g_{\alpha\beta}$ that satisfies the invariance condition (15.2.4).

b: There is a basis for the Lie algebra (that is, a set of generators $\widetilde{t}_\alpha = \mathcal{S}_{\alpha\beta}t_\beta$, with $\mathcal{S}$ a real non-singular matrix) for which the structure constants $\widetilde{C}^\alpha{}_{\beta\gamma}$ are antisymmetric not only in the lower indices β and γ but in all three indices α, β and γ. (In this basis it is convenient to drop the distinction between upper and lower indices α, β, etc., and write $\widetilde{C}_{\alpha\beta\gamma}$ in place of $\widetilde{C}^\alpha{}_{\beta\gamma}$.)

c: The Lie algebra is the direct sum of commuting compact simple and $U(1)$ subalgebras.[4]

Appendix A of this chapter presents a proof of the equivalence of the conditions **a**, **b**, and **c** [7].

Before going on to discuss the physical implications of this result, it will be useful to say a bit more about the condition of compactness. We will not use this here, but a compact Lie algebra consists of the generators of a compact Lie group: one for which the invariant volume of the group is finite. For instance, the rotation group is compact; the Lorentz group is not. As a simple example of a simple Lie algebra that is not compact, consider the commutation relations

$$[t_1, t_2] = -it_3, \qquad [t_2, t_3] = it_1, \qquad [t_3, t_1] = it_2.$$

The structure constant here is real, but not completely antisymmetric; its non-vanishing components are

$$C^3{}_{12} = -C^3{}_{21} = -1, \qquad C^1{}_{23} = -C^1{}_{32} = 1, \qquad C^2{}_{31} = -C^2{}_{13} = 1.$$

The metric given by Eq. (15.A.10) is here diagonal, with elements:

$$g_{11} = g_{22} = -g_{33} = -2.$$

This is not a positive matrix, so the Lie algebra is not compact. It is in fact the Lie algebra of the non-compact group $O(2,1)$, the Lorentz group in two space and one time dimensions.

Two sets of generators that differ by a real non-singular linear transformation are considered to span the same Lie algebra, and generate the same group. This is not true for complex linear transformations of generators. In particular, any simple Lie algebra can be put into a compact form by a change of phase of the generators in a suitable basis. For instance, for the Lie algebra of the above example, it is only necessary to define new generators $t'_1 = it_1$, $t'_2 = it_2$, $t'_3 = t_3$, for which the commutation relations are

$$[t'_1, t'_2] = it'_3, \qquad [t'_2, t'_3] = it'_1, \qquad [t'_3, t'_1] = it'_2.$$

[4]Some definitions: A subalgebra $\mathcal{H}$ of a Lie algebra $\mathcal{G}$ is a linear space, spanned by certain real linear combinations $t_i = \mathcal{S}_{i\alpha} t_\alpha$ of the generators t_α of $\mathcal{G}$, such that $\mathcal{H}$ is itself a Lie algebra, in the sense that the commutators of the t_i with each other are of the form $[t_i, t_j] = ic^k{}_{ij} t_k$. A subalgebra $\mathcal{H}$ is called invariant if the commutator of any element of the whole algebra $\mathcal{G}$ with any element of the subalgebra $\mathcal{H}$ is in the subalgebra $\mathcal{H}$. A simple Lie algebra is one without invariant subalgebras. A $U(1)$ subalgebra of $\mathcal{G}$ is one with just a single generator that commutes with all generators of the whole algebra $\mathcal{G}$. A semi-simple Lie algebra is one that has no invariant Abelian subalgebras, i.e., invariant subalgebras whose generators all commute with each other. Semi-simple Lie algebras are direct sums of simple (but not $U(1)$) Lie algebras. A simple or semi-simple Lie algebra is said to be compact if the matrix $\mathrm{Tr}\{t^A_\alpha t^A_\beta\} = -C^\gamma{}_{\alpha\delta} C^\delta{}_{\beta\gamma}$ is positive-definite. The meaning and importance of the properties of simplicity and compactness will be discussed further below. In saying that a Lie algebra $\mathcal{G}$ is a direct sum of subgroups $\mathcal{H}_n$, it is meant that it is possible to find a basis for $\mathcal{G}$ with generators t_{na} for which the structure constants take the form

$$C^{\ell c}{}_{na\, mb} = \delta_{\ell m} \delta_{mn} C^{(n)c}{}_{ab},$$

where $C^{(n)c}{}_{ab}$ is the structure constant of the subalgebra $\mathcal{H}_n$.

The structure constant is now real and totally antisymmetric: $C^a{}_{bc} = \epsilon_{abc}$. Here $g_{ab} = 2\delta_{ab}$, and the algebra is compact. We recognize this, of course, as the familiar algebra of the compact group $O(3)$ of rotations in three dimensions. To see that this is always possible for any simple Lie algebra, note that the matrix g_{ab} defined by Eq. (15.A.10) is real, symmetric, and non-singular, so that by a real orthogonal transformation it may be put in a diagonal form with non-zero elements along the main diagonal. It is then only necessary to multiply all the generators that correspond in this basis to the negative diagonal elements of g_{ab} by factors i.

We note without proof that the finite-dimensional representations of compact Lie groups are all unitary, and the finite-dimensional representations of compact Lie algebras are correspondingly all Hermitian. Furthermore, it is easy to see that the only Lie algebras that can have any non-trivial representation by independent finite-dimensional Hermitian matrices t_α are direct sums of $U(1)$ and compact simple Lie algebras. To show this, we may simply define

$$g_{\alpha\beta} = \mathrm{Tr}\{t_\alpha t_\beta\}.$$

This matrix is obviously positive-definite, because $g_{\alpha\beta}u^\alpha u^\beta = \mathrm{Tr}\{(u^\alpha t_\alpha)^2\}$ is positive for any real u^α and vanishes only if $u^\alpha t_\alpha = 0$, which is not possible unless all u^α vanish because the t^α are assumed independent. Furthermore this $g_{\alpha\beta}$ satisfies the invariance condition (15.2.4), as can be seen by multiplying the commutation relation (15.1.2) with t_δ and taking the trace; this gives

$$iC^\gamma{}_{\alpha\beta}\,\mathrm{Tr}\{t_\gamma t_\delta\} = \mathrm{Tr}\{[t_\alpha, t_\beta]t_\delta\} = \mathrm{Tr}\{t_\delta t_\alpha t_\beta - t_\beta t_\alpha t_\delta\},$$

which is obviously antisymmetric in β and δ. Having verified **a**, we can rely on the above theorem to infer condition **c**, so that the Lie algebra must be a direct sum of compact simple and $U(1)$ subalgebras.

Let's now return to the physics of gauge theories. In this section we have inferred the existence of a positive symmetric real matrix $g_{\alpha\beta}$, that satisfies the invariance condition (15.2.4), from the necessity of constructing a suitable kinematic term in the Lagrangian for the gauge field, and in Appendix A of this chapter we have shown that this result is equivalent to a condition on the Lie algebra, that it is a direct sum of compact simple and $U(1)$ subalgebras. For our purposes, the important thing about this result is that the simple Lie algebras are all of certain limited types and dimensionalities. For instance, it is easy to see that these is no simple Lie algebra with less than three generators, because in one or two dimensions there can be no non-zero totally antisymmetric structure constants with three indices. With three generators, an invariant subalgebra can be avoided by taking $C^3{}_{12}$, $C^2{}_{31}$, and $C^1{}_{23}$ all non-zero. In the basis in which the structure constant is real and totally antisymmetric, there is obviously only one possibility:

$$C_{\alpha\beta\gamma} = c\epsilon_{\alpha\beta\gamma}.$$

Here c is an arbitrary non-zero real constant, which can be eliminated by a change of scale of the generators, $t_\alpha \longrightarrow t_\alpha/c$, so the Lie algebra is

$$[t_\alpha, t_\beta] = i\epsilon_{\alpha\beta\gamma}t_\gamma.$$

This may be recognized as the Lie algebra of the three-dimensional rotation group $O(3)$, and also of the group $SU(2)$ of unitary unimodular matrices in two dimensions, and was used as a basis for the original non-Abelian gauge theory of Yang and Mills. Continuing in the same way, it can be shown that there are no simple Lie algebras with 4, 5, 6, or 7 generators, one with 8 generators, and so on. Mathematicians (notably Killing and E. Cartan) have been able to catalog all simple Lie algebras. The compact forms of the simple Lie algebras form several infinite classes of algebras of the 'classical' Lie groups—the unitary unimodular, unitary orthogonal, and unitary symplectic groups—plus just five exceptional Lie algebras. This catalog is presented in Appendix B of this chapter.

It is also shown in Appendix A that under the equivalent conditions **a**, **b**, or **c**, the metric takes the form

$$g_{ma,nb} = g_m^{-2}\delta_{mn}\delta_{ab} \tag{15.2.5}$$

with real g_m, where m and n label the simple or $U(1)$ subalgebras, and a and b label the individual generators of these subalgebras. We can eliminate the constants g_m^{-2} by a rescaling of the gauge fields

$$A^\mu{}_{ma} \longrightarrow \widetilde{A}^\mu{}_{ma} \equiv g_m^{-1} A^\mu{}_{ma}, \tag{15.2.6}$$

but then in order to keep the same formulas (15.1.10) and (15.1.13) for $D_\mu\psi$ and $F_\alpha{}^{\mu\nu}$, we must also redefine the matrices t_α and the structure constants

$$t_{ma} \longrightarrow \widetilde{t}_{ma} \equiv g_m t_{ma}, \tag{15.2.7}$$

$$C^{(m)c}{}_{ab} \longrightarrow \widetilde{C}^{(m)c}{}_{ab} \equiv g_m C^{(m)c}{}_{ab}. \tag{15.2.8}$$

That is, we can always define the scale of the gauge fields (now dropping the tildes) so that g_m in Eq. (15.2.5) is unity:

$$g_{\alpha\beta} = \delta_{\alpha\beta}, \tag{15.2.9}$$

but then the transformation matrices t_α and the structure constants $C_{\alpha\beta\gamma}$ contain an unknown multiplicative factor g_m for each simple or $U(1)$ subalgebra. These factors are the coupling constants of the gauge theory. Alternatively, it is sometimes more convenient to adopt some fixed though arbitrary normalization for the t_α and structure constants within each simple or $U(1)$ subalgebra, in which case the coupling constants appear in the gauge-field Lagrangian (15.2.3) as the factors g_m^{-2} in Eq. (15.2.5).

15.3 Field Equations and Conservation Laws

Using Eq. (15.2.9) for the matrix $g_{\alpha\beta}$ in Eq. (15.2.3), the full Lagrangian density is

$$\mathcal{L} = -\frac{1}{4}F_{\alpha\mu\nu}F_\alpha{}^{\mu\nu} + \mathcal{L}_M(\psi, D_\mu\psi), \tag{15.3.1}$$

where in the absence of gauge fields $\mathcal{L}_M(\psi, \partial_\mu\psi)$ would be the 'matter' Lagrangian density. We could, in principle, include a dependence of $\mathcal{L}_M$ on $F_{\alpha\mu\nu}$ as well as higher covariant derivatives $D_\nu D_\mu\psi$, $D_\lambda F_{\alpha\mu\nu}$, etc., but we exclude these non-renormalizable terms here for the same reason as in electrodynamics: as discussed in Section 12.3, such terms would be highly suppressed at ordinary energies by negative powers of some very large mass. For this reason the standard model of the weak, electromagnetic and strong interactions has a Lagrangian of the general form (15.3.1).

The equations of motion of the gauge field are here

$$\partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu A_{\alpha\nu})} = -\partial_\mu F_\alpha{}^{\mu\nu} = \frac{\partial \mathcal{L}}{\partial A_{\alpha\nu}}$$
$$= -F_\gamma{}^{\nu\mu} C_{\gamma\alpha\beta} A_{\beta\mu} - i \frac{\partial \mathcal{L}_M}{\partial D_\nu \psi} t_\alpha \psi$$

and so

$$\partial_\mu F_\alpha{}^{\mu\nu} = -\mathcal{J}_\alpha{}^\nu, \tag{15.3.2}$$

where $\mathcal{J}_\alpha{}^\nu$ is the current:

$$\mathcal{J}_\alpha{}^\nu \equiv -F_\gamma{}^{\nu\mu} C_{\gamma\alpha\beta} A_{\beta\mu} - i \frac{\partial \mathcal{L}_M}{\partial D_\nu \psi} t_\alpha \psi. \tag{15.3.3}$$

The current $\mathcal{J}_\alpha{}^\nu$ is conserved in the ordinary sense

$$\partial_\nu \mathcal{J}_\alpha{}^\nu = 0, \tag{15.3.4}$$

as can be seen either from the Euler–Lagrange equations for ψ and the invariance equivalent (15.2.2) or, more easily, directly from the field equations (15.3.2).

The derivatives in Eqs. (15.3.2) and (15.3.4) are ordinary derivatives, not the gauge-covariant derivatives D_ν, so the gauge invariance of these equations is somewhat obscure. It can be made manifest by rewriting Eq. (15.3.2) in terms of the gauge-covariant derivative of the field strength

$$D_\lambda F_\alpha{}^{\mu\nu} \equiv \partial_\lambda F_\alpha{}^{\mu\nu} - i(t_\beta^A)_{\alpha\gamma} A_{\beta\lambda} F_\gamma{}^{\mu\nu}$$
$$= \partial_\lambda F_\alpha{}^{\mu\nu} - C_{\alpha\gamma\beta} A_{\beta\lambda} F_\gamma{}^{\mu\nu}. \tag{15.3.5}$$

Then Eq. (15.3.2) reads

$$D_\mu F_\alpha{}^{\mu\nu} = -J_\alpha{}^\nu, \tag{15.3.6}$$

where $J_\alpha{}^\nu$ is the current of the matter fields alone

$$J_\alpha{}^\nu \equiv -i \frac{\partial \mathcal{L}_M}{\partial D_\nu \psi} t_\alpha \psi. \tag{15.3.7}$$

This is gauge-covariant, if $\mathcal{L}_M$ is gauge-invariant. Also, by operating on Eq. (15.3.6) with D_ν, using the commutation relation

$$[D_\nu, D_\mu] F_\alpha{}^{\mu\nu} = -i(t_\gamma^A)_{\alpha\beta} F_{\gamma\nu\mu} F_\beta{}^{\mu\nu} = -C_{\gamma\alpha\beta} F_{\gamma\nu\mu} F_\beta{}^{\mu\nu},$$

we see that $J_\alpha{}^\nu$ satisfies a gauge-covariant conservation law

$$D_\nu J_\alpha{}^\nu = 0, \tag{15.3.8}$$

rather than the ordinary conservation law (15.3.4) obeyed by the full current $\mathcal{J}_\alpha{}^\nu$. Also, it is straightforward (using Eq. (15.1.5)) to derive the identities:

$$D_\mu F_{\alpha\nu\lambda} + D_\nu F_{\alpha\lambda\mu} + D_\lambda F_{\alpha\mu\nu} = 0, \tag{15.3.9}$$

which hold whether or not the gauge fields satisfy the field equations.

These results serve to underscore the profound analogy mentioned in Section 15.1 between non-Abelian gauge theories and general relativity. In general relativity there is a matter energy-momentum tensor $T^\nu{}_\mu$, analogous to $J_\alpha{}^\mu$, which satisfies a generally covariant conservation law $T^\nu{}_{\mu;\nu} = 0$, and stands on the right-hand side of the Einstein field equations in their generally covariant form, $R^\nu{}_\mu - \frac{1}{2}\delta^\nu{}_\mu R = -8\pi G T^\nu{}_\mu$. However, $T^\nu{}_\mu$ is not conserved in the ordinary sense: $\partial_\nu T^\nu{}_\mu$ does not vanish. On the other hand, moving the non-linear terms on the left-hand side of the Einstein equation to the right-hand side gives a field equation [8]

$$\left(R^\nu{}_\mu - \frac{1}{2}\delta^\nu{}_\mu R\right)_{\text{LINEAR}} = -8\pi G \tau^\nu{}_\mu,$$

where $\tau^\nu{}_\mu$ is the non-tensor

$$\tau^\nu{}_\mu \equiv T^\nu{}_\mu + \frac{1}{8\pi G}\left(R^\nu{}_\mu - \frac{1}{2}\delta^\nu{}_\mu R\right)_{\text{NONLINEAR}},$$

analogous to $\mathcal{J}_\alpha{}^\nu$. Like $\mathcal{J}_\alpha{}^\nu$, $\tau^\nu{}_\mu$ is conserved in the ordinary sense

$$\partial_\nu \tau^\nu{}_\mu = 0$$

and may be regarded as the current of energy and momentum:

$$P_\mu = \int \tau^0{}_\mu \, d^3x.$$

It contains a purely gravitational term, because gravitational fields carry energy and momentum; without this term, $\tau^\nu{}_\mu$ could not be conserved. Similarly, $\mathcal{J}_\alpha{}^\nu$ contains a gauge-field term (the first term on the right in Eq. (15.3.3)) because for non-Abelian groups (those with $C^\gamma{}_{\alpha\beta} \neq 0$) the gauge fields carry the quantum numbers with which they interact. Because $\mathcal{J}_\alpha{}^\nu$ is conserved in the ordinary sense, it can be regarded as the current of these quantum numbers, with the symmetry generators given by the time-independent quantities

$$T_\alpha = \int \mathcal{J}_\alpha{}^0 \, d^3x. \tag{15.3.10}$$

(Also, the homogeneous equations (15.3.9) involve covariant derivatives, just as do the Bianchi identities of general relativity.) In contrast, none of these complications arises in quantum electrodynamics, because photons do not carry the quantum number, electric charge, with which they interact.

15.4 Quantization

We now proceed to quantize the gauge theories described in the previous two sections. The Lagrangian density is taken in the form (15.3.1):

$$\mathcal{L} = -\frac{1}{4}F_{\alpha\mu\nu}F_\alpha{}^{\mu\nu} + \mathcal{L}_M(\psi, D_\mu\psi), \tag{15.4.1}$$

with

$$\begin{aligned} F_{\alpha\mu\nu} &\equiv \partial_\mu A_{\alpha\nu} - \partial_\nu A_{\alpha\mu} + C_{\alpha\beta\gamma}A_{\beta\mu}A_{\gamma\nu}, \\ D_\mu\psi &\equiv \partial_\mu\psi - it_\alpha A_{\alpha\mu}\psi. \end{aligned}$$

We cannot immediately quantize this theory by setting commutators equal to i times the corresponding Poisson brackets. The problem is one of constraints. In the terminology of Dirac, described in Section 7.6, there is a primary constraint that

$$\Pi_{\alpha 0} \equiv \frac{\partial\mathcal{L}}{\partial(\partial_0 A_\alpha{}^0)} = 0 \tag{15.4.2}$$

and a secondary constraint provided by the field equation for $A_\alpha{}^0$:

$$\begin{aligned} -\partial_\mu\frac{\partial\mathcal{L}}{\partial(\partial_\mu A_{\alpha 0})} + \frac{\partial\mathcal{L}}{\partial A_{\alpha 0}} &= \partial_\mu F_\alpha{}^{\mu 0} + F_\gamma{}^{\mu 0}C_{\gamma\alpha\beta}A_{\beta\mu} + J_\alpha{}^0 \\ &= \partial_k\Pi_\alpha{}^k + \Pi_\gamma{}^k C_{\gamma\alpha\beta}A_{\beta k} + J_\alpha{}^0 = 0, \end{aligned} \tag{15.4.3}$$

where $\Pi_\alpha{}^k \equiv \partial\mathcal{L}/\partial(\partial_0 A_{\alpha k}) = F_\alpha{}^{k0}$ is the 'momentum' conjugate to $A_{\alpha k}$, with k running over the values 1, 2, 3. The Poisson brackets of $\Pi_{\alpha 0}$ and $\partial_k\Pi_\alpha{}^k + \Pi_\gamma{}^k C_{\gamma\alpha\beta}A_{\beta k} + J_\alpha{}^0$ vanish (because the latter quantity is independent of $A_\alpha{}^0$), so these are first class constraints, which cannot be dealt with by replacing Poisson brackets with Dirac brackets.

As in the case of electrodynamics, we deal with these constraints by choosing a gauge. The Coulomb gauge adopted for electrodynamics would lead to painful complications here,[5] so instead we will work in what is known as *axial gauge*, based on the condition

$$A_{\alpha 3} = 0. \tag{15.4.4}$$

The canonical variables of the gauge field are then $A_{\alpha i}$, with i now running over the values 1 and 2, together with their canonical conjugates

$$\Pi_{\alpha i} \equiv \frac{\partial\mathcal{L}}{\partial(\partial_0 A_{\alpha i})} = -F_\alpha{}^{0i} = \partial_0 A_{\alpha i} - \partial_i A_{\alpha 0} + C_{\alpha\beta\gamma}A_\beta{}^0 A_{\gamma i}. \tag{15.4.5}$$

The field $A_{\alpha 0}$ is not an independent canonical variable, but rather is defined in terms of the other variables by the constraint (15.4.3). To see this, note that the 'electric' field strengths $F_\alpha{}^{\mu 0}$ are

$$F_\alpha{}^{i0} = \Pi_{\alpha i}, \qquad F_\alpha{}^{30} = \partial_3 A_\alpha{}^0, \tag{15.4.6}$$

[5] In addition to purely algebraic complications, Coulomb gauge (like many other gauges) has a problem known as the Gribov ambiguity [9]: even with the condition that A_α vanishes at spatial infinity, for each solution of the Coulomb gauge condition $\boldsymbol{\nabla}\cdot\mathbf{A}_\alpha = 0$ there are other solutions that differ by finite gauge transformations. The Gribov ambiguity will not bother us here, because we quantize in axial gauge where it is absent, and we shall use other gauges like Lorentz gauge only to generate a perturbation series.

so the constraint (15.4.3) reads

$$-(\partial_3)^2 A_\alpha{}^0 = \partial_i \Pi_{\alpha i} + \Pi_{\gamma i} C_{\gamma\alpha\beta} A_{\beta i} + J_\alpha{}^0, \tag{15.4.7}$$

which can easily be solved (with reasonable boundary conditions) to give $A_\alpha{}^0$ as a functional of $\Pi_{\gamma i}$, $A_{\beta i}$, and $J_\alpha{}^0$. (We are using a summation convention, with indices i, j, etc. summed over the values 1 and 2.) It should be noted that the canonical conjugate to the matter field ψ_ℓ is

$$\pi_\ell = \frac{\partial \mathcal{L}}{\partial(\partial_0 \psi_\ell)} = \frac{\partial \mathcal{L}_M}{\partial(D_0 \psi_\ell)}, \tag{15.4.8}$$

so the time component of the matter current can be expressed in terms of the canonical variables of the matter fields *alone*

$$J_\alpha{}^0 = -i \frac{\partial \mathcal{L}_M}{\partial D_0 \psi_\ell} (t_\alpha)_{\ell m} \psi_m = -i \pi_\ell (t_\alpha)_{\ell m} \psi_m. \tag{15.4.9}$$

Hence Eq. (15.4.7) defines $A_\alpha{}^0$ at a given time as a functional of the canonical variables $\Pi_{\gamma i}$, $A_{\beta i}$, π_ℓ, and ψ_m at the same time.

Now that we have identified the canonical variables in this gauge, we can proceed to the construction of a Hamiltonian. The Hamiltonian density is

$$\begin{aligned}\mathcal{H} &= \Pi_{\alpha i} \partial_0 A_{\alpha i} + \pi_\ell \partial_0 \psi_\ell - \mathcal{L} \\ &= \Pi_{\alpha i} \big(F_{\alpha 0 i} + \partial_i A_{\alpha 0} - C_{\alpha\beta\gamma} A_{\beta 0} A_{\gamma i} \big) + \pi_\ell \partial_0 \psi_\ell \\ &\quad - \frac{1}{2} F_{\alpha 0 i} F_{\alpha 0 i} + \frac{1}{2} F_{\alpha ij} F_{\alpha ij} + \frac{1}{2} F_{\alpha i 3} F_{\alpha i 3} - \frac{1}{2} F_{\alpha 03} F_{\alpha 03} - \mathcal{L}_M. \end{aligned} \tag{15.4.10}$$

Using Eqs. (15.4.4) and (15.4.6), this is

$$\begin{aligned}\mathcal{H} = \mathcal{H}_M &+ \Pi_{\alpha i} \big(\partial_i A_{\alpha 0} - C_{\alpha\beta\gamma} A_{\beta 0} A_{\gamma i} \big) + \frac{1}{2} \Pi_{\alpha i} \Pi_{\alpha i} \\ &+ \frac{1}{2} F_{\alpha ij} F_{\alpha ij} + \frac{1}{2} \partial_3 A_{\alpha i} \partial_3 A_{\alpha i} - \frac{1}{2} \partial_3 A_{\alpha 0} \partial_3 A_{\alpha 0}, \end{aligned} \tag{15.4.11}$$

where $\mathcal{H}_M$ is the matter Hamiltonian density:

$$\mathcal{H}_M \equiv \pi_\ell \partial_0 \psi_\ell - \mathcal{L}_M. \tag{15.4.12}$$

Following the general rules derived in Section 9.2, we can now use this Hamiltonian density to calculate matrix elements as path integrals over $A_{\alpha i}$, $\Pi_{\alpha i}$, ψ_ℓ, and π_ℓ, with weighting factor $\exp(iI)$, where

$$I = \int d^4x \left[\Pi_{\alpha i} \partial_0 A_{\alpha i} + \pi_\ell \partial_0 \psi_\ell - \mathcal{H} + \epsilon \text{ terms} \right], \tag{15.4.13}$$

in which the 'ϵ terms' serve only to supply the correct imaginary infinitesimal terms in propagator denominators. (See Section 9.2.) We note that Eqs. (15.4.7) and (15.4.9) give $A_\alpha{}^0$ as a functional of the canonical variables, linear in $\Pi_{\alpha i}$ and π_ℓ. Inspection of Eq. (15.4.11) shows then (assuming $\mathcal{L}_M$ to be no more than quadratic in $D_\mu \psi$) that the integrand of the complete action (15.4.13) is no more than quadratic in $\Pi_{\alpha i}$ and π_ℓ. We could therefore carry out the path integral over these canonical 'momenta' by the usual rules of Gaussian integration. The trouble with this procedure is that the coefficients of the terms in Eq. (15.4.13) of second order in $\Pi_{\alpha i}$ are functions of the $A_{\alpha i}$, so the Gaussian integral would yield an awkward field-dependent determinant factor. Also, the whole formalism at this point looks hopelessly non-Lorentz-invariant.

Instead of proceeding in this way, we will apply a trick like that used in the path integral formulation of electrodynamics in Section 9.6. Note that if for a moment we think of $A_{\alpha 0}$ as an independent variable, then the action (15.4.13) is evidently quadratic in $A_{\alpha 0}$, with the coefficient of the second-order term $A_{\alpha 0}(x)A_{\beta 0}(y)$ equal to the field-independent kernel $(\partial_3)^2\delta^4(x-y)$. As we saw in the appendix to Chapter 9, the integral of such a Gaussian over $A_{\alpha 0}(x)$ is, up to a constant factor, equal to the value of the integrand at the stationary 'point' of the argument of the exponential. But the variational derivative of the action here is

$$\frac{\delta I}{\delta A_{\alpha 0}} = -\frac{\partial \mathcal{H}}{\partial A_{\alpha 0}} = J_\alpha{}^0 + \partial_i \Pi_{\alpha i} + C_{\beta\alpha\gamma}\Pi_{\beta i}A_{\gamma i} - \partial_3^2 A_\alpha{}^0,$$

so the stationary 'point' of the action is the solution of the constraint equation (15.4.7). Hence, instead of using for $A_{\alpha 0}$ the solution of Eq. (15.4.7), we can just as well treat it as an independent variable of integration.

With $A_{\alpha 0}$ now regarded as an independent variable, the Hamiltonian $\int d^3x\,\mathcal{H}$ is evidently quadratic in $\Pi_{\alpha i}$, with the coefficient of the second-order term $\Pi_{\alpha i}(x)\Pi_{\beta j}(y)$ given by the field-independent kernel $\frac{1}{2}\delta^4(x-y)\delta_{ij}$. Assuming that the same is true for the matter variable π_ℓ, we can evaluate path integrals over π_ℓ and $\Pi_{\alpha i}$ up to a constant factor by simply setting π_ℓ and $\Pi_{\alpha i}$ at the stationary 'points' of the action corresponding to Eq. (15.4.1):

$$\begin{aligned} 0 &= \frac{\delta I}{\delta \pi_\ell} = \partial_0\psi_\ell - \frac{\partial \mathcal{H}_M}{\partial \pi_\ell}, \\ 0 &= \frac{\delta I}{\delta \Pi_{\alpha i}} = \partial_0 A_{\alpha i} - \Pi_{\alpha i} - \partial_i A_{\alpha 0} + C_{\alpha\beta\gamma}A_{\beta 0}A_{\gamma i} = F_{\alpha 0 i} - \Pi_{\alpha i}. \end{aligned}$$

Inserting these back into Eq. (15.4.13) gives

$$\begin{aligned} I &= \int d^4x \left[\mathcal{L}_M + \frac{1}{2}F_{\alpha 0 i}F_{\alpha 0 i} - \frac{1}{2}F_{\alpha ij}F_{\alpha ij} - \frac{1}{2}\partial_3 A_{\alpha i}\partial_3 A_{\alpha i} + \frac{1}{2}(\partial_3 A_{\alpha 0})^2\right] \\ &= \int d^4x\,\mathcal{L}, \end{aligned} \tag{15.4.14}$$

where $\mathcal{L}$ is the Lagrangian (15.3.1) with which we started! In other words, *we are to do path integrals over $\psi_\ell(x)$ and all four components of $A_{\alpha\mu}(x)$, with a manifestly covariant weighting factor* $\exp(iI)$ *given by Eqs.* (15.4.14) *and* (15.3.1), *but with the axial-gauge condition* enforced by inserting a factor

$$\prod_{x,\alpha}\delta\big(A_{\alpha 3}(x)\big). \tag{15.4.15}$$

As long as $\mathcal{O}_A$, $\mathcal{O}_B$, ... are gauge-invariant, we have

$$\Big\langle T\{\mathcal{O}_A\mathcal{O}_B\cdots\}\Big\rangle_{\rm VACUUM} \propto \int \left[\prod_{\ell,x} d\psi_\ell(x)\right]\left[\prod_{\alpha,\mu,x} dA_{\alpha\mu}(x)\right] \cdot \mathcal{O}_A\mathcal{O}_B\cdots\exp\{iI+\epsilon \text{ terms}\}\prod_{x,\alpha}\delta\big(A_{\alpha 3}(x)\big), \tag{15.4.16}$$

with Lorentz- and gauge-invariant action I given by Eq. (15.4.14).

* * *

For future reference, we note that the volume element $\prod_{\alpha,\mu,x} dA_{\alpha\mu}(x)$ for the integration over gauge fields in (15.4.16) is gauge-invariant, in the sense that

$$\prod_{\alpha,\mu,x} dA_{\Lambda\,\alpha\mu}(x) = \prod_{\alpha,\mu,x} dA_{\alpha\mu}(x), \tag{15.4.17}$$

where $A_{\Lambda\,\alpha\mu}(x)$ is the result of acting on $A_{\alpha\mu}(x)$ with a gauge transformation having transformation parameters $\Lambda_\alpha(x)$. It will be enough to show that this is true for transformations near the identity, say with infinitesimal transformation parameters $\lambda_\alpha(x)$. In this case,

$$A_{\lambda\,\alpha}{}^\mu = A_\alpha{}^\mu + \partial^\mu\lambda_\alpha + C_{\alpha\beta\gamma}A_\beta{}^\mu\lambda_\gamma,$$

so the volume elements are related by

$$\prod_{\alpha,\mu,x} dA_{\lambda\,\alpha\mu}(x) = \text{Det}(\mathcal{N})\prod_{\alpha,\mu,x} dA_{\alpha\mu}(x),$$

where $\mathcal{N}$ is the 'matrix'

$$\mathcal{N}_{\alpha\mu x,\beta\nu y} \equiv \frac{\delta A_{\lambda\,\alpha\mu}(x)}{\delta A_{\beta\nu}(y)} = \delta^4(x-y)\delta_\mu{}^\nu\big[\delta_{\alpha\beta}+C_{\alpha\beta\gamma}\lambda_\gamma(x)\big].$$

The determinant of $\mathcal{N}$ is unity to first order in λ_γ, because the trace $C_{\alpha\alpha\gamma}$ vanishes.

In this chapter we shall assume that the volume element $\prod_{n,x} d\psi_n(x)$ for the integration over matter fields is also gauge-invariant. There are important subtleties here, to which we shall return in Chapter 22, but as shown there this assumption turns out to be valid in our present non-Abelian gauge theories of strong and electroweak interactions.

15.5 The De Witt–Faddeev–Popov Method

Our formula (15.4.16) for the path integral was derived in a gauge that is convenient for canonical quantization, but the Feynman rules that would be derived from this formula would hide the underlying rotational and Lorentz invariance of the theory. In order to derive manifestly Lorentz-invariant Feynman rules, we need to change the gauge.

We first note that Eq. (15.4.16) is (up to an unimportant constant factor) a special case of a general class of functional integrals, of the form:

$$\mathcal{I} = \int\left[\prod_{n,x} d\phi_n(x)\right]\mathcal{G}[\phi]B\big[f[\phi]\big]\,\text{Det}\,\mathcal{F}[\phi], \tag{15.5.1}$$

where $\phi_n(x)$ are a set of gauge and matter fields; $\prod_{n,x} d\phi_n(x)$ is a volume element; and $\mathcal{G}[\phi]$ is a functional of the $\phi_n(x)$, satisfying the gauge-invariance condition:

$$\mathcal{G}[\phi_\lambda] \prod_{n,x} d\phi_{\lambda n}(x) = \mathcal{G}[\phi] \prod_{n,x} d\phi_n(x), \tag{15.5.2}$$

where $\phi_{\lambda n}(x)$ is the result of operating on ϕ with a gauge transformation having parameters $\lambda_\alpha(x)$. (Usually when this is satisfied both the functional $\mathcal{G}$ and the volume element are separately invariant, but Eq. (15.5.2) is all we need here.) Also, $f_\alpha[\phi; x]$ is a non-gauge-invariant 'gauge-fixing functional' of these fields that also depends on x and α; $B[f]$ is some numerical functional defined for general functions $f_\alpha(x)$ of x and α; and $\mathcal{F}$ is the 'matrix':

$$\mathcal{F}_{\alpha x, \beta y}[\phi] \equiv \left. \frac{\delta f_\alpha[\phi_\lambda; x]}{\delta \lambda^\beta(y)} \right|_{\lambda=0}. \tag{15.5.3}$$

(In accordance with our usual notation for functionals of functions or of functionals, $B\big[f[\phi]\big]$ is understood to depend on the values taken by $f_\alpha[\phi; x]$ for all values of the undisplayed variables α and x, with the displayed variable, the function $\phi_n(x)$, held fixed.) Eq. (15.5.1) does not represent the widest possible generalization of Eq. (15.4.16); we will see in Section 15.7 that there is a further generalization that is needed for some purposes. We start here with Eq. (15.5.1) because it will help to motivate the formalism of Section 15.7, and it is adequate for dealing with non-Abelian gauge theories in the most convenient gauges.

We now must check that the path integral (15.4.16) is in fact a special case of Eq. (15.5.1). In Eq. (15.4.16) the fields $\phi_n(x)$ consist of both $A_{\alpha\mu}(x)$ and matter fields $\psi_\ell(x)$, and

$$f_\alpha[A, \psi; x] = A_{\alpha 3}(x), \tag{15.5.4}$$

$$B[f] = \prod_{x,\alpha} \delta\big(f_\alpha(x)\big), \tag{15.5.5}$$

$$\mathcal{G}[A, \psi] = \exp\{iI + \epsilon \text{ terms}\} \mathcal{O}_A \mathcal{O}_B \cdots, \tag{15.5.6}$$

$$\prod_{n,x} d\phi_n(x) = \left[\prod_{\ell,x} d\psi_\ell(x)\right] \left[\prod_{\alpha,\mu,x} dA_\alpha{}^\mu(x)\right]. \tag{15.5.7}$$

(We are now dropping the distinction between upper and lower indices α, β, ...) Comparison of Eq. (15.4.16) with Eqs. (15.5.1)–(15.5.3) shows that these path integrals are indeed the same, aside from the factor $\operatorname{Det} \mathcal{F}[\phi]$. For the particular gauge-fixing functional (15.5.4), this factor is field-independent: if $A_\alpha{}^3(x) = 0$, then the change in $A_\alpha{}^3(x)$ under a gauge transformation with parameters $\lambda_\alpha(x)$ is

$$A_{\lambda\alpha}{}^3(x) = \partial_3 \lambda_\alpha(x) = \int d^4y\, \lambda_\alpha(y) \partial_3 \delta^4(x-y),$$

so that here Eq. (15.5.3) is the field-independent 'matrix'

$$\mathcal{F}_{\alpha x, \beta y}[\phi] = \delta_{\alpha\beta} \partial_3 \delta^4(x-y).$$

The determinant in Eq. (15.5.1) is therefore also field-independent in this gauge. As discussed in Chapter 9, field-independent factors in the functional integral affect only the vacuum-fluctuation part of expectation values and S-matrix elements, and so are irrelevant to the calculation of the connected parts of the S-matrix.

The point of recognizing the functional integral (15.4.16) for non-Abelian gauge theories as a special case of the general path integral (15.5.1) is that in this form we may freely change the gauge. Specifically, we have a theorem, that *the integral* (15.5.1) *is actually independent (within broad limits) of the gauge-fixing functional* $f_\alpha[\phi; x]$*, and depends on the choice of the functional* $B[f]$ *only through an irrelevant constant factor.*

Proof: Replace the integration variable ϕ everywhere in Eq. (15.5.1) with a new integration variable ϕ_Λ, with $\Lambda^\alpha(x)$ any arbitrary (but fixed) set of gauge transformation parameters:

$$\mathcal{I} = \int \left[\prod_{n,x} d\phi_{\Lambda n}(x)\right] \mathcal{G}[\phi_\Lambda] B\big[f[\phi_\Lambda]\big] \,\mathrm{Det}\, \mathcal{F}[\phi_\Lambda]. \tag{15.5.8}$$

(This step is a mathematical triviality, like changing an integral $\int_{-\infty}^{\infty} f(x)dx$ to read $\int_{-\infty}^{\infty} f(y)dy$, and does not yet make use of our assumptions regarding gauge invariance.) Now use the assumed gauge invariance (15.5.2) of the measure $\prod d\phi$ times the functional $\mathcal{G}[\phi]$ to rewrite this as

$$\mathcal{I} = \int \left[\prod_{n,x} d\phi_n(x)\right] \mathcal{G}[\phi] B\big[f[\phi_\Lambda]\big] \,\mathrm{Det}\, \mathcal{F}[\phi_\Lambda]. \tag{15.5.9}$$

Since $\Lambda^\alpha(x)$ was arbitrary, the left-hand side here cannot depend on it. Integrating over $\Lambda^\alpha(x)$ with some suitable weight-functional $\rho[\Lambda]$ (to be chosen below) thus gives

$$\mathcal{I} \int \left[\prod_{\alpha,x} d\Lambda^\alpha(x)\right] \rho[\Lambda] = \int \left[\prod_{n,x} d\phi_n(x)\right] \mathcal{G}[\phi] C[\phi], \tag{15.5.10}$$

where

$$C[\phi] \equiv \int \left[\prod_{\alpha,x} d\Lambda^\alpha(x)\right] \rho[\Lambda] B\big[f[\phi_\Lambda]\big] \,\mathrm{Det}\, \mathcal{F}[\phi_\Lambda]. \tag{15.5.11}$$

Now, Eq. (15.5.3) gives

$$\mathcal{F}_{\alpha x, \beta y}[\phi_\Lambda] \equiv \left.\frac{\delta f_\alpha[(\phi_\Lambda)_\lambda; x]}{\delta \lambda^\beta(y)}\right|_{\lambda=0}. \tag{15.5.12}$$

We are assuming that these transformations form a group; that is, we may write the result of performing the gauge transformation with parameters $\Lambda^\alpha(x)$ followed by the gauge transformation with parameters $\lambda^\alpha(x)$ as the action of a single 'product' gauge transformation with parameters $\widetilde{\Lambda}^\alpha(x; \Lambda, \lambda)$,

$$(\phi_\Lambda)_\lambda = \phi_{\widetilde{\Lambda}(\Lambda,\lambda)}. \tag{15.5.13}$$

Using the chain rule of partial (functional) differentiation, we have then

$$\mathcal{F}_{\alpha x, \beta y}[\phi_\Lambda] = \int \mathcal{J}_{\alpha x, \gamma z}[\phi, \Lambda] \mathcal{R}^{\gamma z}{}_{\beta y}[\Lambda] d^4 z, \tag{15.5.14}$$

where

$$\mathcal{J}_{\alpha x,\gamma z}[\phi,\Lambda] \equiv \left.\frac{\delta f_\alpha[\phi_{\widetilde{\Lambda}};x]}{\delta\widetilde{\Lambda}^\gamma(z)}\right|_{\widetilde{\Lambda}=\Lambda} = \frac{\delta f_\alpha[\phi_\Lambda;x]}{\delta\Lambda^\gamma(z)} \tag{15.5.15}$$

and

$$\mathcal{R}^{\gamma z}{}_{\beta y}[\Lambda] \equiv \left.\frac{\delta\widetilde{\Lambda}^\gamma(z;\Lambda,\lambda)}{\delta\lambda^\beta(y)}\right|_{\lambda=0}. \tag{15.5.16}$$

It follows that

$$\operatorname{Det}\mathcal{F}[\phi_\Lambda] = \operatorname{Det}\mathcal{J}[\phi,\Lambda]\operatorname{Det}\mathcal{R}[\Lambda]. \tag{15.5.17}$$

We note that $\operatorname{Det}\mathcal{J}[\phi,\Lambda]$ is nothing but the Jacobian of the transformation of integration variables from the $\Lambda^\alpha(x)$ to (for a fixed ϕ) the $f_\alpha[\phi_\Lambda;x]$. Hence, if we choose the weight-function $\rho(\Lambda)$ as

$$\rho(\Lambda) = \frac{1}{\operatorname{Det}\mathcal{R}[\Lambda]} \tag{15.5.18}$$

then

$$\begin{aligned} C[\phi] &= \int \left[\prod_{\alpha,x} d\Lambda^\alpha(x)\right] \operatorname{Det}\mathcal{J}[\phi,\Lambda] B\big[f[\phi_\Lambda]\big] \\ &= \int \left[\prod_{\alpha,x} df_\alpha(x)\right] B[f] = C, \end{aligned} \tag{15.5.19}$$

which is clearly independent of ϕ. (Eq. (15.5.18) may be recognized by the reader as giving the invariant (Haar) measure on the space of group parameters.) We have then at last

$$\mathcal{I} = \frac{C\displaystyle\int \left[\prod_{n,x} d\phi_n(x)\right] \mathcal{G}[\phi]}{\displaystyle\int \left[\prod_{\alpha,x} d\Lambda^\alpha(x)\right] \rho[\Lambda]}. \tag{15.5.20}$$

This is clearly independent of our choice of $f_\alpha[\phi;x]$, which has been reduced to a mere variable of integration, and it depends on $B[f]$ only through the constant C, as was to be proved.

Before proceeding with the applications of this theorem, we should pause to note a tricky point in the derivation. The integrals in the numerator and denominator of Eq. (15.5.20) are both ill-defined for the same reason. Since $\mathcal{G}[\phi]$ is assumed to be gauge-invariant, its integral over ϕ cannot possibly converge; the integrand is constant along all 'orbits,' obtained by sending ϕ into ϕ_λ with all possible $\lambda^\alpha(x)$. Likewise, the integrand in the denominator is divergent, because $\rho(\Lambda)\prod d\Lambda$ is nothing but the usual invariant volume element for integrating over the group, and this is also constant along 'orbits' $\Lambda \longrightarrow \widetilde{\Lambda}(\Lambda,\lambda)$. This divergence can be eliminated in both the numerator and denominator of Eq. (15.5.20) by formulating the theory on a finite spacetime lattice, in which case the volume of the gauge group is just the volume of the global Lie group itself times the number of lattice sites. Because the gauge-fixing factor $B[f]$ eliminates this divergence in the original definition (15.5.1) of the left-hand side of Eq. (15.5.20), we may presume that, as the number of lattice sites goes to infinity, it cancels between the numerator and denominator of the right-hand side of Eq. (15.5.20).

Now to the point. We have seen that the vacuum expectation value (15.4.16) in axial gauge is given by a functional integral of the general form (15.5.1). Armed with the above theorem, we conclude then that

$$\begin{aligned}\left\langle T\{\mathcal{O}_A\mathcal{O}_B\cdots\}\right\rangle_V \propto \int\left[\prod_{\ell,x} d\psi_\ell(x)\right]\left[\prod_{\alpha,\mu,x} dA_\alpha{}^\mu(x)\right]\\ \cdot\,\mathcal{O}_A\mathcal{O}_B\cdots\exp\{iI+\epsilon\text{ terms}\}B\big[f[A,\psi]\big]\,\mathrm{Det}\,\mathcal{F}[A,\psi]\end{aligned} \tag{15.5.21}$$

for (almost) any choice of $f_\alpha[A,\psi;x]$ and $B[f]$. We are now therefore free to use Eq. (15.5.21) to derive the Feynman rules in a more convenient gauge.

The path integrals that we understand how to calculate are of Gaussians times polynomials, so we will generally take

$$B[f]=\exp\left(-\frac{i}{2\xi}\int d^4x\, f_\alpha(x)f_\alpha(x)\right) \tag{15.5.22}$$

with arbitrary real parameter ξ. With this choice, the effect of the factor B in Eq. (15.5.21) is just to add a term to the effective Lagrangian

$$\mathcal{L}_{\rm EFF}=\mathcal{L}-\frac{1}{2\xi}f_\alpha f_\alpha. \tag{15.5.23}$$

The simplest Lorentz-invariant choice of the gauge-fixing function f_α is the same as in electrodynamics:

$$f_\alpha=\partial_\mu A_\alpha{}^\mu. \tag{15.5.24}$$

The bare gauge-field propagator can then be calculated just as in quantum electrodynamics. The free-vector-boson part of the effective action can be written

$$\begin{aligned}I_{0A}&=-\int d^4x\left[\frac{1}{4}(\partial_\mu A_{\alpha\nu}-\partial_\nu A_{\alpha\mu})(\partial^\mu A_\alpha{}^\nu-\partial^\nu A_\alpha{}^\mu)+\frac{1}{2\xi}(\partial_\mu A_\alpha{}^\mu)(\partial_\nu A_\alpha{}^\nu)+\epsilon\text{ terms}\right]\\ &=-\frac{1}{2}\int d^4x\,\mathcal{D}_{\alpha\mu x,\beta\nu y}A_\alpha{}^\mu(x)A_\beta{}^\nu(y),\end{aligned}$$

where

$$\begin{aligned}\mathcal{D}_{\alpha\mu x,\beta\nu y}&=\eta_{\mu\nu}\frac{\partial^2}{\partial x^\lambda\partial y_\lambda}\delta^4(x-y)-\left(1-\frac{1}{\xi}\right)\frac{\partial^2}{\partial x^\mu\partial y^\nu}\delta^4(x-y)+\epsilon\text{ terms}\\ &=(2\pi)^{-4}\int d^4p\left[\eta_{\mu\nu}(p^2-i\epsilon)-\left(1-\frac{1}{\xi}\right)p_\mu p_\nu\right]e^{ip\cdot(x-y)}.\end{aligned}$$

Taking the reciprocal of the matrix in square brackets, we find the propagator:

$$\begin{aligned}\Delta_{\alpha\mu,\beta\nu}(x,y)&=(\mathcal{D}^{-1})_{\alpha\mu x,\beta\nu y}\\ &=(2\pi)^{-4}\int d^4p\left[\eta_{\mu\nu}+(\xi-1)\frac{p_\mu p_\nu}{p^2}\right]\frac{e^{ip\cdot(x-y)}}{p^2-i\epsilon}.\end{aligned} \tag{15.5.25}$$

This is a generalization of both Landau and Feynman gauges, which are recovered by taking $\xi = 0$ and $\xi = 1$, respectively. For $\xi \to 0$, the functional (15.5.22) oscillates very rapidly except near $f_\alpha = 0$, so this functional acts like a delta-function imposing the Landau gauge condition $\partial_\mu A^\mu = 0$, leading naturally to a propagator satisfying the corresponding condition $\partial^\mu \Delta_{\alpha\mu,\beta\nu} = 0$. For non-zero values of ξ the functional $B[f]$ does not pick out gauge fields satisfying any specific gauge condition on the field $A_{\alpha\mu}$, but it is common to refer to the propagator (15.5.25) as being in a 'generalized Feynman gauge' or 'generalized ξ-gauge.' It is often a good strategy to calculate physical amplitudes with ξ left arbitrary, and then at the end of the calculation check that the results are ξ-independent.

With one qualification, the Feynman rules are now obvious: the contributions of vertices are to be read off from the interaction terms in the original Lagrangian $\mathcal{L}$, with gauge-field propagators given by Eq. (15.5.25), and matter-field propagators calculated as before. To be specific, the trilinear interaction term in $\mathcal{L}$

$$-\frac{1}{2} C_{\alpha\beta\gamma}(\partial_\mu A_{\alpha\nu} - \partial_\nu A_{\alpha\mu}) A_\beta{}^\mu A_\gamma{}^\nu$$

corresponds to a vertex to which are attached three vector boson lines. If these lines carry (incoming) momenta p, q, k and Lorentz and gauge-field indices $\mu\alpha$, $\nu\beta$, $\lambda\gamma$, then according to the momentum-space Feynman rules, the contribution of such a vertex to the integrand is

$$i(2\pi)^4 \delta^4(p+q+k)[-iC_{\alpha\beta\gamma}]\big[p_\nu \eta_{\mu\lambda} - p_\lambda \eta_{\mu\nu} + q_\lambda \eta_{\nu\mu} - q_\mu \eta_{\nu\lambda} + k_\mu \eta_{\lambda\nu} - k_\nu \eta_{\lambda\mu}\big]. \quad (15.5.26)$$

Also, the A^4 interaction term in $\mathcal{L}$,

$$-\frac{1}{4} C_{\epsilon\alpha\beta} C_{\epsilon\gamma\delta} A_{\alpha\mu} A_{\beta\nu} A_\gamma{}^\mu A_\delta{}^\nu,$$

corresponds to a vertex to which are attached four vector boson lines. If these lines carry (incoming) momenta p, q, k, ℓ, and Lorentz and gauge indices $\mu\alpha$, $\nu\beta$, $\rho\gamma$, and $\sigma\delta$, then the contribution of such a vertex to the integrand is

$$\begin{aligned} i(2\pi)^4 \delta^4(p+q+k+\ell) \cdot \big[& - C_{\epsilon\alpha\beta} C_{\epsilon\gamma\delta}(\eta_{\mu\rho}\eta_{\nu\sigma} - \eta_{\mu\sigma}\eta_{\nu\rho}) \\ & - C_{\epsilon\alpha\gamma} C_{\epsilon\delta\beta}(\eta_{\mu\sigma}\eta_{\rho\nu} - \eta_{\mu\nu}\eta_{\sigma\rho}) \\ & - C_{\epsilon\alpha\delta} C_{\epsilon\beta\gamma}(\eta_{\mu\nu}\eta_{\rho\sigma} - \eta_{\mu\rho}\eta_{\sigma\nu})\big]. \end{aligned} \quad (15.5.27)$$

(Recall that the structure constants $C_{\alpha\beta\gamma}$ contain coupling constant factors, so the factors (15.5.26) and (15.5.27) are respectively of first and second order in coupling constants.)

The one complication in the Feynman rules with which we have not yet dealt is the presence in Eq. (15.5.21) of the factor $\mathrm{Det}\,\mathcal{F}$, which for general gauges is *not* a constant. We now turn to a consideration of this factor.

15.6 Ghosts

We now consider the effect of the factor $\mathrm{Det}\,\mathcal{F}$ in Eq. (15.5.22) on the Feynman rules for a non-Abelian gauge theory. In order to be able to treat this effect as a modification of the

Feynman rules, recall that as shown in Section 9.5, the determinant of any matrix $\mathcal{F}_{\alpha x,\beta y}$ may be expressed as a path integral

$$\text{Det}\,\mathcal{F} \propto \int \left[\prod_{\alpha,x} d\omega^*_\alpha(x)\right] \left[\prod_{\alpha,x} d\omega_\alpha(x)\right] \exp(iI_{GH}), \tag{15.6.1}$$

where

$$I_{GH} \equiv \int d^4x\, d^4y\, \omega^*_\alpha(x)\omega_\beta(y)\mathcal{F}_{\alpha x,\beta y}. \tag{15.6.2}$$

Here ω^*_α and ω_α are a set of independent anticommuting classical variables, and the constant of proportionality is field-independent. (We have to choose the ω_α and ω^*_α field variables to be fermionic in order to reproduce the factor $\text{Det}\,\mathcal{F}$; had we chosen these field variables to be bosonic, the path integral (15.6.1) would have been proportional to $(\text{Det}\,\mathcal{F})^{-1}$.) The fields ω^*_α and ω_α are not necessarily related by complex conjugation; indeed, in Section 15.7 we shall see that for some purposes we need to assume that ω^*_α and ω_α are independent real variables. The whole effect of the factor $\text{Det}\,\mathcal{F}$ is the same as that of including $I_{GH}(\omega, \omega^*)$ in the full effective action, and integrating over 'fields' ω and ω^*. That is, for arbitrary gauge-fixing functionals $f_\alpha(x)$,

$$\begin{aligned}\langle T\{\mathcal{O}_A \cdots\}\rangle_V \propto \int & \left[\prod_{n,x} d\psi_n(x)\right] \left[\prod_{\alpha,\mu,x} dA_{\alpha\mu}(x)\right] \\ & \cdot \left[\prod_{\alpha,x} d\omega_\alpha(x) d\omega^*_\alpha(x)\right] \exp\big(iI_{\text{MOD}}[\psi, A, \omega, \omega^*]\big)\mathcal{O}_A \cdots ,\end{aligned} \tag{15.6.3}$$

where I_{MOD} is a modified action

$$I_{\text{MOD}} = \int d^4x \left[\mathcal{L} - \frac{1}{2\xi} f_\alpha f_\alpha\right] + I_{GH}. \tag{15.6.4}$$

The fields ω_α and ω^*_α are Lorentz scalars (at least in covariant gauges) but satisfy Fermi statistics. The connection between spin and statistics is not really violated here, because there are no particles described by these fields that can appear in initial or final states. For that reason, ω_α and ω^*_α are called the fields of 'ghost' and 'antighost' particles. Inspection of Eq. (15.6.2) shows that the action respects the conservation of a quantity known as 'ghost number,' equal to $+1$ for ω_α, -1 for ω^*_α, and zero for all other fields.

The Feynman rules for the ghosts are simplest in the case in which the 'matrix' $\mathcal{F}$ may be expressed as

$$\mathcal{F} = \mathcal{F}_0 + \mathcal{F}_1, \tag{15.6.5}$$

where $\mathcal{F}_0$ is field-independent and of zeroth order in coupling constants, while $\mathcal{F}_1$ is field-dependent and proportional to one or more coupling constant factors. In this case, the ghost propagator is just

$$\Delta_{\alpha\beta}(x, y) = (\mathcal{F}_0^{-1})_{\alpha x,\beta y} \tag{15.6.6}$$

and the ghost vertices are to be read off from the interaction term

$$I'_{GH} = \int d^4x\, d^4y\, \omega^*_\alpha(x)\omega_\beta(y)(\mathcal{F}_1)_{\alpha x,\beta y}. \tag{15.6.7}$$

For instance, in the generalized ξ-gauge discussed in the previous section, we have

$$f_\alpha = \partial_\mu A_\alpha{}^\mu \tag{15.6.8}$$

and for infinitesimal gauge parameters λ_α, Eq. (15.1.9) gives:

$$A_{\lambda\alpha}{}^\mu = A_\alpha{}^\mu + \partial^\mu \lambda_\alpha + C_{\alpha\gamma\beta}\lambda_\beta A_\gamma{}^\mu,$$

so that

$$\begin{aligned}\mathcal{F}_{\alpha x,\beta y} &\equiv \left.\frac{\delta\, \partial_\mu A_{\lambda\alpha}{}^\mu(x)}{\delta\lambda_\beta(y)}\right|_{\lambda=0} \\ &= \Box\delta^4(x-y)\delta_{\alpha\beta} + C_{\alpha\gamma\beta}\frac{\partial}{\partial x^\mu}\left[A_\gamma{}^\mu(x)\delta^4(x-y)\right].\end{aligned} \tag{15.6.9}$$

This is of the form (15.6.5), with

$$(\mathcal{F}_0)_{\alpha x,\beta y} = \Box\delta^4(x-y)\delta_{\alpha\beta}, \tag{15.6.10}$$

$$(\mathcal{F}_1)_{\alpha x,\beta y} = -C_{\alpha\beta\gamma}\frac{\partial}{\partial x^\mu}\left[A_\gamma{}^\mu(x)\delta^4(x-y)\right]. \tag{15.6.11}$$

From Eqs. (15.6.6) and (15.6.10), we see that the ghost propagator is

$$\Delta_{\alpha\beta}(x,y) = \delta_{\alpha\beta}(2\pi)^{-4}\int d^4p\,(p^2 - i\epsilon)^{-1}e^{ip\cdot(x-y)}, \tag{15.6.12}$$

so in this gauge the ghosts behave like spinless fermions of zero mass, transforming according to the adjoint representation of the gauge group. Using Eqs. (15.6.7) and (15.6.11) and integrating by parts, we find that the ghost interaction term in the action is now

$$I'_{GH} = \int d^4x\, C_{\alpha\beta\gamma}\frac{\partial\omega^*_\alpha}{\partial x^\mu}A_\gamma{}^\mu\omega_\beta. \tag{15.6.13}$$

This interaction corresponds to vertices to which are attached one outgoing ghost line, one incoming ghost line, and one vector boson line. If these lines carry (incoming) momenta p, q, k respectively and gauge group indices α, β, γ respectively, and the gauge field carries a vector index μ, then the contribution of such a vertex to the integrand is given by the momentum-space Feynman rules as

$$i(2\pi)^4\delta^4(p+q+k)\cdot ip_\mu C_{\alpha\beta\gamma}. \tag{15.6.14}$$

The ghosts propagate around loops, with single vector boson lines attached at each vertex along the loops, and with an extra minus sign supplied for each loop as is usual for fermionic field variables.

The extra minus sign for ghost loops suggests that each ghost field ω_α, together with the associated antighost field ω^*_α, represents something like a negative degree of freedom. These negative degrees of freedom are necessary because in using covariant gauge field propagators we are really over-counting; the physical degrees of freedom are the components of $A_\alpha{}^\mu(x)$, less the parameters $\Lambda_\alpha(x)$ needed to describe a gauge transformation.

In summary, the modified action (15.6.4) may be written in generalized ξ-gauge as

$$I_{\text{MOD}} = \int d^4x\, \mathcal{L}_{\text{MOD}} \tag{15.6.15}$$

with a modified Lagrangian density:

$$\begin{aligned}\mathcal{L}_{\text{MOD}} = \mathcal{L}_M - \frac{1}{4} F_\alpha{}^{\mu\nu} F_{\alpha\mu\nu} - \frac{1}{2\xi}(\partial_\mu A_\alpha{}^\mu)(\partial_\nu A_\alpha{}^\nu) \\ - \partial_\mu \omega^*_\alpha \partial^\mu \omega_\alpha + C_{\alpha\beta\gamma}(\partial_\mu \omega^*_\alpha) A_\gamma{}^\mu \omega_\beta.\end{aligned} \tag{15.6.16}$$

It is important that this Lagrangian is *renormalizable* (if the matter Lagrangian $\mathcal{L}_M$ is), in the elementary sense that its terms involve products of fields and their derivatives of total dimensionality (in powers of mass) four or less. (The kinematic term $-\partial_\mu \omega^*_\alpha \partial^\mu \omega_\alpha$ in Eq. (15.6.16) fixes the dimensionality of the fields ω and ω^* to be mass to the power unity, just like ordinary scalar and gauge fields.) However, there is more to renormalizability than power counting; it is necessary also that there be a counterterm to absorb every divergence. In the next section we shall consider a remarkable symmetry that will be used in Section 17.2 to show that non-Abelian gauge theories are indeed renormalizable in this sense, and that can even take the place of the Faddeev–Popov–De Witt approach that we have been following.

15.7 BRST Symmetry

Although the Faddeev–Popov–De Witt method described in the previous two sections makes the Lorentz invariance of the theory manifest, it still rests on a choice of gauge, and hence naturally it hides the underlying gauge invariance of the theory. This is a serious problem in trying to prove the renormalizability of the theory—gauge invariance restricts the form of the terms in the Lagrangian that are available as counterterms to absorb ultraviolet divergences, but once we choose a gauge, how do we know that gauge invariance still restricts the ways that the infinities can appear?

Remarkably, however, even after we choose a gauge, the path integral still does have a symmetry related to gauge invariance. This symmetry was discovered by Becchi, Rouet, and Stora [10] (and independently by Tyutin [11]) in 1975, several years after the work of Faddeev and Popov and De Witt, and is known in honor of its discoverers as BRST symmetry. This symmetry will be presented more-or-less as it was originally discovered, as a by-product of the method of Faddeev, Popov, and De Witt, but as we shall see it can also be regarded as a replacement for the Faddeev–Popov–De Witt approach.

We have seen in Eqs. (15.6.3) and (15.6.4) that the Feynman rules for a non-Abelian gauge theory may be obtained from a path integral over matter, gauge, and ghost fields, with a modified action, which we may write

$$I_{\text{MOD}} = I_{\text{EFF}} + I_{GH} = \int d^4x\, \mathcal{L}_{\text{MOD}}, \tag{15.7.1}$$

$$\mathcal{L}_{\text{MOD}} = \mathcal{L} - \frac{1}{2\xi} f_\alpha f_\alpha + \omega^*_\alpha \Delta_\alpha, \tag{15.7.2}$$

where we have now introduced the quantity

$$\Delta_\alpha(x) = \int d^4y\, \mathcal{F}_{\alpha x,\beta y}[A,\psi]\, \omega_\beta(y). \tag{15.7.3}$$

This is for the choice

$$B[f] \propto \exp\left(-\frac{i}{2\xi}\int d^4x\, f_\alpha f_\alpha\right) \tag{15.7.4}$$

of the gauge-fixing functional in Eq. (15.5.21). For our present purposes, it will be helpful to rewrite $B[f]$ as a Fourier integral:

$$B[f] = \int \left[\prod_{\alpha,x} dh_\alpha(x)\right] \exp\left[\frac{i\xi}{2}\int h_\alpha h_\alpha\right] \exp\left[i\int d^4x\, f_\alpha h_\alpha\right]. \tag{15.7.5}$$

We must now do our path integrals over the field h_α (often known as a 'Nakanishi–Lautrup' field [11a]) as well as over matter, gauge, ghost and antighost fields, with a new modified action

$$I_{\text{NEW}} = \int d^4x \left(\mathcal{L} + \omega^*_\alpha \Delta_\alpha + h_\alpha f_\alpha + \frac{1}{2}\xi h_\alpha h_\alpha\right). \tag{15.7.6}$$

This modified action is not gauge-invariant—indeed, it had better not be, if we are to be able to use it in path integrals. However, it is invariant under a 'BRST' symmetry transformation, parameterized by an infinitesimal constant θ that *anticommutes* with ω_α, ω^*_α, and all fermionic matter fields. For a given θ, the BRST transformation is

$$\delta_\theta \psi = i t_\alpha \theta \omega_\alpha \psi, \tag{15.7.7}$$

$$\delta_\theta A_{\alpha\mu} = \theta D_\mu \omega_\alpha = \theta\big[\partial_\mu \omega_\alpha + C_{\alpha\beta\gamma} A_{\beta\mu} \omega_\gamma\big], \tag{15.7.8}$$

$$\delta_\theta \omega^*_\alpha = -\theta h_\alpha, \tag{15.7.9}$$

$$\delta_\theta \omega_\alpha = -\frac{1}{2}\theta C_{\alpha\beta\gamma}\omega_\beta\omega_\gamma, \tag{15.7.10}$$

$$\delta_\theta h_\alpha = 0. \tag{15.7.11}$$

(Recall that in fermionic path integrals, there is no connection between ω_α and ω^*_α, so that Eq. (15.7.9) does not need to be the adjoint of Eq. (15.7.10).) Because h_α is BRST-invariant, we could if we like replace the Gaussian factor $\exp(\frac{1}{2}i\xi\int h_\alpha h_\alpha)$ in Eq. (15.7.5) with an arbitrary smooth functional of h_α, yielding an arbitrary functional $B[f]$, without affecting the BRST invariance of the action. However, for the purposes of diagrammatic calculation and renormalization it will help to leave $B[f]$ as a Gaussian.

In checking the invariance of the action (15.7.1), it will be very useful first to note that the transformation (15.7.7)–(15.7.11) is *nilpotent*; that is, if F is any functional of ψ, A, ω, ω^*, and h, and we define sF by

$$\delta_\theta F \equiv \theta s F, \tag{15.7.12}$$

then[6]

$$\delta_\theta(sF) = 0 \tag{15.7.13}$$

[6] In the original work on BRST symmetry the functional $B[f]$ was left in the form (15.7.4), so that h_α was replaced in Eq. (15.7.9) with $-f_\alpha/\xi$, and the BRST transformation was only nilpotent when acting on functions of ω_α and the gauge and matter fields, but not of ω^*_α.

or equivalently

$$s(sF) = 0. \tag{15.7.14}$$

It is straightforward to verify this nilpotence when δ_θ acts on a single field. First, acting on a matter field,

$$\begin{aligned}\delta_\theta s\psi = it_\alpha \delta_\theta(\omega_\alpha \psi) &= -\frac{1}{2}iC_{\alpha\beta\gamma}t_\alpha\theta\omega_\beta\omega_\gamma\psi - t_\alpha t_\beta\omega_\alpha\theta\omega_\beta\psi \\ &= -\frac{1}{2}iC_{\alpha\beta\gamma}t_\alpha\theta\omega_\beta\omega_\gamma\psi + t_\alpha t_\beta\theta\omega_\alpha\omega_\beta\psi.\end{aligned}$$

The product $\omega_\alpha\omega_\beta$ in the second term on the right is antisymmetric in α and β, so we can replace $t_\alpha t_\beta$ in this term with $\frac{1}{2}[t_\alpha, t_\beta]$, and this term thus cancels the first term:

$$ss\psi = 0. \tag{15.7.15}$$

Next, acting on a gauge field, we have

$$\begin{aligned}\delta_\theta sA_{\alpha\mu} &= \delta_\theta D_\mu\omega_\alpha \\ &= \partial_\mu\delta_\theta\omega_\alpha + C_{\alpha\beta\gamma}\delta_\theta A_{\beta\mu}\omega_\gamma + C_{\alpha\beta\gamma}A_{\beta\mu}\delta_\theta\omega_\gamma \\ &= \theta\Big(-\frac{1}{2}C_{\alpha\beta\gamma}\partial_\mu(\omega_\beta\omega_\gamma) + C_{\alpha\beta\gamma}(\partial_\mu\omega_\beta)\omega_\gamma + C_{\alpha\beta\gamma}C_{\beta\delta\epsilon}A_{\delta\mu}\omega_\epsilon\omega_\gamma - \frac{1}{2}C_{\alpha\beta\gamma}C_{\gamma\delta\epsilon}A_{\beta\mu}\omega_\delta\omega_\epsilon\Big) \\ &= \theta\Big(\frac{1}{2}C_{\alpha\beta\gamma}(\partial_\mu\omega_\beta)\omega_\gamma + \frac{1}{2}C_{\alpha\beta\gamma}(\partial_\mu\omega_\gamma)\omega_\beta - C_{\alpha\beta\gamma}C_{\gamma\delta\epsilon}A_{\delta\mu}\omega_\epsilon\omega_\beta - \frac{1}{2}C_{\alpha\beta\gamma}C_{\gamma\delta\epsilon}A_{\beta\mu}\omega_\delta\omega_\epsilon\Big).\end{aligned}$$

The first two terms of the final expression cancel because $C_{\alpha\beta\gamma}$ is antisymmetric in β and γ, and the third and fourth terms cancel because of the Jacobi identity (15.1.5), so

$$ssA_{\alpha\mu} = 0. \tag{15.7.16}$$

Eqs. (15.7.9) and (15.7.11) show immediately that

$$ss\omega^*_\alpha = 0 \tag{15.7.17}$$

and

$$ssh_\alpha = 0. \tag{15.7.18}$$

Finally,

$$\begin{aligned}\delta_\theta s\omega_\alpha &= -\frac{1}{2}C_{\alpha\beta\gamma}\delta_\theta(\omega_\beta\omega_\gamma) \\ &= \frac{1}{4}\theta\left(C_{\alpha\beta\gamma}C_{\beta\delta\epsilon}\omega_\delta\omega_\epsilon\omega_\gamma + C_{\alpha\beta\gamma}C_{\gamma\delta\epsilon}\omega_\beta\omega_\delta\omega_\epsilon\right) \\ &= \frac{1}{4}\theta C_{\alpha\beta\gamma}C_{\gamma\delta\epsilon}\left[-\omega_\delta\omega_\epsilon\omega_\beta + \omega_\beta\omega_\delta\omega_\epsilon\right].\end{aligned}$$

But ω_β commutes with $\omega_\delta\omega_\epsilon$, so this too vanishes

$$ss\omega_\alpha = 0. \tag{15.7.19}$$

Now consider a product of two fields ϕ_1 and ϕ_2, either or both of which may be ψ, A, ω, ω^*, or h, not necessarily at the same point in spacetime. Then

$$\delta_\theta(\phi_1\phi_2) = \theta(s\phi_1)\phi_2 + \phi_1\theta(s\phi_2) = \theta\big[(s\phi_1)\phi_2 \pm \phi_1 s\phi_2\big],$$

where the sign $\pm$ is plus if ϕ_1 is bosonic, minus if ϕ_1 is fermionic. That is,

$$s(\phi_1\phi_2) = (s\phi_1)\phi_2 \pm \phi_1 s\phi_2.$$

Since as we have seen $\delta_\theta(s\phi_1) = \delta_\theta(s\phi_2) = 0$, the effect of a BRST transformation on $s(\phi_1\phi_2)$ is

$$\delta_\theta s(\phi_1\phi_2) = (s\phi_1)\theta(s\phi_2) \pm \theta(s\phi_1)(s\phi_2).$$

But $s\phi$ always has statistics opposite to ϕ, so moving θ to the left in the first term on the right-hand side introduces a sign factor $\mp$:

$$\delta_\theta s(\phi_1\phi_2) = \theta\big[\mp(s\phi_1)(s\phi_2) \pm (s\phi_1)(s\phi_2)\big] = 0.$$

Continuing in this way, we see that BRST transformations are nilpotent acting on any product of fields at arbitrary spacetime points:

$$\delta_\theta s(\phi_1\phi_2\phi_3\cdots) = 0.$$

Any functional $F[\phi]$ can be written as a sum of multiple integrals of such products with c-number coefficients, so likewise

$$\delta_\theta sF[\phi] = \theta s sF[\phi] = 0. \tag{15.7.20}$$

This completes the proof of the nilpotency of the BRST transformation.

Now let us return to the verification of the BRST invariance of the action (15.7.6). First note that for any functional of matter and gauge fields alone, the BRST transformation is just a gauge transformation with infinitesimal gauge parameter

$$\lambda_\alpha(x) = \theta\omega_\alpha(x). \tag{15.7.21}$$

Therefore the first term in Eq. (15.7.6) is automatically BRST-invariant:

$$\delta_\theta \int d^4x\,\mathcal{L} = 0. \tag{15.7.22}$$

To calculate the effect of a BRST transformation on the rest of the action (15.7.6), note that its effect on the gauge-fixing function is just the gauge transformation (15.7.21), so

$$\begin{aligned}\delta_\theta f_\alpha[x;A,\psi] &= \int \left.\frac{\delta f_\alpha[x;A_\lambda,\psi_\lambda]}{\delta\lambda^\beta(y)}\right|_{\lambda=0} \theta\omega_\beta(y)\,d^4y \\ &= \theta\int \mathcal{F}_{\alpha x,\beta y}[A,\psi]\,\omega_\beta(y)\,d^4y\end{aligned}$$

or in terms of the quantity (15.7.3)

$$\delta_\theta f_\alpha[x;A,\psi] = \theta\Delta_\alpha(x;A,\psi,\omega). \tag{15.7.23}$$

(Note that $\mathcal{F}$ is a bosonic quantity, so there is no sign change in moving θ to the left here.) Also recall that $\delta_\theta \omega^*_\alpha = -\theta h_\alpha$ and $\delta_\theta h_\alpha = 0$. Therefore the terms in the integrand of the 'new' action (15.7.6) other than $\mathcal{L}$ may be written

$$\omega^*_\alpha \Delta_\alpha + h_\alpha f_\alpha + \frac{1}{2}\xi h_\alpha h_\alpha = s\left(\omega^*_\alpha f_\alpha + \frac{1}{2}\xi \omega^*_\alpha h_\alpha\right) \tag{15.7.24}$$

or in other words

$$I_{\text{NEW}} = \int d^4x\, \mathcal{L} + s\Psi, \tag{15.7.25}$$

where

$$\Psi \equiv \int d^4x \left(\omega^*_\alpha f_\alpha + \frac{1}{2}\xi \omega^*_\alpha h_\alpha\right). \tag{15.7.26}$$

The nilpotence of the BRST transformation tells us immediately that the term $s\Psi$ as well as $\int d^4x\, \mathcal{L}$ is BRST-invariant.

In a sense the converse of this result also applies: we shall see in Section 17.2 that a renormalizable Lagrangian that obeys BRST invariance and the other symmetries of the Lagrangian (15.7.25) must take the form of Eq. (15.7.25), aside from changes in the values of various constant coefficients. But this is not enough to establish the renormalizability of these theories. BRST symmetry transformations act non-linearly on the fields, and in this case there is no simple connection between the symmetries of the Lagrangian and the symmetries of matrix elements and Greens functions. Using the external field methods developed in the next chapter, it will be shown in Section 17.2 that the ultraviolet divergent terms in Feynman amplitudes (though not the finite parts) do obey a sort of renormalized BRST invariance, which allows the proof of renormalizability to be completed.

Eq. (15.7.25) shows that the physical content of any gauge theory is contained in the *kernel* of the BRST operator (that is, in a general BRST-invariant term $\int d^4x\, \mathcal{L} + s\Psi$), modulo terms in the *image* of the BRST transformation (that is, terms of the form $s\Psi$). The kernel modulo the image of any nilpotent transformation is said to form the *cohomology* of the transformation. There is another sense in which the physical content of a gauge theory may be identified with the cohomology of the BRST operator [12]. It is a fundamental physical requirement that matrix elements between physical states should be independent of our choice of the gauge-fixing function f_α, or in other words, of the functional Ψ in Eq. (15.7.25). The change in any matrix element $\langle\alpha\|\beta\rangle$ due to a change $\widetilde{\delta}\Psi$ in Ψ is

$$\widetilde{\delta}\langle\alpha\|\beta\rangle = i\langle\alpha\|\widetilde{\delta}I_{\text{NEW}}\|\beta\rangle = i\langle\alpha\|s\widetilde{\delta}\Psi\|\beta\rangle. \tag{15.7.27}$$

(We use a tilde here to distinguish this arbitrary change in the gauge-fixing function from a BRST transformation or a gauge transformation.) We can introduce a fermionic BRST 'charge' Q, defined so that for any field operator Φ,

$$\delta_\theta \Phi = i[\theta Q, \Phi] = i\theta[Q, \Phi]_\mp,$$

or in other words,

$$[Q, \Phi]_\mp = is\Phi, \tag{15.7.28}$$

the sign being $-$ or $+$ according as Φ is bosonic or fermionic. The nilpotence of the BRST transformation then gives

$$0 = -ss\Phi = [Q, [Q, \Phi]_\mp]_+ = [Q^2, \Phi]_- .$$

For this to be satisfied for all operators Φ, it is necessary for Q^2 either to vanish or be proportional to the unit operator. But Q^2 cannot be proportional to the unit operator, since it has a non-vanishing ghost quantum number[7], so it must vanish:

$$Q^2 = 0. \tag{15.7.29}$$

From Eqs. (15.7.27) and (15.7.28), we have

$$\widetilde{\delta}\langle\alpha\|\beta\rangle = \langle\alpha\|[Q, \widetilde{\delta}\Psi]\|\beta\rangle. \tag{15.7.30}$$

In order for this to vanish for all changes $\widetilde{\delta}\Psi$ in Ψ, it is necessary that

$$\langle\alpha\|Q = Q\|\beta\rangle = 0. \tag{15.7.31}$$

Thus physical states are in the kernel of the nilpotent operator Q. Two physical states that differ only by a state vector in the image of Q, that is, of form $Q\|\cdots\rangle$, evidently have the same matrix element with all other physical states, and are therefore physically equivalent. Hence independent physical states correspond to states in the kernel of Q, modulo the image of Q—that is, they correspond to the cohomology of Q.

To see how this works in practice, let us consider the simple example of pure electrodynamics.[8] Taking the gauge-fixing function as $f = \partial_\mu A^\mu$ and integrating over the auxiliary field h, the BRST transformation (15.7.8)–(15.7.10) is here

$$sA_\mu = \partial_\mu\omega, \qquad s\omega^* = \partial_\mu A^\mu/\xi, \qquad s\omega = 0. \tag{15.7.32}$$

We expand the fields in normal modes[9]

$$\begin{aligned} A^\mu(x) &= (2\pi)^{-3/2}\int \frac{d^3p}{\sqrt{2p^0}}\left[a^\mu(\mathbf{p})e^{ip\cdot x} + a^{\mu *}(\mathbf{p})e^{-ip\cdot x}\right], \\ \omega(x) &= (2\pi)^{-3/2}\int \frac{d^3p}{\sqrt{2p^0}}\left[c(\mathbf{p})e^{ip\cdot x} + c^*(\mathbf{p})e^{-ip\cdot x}\right], \\ \omega^*(x) &= (2\pi)^{-3/2}\int \frac{d^3p}{\sqrt{2p^0}}\left[b(\mathbf{p})e^{ip\cdot x} + b^*(\mathbf{p})e^{-ip\cdot x}\right]. \end{aligned} \tag{15.7.33}$$

[7] Recall that the ghost quantum number is defined as $+1$ for ω_α, -1 for ω^*_α, and 0 for all gauge and matter fields.

[8] Eqs. (15.6.11) and (15.6.7) show that because the structure constants vanish in electrodynamics, the ghosts here are not coupled to other fields. Nevertheless, electrodynamics provides a good example of the use of BRST symmetry in identifying physical states. Indeed, in analyzing the physicality conditions on 'in' and 'out' states we ignore interactions, so for this purpose a non-Abelian gauge theory is treated like several copies of quantum electrodynamics.

[9] Just as $\omega^*(x)$ is not to be thought of as the Hermitian adjoint of $\omega(x)$, b^* and c^* are not the adjoints of c and b. But since $A^\mu(x)$ is Hermitian, $\omega(x)$ is Hermitian if Q is.

Matching coefficients of $e^{\pm ip\cdot x}$ on both sides of Eq. (15.7.28) yields

$$\begin{aligned} &[Q, a^\mu(\mathbf{p})]_- = -p^\mu c(\mathbf{p}), && [Q, a^{\mu *}(\mathbf{p})]_- = p^\mu c^*(\mathbf{p}), \\ &[Q, b(\mathbf{p})]_+ = p^\mu a_\mu(\mathbf{p})/\xi, && [Q, b^*(\mathbf{p})]_+ = p^\mu a^*_\mu(\mathbf{p})/\xi, \\ &[Q, c(\mathbf{p})]_+ = [Q, c^*(\mathbf{p})]_+ = 0. && \end{aligned} \tag{15.7.34}$$

Consider any state $\|\psi\rangle$ satisfying the physicality condition (15.7.31):

$$Q\|\psi\rangle = 0. \tag{15.7.35}$$

The states $\|e,\psi\rangle = e_\mu a^{\mu *}(\mathbf{p})\|\psi\rangle$ with one additional photon then satisfy the physicality condition $Q\|e,\psi\rangle = 0$ if $e_\mu p^\mu = 0$. Also, the state $\|\psi\rangle' = b^*(\mathbf{p})\|\psi\rangle$ satisfies

$$Q\|\psi\rangle' = p^\mu a^*_\mu(\mathbf{p})\|\psi\rangle/\xi, \tag{15.7.36}$$

so $\|e+\alpha p,\psi\rangle = \|e,\psi\rangle + \xi\alpha Q\|\psi\rangle'$, and is therefore physically equivalent to $\|e,\psi\rangle$. From this we conclude that e^μ is physically equivalent to $e^\mu + \alpha p^\mu$, which is the usual 'gauge-invariance' condition on photon polarization vectors. On the other hand,

$$Qb^*(\mathbf{p})\|\psi\rangle = p^\mu a^*_\mu(\mathbf{p})\|\psi\rangle \neq 0,$$

so $b^*\|\psi\rangle$ does not satisfy the physicality condition (15.7.31). Also, for any e_μ with $e\cdot p \neq 0$,

$$c^*(\mathbf{p})\|\psi\rangle = Qe_\mu a^{\mu *}(\mathbf{p})\|\psi\rangle/e\cdot p,$$

so $c^*\|\psi\rangle$ is BRST-exact, and hence equivalent to zero. *Thus the physical Hilbert space is free of ghosts and antighosts.*

To maintain Lorentz invariance, we must interpret all four components of $a^\mu(\mathbf{p})$ as annihilation operators, in the sense that

$$0 = a_\mu(\mathbf{p})\|\Omega\rangle, \tag{15.7.37}$$

where $\|\Omega\rangle$ is the BRST-invariant vacuum state. But the canonical commutation relations derived from the BRST-invariant action (say, with $\xi = 1$) give

$$[a_\mu(\mathbf{p}), a^*_\nu(\mathbf{p}')]_- = \eta_{\mu\nu}\delta^3(\mathbf{p}-\mathbf{p}'), \tag{15.7.38}$$

corresponding to the propagator in Feynman gauge. This violates the usual positivity rules of quantum mechanics, because Eqs. (15.7.37) and (15.7.38) yield [13]

$$\langle\Omega\|a_0(\mathbf{p})a^*_0(\mathbf{p}')\|\Omega\rangle = -\langle\Omega\|\Omega\rangle. \tag{15.7.39}$$

Nevertheless we can rest assured that all amplitudes among physical states satisfy the usual positivity conditions, because these states satisfy Eq. (15.7.31), and for such states the transition amplitudes are the same as they would be in a more physical gauge like Coulomb or axial gauge, where there is no problem of positivity or unitarity.

The Faddeev–Popov–De Witt formalism described so far necessarily yields an action that is bilinear in the ghost fields ω^*_α and ω_α. This is adequate for renormalizable Yang–Mills theories with the gauge-fixing function $f_\alpha = \partial_\mu A^\mu_\alpha$, but not in more general cases. For instance, as we shall see in Section 17.2, in other gauges renormalizable Yang–Mills theories need $\omega^*\omega^*\omega\omega$ terms in the action Lagrangian density to serve as counterterms for the ultraviolet divergences in loop graphs with four external ghost lines.

Fortunately the Faddeev–Popov–De Witt formalism represents only one way of generating a class of equivalent Lagrangians that yield the same unitary S-matrix. The BRST formalism provides a more general approach, that dispenses altogether with the Faddeev–Popov–De Witt formalism. In this approach, one takes the action to be the most general local functional of matter, gauge, ω^A, ω^{*A} and h^A fields with ghost number zero that is invariant under the BRST transformation (15.7.7)–(15.7.11) and under any other global symmetries of the theory. (For renormalizable theories one would also limit the Lagrangian density to operators of dimensionality four or less, but this restriction plays no role in the following discussion.) In the next section we shall prove, in a context more general than Yang–Mills theories, that the most general action of this sort is the sum of a functional of the matter and gauge fields (collectively called ϕ) alone, plus a term given by the action of the BRST operator s on an arbitrary functional Ψ of ghost number -1:

$$I_{\rm NEW}[\phi,\omega,\omega^*,h] = I_0[\phi] + s\Psi[\phi,\omega,\omega^*,h], \tag{15.7.40}$$

as for instance in the Faddeev–Popov–De Witt action (15.7.25), but with $s\Psi$ now not necessarily bilinear in ghost and antighost fields.

By the same argument as before, the S-matrix elements for states that are annihilated by the BRST generator Q are independent of the choice of Ψ in Eq. (15.7.40), so if there is any choice of Ψ for which the ghosts decouple, then the ghosts decouple in general. In Yang–Mills theories, such a Ψ is provided by quantization of the theory in axial gauge, so in such theories ghosts decouple for arbitrary choices of the functional $\Psi[\phi,\omega,\omega^*,h]$, not just those choices like (15.7.25) that are generated by the Faddeev–Popov–De Witt formalism.

We can go further, and free ourselves of all dependence on canonical quantization in Lorentz-non-invariant gauges like axial gauge. Again, take the action to be the most general functional of gauge, matter, ω^A, ω^{*A} and h^A fields with ghost number zero, that is invariant under the BRST transformation (15.7.7)–(15.7.11) and under any other global symmetries of the theory, including Lorentz invariance. From the BRST invariance of the action we can infer the existence of a conserved nilpotent BRST generator Q. With the ghost and antighost fields treated as Hermitian, Q is also Hermitian. The space of physical states is defined as above as consisting of states annihilated by Q, with two states treated as equivalent if their difference is Q acting on another state. It has been shown that for Yang–Mills theories this space is free of ghosts and antighosts and has a positive-definite norm, and that the S-matrix in this space is unitary [13a].

This procedure is known as *BRST quantization*. It has been extended to theories with other local symmetries, such as general relativity and string theories. Unfortunately, it seems so far to be necessary to give separate proofs in each case that the BRST-cohomology is ghost-free and that the S-matrix acting in this space is unitary. The key point in these proofs is that, for each negative-norm degree of freedom, such as the time components of the gauge fields in Yang–Mills theories, there is one local symmetry that allows this degree of freedom to be transformed away.

* * *

Although we shall not use it here, there is a beautiful geometric interpretation [14] of the ghosts and the BRST symmetry that should be mentioned. The gauge fields A_α^μ may be written as one-forms $A_\alpha = A_{\alpha\mu}dx^\mu$, where dx^μ are a set of anticommuting c-numbers. (See Section 5.8.) This can be combined with the ghost to compose a one-form $\mathcal{A}_\alpha = A_\alpha + \omega_\alpha$ in an extended space. Also, the ordinary exterior derivative $d = dx^\mu \partial/\partial x^\mu$ may be combined with the BRST operator s to form an exterior derivative $\mathcal{D} = d + s$ in this space, which is nilpotent because $s^2 = d^2 = sd + ds = 0$.

The next chapter will introduce external field methods, which will be used along with the BRST symmetry in Chapter 17 to complete the proof of the renormalizability of non-Abelian gauge theories.

15.8 Generalizations of BRST Symmetry[10]

The BRST symmetry described in the previous section has a useful generalization to the quantization of a wide class of theories, including general relativity and string theories. In all these cases, we deal with an action $I[\phi]$ and measure $[d\phi] \equiv \prod_r d\phi^r$ that are invariant under a set of infinitesimal transformations

$$\phi^r \longrightarrow \phi^r + \epsilon^A \delta_A \phi^r. \tag{15.8.1}$$

This is an abbreviated 'De Witt' notation, with r and A including spacetime coordinates as well as discrete labels, and sums including integrals over these coordinates. For instance, for the gauge transformation (15.1.9), the index A consists of a group index α and a spacetime coordinate x, with $\epsilon^{\alpha x} \equiv \epsilon^\alpha(x)$, while the index r consists of a vector index μ as well as a group index α and a spacetime coordinate x, with $\phi^{\mu\alpha x} \equiv A_\mu^\alpha(x)$; in the notation of Eq. (15.8.1), the variation $\delta_A\phi^r$ in the transformation (15.1.9) reads

$$\delta_{\beta y}\phi^{\mu\alpha x} = \delta_\alpha^\beta \frac{\partial}{\partial x^\mu}\delta^4(x-y) + C^\beta{}_{\gamma\alpha}\phi^{\mu\gamma x}\delta^4(x-y).$$

As in the special case of Yang–Mills theories discussed in the previous section, BRST invariance can be used as a substitute for the Faddeev–Popov–De Witt formulation of these theories, one that is applicable even where the Faddeev–Popov–De Witt approach fails. Nevertheless, in order to motivate the introduction of BRST invariance, we shall begin here with the Faddeev–Popov–De Witt formulation of theories with general local symmetries, and then go on to consider further generalizations.

[10]This section lies somewhat out of the book's main line of development, and may be omitted in a first reading.

By following the same arguments used to derive Eq. (15.5.21), we obtain the general Faddeev–Popov–De Witt theorem:

$$\frac{C}{\Omega}\int[d\phi]e^{iI[\phi]}V[\phi]=\int[d\phi]e^{iI[\phi]}B\{f[\phi]\}\,\mathrm{Det}(\delta_A f_B[\phi])V[\phi], \tag{15.8.2}$$

where $V[\phi]$ is an arbitrary functional of ϕ^r that is invariant under the gauge transformations (15.8.1); $f_A[\phi]$ are a set of gauge-fixing functionals[11] of the ϕ^r, chosen so that the 'matrix' $\delta_A f_B[\phi]$ has a non-vanishing determinant, and $B[f]$ is a more-or-less arbitrary functional of the f_A (as for instance $\prod_A \delta(f_A)$). The constant Ω is the volume of the gauge group, and the constant C is defined (as in Eq. (15.5.19)) by

$$C\equiv\int[df]B[f]. \tag{15.8.3}$$

As we have seen, in gauge theories the importance of Eq. (15.8.2) is that it tells us that the integral on the right-hand side is independent of the choice of the gauge-fixing functionals f_A, and depends on $B[f]$ only through the constant C. Where some meaning can be given to the usually infinite group volume Ω, as in gauge theories on a finite spacetime lattice, Eq. (15.8.2) can also have value as a formula for the integral on the left-hand side.

To define a nilpotent BRST transformation, we must first express the functional $B[f]$ as a Fourier transform

$$B[f]=\int[dh]\exp(ih^A f_A)\mathcal{B}[h], \tag{15.8.4}$$

where $[dh]\equiv\prod_A dh^A$. Also, the determinant can be expressed as an integral over fermionic c-number fields[12] ω^{*A} and ω^A:

$$\mathrm{Det}(\delta_A f_B[\phi])\propto\int[d\omega^*][d\omega]\exp\left(i\omega^{*B}\omega^A\delta_A f_B\right), \tag{15.8.5}$$

where $[d\omega^*]\equiv\prod_A d\omega^{*A}$ and $[d\omega]\equiv\prod_A d\omega^A$, and as usual '$\propto$' means proportional with field-independent factors. Inserting these in Eq. (15.8.2) gives a general formula for the gauge-fixed path integral

$$\begin{aligned}&\int[d\phi]\exp\bigl(iI[\phi]\bigr)B\{f[\phi]\}\,\mathrm{Det}(\delta_A f_B[\phi])V[\phi]\\&\qquad\propto\int[d\phi][dh][d\omega^*][d\omega]\exp\bigl(iI_{\mathrm{NEW}}[\phi,h,\omega,\omega^*]\bigr)\mathcal{B}[h]V[\phi],\end{aligned} \tag{15.8.6}$$

where I_{NEW} is the new total action:

$$I_{\mathrm{NEW}}[\phi,h,\omega,\omega^*]=I[\phi]+h^A f_A[\phi]+\omega^{*B}\omega^A\delta_A f_B[\phi]. \tag{15.8.7}$$

[11] We use the same letters A, B, etc. to label the f_A as the gauge variations δ_A, in order to emphasize that there must be as many gauge-fixing functionals as there are independent gauge transformations. However, in some cases like string theory it is natural to use gauge-fixing functionals f^a for which, although the index a runs over 'as many' values as the index A on the gauge variations δ_A, the values taken by these indices are quite different. No change is needed in the present formalism as long as we can define $f_A=c_{Aa}f^a$, with c_{Aa} field-independent and non-singular.

[12] It is common in string theories and elsewhere to find the ghost fields ω^{*A} and ω^A written b^A (or b_A) and c^A, respectively.

As mentioned in Section 15.6, we can think of the ghost fields as compensation for the fact that we are integrating over all ϕ^r, including those ϕ^r that differ only by gauge transformations (15.8.1). Because ghosts are fermions, loops of ghost lines carry extra minus signs that allow these loops to compensate for the integration over gauge-equivalent ϕs. But for this to work, there must be just as many ghost fields ω^A as there are independent gauge transformations. That is, since the ω^A are independent, the gauge transformations (15.8.1) must all be independent. This is the case for gauge transformations in Yang–Mills theory and coordinate transformations in general relativity, but not always. The classic example of a theory with non-independent gauge transformations is the theory of p-form gauge fields, described in Section 8.8. A p-form A (an antisymmetric tensor of rank p) undergoes a gauge transformation $A \longrightarrow A + d\phi$, where ϕ is a $(p-1)$-form, and $d\phi$ is its exterior derivative (the antisymmetrized derivative). Because d is nilpotent, for $p \geq 2$ we can shift ϕ by an amount $d\psi$ without changing the gauge transformation, so there is a sort of invariance under gauge transformations of gauge transformations, in which the transformation parameters are the $(p-2)$-forms ψ. In such cases we must compensate for introducing too many ghosts by also introducing 'ghosts of ghosts' [15]. For $p \geq 3$ we need to compensate further by introducing 'ghosts of ghosts of ghosts,' and so on. In what follows we shall assume that the gauge transformations (15.8.1) are all independent, so that the ghost fields ω^A (and the antighosts ω^{*A}) are all we need.

Although the original symmetry (15.8.1) has been eliminated by the insertion of the non-gauge-invariant functional $B[f]$, the new total action has an exact symmetry under the infinitesimal BRST transformations

$$\chi \longrightarrow \chi + \theta s \chi, \tag{15.8.8}$$

where χ is any of the ϕ^r, ω^A, ω^{*A}, or h^A; θ is an infinitesimal anticommuting c-number; and s is the *Slavnov operator*

$$s = \omega^A \delta_A \phi^r \frac{\delta_L}{\delta \phi^r} - \frac{1}{2}\omega^B \omega^C f^A{}_{BC} \frac{\delta_L}{\delta \omega^A} - h^A \frac{\delta_L}{\delta \omega^{*A}}. \tag{15.8.9}$$

In Eq. (15.8.9) the subscript L denotes left differentiation, defined so that if $\delta F = \delta\chi G$, then $\delta_L F/\delta\chi = G$, and $f^A{}_{BC}$ is the structure constant[13] appearing in the commutation relation

$$[\delta_B, \delta_C] = f^A{}_{BC} \delta_A. \tag{15.8.10}$$

The $f^A{}_{BC}$ are field-independent in non-Abelian gauge theories and in string theories, though not always, but the BRST formalism is not limited to this case. A straightforward calculation gives

$$\begin{aligned} s^2 = {} & \frac{1}{2}\omega^A \omega^B \left[\delta_A \phi^s \frac{\delta_L(\delta_B \phi^r)}{\delta \phi^s} - \delta_B \phi^s \frac{\delta_L(\delta_A \phi^r)}{\delta \phi^s} - f^C{}_{AB} \delta_C \phi^r \right] \frac{\delta_L}{\delta \phi^r} \\ & - \frac{1}{2}\omega^B \omega^C \omega^D \left[f^E{}_{BC} f^A{}_{DE} + \delta_D \phi^r \frac{\delta_L f^A{}_{BC}}{\delta \phi^r} \right] \frac{\delta_L}{\delta \omega^A}. \end{aligned} \tag{15.8.11}$$

[13] For instance, for a gauge transformation acting on a matter field $\psi(x)$ we have $\delta_{\beta y}\psi(x) = i t_\beta \psi(x) \delta^4(x-y)$, and so $\delta_{\alpha z}\delta_{\beta y}\psi(x) = -t_\alpha t_\beta \psi(x)\delta^4(x-y)\delta^4(x-z)$. Hence in this case we have $f^{\gamma x}{}_{\beta y, \alpha z} = C^\gamma{}_{\beta\alpha}\delta^4(x-y)\delta^4(x-z)$.

Hence the condition that the BRST transformation be nilpotent is equivalent to the commutation relation (15.8.10), together with a consistency condition

$$f^{E}{}_{[BC}f^{A}{}_{D]E} + \delta_{[D}\phi^{r}\left(\delta_{L}f^{A}{}_{BC]}/\delta\phi^{r}\right) = 0, \tag{15.8.12}$$

where the brackets in subscripts indicate antisymmetrization with respect to the enclosed indices B, C, and D. Eq. (15.8.12) may be derived from the commutation relation (15.8.10) in the same way as the usual Jacobi identity, and takes the place of the Jacobi identity for symmetries with field-dependent structure constants.

To show that the transformation (15.8.8) is a symmetry of $I_{\rm NEW}$, we note (recalling that θ anticommutes with ω^{*A}) that Eq. (15.8.7) may be rewritten

$$I_{\rm NEW}[\phi, h, \omega, \omega^*] = I[\phi] - s(\omega^{*A}f_A). \tag{15.8.13}$$

The term $I[\phi]$ is BRST-invariant, because on the fields ϕ^r a BRST transformation is just a gauge transformation (15.8.1) with ϵ^A replaced with $\theta\omega^A$, which commutes with all ϕ^r. The term $s(\omega^{*A}f_A)$ is BRST-invariant because BRST transformations are nilpotent.

For several reasons we may need to consider a wider class of actions than those that can be constructed by the Faddeev–Popov–De Witt approach, by simply requiring that the action is invariant under the BRST transformation (15.8.8). As a step toward showing that such an action yields physically sensible results, we shall now prove the general result (already used in the previous section) that the most general BRST-invariant functional of ghost number zero is the sum of a functional of the ϕ alone, plus a term given by the action of the BRST operator s on an arbitrary functional Ψ of ghost number -1:

$$I_{\rm NEW}[\phi, \omega, \omega^*, h] = I_0[\phi] + s\Psi[\phi, \omega, \omega^*, h] \tag{15.8.14}$$

as for instance in the Faddeev–Popov–De Witt action (15.8.13). In brief, the BRST cohomology consists of gauge-invariant functionals $I[\phi]$ of the fields ϕ^r alone.

To prove Eq. (15.8.14), we note that the BRST transformation (15.8.8)–(15.8.9) does not change the total number of h^A and ω^{*A} fields, so if we expand I in a series of terms I_N that contain definite total numbers N of h^A and ω^{*A} fields, then there can be no cancellations in sI between terms with different N, so each term must be separately BRST-invariant:

$$sI_N = 0. \tag{15.8.15}$$

We next introduce what is called a Hodge operator:

$$t \equiv \omega^{*A}\frac{\delta}{\delta h^A}. \tag{15.8.16}$$

It is straightforward to check the anticommutation relation

$$\{s, t\} = -\omega^{*A}\frac{\delta_L}{\delta\omega^{*A}} - h^A\frac{\delta}{\delta h^A}. \tag{15.8.17}$$

Applying the operator $\{s, t\}$ to I_N and using Eq. (15.8.15) then gives

$$stI_N = -NI_N, \tag{15.8.18}$$

so each I_N except for I_0 is BRST-exact, in the sense that it may be written as the operator s acting on some other functional. The complete functional I may therefore be written in the form $I_0 + s\Psi$, with

$$\Psi = -\sum_{N=1}^{\infty} \frac{tI_N}{N}. \tag{15.8.19}$$

The term I_0 is by definition independent of ω^{*A} and h^A, and since we assume it has zero ghost number it must also be independent of ω^A, as was to be proved.

To show the invariance of physical matrix elements under changes in the definition of the gauge-fixing functional Ψ, we define a fermionic 'charge' Q, such that the change under a BRST transformation of any operator Φ is

$$\delta_\theta \Phi = i[\theta Q, \Phi] = i\theta[Q, \Phi]_\mp \tag{15.8.20}$$

with the top or bottom sign in $[x, y]_\mp \equiv xy \mp yx$ according as Φ is bosonic or fermionic. Just as in the previous section, the nilpotence of the BRST transformation then tells us that $Q^2 = 0$. Matrix elements of gauge-invariant operators between physical states will be independent of the choice of Ψ if and only if the physical states $\|\alpha\rangle$ and $\langle\beta\|$ satisfy

$$Q\|\alpha\rangle = \langle\beta\|Q = 0, \tag{15.8.21}$$

so that physically distinguishable physical states are again in one-to-one correspondence with elements of the cohomology of Q. The general BRST-invariant action (15.8.3) will therefore yield physically sensible results for any gauge-fixing functional Ψ if we can find *some* Ψ, like axial gauge in Yang–Mills theories, in which the ghosts do not interact with other fields. If this ghost-free choice of Ψ is inconvenient for actual calculation, as for instance axial gauge is inconvenient because it violates Lorentz invariance, we can adopt any gauge-fixing Ψ we like, and still be confident that there is a unitary S-matrix with no ghosts in initial or final states.

This approach works well in string theories, where so-called light-cone quantization takes the place of axial gauge. But in other theories like general relativity there is no way of choosing a coordinate system in which the ghosts decouple. Such theories may be dealt with by the BRST-quantization method described at the end of the previous section, using BRST invariance to prove that the S-matrix in a physical ghost-free Hilbert space is unitary.

The discovery [17] of invariance under an 'anti-BRST' symmetry [18] showed that, despite appearances, there is a similarity between the roles of ω^A and ω^{*A}, which remains somewhat mysterious.

15.9 The Batalin–Vilkovisky Formalism[14]

This section will describe a powerful formalism, widely known as the Batalin–Vilkovisky [19] method. It is developed in the Lagrangian framework, but has its roots in the earlier Batalin–Fradkin–Vilkovisky formalism [20], which had been derived in the Hamiltonian

[14] This section lies somewhat out of the book's main line of development, and may be omitted in a first reading.

framework. (The two schemes have been proved perturbatively equivalent [21].) As we shall see in Section 17.1, the same formal machinery had been developed even earlier by Zinn–Justin [22] in order to deal with the renormalization of gauge theories. There are at least three areas where this formalism has proved invaluable:

(i) Up to this point, we have considered only irreducible symmetries with an algebra that closes in the sense of Eq. (15.8.10). In some theories, such as supergravity (without auxiliary fields) [23], the algebra is open: it closes only when the field equations are satisfied, so that terms appear in Eq. (15.8.10) proportional to $\delta I/\delta\chi^n$. Similar terms will also then appear in the consistency conditions (15.8.12). Equation (15.8.11) then shows that in such theories s^2 will not vanish, but rather will equal a linear combination of the derivatives $\delta I/\delta\chi^n$. As we will see in this section, the Batalin–Vilkovisky method allows us to deal with very general gauge theories, including those with open or reducible gauge symmetry algebras.

(ii) As mentioned above, essential aspects of the Batalin–Vilkovisky formalism were originally developed by Zinn–Justin in order to prove the renormalizability of gauge theories. The crucial point, to be explained in Section 17.1, is that although the sum of all one-particle-irreducible diagrams in a background field does not obey the BRST symmetries of the original action, it does share one of the key properties of the action, known as the master equation.

(iii) The Batalin–Vilkovisky method provides a convenient way of analyzing the possible violations of symmetries of the action by quantum effects. It is used for this purpose in Section 22.6.

The starting point of the Batalin–Vilkovisky formalism is the introduction of what are called "antifields," one for each field in the theory. We let χ^n run over all the fields ϕ^r, ω^A, ω^{*A}, and h^A, and for each χ^n we introduce an external antifield $\chi^\ddagger_n$, with the same Bose or Fermi statistics and opposite ghost number as that of the BRST-transformed field $s\chi^n$. [15] That is, $\chi^\ddagger_n$ has the opposite statistics to χ^n, and a ghost number equal to $-\operatorname{gh}(\chi^n)-1$, where $\operatorname{gh}(\chi^n)$ is the ghost number of χ^n. In the simplest cases, including Yang–Mills theories and quantum gravity, the original gauge-invariant action $I[\phi]$ is supplemented with a term coupling the antifields $\chi^\ddagger_n$ to the $s\chi^n$, giving an action

$$S[\chi,\chi^\ddagger]=I[\phi]+(s\chi^n)\chi^\ddagger_n. \tag{15.9.1}$$

This satisfies what is called the *master equation*

$$0=\frac{\delta_R S}{\delta\chi^\ddagger_n}\frac{\delta_L S}{\delta\chi^n}, \tag{15.9.2}$$

[15] The symbol $\ddagger$ is used here in place of the more usual $*$ in order to emphasize that it has nothing whatever to do with complex conjugation or charge conjugation. In particular, the antighost field ω^{*A} is *not* the same as the antifield $\omega^\ddagger_A$ of ω^A.

with 'R' and 'L' here denoting right- and left-differentiation. To check this, note that the terms in Eq. (15.9.2) of zeroth order in antifields just yield the condition of gauge invariance

$$0 = (s\phi^r)\frac{\delta_L I}{\delta\phi^r} = \omega^A \delta_A I[\phi], \tag{15.9.3}$$

while the terms linear in the antifields $\phi^\ddagger_r$ provide the condition of nilpotence:

$$0 = (s\chi^m)\frac{\delta_L(s\chi^n)}{\delta\chi^m} = s^2\chi^n. \tag{15.9.4}$$

The $\chi^\ddagger_n$ are external fields, and must be given suitable values before we use $S[\chi, \chi^\ddagger]$ to calculate the S-matrix. For this purpose, we introduce an arbitrary fermionic functional $\Psi[\chi]$ with ghost number -1, and set [16]

$$\chi^\ddagger_n = \frac{\delta\Psi[\chi]}{\delta\chi^n}. \tag{15.9.5}$$

Then Eq. (15.9.1) becomes

$$S[\phi, \delta\Psi/\delta\chi] = I[\phi] + (s\chi^n)\frac{\delta\Psi[\chi]}{\delta\chi^n} = I[\phi] + s\Psi[\chi]. \tag{15.9.6}$$

Comparison with Eq. (15.8.14) shows that this is the same as the gauge-fixed action $I_{\rm NEW}[\chi]$. Thus, using the same arguments as in the previous section, physical matrix elements are not affected by small changes in Ψ. The action (15.8.7) constructed by the Faddeev–Popov–De Witt method corresponds to the choice $\Psi = -\omega^{*A} f_A$, for which $\phi^\ddagger_r = \omega^{*A}\delta f_A/\delta\phi^r$, $\omega^\ddagger_C = 0$, and $\omega^{*\ddagger}_A = -f_A$.

So far, nothing new has been accomplished. The first new point is that the master equation (15.9.2) can be used for more general theories by letting $S[\chi, \chi^\ddagger]$ be a non-linear functional of the antifields $\chi^\ddagger_n$. (For reducible theories we must also include ghosts for ghosts among the χ^n, as discussed in the previous section, along with their antifields.) As above, we take the statistics of $\chi^\ddagger_n$ to be opposite to that of χ^n, and its ghost number equal to $-\,\mathrm{gh}(\chi^n) - 1$, and we require $S[\chi, \chi^\ddagger]$ to be a bosonic operator of ghost number zero. Because ω^{*A} and h_A have linear BRST transformations they are unaffected by the complications affecting the other χ^n (in this connection, see Section 16.4), so they and their antifields enter in the action $S[\chi, \chi^\ddagger]$ in the same way as in Eq. (15.9.1). That is,

$$S = S_{\rm min}[\phi, \omega, \phi^\ddagger, \omega^\ddagger] - h^A\omega^{*\ddagger}_A, \tag{15.9.7}$$

where $\phi^\ddagger_r$, $\omega^\ddagger_A$, and $\omega^{*\ddagger}_A$ are the antifields of ϕ^r, ω^A, and ω^{*A}, with ghost numbers -1, -2, and 0, respectively, and $S_{\rm min}[\phi, \omega, \phi^\ddagger, \omega^\ddagger]$ is some bosonic functional of ghost number zero. The last term in (15.9.7) has no effect on the master equation, so $S_{\rm min}$ satisfies the master equation by itself. [17]

[16] It is not necessary to distinguish between left- and right-differentiation here, because either χ^n or $\delta\Psi/\delta\chi^n$ must be bosonic.

[17] The fields ϕ^r, ω^A, $\phi^\ddagger_r$, $\omega^\ddagger_A$ are sometimes called *minimal variables*, while fields like ω^{*A} and h_A which together with their antifields enter bilinearly as in Eq. (15.9.7), are called *trivial pairs*.

Because $S_{\rm min}$ has ghost number zero, its expansion in powers of antifields must take the form

$$
\begin{aligned}
S_{\rm min} = I[\phi] + \omega^A f^r_A[\phi]\phi^\ddagger_r + \frac{1}{2}\omega^A\omega^B f^C{}_{AB}[\phi]\omega^\ddagger_C \\
+ \frac{1}{2}\omega^A\omega^B f^{rs}_{AB}[\phi]\phi^\ddagger_r\phi^\ddagger_s + \omega^A\omega^B\omega^C f^{rD}_{ABC}[\phi]\phi^\ddagger_r\omega^\ddagger_D \\
+ \frac{1}{2}\omega^A\omega^B\omega^C\omega^D f^{EF}_{ABCD}[\phi]\omega^\ddagger_E\omega^\ddagger_F + \cdots .
\end{aligned}
\tag{15.9.8}
$$

The term in the master equation (15.9.2) of zeroth order in antifields (and hence first order in ω^A) yields

$$
0 = f^r_A[\phi]\frac{\delta I[\phi]}{\delta\phi^r}, \tag{15.9.9}
$$

which is just the statement that $I[\phi]$ is invariant under the transformation

$$
\phi^r \longrightarrow \phi^r + \epsilon^A f^r_A[\phi] \tag{15.9.10}
$$

with ϵ^A arbitrary infinitesimals. The term in the master equation proportional to $\phi^\ddagger_s$ on the right and $\omega^A\omega^B$ on the left yields

$$
0 = f^r_A[\phi]\frac{\delta f^s_B[\phi]}{\delta\phi^r} - f^r_B[\phi]\frac{\delta f^s_A[\phi]}{\delta\phi^r} + f^C{}_{AB}[\phi]f^s_C[\phi] + \frac{\delta I[\phi]}{\delta\phi^r}f^{rs}_{AB}[\phi], \tag{15.9.11}
$$

which, when the field equations $\delta I/\delta\phi = 0$ are satisfied, becomes the commutation relation (with structure constant $f^C{}_{AB}[\phi]$) for the transformation (15.9.10). The other term in the master equation linear in antifields is proportional to $\omega^A\omega^B\omega^C$ on the left and $\omega^\ddagger_D$ on the right; it yields

$$
0 = f^r_{[A}[\phi]\frac{\delta f^D{}_{BC]}[\phi]}{\delta\phi^r} - f^E{}_{[AB}[\phi]f^D{}_{C]E}[\phi] + f^{rD}_{ABC}[\phi]\frac{\delta I[\phi]}{\delta\phi^r}, \tag{15.9.12}
$$

where square brackets in the subscripts indicate antisymmetrization with respect to the enclosed indices A, B, and C. When the field equations are satisfied this becomes the generalized Jacobi identity (15.8.12). Equation (15.9.11) is necessary for the consistency of the symmetry condition (15.9.9) (assuming that the f^r_A furnish a complete set of gauge symmetries), while Eq. (15.9.12) is necessary for the consistency of the commutation relations (15.9.11). Note that the terms in Eqs. (15.9.11) and (15.9.12) that arise from the terms in $S_{\rm min}$ quadratic in antifields are proportional to $\delta I[\phi]/\delta\chi$, so they vanish when the field equations are satisfied, and in this sense are characteristic of open symmetry algebras. Terms in the master equation of second or higher order in antifields involve terms in $S_{\rm min}$ that are of third and/or higher order in antifields. These provide consistency conditions for Eqs. (15.9.11) and (15.9.12), consistency conditions for these consistency conditions, and so on. It is a virtue of the Batalin–Vilkovisky formalism that all these consistency conditions are incorporated in the one master equation.

The master equation may be reinterpreted as a statement of invariance of S under a generalized BRST transformation. In order to see this, and for future purposes, it is useful to introduce a formal device known as the *antibracket*. Returning now to our previous notation, the antibracket of two general functionals $F[\chi, \chi^\ddagger]$ and $G[\chi, \chi^\ddagger]$ is defined by

$$(F, G) \equiv \frac{\delta_R F}{\delta \chi^n} \frac{\delta_L G}{\delta \chi^\ddagger_n} - \frac{\delta_R F}{\delta \chi^\ddagger_n} \frac{\delta_L G}{\delta \chi^n}. \tag{15.9.13}$$

Note that right- and left-functional derivatives of a bosonic functional like S with respect to a bosonic or fermionic field variable are equal to each other or negatives of each other, respectively. Since just one of either $\chi^\ddagger_n$ or χ^n is always fermionic and the other bosonic, it follows that for the antibracket (S, S) the second term on the right-hand side in Eq. (15.9.13) changes sign if we reverse left- and right-differentiation

$$\frac{\delta_R S}{\delta \chi^\ddagger_n} \frac{\delta_L S}{\delta \chi^n} = -\frac{\delta_L S}{\delta \chi^\ddagger_n} \frac{\delta_R S}{\delta \chi^n} = -\frac{\delta_R S}{\delta \chi^n} \frac{\delta_L S}{\delta \chi^\ddagger_n}.$$

(The last step is allowed because, since one of the factors here is bosonic, their order is immaterial.) We see that the second term on the right in Eq. (15.9.13) for (S, S) is the negative of the first. The master equation (15.9.2) may therefore be written as the requirement that the antibracket of S with itself vanishes:

$$(S, S) = 0. \tag{15.9.14}$$

This is a non-trivial requirement, because the antibracket has the general symmetry property

$$(F, G) = \pm (G, F), \tag{15.9.15}$$

the sign being $+1$ where F and G are both bosonic, and -1 otherwise. In particular, (F, F) automatically vanishes if F is fermionic, but not if F is bosonic.

The generalized BRST transformation is defined by

$$\widehat{\delta}_\theta \chi^n = \theta \frac{\delta_R S}{\delta \chi^\ddagger_n} = -\theta (S, \chi^n), \tag{15.9.16}$$

$$\widehat{\delta}_\theta \chi^\ddagger_n = -\theta \frac{\delta_R S}{\delta \chi^n} = -\theta (S, \chi^\ddagger_n), \tag{15.9.17}$$

where θ is a fermionic infinitesimal constant. (When S is of the form (15.9.1), the transformation of χ is the same as the original BRST transformation $\delta_\theta \chi^n = \theta s \chi^n$.) To evaluate the effect of this transformation on general functionals, we note that the antibracket acts as a derivative, in the sense that

$$(F, GH) = (F, G) H \pm G (F, H), \tag{15.9.18}$$

where the sign is -1 if G is fermionic and F is bosonic, and $+1$ otherwise. Hence if G and H are arbitrary functionals of χ and $\chi^\ddagger$ with $\widehat{\delta}_\theta G = -\theta (S, G)$ and $\widehat{\delta}_\theta H = -\theta (S, H)$, then

$$\widehat{\delta}_\theta (GH) = -\theta (S, G) H - G \theta (S, H) = \theta \big[(S, G) H \pm G (S, H) \big],$$

where the sign is $+$ or $-$ if G is bosonic or fermionic. Taking F in Eq. (15.9.18) equal to the bosonic functional S, we see then that

$$\widehat{\delta}_\theta(GH) = -\theta(S, GH).$$

Together with Eqs. (15.9.16) and (15.9.17), this shows that for any functional F formed as a sum of products of fields and antifields

$$\widehat{\delta}_\theta F = -\theta(S, F). \tag{15.9.19}$$

The master equation (15.9.14) may be interpreted as the statement that these generalized BRST transformations leave S invariant

$$\widehat{\delta}_\theta S = -\theta(S, S) = 0. \tag{15.9.20}$$

Like the original BRST transformation, this symmetry transformation is nilpotent. To see this, we use the Jacobi identity for the antibracket:

$$\pm(F, (G, H)) + \text{cyclic permutations} = 0, \tag{15.9.21}$$

where the sign in the first term is $-$ if F and H are bosonic and $+$ otherwise, with corresponding signs for the other two cyclic permutations of F, G, and H. Taking $F = G = S$, Eq. (15.9.21) becomes

$$0 = \mp(S, (S, H)) \mp (H, (S, S)) + (S, (H, S)) = \mp 2(S, (S, H)) \mp (H, (S, S)),$$

where the sign is $-$ or $+$ if H is bosonic or fermionic. The master equation (15.9.14) then yields the nilpotency condition

$$(S, (S, H)) = 0. \tag{15.9.22}$$

Because of this symmetry, the solution of the master equation is not unique. For instance, Eq. (15.9.22) shows that for any given solution S we can find another solution given by the infinitesimal transformation

$$S' = S + (\delta F, S), \tag{15.9.23}$$

with δF an infinitesimal functional of χ and $\chi^\ddagger$ that is arbitrary, except that it must be fermionic and have ghost number -1 in order that S' should be bosonic and have ghost number zero. In particular, taking δF to be a fermionic functional $\epsilon\Psi$ of χ^n alone gives

$$S'[\chi, \chi^\ddagger] = S[\chi, \chi^\ddagger] + \epsilon \frac{\delta\Psi[\chi]}{\delta\chi^n} \frac{\delta_R S}{\delta\chi^\ddagger_n} = S\left[\chi, \chi^\ddagger + \epsilon\frac{\delta\Psi}{\delta\chi}\right]. \tag{15.9.24}$$

These infinitesimal transformations may be trivially integrated, and show that the master equation is still satisfied if we shift the antifields to new variables $\chi^{\ddagger\prime}_n \equiv \chi^\ddagger_n - \delta\Psi/\delta\chi^n$.

The transformation (15.9.23) is a special case of what are usually called canonical transformations, and will here be called "anticanonical transformations" to distinguish them from the canonical transformations of Chapter 7. An anticanonical transformation is any transformation of fields and antifields, finite or infinitesimal, that leaves unchanged the fundamental antibracket relations:

$$(\chi^n, \chi^\ddagger_m) = \delta^n_m, \qquad (\chi^n, \chi^m) = (\chi^\ddagger_n, \chi^\ddagger_m) = 0. \tag{15.9.25}$$

For instance, consider the infinitesimal anticanonical transformation generated by an infinitesimal fermion generator δF, under which any bosonic or fermionic functional G is transformed into

$$G \longrightarrow G' = G + (\delta F, G). \tag{15.9.26}$$

It is easy to show that this does not change the fundamental antibrackets (15.9.25). For this purpose, note that the antibracket (G, H) of two functionals G and H is transformed into (G', H'), which to first order in infinitesimals is

$$(G', H') = (G, H) + ((\delta F, G), H) + (G, (\delta F, H)).$$

Using the Jacobi identity (15.9.21), this is

$$(G', H') = (G, H) \pm (\delta F, (G, H)), \tag{15.9.27}$$

with the sign $+$ if G and H are both bosonic, and $-$ otherwise. In particular, if (G, H) is a c-number then it is unchanged by an anticanonical transformation. (This is another way of seeing that the transformation (15.9.23) leaves the master equation unchanged.) The fields and antifields have c-number antibrackets (15.9.25), so the same must be true for the transformed fields and antifields.

To calculate the S-matrix we must give some definite value to the antifields. As in the simple case of closed gauge algebras where S is of the linear form (15.9.5), we can do this by taking the antifields in the form (15.9.5); that is, we calculate the S-matrix using the "gauge-fixed" action

$$I_\Psi[\chi] = S\left[\chi, \frac{\delta \Psi[\chi]}{\delta \chi}\right], \tag{15.9.28}$$

where $\Psi[\chi]$ is a fermionic functional of ghost number -1. According to the remarks following Eq. (15.9.24), this is the same as taking the canonically transformed antifields $\chi^{\ddagger\prime}$ equal to zero.

The gauge-fixed action is invariant under a BRST transformation that acts on fields χ^n alone:

$$\delta_\theta \chi^n = \theta s \chi^n, \qquad s\chi^n = \left.\frac{\delta_R S[\chi, \chi^\ddagger]}{\delta \chi^\ddagger_n}\right|_{\chi^\ddagger = \delta\Psi/\delta\chi}. \tag{15.9.29}$$

To check this, note that

$$\begin{aligned} sI_\Psi[\chi] = &\left.\left(\frac{\delta_R S[\chi, \chi^\ddagger]}{\delta \chi^\ddagger_n}\frac{\delta_L S[\chi, \chi^\ddagger]}{\delta \chi^n}\right)\right|_{\chi^\ddagger = \delta\Psi/\delta\chi} \\ &+ \left.\left(\frac{\delta_R S[\chi, \chi^\ddagger]}{\delta \chi^\ddagger_m}\frac{\delta_L^2 \Psi[\chi]}{\delta \chi^m \delta \chi^n}\frac{\delta_L S[\chi, \chi^\ddagger]}{\delta \chi^\ddagger_n}\right)\right|_{\chi^\ddagger = \delta\Psi/\delta\chi}. \end{aligned}$$

The first term on the right-hand side vanishes as a result of the master equation, and the second because the summand is antisymmetric [18] in m and n.

For closed algebras with S of the form (15.9.1), the transformation (15.9.29) is the original BRST transformation $\delta_\theta \chi^n = \theta s \chi^n$. But for general open algebras, the transformation (15.9.29) is not the same as the original BRST transformation, and in general is not even nilpotent unless the field equations are satisfied. Instead, from the terms in the master equation of first order in the shifted antifields $\chi^{\ddagger\prime}_n = \chi^\ddagger_n - \delta\Psi[\chi]/\delta\chi^n$, we find

$$s^2\chi^m = \mp \left(\frac{\delta_L}{\delta\chi^\ddagger_m}\frac{\delta_R}{\delta\chi^\ddagger_n} I[\chi,\chi^\ddagger]\right)\Bigg|_{\chi^\ddagger=\delta\Psi/\delta\chi} \frac{\delta I_\Psi[\chi]}{\delta\chi^n}, \tag{15.9.30}$$

with the sign $-$ or $+$ if χ^m is bosonic or fermionic, respectively. Again, we see that the dependence on the field equations characteristic of open gauge algebras is associated with terms in S quadratic in the antifields.

Until now we have been considering only the formulation of a classical field theory based on an open or closed gauge algebra. Now we must consider how quantum mechanical calculations are done in such theories. Physical matrix elements may be calculated by functional integrals weighted with $\exp(iI_\Psi[\chi])$, where as explained above, $I_\Psi[\chi]$ is obtained from $S[\chi,\chi^\ddagger]$ by setting $\chi^\ddagger_n = \delta\Psi[\chi]/\delta\chi^n$, or in other words $\chi^{\ddagger\prime}_n = 0$. We wish to evaluate the effect of a change in $\Psi[\chi]$ on these matrix elements. First consider the vacuum–vacuum amplitude

$$Z_\Psi = \int \left[\prod d\chi\right] \exp(iI_\Psi[\chi]). \tag{15.9.31}$$

Under a shift $\delta\Psi[\chi]$ in $\Psi[\chi]$, this changes by an amount

$$\delta Z = i\int \left[\prod d\chi\right] \exp(iI_\Psi[\chi]) \left(\frac{\delta_R S[\chi,\chi^\ddagger]}{\delta\chi^\ddagger_n}\right)\Bigg|_{\chi^\ddagger=\delta\Psi/\delta\chi} \left(\frac{\delta(\delta\Psi[\chi])}{\delta\chi^n}\right). \tag{15.9.32}$$

Integrating by parts in field space, this becomes

$$\delta Z = \int \left[\prod d\chi\right] \exp(iI_\Psi[\chi]) \left\{\frac{\delta_R S[\chi,\chi^\ddagger]}{\delta\chi^\ddagger_n}\frac{\delta_L I_\Psi[\chi]}{\delta\chi^n} - i\Delta S[\chi,\chi^\ddagger]\right\}\Bigg|_{\chi^\ddagger=\delta\Psi/\delta\chi} \delta\Psi[\chi], \tag{15.9.33}$$

where

$$\Delta \equiv \frac{\delta_R}{\delta\chi^\ddagger_n}\frac{\delta_L}{\delta\chi^n}. \tag{15.9.34}$$

We see that the condition for the Ψ independence of the vacuum–vacuum amplitude is in general *not* the master equation (15.9.2), but rather what is called the *quantum master equation*

$$(S,S) - 2i\Delta S = 0 \qquad \text{at} \qquad \chi^\ddagger = \delta\Psi/\delta\chi. \tag{15.9.35}$$

[18] For χ^m and χ^n both bosonic this is because $\delta S/\delta\chi^\ddagger_m$ and $\delta S/\delta\chi^\ddagger_n$ anticommute. For one of χ^m and χ^n fermionic and the other bosonic, this is because the right- and left-derivatives of S with respect to whichever χ is fermionic have opposite signs. The terms with χ^m and χ^n both fermionic are antisymmetric because for these terms $\delta^2_L\Psi/\delta\chi^m\delta\chi^n$ is antisymmetric.

In cgs units a factor $1/\hbar$ would accompany each factor of $S[\chi,\chi^\ddagger]$, so the second term in Eq. (15.9.35) would have a coefficient $-2i\hbar$ in place of $-2i$. Thus whenever the quantum master equation (15.9.35) is satisfied, the term in S of zeroth order in $\hbar$ satisfies the original master equation (15.9.2). Usually it is easy to construct an action that satisfies the classical master equation, so that the first term in Eq. (15.9.35) vanishes, and the question is then whether the second term also vanishes. The case where the quantum master equation is not satisfied by a local action is considered in Chapter 22, on anomalies.

Assuming the quantum master equation (15.9.35) to be satisfied, the change in the vacuum expectation value of an operator $\mathcal{O}[\chi]$ due to a change $\delta\Psi$ in Ψ is

$$\delta\langle\mathcal{O}\rangle = -\frac{i}{Z_\Psi}\int\left[\prod d\chi\right]\exp(iI_\Psi[\chi])\frac{\delta_R\mathcal{O}[\chi]}{\delta\chi^n}\left(-\frac{\delta_R S(\chi,\chi^\ddagger)}{\delta\chi^\ddagger_n}\right)\Bigg|_{\chi^\ddagger=\delta\Psi/\delta\chi}\delta\Psi[\chi]. \quad (15.9.36)$$

The coefficient of the exponential in the integrand in Eq. (15.9.36) is just $s\mathcal{O}[\chi]$. We see that expectation values of operators that are invariant [19] under the generalized BRST transformation (15.9.16)–(15.9.17) are unaffected by a change in the gauge-fixing fermion Ψ. Corresponding results hold for vacuum expectation values of two or more operators.

Appendix A A Theorem Regarding Lie Algebras

In this appendix we consider a general Lie algebra $\mathcal{G}$, with generators t_α and structure constants $C^\alpha{}_{\beta\gamma}$, and prove the equivalence of three conditions:

a: There exists a real symmetric positive-definite matrix $g_{\alpha\beta}$ that satisfies the invariance condition

$$g_{\alpha\beta}C^\beta{}_{\gamma\delta} = -g_{\gamma\beta}C^\beta{}_{\alpha\delta}. \quad (15.A.1)$$

(This is the condition (15.2.4) shown in Section 15.2 to be necessary on physical grounds.)

b: There is a basis for the Lie algebra (that is, a set of generators $\tilde{t}_\alpha = \mathcal{S}_{\alpha\beta}t_\beta$, with $\mathcal{S}$ a real non-singular matrix) for which the structure constants $\tilde{C}^\alpha{}_{\beta\gamma}$ are antisymmetric not only in the lower indices β and γ but in all three indices α, β and γ.

c: The Lie algebra $\mathcal{G}$ is the direct sum of commuting compact simple and $U(1)$ subalgebras $\mathcal{H}_m$.

We shall prove the equivalence of statements **a**, **b**, and **c** by showing that **a** implies **b**, **b** implies **c**, and **c** implies **a**. As a by-product, we shall also show that if these conditions are satisfied, then it is possible to choose the generators of $\mathcal{G}$ as t_{ma}, with m labelling the simple

[19] For open gauge theories there are generally no operators other than constants that are invariant under the transformations (15.9.16)–(15.9.17). Instead one should consider operators $\mathcal{O}(\chi,\chi^\ddagger)$ that are invariant under the nilpotent "quantum BRST operator" σ, defined by $\sigma\mathcal{O} = (\mathcal{O},S) - i\Delta\mathcal{O}$. Where $\mathcal{O}$ depends only on χ, this condition reduces to Eq. (15.9.36). If $\sigma\mathcal{O} = 0$ then the expectation value of $\mathcal{O}(\Psi,\delta\Psi/\delta\chi)$ is unaffected by a small change in the gauge-fixing fermion Ψ [24].

or $U(1)$ subalgebra $\mathcal{H}_m$ to which t_{ma} belongs and a labelling the individual generators in this subalgebra, and with the matrix $g_{ma,nb}$ that satisfies Eq. (15.A.1) taking the form

$$g_{ma,nb} = g_m^{-2}\delta_{mn}\delta_{ab}, \tag{15.A.2}$$

where the g_m^{-2} are arbitrary real positive constants.

First, let us assume **a**, the existence of a real symmetric positive-definite $g_{\alpha\beta}$ satisfying the invariance condition (15.A.1). We may then define new generators

$$\widetilde{t}_\alpha \equiv (g^{-1/2})_{\alpha\beta} t_\beta, \tag{15.A.3}$$

in which the existence of a real inverse square-root matrix $g^{-1/2}$ is guaranteed by the positive-definiteness of $g_{\alpha\beta}$. These satisfy a Lie algebra

$$[\widetilde{t}_\alpha, \widetilde{t}_\beta] = i\widetilde{C}_{\alpha\beta\gamma}\widetilde{t}_\gamma, \tag{15.A.4}$$

where

$$\widetilde{C}_{\gamma\alpha\beta} \equiv (g^{-1/2})_{\alpha\alpha'}(g^{-1/2})_{\beta\beta'}(g^{+1/2})_{\gamma\gamma'}C^{\gamma'}{}_{\alpha'\beta'}. \tag{15.A.5}$$

(In this basis it is convenient to drop the distinction between upper and lower indices α, β, etc., and write $\widetilde{C}_{\gamma\alpha\beta}$ in place of $\widetilde{C}^\gamma{}_{\alpha\beta}$.) Then Eq. (15.A.1) tells us that $\widetilde{C}_{\alpha\gamma\delta}$ is antisymmetric in α and γ as well as in γ and δ, and hence is totally antisymmetric, verifying **b**.

Next let us assume **b**, the existence of a basis for the Lie algebra for which the structure constants are totally antisymmetric. In this basis, the matrices $(\widetilde{t}^A_\alpha)_{\beta\gamma} = -i\widetilde{C}_{\beta\gamma\alpha}$ of the adjoint representation are imaginary and antisymmetric, and hence Hermitian. According to a general theorem about Hermitian matrices,[20] the $\widetilde{t}^A_\alpha$ are then either irreducible or totally reducible.

By an irreducible set of $N \cdot N$ matrices $\widetilde{t}^A_\alpha$ is meant a set for which there exists no subspace of dimensionality $< N$ that is left invariant by all the $\widetilde{t}^A_\alpha$—that is, no set of less than N non-zero vectors $(u_r)_\beta$ for which $(\widetilde{t}^A_\alpha)_{\beta\gamma}(u_r)_\gamma$ is for each α and r a linear combination of vectors $(u_s)_\beta$. Since the matrices $(\widetilde{t}^A_\alpha)_{\beta\gamma}$ are proportional to the structure constants in this basis, this is equivalent to the statement that there is no set of linear combinations $\mathcal{T}_r \equiv (u_r)_\gamma \widetilde{t}^A_\gamma$ that is closed under commutation with all the $\widetilde{t}^A_\alpha$, that is, for which $[\widetilde{t}^A_\alpha, \mathcal{T}_r]$ is for each α and r a linear combination of the $\mathcal{T}_s$. Such a set of matrices $\mathcal{T}_r$ would furnish the generators of an invariant subalgebra of the full Lie algebra; the absence of such a set means that the Lie algebra is simple.

[20] If a set of matrices H_α is not irreducible, then by definition there must be a set of vectors u_n that span a subspace (other than the whole space) that is left invariant by the H_α: that is, for all α and n, $H_\alpha u_n = \sum_m (C_\alpha)_{mn} u_m$. In this case we can adopt a basis consisting of the vectors u_n together with a set v_k that span the space orthogonal to all the u_n. If the H_α are Hermitian then $(u_m, H_\alpha v_k) = \sum_n (C_\alpha)_{mn}(u_n, v_k) = 0$, so the space spanned by the v_k is also invariant under the H_α: $H_\alpha v_k = \sum_l (D_\alpha)_{lk} v_l$. In this basis the matrices H_α are simultaneously reduced to a block-diagonal form:

$$H_\alpha = \begin{pmatrix} C_\alpha & 0 \\ 0 & D_\alpha \end{pmatrix}.$$

Continuing in this way, we can completely reduce the matrices H_α to a block-diagonal form with irreducible matrices in the blocks.

By a totally reducible set of matrices $\widetilde{t}^A_\alpha$ is meant one that by a suitable choice of basis may be written as block-diagonal supermatrices

$$(\widetilde{t}^A_\alpha)_{ma,nb} = [\widetilde{t}^{A(m)}_\alpha]_{ab}\delta_{mn}, \tag{15.A.6}$$

where the submatrices $\widetilde{t}^{A(m)}_\alpha$ are either irreducible or vanish.[21] Adopting this basis also for the Lie algebra itself, the structure constants are then

$$\widetilde{C}_{\ell c,ma,nb} = i(\widetilde{t}^A_{\ell c})_{ma,nb} = i(\widetilde{t}^{A(m)}_{\ell c})_{ab}\delta_{mn}. \tag{15.A.7}$$

But since this is totally antisymmetric, and proportional to δ_{mn}, it must also be proportional to $\delta_{\ell n}$ and $\delta_{\ell m}$:

$$\widetilde{C}_{\ell c,ma,nb} = \delta_{\ell n}\delta_{\ell m}C^{(\ell)}_{cab}. \tag{15.A.8}$$

In other words, for any representation $t^{(m)}_a \equiv t_{ma}$ of the Lie algebra in this basis, we have

$$[t^{(m)}_a, t^{(n)}_b] = i\delta_{mn}C^{(m)}_{cab}t^{(m)}_c, \tag{15.A.9}$$

with $C^{(m)}_{cab}$ real and totally antisymmetric in the indices a, b, c. The fact that it is possible to construct a basis in which the generators fall into sets $t^{(m)}$, with the commutators of the generators in one set with each other given by a linear combination of generators in that set, and with all members of one set commuting with all members of any other set, is what we mean when we say that the Lie algebra is a *direct sum* of the subalgebras $t^{(m)}$. For each m, the set of matrices of the adjoint representation $t^{(m)A}$ is either irreducible or zero, corresponding to a subalgebra that is either simple or consists of so-called $U(1)$ generators, which commute with all generators of the whole algebra.

We have thus shown that the most general Lie algebra with totally antisymmetric real structure constants is a direct sum of one or more simple and/or $U(1)$ Lie algebras. Furthermore, the simple subalgebras are compact, in the sense that each matrix $-C^{(m)}_{acd}C^{(m)}_{bdc}$ is positive-definite, because for any real vector u_a, $-C^{(m)}_{acd}C^{(m)}_{bdc}u_a u_b = \sum_{cd}[\sum_a u_a C^{(m)}_{acd}]^2$ is a sum of positive quantities, which cannot vanish unless $u_a = 0$, since for $u_a \neq 0$ the condition that $\sum_a u_a C^{(m)}_{acd} = 0$ would imply that $\sum_a u_a t^{(m)}_a$ is itself an invariant Abelian subalgebra, in contradiction with the fact that the $t^{(m)}_a$ form a simple Lie algebra. This completes the proof of **c**.

Finally, let us assume **c**, that we have a Lie algebra that is the direct sum of a set of simple or $U(1)$ Lie algebras; that is, in some basis $t^{(m)}_a = \mathcal{S}_{ma,\alpha}t_\alpha$ with real non-singular $\mathcal{S}$, we have

$$[t^{(m)}_a, t^{(n)}_b] = i\delta_{mn}C^{(m)c}{}_{ab}t^{(m)}_c,$$

where each subalgebra $t^{(m)}$ is either simple or commutes with everything. We furthermore assume that the simple subalgebras are compact, in the sense that the matrices

$$g^{(m)}_{ab} \equiv -C^{(m)c}{}_{ad}C^{(m)d}{}_{bc} \tag{15.A.10}$$

[21] Here m and n label the blocks along the main diagonal, and a and b label rows and columns within these blocks. Also, m is not summed, and the range of the indices a, b, etc. in general depends on m.

are positive-definite. To construct a real positive-definite matrix $g_{\alpha\beta}$ satisfying Eq. (15.A.1), in the basis of generators $t_{ma} = t_a^{(m)}$, we take

$$g_{ma,nb} = g_{ab}^{(m)} \delta_{mn}, \tag{15.A.11}$$

where $g_{ab}^{(m)}$ is taken as the matrix (15.A.10) when $t^{(m)}$ is a simple subalgebra, and is taken as an arbitrary real symmetric positive-definite matrix when $t^{(m)}$ is a direct sum of one or more $U(1)$ subalgebras. The matrix (15.A.11) is obviously real, symmetric, and positive-definite because each $g_{ab}^{(m)}$ is. To check the requirement (15.A.1), recall the Jacobi identity for the structure constants:

$$C^{(m)c}{}_{ad} C^{(m)d}{}_{be} + C^{(m)c}{}_{bd} C^{(m)d}{}_{ea} + C^{(m)c}{}_{ed} C^{(m)d}{}_{ab} = 0 \tag{15.A.12}$$

and contract with $C^{(m)e}{}_{fc}$. After renaming the summation indices in the third term so that $c \longrightarrow d \longrightarrow e \longrightarrow c$ and using Eq. (15.A.10), the result for the simple subalgebras may be written:

$$g_{df}^{(m)} C^{(m)d}{}_{ab} = C^{(m)c}{}_{ad} C^{(m)d}{}_{be} C^{(m)e}{}_{cf} - C^{(m)c}{}_{fd} C^{(m)d}{}_{be} C^{(m)e}{}_{ca}.$$

The important point is that this shows that the left-hand side is antisymmetric in a and f:

$$g_{df}^{(m)} C^{(m)d}{}_{ab} = -g_{da}^{(m)} C^{(m)d}{}_{fb}. \tag{15.A.13}$$

The same result holds trivially for the $U(1)$ subalgebras, where the structure constants vanish. The symmetry condition (15.A.1) follows immediately from Eq. (15.A.13), thus completing the proof of **a**. This concludes our proof of the equivalence of statements **a**, **b**, and **c**.

Now we come back to the matrix $g_{\alpha\beta}$. With totally antisymmetric structure constants, the invariance condition (15.2.4) may be expressed as the statement that this matrix commutes with all the matrices in the adjoint representation of the Lie algebra

$$[g, \widetilde{t}_\gamma^A] = 0. \tag{15.A.14}$$

We have seen that all $(\widetilde{t}_\gamma^A)_{\alpha\beta}$ can be put in block-diagonal form, with irreducible (or zero) submatrices along the main diagonal. A well-known theorem [25] tells us that then $g_{\alpha\beta}$ must also be block-diagonal, with blocks of the same size and position as in the $\widetilde{t}_\gamma^A$, and with the submatrix in each block proportional to the unit matrix. (Where two of the submatrices in the $\widetilde{t}_\gamma^A$ are equivalent, in the sense that they can be related by a similarity transformation, it may be necessary to make a suitable change of basis in order to bring the submatrices of $g_{\alpha\beta}$ into a form proportional to unit matrices.) In the notation of Eq. (15.A.11), the metric is then given by Eq. (15.A.2).

Appendix B The Cartan Catalog

We present here without proof the complete catalog of simple Lie algebras, worked out in its final form by E. Cartan [26]. These will be presented here in their 'compact' form—that is, with generators that can be faithfully represented by finite-dimensional Hermitian matrices.

The Lie algebras will be labelled with a subscript $n \geq 1$ indicating their 'rank'—the number of independent commuting linear combinations of generators.

A_n: This is the algebra of the special unitary group $SU(n+1)$, the group of all unitary ($U^\dagger = U^{-1}$) unimodular ($\mathrm{Det}\, U = 1$) matrices in $n+1$ dimensions. Any such matrix that is infinitesimally close to the identity may be expressed as

$$U = \mathbf{1} + iH,$$

with infinitesimal H satisfying the conditions

$$H^\dagger = H, \qquad \mathrm{Tr}\, H = 0,$$

so A_n is the Lie algebra of all Hermitian traceless matrices in $n+1$ dimensions. Any set of commuting Hermitian matrices may be simultaneously diagonalized, and the maximum number of independent diagonal traceless matrices in $n+1$ dimensions is n, so this is the rank of A_n. Any Hermitian matrix in $n+1$ dimensions has $(n+1)^2$ independent real parameters ($n+1$ real numbers on the main diagonal, and $n(n+1)/2$ complex numbers above the main diagonal, equal to the complex conjugates of those below it), and one of these is eliminated by the tracelessness condition, so the dimensionality of A_n is

$$d(A_n) = (n+1)^2 - 1 = n(n+2).$$

All of the A_n are simple.

B_n: This is the algebra of the unitary orthogonal group $O(2n+1)$, consisting of all unitary ($\mathcal{O}^\dagger = \mathcal{O}^{-1}$) and orthogonal ($\mathcal{O}^T = \mathcal{O}^{-1}$) and hence real matrices in $2n+1$ dimensions. (The restriction to unitary matrices is sometimes indicated by calling the group $UO(2n+1)$.) Any such matrix $\mathcal{O}$ that is infinitesimally close to the identity may be expressed as

$$\mathcal{O} = \mathbf{1} + iA,$$

where A is an infinitesimal matrix satisfying the conditions

$$A^* = -A = A^T.$$

(It would make no difference here if we restricted ourselves to the subgroup $SO(2n+1)$, for which $\mathcal{O}$ is subject to the further condition $\mathrm{Det}\, \mathcal{O} = 1$, since any orthogonal matrix $\mathcal{O}$ that is close to the identity would have $\mathrm{Det}\, \mathcal{O} = 1$ anyway.) Any set of commuting imaginary antisymmetric ($2n+1$)-dimensional matrices may (by a common orthogonal transformation) be put in the form of a supermatrix

$$\begin{pmatrix} a_1\sigma_2 & 0 & \cdots & 0 & 0 \\ 0 & a_2\sigma_2 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & a_n\sigma_2 & 0 \\ 0 & 0 & \cdots & 0 & 0 \end{pmatrix}$$

where σ_2 is the usual $2 \cdot 2$ matrix

$$\sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

and $a_1, \ldots, a_n$ are real. The rank of B_n is thus evidently n. An imaginary antisymmetric matrix is completely specified by the imaginary numbers above the real diagonal, so its dimensionality is

$$d(B_n) = \frac{(2n+1)(2n)}{2} = n(2n+1).$$

All of the B_n are simple.

There is an alternative definition of $O(N)$ that will help in understanding the motivation for the next large set of simple Lie algebras. Instead of defining $O(N)$ as the group of N-dimensional real matrices that satisfy the orthogonality condition $\mathcal{O}^T\mathcal{O} = \mathbf{1}$, it can just as well be defined as the group of N-dimensional real matrices $\mathcal{M}$ that satisfy the condition

$$\mathcal{M}^T \mathcal{P} \mathcal{M} = \mathcal{P},$$

where $\mathcal{P}$ is an arbitrary positive-definite real symmetric matrix. This is because any such $\mathcal{P}$ may always be expressed as $\mathcal{P} = \mathcal{R}^T\mathcal{R}$ with $\mathcal{R}$ some real non-singular matrix, so there is a similarity transformation that takes any matrix $\mathcal{M}$ satisfying the above condition into a real orthogonal matrix, $\mathcal{R}\mathcal{M}\mathcal{R}^{-1}$. Thus we may let $\mathcal{P}$ be various different real symmetric positive-definite matrices without changing the group.

C_n: This is the algebra of the unitary symplectic group $USp(2n)$, the group of unitary matrices $\mathcal{M}$ that leave an antisymmetric non-singular matrix $\mathcal{A}$ invariant:

$$\mathcal{M}^T \mathcal{A} \mathcal{M} = \mathcal{A},$$

$$\mathcal{A}^T = -\mathcal{A}, \qquad \mathrm{Det}\,\mathcal{A} \neq 0.$$

(Note that in d dimensions, $\mathrm{Det}\,\mathcal{A} = \mathrm{Det}\,\mathcal{A}^T = (-)^d\,\mathrm{Det}\,\mathcal{A}$, so if d is odd then $\mathrm{Det}\,\mathcal{A}$ must vanish. Thus there is no $USp(d)$ unless d is even.) Any such antisymmetric non-singular (perhaps complex) matrix $\mathcal{A}$ may be written in the standard form

$$\mathcal{A} = \mathcal{R}^T \mathcal{A}_0 \mathcal{R},$$

where $\mathcal{R}$ is unitary and $\mathcal{A}_0$ is the supermatrix:

$$\mathcal{A}_0 = \begin{pmatrix} 0 & \mathbf{1} \\ -\mathbf{1} & 0 \end{pmatrix}.$$

(Alternatively, we could take $\mathcal{A}_0$ as a block-diagonal supermatrix, with σ_1s along the main diagonal.) Hence $USp(2n)$ may be described as the group of unitary matrices $\mathcal{S}$ satisfying

$$\mathcal{S}^T \mathcal{A}_0 \mathcal{S} = \mathcal{A}_0$$

since by a unitary transformation any such $\mathcal{S}$ can be transformed into a unitary matrix $\mathcal{M} = \mathcal{R}^{-1}\mathcal{S}\mathcal{R}$ that satisfies the previous condition $\mathcal{M}^T\mathcal{A}\mathcal{M} = \mathcal{A}$. Any such matrix $\mathcal{S}$ that is infinitesimally far from the identity may be written

$$\mathcal{S} = \mathbf{1} + i\mathcal{H},$$

where $\mathcal{H}$ is an infinitesimal matrix satisfying the conditions

$$\mathcal{H}^\dagger = \mathcal{H}, \qquad \mathcal{H}^T \mathcal{A}_0 + \mathcal{A}_0 \mathcal{H} = 0.$$

The most general $2n$-dimensional matrix $\mathcal{H}$ satisfying these conditions may be written as a supermatrix

$$\mathcal{H} = \begin{pmatrix} \mathcal{A} & \mathcal{B} \\ \mathcal{B}^* & -\mathcal{A}^* \end{pmatrix},$$

where the n-dimensional complex submatrices $\mathcal{A}$, $\mathcal{B}$ satisfy

$$\mathcal{A}^\dagger = \mathcal{A}, \qquad \mathcal{B}^T = \mathcal{B}.$$

A maximal set of commuting generators have $\mathcal{A}$ diagonal and $\mathcal{B}$ zero, and so are of the form

$$\mathcal{H} = \mathrm{diag}(a_1, \ldots, a_n, -a_1, \ldots, -a_n)$$

with $a_1, \ldots, a_n$ real. The rank of C_n is thus evidently n. The dimensionality of C_n is the number n^2 of independent real parameters in the Hermitian matrix $\mathcal{A}$, plus the number $2n(n+1)/2$ of independent real parameters in the complex symmetric matrix $\mathcal{B}$

$$d(C_n) = n^2 + 2 \cdot n(n+1)/2 = n(2n+1).$$

All of the C_n algebras are simple.
D_n: This is the algebra of the unitary orthogonal group $O(2n)$, consisting of all unitary orthogonal matrices in $2n$ dimensions. The discussion of B_n can be carried over to D_n, except that here any set of commuting generators can be put in the form

$$\begin{pmatrix} a_1\sigma_2 & 0 & \cdots & 0 \\ 0 & a_2\sigma_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_n\sigma_2 \end{pmatrix}$$

so the rank is still n. Also, the dimensionality here is

$$d(D_n) = \frac{(2n)(2n-1)}{2} = n(2n-1).$$

All of the D_n are simple except for D_1, which is the Abelian algebra with just one generator, and D_2, which is the direct sum $B_1 + B_1$.
Exceptional Lie Algebras: In addition to the above classical Lie algebras, there are just five special cases, the algebras $G_2(d = 14)$; $F_4(d = 52)$; $E_6(d = 78)$; $E_7(d = 133)$; $E_8(d = 248)$.

Not all of the classical Lie algebras are really different. There are just four isomorphisms

$$A_1 = B_1 = C_1, \qquad C_2 = B_2, \qquad A_3 = D_3.$$

These correspond to isomorphisms among Lie groups of which these are the algebras. However, isomorphisms like $B_1 = A_1$, $B_2 = C_2$, and $D_3 = A_3$ do not mean that $SO(3)$ is isomorphic to $SU(2)$, or that $SO(5)$ is isomorphic to $USp(4)$, or that $SO(6)$ is isomorphic to $SU(4)$. Instead, $SU(2)$, $USp(4)$, and $SU(4)$ are the simply connected covering groups of $SO(3)$, $SO(5)$, and $SO(6)$. (Covering groups are discussed in Chapter 2.) Nevertheless, the isomorphisms of the Lie algebras make it especially easy to construct the double-valued fundamental spinor representations of $SO(3)$, $SO(5)$, and $SO(6)$; they are just the defining representations of $SU(2)$, $USp(4)$, and $SU(4)$, respectively. Also $SO(4)$ is isomorphic to $SO(3)\cdot SO(3)$, so its double-valued spinor representation is just the defining representation of $SU(2)\cdot SU(2)$. For $d \geq 7$ the double-valued spinor representations of $SO(d)$ must be constructed by other means. The simplest technique uses Clifford algebras, discussed in Section 5.4.

Problems

1. Derive the Bianchi identity

$$D_\mu F_{\alpha\nu\lambda} + D_\nu F_{\alpha\lambda\mu} + D_\lambda F_{\alpha\mu\nu} = 0.$$

2. Suppose we use generalized Coulomb gauge in a non-Abelian gauge theory, taking the gauge-fixing function as $f_\alpha = \boldsymbol{\nabla}\cdot\mathbf{A}_\alpha$. Derive the ghost Lagrangian. What is the ghost propagator? (Take $B[f] = \exp(-i\int d^4x\, f_\alpha f_\alpha/2\xi)$.)

3. Suppose that in electrodynamics we use a gauge-fixing function $f = \partial_\mu A^\mu + cA_\mu A^\mu$, with c an arbitrary constant. Derive the ghost Lagrangian.
(Take $B[f] = \exp(-i\int d^4x\, f_\alpha f_\alpha/2\xi)$.) What is the ghost propagator?

4. Show that there is no simple Lie algebra with just four generators.

5. Show that if a field $\psi_\ell(x)$ belonging to a representation of a gauge group with generator matrices t_α varies along a path $x^\mu = x^\mu(\tau)$ according to the differential equation

$$\frac{d\psi(\tau)}{d\tau} = it_\alpha\psi(\tau)A_{\alpha\mu}(x(\tau))\frac{dx^\mu(\tau)}{d\tau},$$

(with both $\psi_\ell(x)$ and $A_{\alpha\mu}(x)$ classical c-number fields) then the change in ψ around any small closed path γ around a point X^μ is proportional to

$$t_\alpha\psi(X)F_{\alpha\mu\nu}(X)\oint_\gamma x^\mu(\tau)\frac{dx^\nu(\tau)}{d\tau}\,d\tau.$$

Find the constant of proportionality. Use this result to show that if $F_{\alpha\mu\nu}(x)$ vanishes everywhere then it is possible by a gauge transformation to make $A_{\alpha\mu}(x)$ vanish throughout at least a finite region. (Hint: Follow the analogous argument in electrodynamics, or in general relativity, as in Ref. [6].)

6. Applying path-integral methods to a general non-Abelian gauge theory, calculate the propagator of the gauge field $A_{\alpha\mu}(x)$ if we choose a gauge-fixing functional $f_\alpha = n_\mu A^\mu_\alpha$, with n_μ arbitrary constants. (Take $B[f] = \exp(-i\int d^4x\, f_\alpha f_\alpha/2\xi)$.) What is the ghost Lagrangian? What is the ghost propagator? What is the ghost interaction vertex?

7. Suppose we adopt BRST invariance instead of gauge invariance as a fundamental physical principle. Derive the most general Lagrangian density constructed from sums of products of fields and field derivatives of dimensionality $(\text{mass})^d$ with $d \leq 4$, constructed from $A_{\alpha\mu}$, ω_α, ω^*_α, h_α, and/or their derivatives, that is invariant (up to total derivatives) under Lorentz transformations, ghost number phase transformations ($\omega_\alpha \longrightarrow e^{i\theta}\omega_\alpha$, $\omega^*_\alpha \longrightarrow e^{-i\theta}\omega^*_\alpha$), global gauge transformations (ϵ_α constant), and BRST invariance.

8. Show that the antibracket satisfies the symmetry condition (15.9.15) and the Jacobi identity (15.9.21).

9. Show that if a functional O satisfies the condition $(O, S) = i\Delta S$ and the action S satisfies the quantum master equation then the quantum average $\langle O \rangle$ is independent of the gauge-fixing functional Ψ.

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$$\bar{s} = \omega^{*A}\delta_A\phi^r \frac{\delta_L}{\delta\phi^r} - \frac{1}{2}\omega^{*B}\omega^{*C} f^A{}_{BC}\frac{\delta_L}{\delta\omega^{*A}} - \omega^{*B}h^C f^A{}_{BC}\frac{\delta}{\delta h^A} + \left[-f^A{}_{BC}\omega^{*B}\omega^C + h^A\right]\frac{\delta_L}{\delta\omega^A}.$$

It is straightforward to show that this transformation is nilpotent:

$$\bar{s}^2 = 0.$$

Also, the BRST and anti-BRST transformations anticommute. Furthermore, there is a cohomology theorem (unpublished) for the anti-BRST transformations, like that for the BRST transformations: the most general functional I of ϕ^r, ω^A, ω^{*A}, and h^A that satisfies the anti-BRST invariance condition $\bar{s}I = 0$ and has ghost number zero is of the form

$$I[\phi, \omega, \omega^*, h] = I_0[\phi] + \bar{s}\bar{\Psi}[\phi, \omega, \omega^*, h].$$

(Just use the Hodge operator $\bar{t} \equiv \omega^A \delta/\delta h^A$ in place of t.) Anti-BRST symmetry can be an aid in enumerating possible terms in Lagrangians and Greens functions, but I do not know of any case where it is indispensable.

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16 External Field Methods

It is often useful to consider quantum field theories in the presence of a classical external field. One reason is that in many physical situations, there really is an external field present, such as a classical electromagnetic or gravitational field, or a scalar field with a non-vanishing vacuum expectation value. (As we shall see in Chapter 19, such scalar fields can play an important role in the spontaneous breakdown of symmetries of the Lagrangian.) But even where there is no actual external field present in a problem, some calculations are greatly facilitated by considering physical amplitudes in the presence of a fictitious external field. This chapter will show that it is possible to take all multiloop effects into account by summing 'tree' graphs whose vertices and propagators are taken from a *quantum effective action*, which is nothing but the one-particle-irreducible connected vacuum–vacuum amplitude in the presence of an external field. It will turn out in the next chapter that this provides an especially handy way both of completing the proof of the renormalizability of non-Abelian gauge theories begun in Chapter 15, and of calculating the charge renormalization factors that we need in order to establish the crucial property of asymptotic freedom in quantum chromodynamics.

16.1 The Quantum Effective Action

Consider a quantum field theory with action $I[\phi]$, and suppose we 'turn on' a set of classical currents $J_r(x)$ coupled to the fields $\phi^r(x)$ of the theory. The complete vacuum vacuum amplitude in the presence of these currents is then

$$\begin{aligned} Z[J] &\equiv \langle \Omega;\, \text{out} \| \Omega;\, \text{in} \rangle_J \\ &= \int \left[\prod_{s,y} d\phi^s(y) \right] \exp\left(iI[\phi] + i \int d^4x\, \phi^r(x) J_r(x) + \epsilon \text{ terms} \right). \end{aligned} \tag{16.1.1}$$

(The fields $\phi^r(x)$ need not be scalars. They might even be fermionic, though until Section 16.4 we shall not bother to keep track of the signs that would appear in that case.) The Feynman rules for calculating $Z[J]$ are just the same as for calculating the vacuum–vacuum amplitude $Z[0]$ in the absence of the external current, except that the Feynman diagrams now contain vertices of a new kind, to which a *single* ϕ^r-line is attached. Such a vertex labelled with a coordinate x contributes a 'coupling' factor $iJ_r(x)$ to the integrand of the position-space Feynman amplitude. Equivalently, we could say that in the expansion of $Z[J]$ in powers of J, the coefficient of the term proportional to $iJ_r(x)\, iJ_s(y) \cdots$ is just the sum of diagrams with external lines (including propagators) corresponding to the fields $\phi^r(x)$, $\phi^s(y)$, etc. In particular, the first derivative gives the vacuum matrix element of the quantum mechanical operator $\Phi^r(x)$ corresponding to $\phi^r(x)$:

$$\begin{aligned} \left[\frac{\delta}{\delta J_r(y)} Z[J] \right]_{J=0} &= i \int \left[\prod_{r,x} d\phi^r(x) \right] \phi^r(y) \exp\{iI[\phi] + \epsilon \text{ terms}\} \\ &= i \langle \Omega;\, \text{out} \| \Phi^r(x) \| \Omega;\, \text{in} \rangle_{J=0}. \end{aligned} \tag{16.1.2}$$

We sometimes work with functionals $Z[J]$ defined by (16.1.1) where the $\phi^r(x)$ are not elementary fields (that is, fields appearing in the action) but products of such fields. Where $\phi^r(x)$ is a product of N elementary fields, the new vertices in the Feynman rules for $Z[J]$ have N lines attached. Some of the results of this chapter (including Eq. (16.1.2)) apply in this case, but where Feynman diagrams are involved, it will be tacitly assumed that the $\phi^r(x)$ are elementary.

Now, $Z[J]$ is given by the sum of *all* vacuum–vacuum amplitudes in the presence of the current J, including disconnected as well as connected diagrams, but not counting as different those diagrams that differ only by a permutation of vertices in the same or different connected subdiagrams. A general diagram that consists of N connected components will contribute to $Z[J]$ a term equal to the product of the contributions of these components, divided by the number $N!$ of permutations of vertices that merely permute all the vertices in one connected component with all the vertices in another.[22] Hence, the sum of all graphs is

$$Z[J] = \sum_{N=0}^{\infty} \frac{1}{N!}\big(iW[J]\big)^N = \exp\big(iW[J]\big), \tag{16.1.3}$$

where $iW[J]$ is the sum of all *connected* vacuum–vacuum amplitudes, again not counting as different those diagrams that differ only by a permutation of vertices.

For many purposes, it will be useful to go one step further, and in place of $W[J]$ work with the sum of all connected *one-particle-irreducible* graphs. (A one-particle-irreducible graph is one that cannot be disconnected by cutting through any one internal line.) We can give a formal expression for this sum as follows. First, define $\phi^r_J(x)$ as the vacuum expectation value of the operator $\Phi^r(x)$ in the presence of the current J:

$$\phi^r_J(x) \equiv \frac{\langle\Omega;\,\mathrm{out}\|\Phi^r(x)\|\Omega;\,\mathrm{in}\rangle_J}{\langle\Omega;\,\mathrm{out}\|\Omega;\,\mathrm{in}\rangle_J} = -\frac{i}{Z[J]}\frac{\delta}{\delta J_r(x)}Z[J] \tag{16.1.4}$$

or in terms of the sum of connected graphs

$$\phi^r_J(x) = \frac{\delta}{\delta J_r(x)}W[J]. \tag{16.1.5}$$

This formula can be inverted. Define $J_{\phi r}(x)$ as the current for which (16.1.4) has a prescribed value $\phi^r(x)$:

$$\phi^r_J(x) = \phi^r(x) \quad \text{if} \quad J_r(x) = J_{\phi r}(x).$$

The *quantum effective action* [1] $\Gamma[\phi]$ is defined (as a functional of ϕ, not J) by the Legendre transformation

$$\Gamma[\phi] \equiv -\int d^4x\, \phi^r(x) J_{\phi r}(x) + W[J_\phi]. \tag{16.1.6}$$

We will soon show that $\Gamma[\phi]$ is the sum of all connected one-particle-irreducible graphs in the presence of the current J_ϕ. However, let us first take a look at another aspect of its physical significance.

[22] The contribution of a Feynman diagram with N connected components containing $n_1, n_2, \ldots, n_N$ vertices will be proportional to a factor $1/(n_1 + \cdots + n_N)!$ from the Dyson expansion, and a factor $(n_1 + \cdots + n_N)!/N!$ equal to the number of permutations of these vertices, counting as identical those permutations that merely permute all the vertices in one component with all the vertices in another.

Note that the variational derivative of $\Gamma[\phi]$ is

$$\begin{aligned}\frac{\delta\Gamma[\phi]}{\delta\phi^s(y)} &= -\int d^4x\,\phi^r(x)\frac{\delta J_{\phi r}(x)}{\delta\phi^s(y)} - J_{\phi s}(y)\\ &\quad + \int d^4x \left[\frac{\delta W[J]}{\delta J_r(x)}\right]_{J=J_\phi}\frac{\delta J_{\phi r}(x)}{\delta\phi^s(y)}\end{aligned}$$

or, using Eq. (16.1.5),

$$\frac{\delta\Gamma[\phi]}{\delta\phi^s(y)} = -J_{\phi s}(y). \tag{16.1.7}$$

Thus $\Gamma[\phi]$ is the 'effective action', in the sense that the possible values for the external fields $\phi^s(y)$ in the *absence* of a current J are given by the stationary 'points' of Γ:

$$\frac{\delta\Gamma[\phi]}{\delta\phi^s(y)} = 0 \quad \text{for} \quad J = 0. \tag{16.1.8}$$

This may be compared with the classical field equations, which just require that the actual action $I[\phi]$ be stationary. Hence Eq. (16.1.8) may be regarded as the equation of motion for the external field ϕ, taking quantum corrections into account.

Not only does $\Gamma[\phi]$ provide the quantum-corrected field equations; it is an effective action in the sense that $iW[J]$ may be calculated as a sum of connected *tree* graphs for the vacuum–vacuum amplitude, with vertices calculated as if the action were $\Gamma[\phi]$ instead of $I[\phi]$. By a tree graph is meant one that becomes disconnected if we cut any internal line. All the effects of loop diagrams are taken into account by using $\Gamma[\phi]$ in place of $I[\phi]$.

To see this [2], let us consider the quantity $W_\Gamma[J,g]$ that we would get instead of $W[J]$, if we used an action $g^{-1}\Gamma[\phi]$ in place of $I[\phi]$:

$$\exp\{iW_\Gamma[J,g]\} \equiv \int \prod_{r,x} d\phi^r(x)\, \exp\left\{ig^{-1}\left[\Gamma[\phi] + \int d^4x\,\phi^r(x)J_r(x)\right] + \epsilon \text{ terms}\right\}, \tag{16.1.9}$$

with arbitrary constant g. The propagator here is the inverse of the coefficient of the term in $g^{-1}\Gamma[\phi]$ quadratic in ϕ, and is hence proportional to g, while all vertices make a contribution proportional to $1/g$, so a graph with V vertices (including those produced by the current J) and I internal lines (including those attached to the J vertices) is proportional to g^{I-V}. For any connected graph, the number of loops is $L = I - V + 1$, so the L-loop term in $W_\Gamma[J,g]$ has the g dependence

$$\big(W_\Gamma[J,g]\big)_{L\text{ loops}} \propto g^{L-1}. \tag{16.1.10}$$

Equivalently, we may write (at least formally)

$$W_\Gamma[J,g] = \sum_{L=0}^{\infty} g^{L-1}W_\Gamma^{(L)}[J], \tag{16.1.11}$$

where (as can be seen by setting $g = 1$), the quantity $W_\Gamma^{(L)}[J]$ is the L-loop contribution to the connected vacuum amplitude $W[J,1]$ that we would obtain if we used $\Gamma[\phi]$ (without a factor g) in place of the action $I[\phi]$.

Now, we are specially interested here in the sum of tree graphs, those without loops, calculated with vertices and propagators calculated as if the action were $\Gamma[\phi]$ instead of $I[\phi]$. In our present notation, this is $W_\Gamma^{(0)}[J]$. In order to isolate the $L = 0$ term in Eq. (16.1.11), consider the limit $g \to 0$. In this limit, the path integral (16.1.9) is dominated by the point of stationary phase,

$$\exp\{iW_\Gamma[J,g]\} \propto \exp\left\{ig^{-1}\left[\Gamma[\phi_J] + \int d^4x\, \phi_J^r(x) J_r(x)\right]\right\}, \tag{16.1.12}$$

where, because of its definition as the field produced by the current J, the field ϕ_J is the stationary point of the exponent, in the sense that

$$\left.\frac{\delta\Gamma[\phi]}{\delta\phi^r(x)}\right|_{\phi=\phi_J} = -J_r(x). \tag{16.1.13}$$

The proportionality factor in Eq. (16.1.12) is in general a functional of J, but it is a power series in g starting with terms of order g^0. Hence taking the logarithm of both sides and isolating the terms of order g^{-1} gives

$$W_\Gamma^{(0)}[J] = \Gamma[\phi_J] + \int d^4x\, \phi_J^r(x) J_r(x). \tag{16.1.14}$$

By setting $\phi = \phi_J$ in Eq. (16.1.6), we see that the right-hand side of Eq. (16.1.14) is just $W[J]$:

$$W_\Gamma^{(0)}[J] = W[J]. \tag{16.1.15}$$

To recapitulate, this says that $W[J]$ may be calculated by using $\Gamma[\phi]$ in place of $I[\phi]$ (the subscript Γ) and keeping only tree (0-loop) graphs:

$$iW[J] = \int_{\substack{\text{CONNECTED}\\\text{TREE}}} \left[\prod_{r,x} d\phi^r(x)\right] \exp\left\{i\Gamma[\phi] + i\sum_r \int \phi^r(x) J_r(x)\, d^4x\right\}. \tag{16.1.16}$$

Now, any connected graph for $iW[J]$ can be regarded as a tree, whose vertices consist of one-particle-irreducible subgraphs. Thus in order for Eq. (16.1.16) to be correct, $i\Gamma[\phi]$ must be the sum of all one-particle-irreducible connected graphs with arbitrary numbers of external lines, each external line corresponding to a factor ϕ rather than a propagator or wave function. For this reason, the coefficients in an expansion of $\Gamma[\phi]$ in powers of fields and their derivatives around some fixed field ϕ_0 may be regarded as *renormalized* coupling constants, with the renormalization 'point' specified by ϕ_0 rather than by some set of momenta.

Equivalently, $i\Gamma[\phi_0]$ for some fixed field $\phi_0^r(x)$ may be expressed as the sum of one-particle-irreducible graphs for the vacuum–vacuum amplitude, calculated with a shifted action $I[\phi + \phi_0]$:

$$i\Gamma[\phi_0] = \int_{\substack{\text{1PI}\\\text{CONNECTED}}} \left[\prod_{r,x} d\phi^r(x)\right] \exp\{iI[\phi + \phi_0]\}. \tag{16.1.17}$$

This is because any place where ϕ_0 appears in any of the vertices or propagators within the one-particle-irreducible graphs in Eq. (16.1.17) is also a place where an external ϕ-line could be attached. (The restriction to one-particle-irreducible graphs plays an essential role in Eq. (16.1.17); without this restriction we could shift the variable of integration, yielding an integral that would be manifestly independent of ϕ_0.) In place of Eq. (16.1.17), it is often convenient to write

$$\exp\{i\Gamma[\phi_0]\} = \int_{1\text{PI}} \left[\prod_{r,x} d\phi^r(x)\right] \exp\{iI[\phi + \phi_0]\}, \tag{16.1.18}$$

in which we evaluate the path integral including all graphs, connected or not, in which each connected component is one-particle-irreducible.

* * *

This formalism provides a simple method of summing tree graphs. As one example, consider the relation between the complete two-point function $\Delta^{rx,sy}$ and its one-particle-irreducible part $\Pi_{rx,sy}$. From Eqs. (16.1.5) and (16.1.7), we find

$$\Delta^{rx,sy} \equiv \frac{\delta^2 W[J]}{\delta J_r(x)\,\delta J_s(y)} = \frac{\delta \phi^r_J(x)}{\delta J_s(y)}, \tag{16.1.19}$$

$$\Pi_{rx,sy} \equiv \frac{\delta^2 \Gamma[\phi]}{\delta \phi^r(x)\,\delta \phi^s(y)} = -\frac{\delta J_{\phi r}(x)}{\delta \phi^s(y)}. \tag{16.1.20}$$

It follows immediately that the 'matrices' Δ and Π are related by

$$\Delta = -\Pi^{-1}. \tag{16.1.21}$$

This is the counterpart of the familiar relation (10.3.15) between propagators and self-energy parts, with the extra term $q^2 + m^2$ in the denominator in Eq. (10.3.15) representing the zeroth-order term in the one-particle-irreducible two-point function.

16.2 Calculation of the Effective Potential

To see how the formalism of the previous section works in practice, consider a simple example, the renormalizable theory of a single real scalar field $\phi(x)$, with action

$$I[\phi] = -\int d^4x \left[\lambda + \frac{1}{2}\partial_\rho\phi\partial^\rho\phi + \frac{1}{2}m^2\phi^2 + \frac{1}{24}g\phi^4\right]. \tag{16.2.1}$$

(We are here including a 'cosmological constant' $-\lambda$ in the Lagrangian density, for reasons that will become apparent.) Suppose for simplicity that we wish to calculate $\Gamma[\phi_0]$ for a position-independent field $\phi_0(x) = \phi_0$. Then every term in $\Gamma[\phi_0]$ will contain a factor of the volume of spacetime

$$\mathcal{V}_4 = \int d^4x = \delta^4(p - p)(2\pi)^4 \tag{16.2.2}$$

Figure 16.1. Feynman diagrams with zero, one, or two loops for the quantum effective action of the theory of a neutral scalar field ϕ with interaction ϕ^4.

arising from the momentum conservation delta-function. We will therefore write, for ϕ_0 constant

$$\Gamma[\phi_0] = -\mathcal{V}_4 V(\phi_0), \tag{16.2.3}$$

where $V(\phi_0)$ is an ordinary function, known as the *effective potential*. In this section we will calculate the effective potential to one-loop order. This was originally done by Coleman and E. Weinberg [3] in a study of spontaneous symmetry breaking, the subject of Chapters 19 and 21. The results were also used by them in one of the early applications of the renormalization group, to be described in Section 18.2.

Shifting ϕ by ϕ_0, the action in Eq. (16.1.18) is[23]

$$\begin{aligned} I[\phi+\phi_0] = & -\mathcal{V}_4\left[\lambda + \frac{1}{2}m^2\phi_0^2 + \frac{1}{24}g\phi_0^4\right] - \left[m^2\phi_0 + \frac{1}{6}g\phi_0^3\right]\int d^4x\,\phi \\ & - \int d^4x\left[\frac{1}{2}\partial_\rho\phi\partial^\rho\phi + \frac{1}{2}\mu^2(\phi_0)\phi^2\right] - \int d^4x\left[\frac{1}{6}g\phi_0\phi^3 + \frac{1}{24}g\phi^4\right], \end{aligned} \tag{16.2.4}$$

where μ^2 is the field-dependent mass

$$\mu^2(\phi_0) = m^2 + \frac{1}{2}g\phi_0^2. \tag{16.2.5}$$

Note that there now appear new interactions proportional to ϕ (which have no effect on one-particle-irreducible graphs) and also ϕ^3, as well as terms with the same structure as those in the original action.

The Feynman diagrams for $\Gamma[\phi_0]$ with up to two loops are shown in Figure 16.1. The zero-loop term in the vacuum–vacuum amplitude is just given by the constant term in $I[\phi+\phi_0]$

$$i\Gamma^{(0\text{ loop})}[\phi_0] = -i\mathcal{V}_4\left(\lambda + \frac{1}{2}m^2\phi_0^2 + \frac{g}{24}\phi_0^4\right). \tag{16.2.6}$$

The one-loop term is given by

$$\exp\left(i\Gamma^{(1\text{ loop})}[\phi_0]\right) = \int\prod_x d\phi(x)\exp\left\{-\frac{1}{2}i\int d^4x\left[\partial_\rho\phi\partial^\rho\phi + \mu^2(\phi_0)\phi^2\right] + \epsilon\text{ terms}\right\}. \tag{16.2.7}$$

[23]There are limitations on the applicability of perturbation theory for $m^2 < 0$, discussed in the following section.

We learned how to calculate such integrals in Chapter 9; the result is given by Eq. (9.A.18):

$$i\Gamma^{(1\ \text{loop})}[\phi_0] = \ln \operatorname{Det}\left(\frac{iK}{\pi}\right)^{-1/2} = -\frac{1}{2}\operatorname{Tr}\ln\left(\frac{iK}{\pi}\right), \tag{16.2.8}$$

where here

$$K_{x,y} = \left[\frac{\partial^2}{\partial x^\lambda \partial y_\lambda} + \mu^2(\phi_0) - i\epsilon\right]\delta^4(x-y). \tag{16.2.9}$$

As usual, to calculate such traces it is helpful to diagonalize the 'matrix' K by passing to momentum space:

$$\begin{aligned} K_{p,q} &= \int \frac{d^4x}{(2\pi)^2} e^{-ip\cdot x} \frac{d^4y}{(2\pi)^2} e^{iq\cdot y} K_{x,y} \\ &= \left(p^2 + \mu^2(\phi_0) - i\epsilon\right)\delta^4(p-q). \end{aligned} \tag{16.2.10}$$

The logarithm of this diagonal matrix is just the diagonal matrix with logarithms along the main diagonal:

$$\left[\ln\left(\frac{iK}{\pi}\right)\right]_{p,q} = \ln\left[\frac{i}{\pi}\left(p^2 + \mu(\phi_0) - i\epsilon\right)\right]\delta^4(p-q) \tag{16.2.11}$$

and its trace is then

$$\begin{aligned} i\Gamma^{(1\ \text{loop})}[\phi_0] &= -\frac{1}{2}\int d^4p \left[\ln\left(\frac{iK}{\pi}\right)\right]_{p,p} \\ &= -\frac{\mathcal{V}_4}{2(2\pi)^4}\int d^4p\ \ln\left(\frac{i}{\pi}\left(p^2 + \mu^2(\phi_0) - i\epsilon\right)\right). \end{aligned} \tag{16.2.12}$$

Putting together Eqs. (16.2.6) and (16.2.12), we have the effective potential to one-loop order

$$V(\phi_0) = \lambda + \frac{1}{2}m^2\phi_0^2 + \frac{g}{24}\phi_0^4 + \mathcal{I}\big(\mu^2(\phi_0)\big), \tag{16.2.13}$$

where

$$\mathcal{I}(\mu^2) \equiv -\frac{i}{2(2\pi)^4}\int d^4p\ \ln\left(\frac{i}{\pi}\left[p^2 + \mu^2 - i\epsilon\right]\right). \tag{16.2.14}$$

It is painfully obvious that this formula for the effective potential contains ultraviolet divergences. Fortunately, these are naturally absorbed into a renormalization of the parameters of the theory. Although the integral (16.2.14) is divergent, simple power-counting shows that it is made convergent by differentiating three times with respect to μ^2:

$$\mathcal{I}'''(\mu^2) = -\frac{i}{(2\pi)^4}\int d^4p\ \left(p^2 + \mu^2 - i\epsilon\right)^{-3}.$$

Once again, the $-i\epsilon$ term tells us that we must rotate the p^0 contour counterclockwise, so that $p^0 = ip_4$, with p_4 running from $-\infty$ to $+\infty$:

$$\mathcal{I}'''(\mu^2) = \frac{1}{(2\pi)^4}\int_0^\infty \frac{2\pi^2 k^3\, dk}{(k^2+\mu^2)^3} = \frac{1}{32\pi^2\mu^2}.$$

Integrating thrice, we have then

$$\mathcal{I}(\mu^2) = \frac{\mu^4 \ln \mu^2}{64\pi^2} + A + B\mu^2 + C\mu^4.$$

The constants A, B, C are not determined by this method of calculation, which is hardly a serious problem since they are obviously infinite anyway. We eliminate these constants by defining 'renormalized' values for λ, m^2, and g:

$$\begin{aligned} \lambda_R &\equiv \lambda + A + Bm^2 + Cm^4, \\ m_R^2 &\equiv m^2 + gB + 2gm^2C, \\ g_R &\equiv g + 6g^2C. \end{aligned}$$

Our final result for the potential to one-loop order is then

$$V(\phi_0) = \lambda_R + \frac{1}{2}m_R^2\phi_0^2 + \frac{g_R}{24}\phi_0^4 + \frac{\mu^4(\phi_0)\ln\mu^2(\phi_0)}{64\pi^2}, \tag{16.2.15}$$

where $\mu(\phi_0)$ is the field-dependent mass defined by Eq. (16.2.5), which to this order can be calculated using m_R and g_R in place of m and g:

$$\mu^2(\phi) = m_R^2 + \frac{1}{2}g_R\phi^2.$$

Similar results hold if the theory contains a complex spin $\frac{1}{2}$ fermion field $\psi(x)$ that interacts with the scalar ϕ. For instance, if this interaction Hamiltonian density has the simple form $G\phi\bar{\psi}\psi$, then the mass $M(\phi_0)$ of the fermion in the presence of a constant scalar background field ϕ_0 is of the form:

$$M(\phi_0) = M(0) + G\phi_0.$$

It is easy to see that in this case the potential (16.2.1) then receives an additional term:

$$\begin{aligned} V(\phi_0) = \lambda_R + \frac{1}{2}m_R^2\phi_0^2 + \frac{g_R}{24}\phi_0^4 &+ \frac{\mu^4(\phi_0)\ln\mu^2(\phi_0)}{64\pi^2} \\ &- \frac{M^4(\phi_0)\ln M^2(\phi_0)}{32\pi^2}. \end{aligned} \tag{16.2.16}$$

The numerical coefficient in the new term is twice that of the $\mu^4 \ln \mu^2$ term in Eq. (16.2.15), because of the two spin states of the fermion described by ψ, while the sign is opposite because (as shown in Chapter 9) fermionic path integrals of Gaussians give a result proportional to the determinant of the matrix coefficient in the exponent, while bosonic path integrals of Gaussians give a result proportional to the inverse of this determinant.

16.3 Energy Interpretation

The effective action $\Gamma[\phi]$ and potential $V[\phi]$ have an important interpretation in terms of the energy and energy density respectively [4]. To see this, suppose that we turn on a current $\mathcal{J}^n(\mathbf{x}, t)$ that rises smoothly from zero at $t = -\infty$ to a finite value $\mathcal{J}^n(\mathbf{x})$, and remains at that value for a long time T, after which it drops smoothly again to zero at $t = +\infty$. The effect of this perturbation is to convert the vacuum to a state with a definite energy $E[\mathcal{J}]$ (a functional of $\mathcal{J}^n(\mathbf{x})$), in which it remains for a time T, after which it returns to the vacuum. However, although the 'out' vacuum is the same physical state as the 'in' vacuum, the state vectors differ by the phase $\exp(-iE[\mathcal{J}]T)$ accumulated during the time T:

$$\langle \Omega;\, \text{out} \| \Omega;\, \text{in} \rangle_{\mathcal{J}} = \exp\bigl(-iE[\mathcal{J}]T\bigr). \tag{16.3.1}$$

Comparing with Eqs. (16.1.1) and (16.1.3), this gives

$$W[J] = -E[\mathcal{J}]T. \tag{16.3.2}$$

To see the connection between this energy and the effective action, suppose we seek the state $\|\Omega_\phi\rangle$ that minimizes the energy expectation value

$$\langle H \rangle_\Omega = \frac{\langle \Omega \| H \| \Omega \rangle}{\langle \Omega \| \Omega \rangle}, \tag{16.3.3}$$

subject to the condition that the quantum fields $\Phi_n(\mathbf{x}, t)$ have the time-independent expectation value $\phi_n(\mathbf{x})$

$$\frac{\langle \Omega \| \Phi_n(\mathbf{x}, t) \| \Omega \rangle}{\langle \Omega \| \Omega \rangle} = \phi_n(\mathbf{x}). \tag{16.3.4}$$

It is convenient also to impose the condition that $\|\Omega\rangle$ is normalized

$$\langle \Omega \| \Omega \rangle = 1. \tag{16.3.5}$$

To minimize the expectation value (16.3.3) subject to the constraints (16.3.4) and (16.3.5), we use the method of Lagrange multipliers, and instead minimize the quantity

$$\langle \Omega \| H \| \Omega \rangle - \alpha \langle \Omega \| \Omega \rangle - \int d^3x\, \beta^n(\mathbf{x}) \langle \Omega \| \Phi_n(\mathbf{x}) \| \Omega \rangle \tag{16.3.6}$$

with no constraints on $\|\Omega\rangle$. This gives

$$H \|\Omega\rangle = \alpha \|\Omega\rangle + \int d^3x\, \beta^n(\mathbf{x}) \Phi_n(\mathbf{x}) \|\Omega\rangle. \tag{16.3.7}$$

Both α and $\beta^n(\mathbf{x})$ are to be chosen in such a way as to satisfy the constraints (16.3.4) and (16.3.5) and therefore depend functionally on the prescribed expectation value $\phi_n(\mathbf{x})$.

Now, we have said that in the presence of a current $\mathcal{J}^n(\mathbf{x})$, the Hamiltonian $H - \int d^3x\, \mathcal{J}^n(\mathbf{x}) \Phi_n(\mathbf{x})$ has an eigenvalue $E[\mathcal{J}]$:

$$\left[H - \int d^3x\, \mathcal{J}^n(\mathbf{x}) \Phi_n(\mathbf{x}) \right] \|\Psi_{\mathcal{J}}\rangle = E[\mathcal{J}] \|\Psi_{\mathcal{J}}\rangle \tag{16.3.8}$$

with a normalized eigenvector $\|\Psi_{\mathcal{J}}\rangle$. Furthermore, since slowly turning on this current converts the vacuum into this energy eigenstate, we can presume that $E[\mathcal{J}]$ is the lowest energy eigenstate in the presence of this current. Therefore Eqs. (16.3.4), (16.3.5), and (16.3.7) are satisfied by

$$\|\Omega\rangle = \|\Psi_{\mathcal{J}_\phi}\rangle, \tag{16.3.9}$$

$$\alpha = E[\mathcal{J}_\phi], \tag{16.3.10}$$

$$\beta^n(\mathbf{x}) = \mathcal{J}^n_\phi(\mathbf{x}), \tag{16.3.11}$$

where $\mathcal{J}_\phi(\mathbf{x})$ is the current for which $\Phi(\mathbf{x})$ has an expectation value $\phi(\mathbf{x})$ in the state $\|\Psi_{\mathcal{J}}\rangle$.

Setting $\mathcal{J} = \mathcal{J}_\phi$ in Eq. (16.3.8) and taking the scalar product with $\|\Psi_{\mathcal{J}_\phi}\rangle$, the minimum energy of states in which the fields Φ_n are constrained to have the expectation values ϕ_n is seen to be

$$\langle H\rangle_\Omega = E[\mathcal{J}_\phi] + \int d^3x\, \mathcal{J}^n_\phi(\mathbf{x})\phi_n(\mathbf{x}). \tag{16.3.12}$$

Recalling Eq. (16.3.2) and the assumed form of $J(x)$, this is

$$\langle H\rangle_\Omega = \frac{1}{T}\left[-W[J_\phi] + \int d^4x\, J^n_\phi(x)\phi_n(x)\right] = -\frac{1}{T}\Gamma[\phi]. \tag{16.3.13}$$

As noted in the previous section, if the field $\phi(x)$ has a constant value ϕ over a large spacetime volume $\mathcal{V}_4 = \mathcal{V}_3 T$, then we may write the effective action in terms of an effective potential $V(\phi)$:

$$\Gamma[\phi] = -\mathcal{V}_3 T\, V(\phi). \tag{16.3.14}$$

In this case Eq. (16.3.13) tells us that the energy density is

$$\langle H\rangle_\Omega/\mathcal{V}_3 = V(\phi). \tag{16.3.15}$$

This is the main result: $V(\phi)$ is *the minimum of the expectation value of the energy density for all states constrained by the condition that the scalar fields Φ_n have expectation values ϕ_n*. One consequence is that in the absence of external currents the vacuum state will relax to a state in which the potential $V(\phi)$ is not only stationary, which is required by the field equations (16.1.8), but also a minimum.

This result helps to resolve a problem in the interpretation of the quantum effective potential. The Euclidean version of the path-integral formalism (described in Appendix A of Chapter 23) makes it manifest that the two-point function Δ is positive (in the matrix sense), so according to Eq. (16.1.21) the same is true of $-\Pi = \Delta^{-1}$. Together with Eq. (16.2.3), this implies that for a single scalar field the effective potential $V(\phi)$ must have a positive (or zero) second derivative with respect to ϕ. More generally, the effective potential must be convex [5]:

$$V\big(\lambda\phi_1 + (1-\lambda)\phi_2\big) \leq \lambda V(\phi_1) + (1-\lambda)V(\phi_2) \quad \text{for } 0 \leq \lambda \leq 1.$$

But inspection of Eq. (16.2.13) shows that for $m^2 < 0$ and $g > 0$ the zero-loop approximation to the effective potential for the scalar field theory with action (16.2.1) has a *negative-definite* second derivative when ϕ is between the two minima of the effective potential at $\pm\sqrt{6|m^2|/g}$. This contradiction arises because the derivation of perturbation theory implicitly relies on the existence of a stable vacuum, but when $V''(\phi) < 0$, the field ϕ is at a value of $\bar{\phi}$ where $V(\bar{\phi}) - J_\phi \bar{\phi}$ is a maximum rather than a minimum, which means that the vacuum state in the presence of the current J_ϕ is unstable.

So what is the true effective potential for this scalar field theory when $m^2 < 0$ and ϕ lies between the two minima of the potential? The result of this section is that we must find the state of minimum energy in which the expectation value of the operator Φ equals ϕ. As long as ϕ is between the two minima of the potential, we can give Φ an expectation value ϕ by taking the state as a suitable linear combination of the two states where ϕ is at the minima $\phi \simeq \pm\sqrt{6|m^2|/g}$. The energy in this state is equal to the energy at the minima, so this state clearly minimizes the energy. (Interference terms here vanish in the limit of infinite volume, for reasons explained in Section 19.1.) Thus the effective potential between the two minima of the potential is a *constant*, satisfying the requirement that it have a nonpositive second derivative. The same argument shows that in more general theories where the potential has two local minima of unequal energy, the potential between these minima is linear [6].

16.4 Symmetries of the Effective Action

In some but not all cases, the symmetries of the action $I[\phi]$ are automatically also symmetries of the effective action $\Gamma[\phi]$. For instance, in the example of Section 16.2, the action (16.2.1) has a symmetry under the discrete transformation $\phi \to -\phi$. Thus it follows from their definition that $Z[J]$ and $W[J]$ are even under the corresponding reflection $J \to -J$. Eq. (16.1.5) then shows that $\phi_{-J} = -\phi_J$, and hence $J_{-\phi} = -J_\phi$, so Eq. (16.1.6) shows that $\Gamma[\phi]$ is even under $\phi \to -\phi$. This is borne out by the one-loop result (16.2.15). The fermion-loop contribution in (16.2.16) also exhibits the symmetry under $\phi \to -\phi$ in the special case where $M(0) = 0$, because in this case the action is invariant under the combined transformation $\phi \to -\phi$, $\psi \to \gamma^5\psi$.

We encounter problems in establishing the renormalizability of a theory, unless we can show that the symmetries that we impose on the action also apply to the effective action. For instance, in the example above, if $I[\phi]$ were assumed to be even in ϕ but $\Gamma[\phi]$ turned out not to be, then the coefficients of the terms in Γ proportional to $\int d^4x\,\phi$ and $\int d^4x\,\phi^3$ would be divergent, but the symmetry of the action would not allow us to introduce counterterms to absorb these infinities.

With this motivation, let's now turn to the important class of symmetries generated by infinitesimal transformations

$$\chi^n(x) \longrightarrow \chi^n(x) + \epsilon F^n[x;\chi], \tag{16.4.1}$$

where F^n is a function of x^μ that depends functionally on χ^n. (For instance, $F^n[x;\chi]$ may be an ordinary function of the χ^n and their derivatives at the point x.) We are now using the symbol χ^n rather than ϕ^n to denote the different types of fields, to emphasize that these include not only ordinary gauge and matter fields (which in the next chapter will be denoted $\phi^r(x)$), but all other fields appearing in the gauge-fixed action, including ghost fields. We repeat that these $\chi^n(x)$ may be of any type, not necessarily scalars.

We assume that both the action and the measure are invariant under the symmetry transformation (16.4.1):

$$I[\chi+\epsilon F]=I[\chi], \tag{16.4.2}$$

$$\prod_{n,x} d\big(\chi^n(x)+\epsilon F^n[x;\chi]\big)=\prod_{n,x} d\chi^n(x). \tag{16.4.3}$$

(It is actually sufficient for only the product $(\prod_{n,x} d\chi^n(x))\exp(iI)$ to be invariant, but where this is true usually Eqs. (16.4.2) and (16.4.3) both apply.) Replacing the integration variables in Eq. (16.1.1) with $\chi^n(x)+\epsilon F^n[x;\chi]$, we have then

$$\begin{aligned}
Z[J] &= \int \left[\prod_{n,x} d\big(\chi^n(x)+\epsilon F^n[x;\chi]\big)\right] \\
&\qquad \cdot \exp\left\{iI[\chi+\epsilon F]+i\int d^4x\,\big(\chi^n(x)+\epsilon F^n[x;\chi]\big)J_n(x)\right\} \\
&= \int \left[\prod_{n,x} d\chi^n(x)\right] \exp\left\{iI[\chi]+i\int d^4x\,\big(\chi^n(x)+\epsilon F^n[x;\chi]\big)J_n(x)\right\} \\
&= Z[J]+i\epsilon\int\left(\prod_{n,x} d\chi^n(x)\right)\int F^n[y;\chi]J_n(y)\,d^4y \\
&\qquad \cdot \exp\left\{iI[\chi]+i\int d^4x\,\chi^n(x)J_n(x)\right\}
\end{aligned}$$

and hence

$$\int d^4y\,\langle F^n(y)\rangle_J J_n(y)=0, \tag{16.4.4}$$

where $\langle\ \rangle_J$ denotes the quantum average in the presence of the current $J_n(x)$,

$$\begin{aligned}
Z[J]\langle F^n(y)\rangle_J \equiv \int &\left(\prod_{n,x} d\chi^n(x)\right) F^n[y;\chi] \\
&\cdot \exp\left\{iI[\chi]+i\int d^4x\,\chi^n(x)J_n(x)\right\},
\end{aligned} \tag{16.4.5}$$

normalized so that $\langle 1\rangle_J=1$. But recall that $J_n(y)$ is given in terms of the effective action $\Gamma[\chi]$ by Eq. (16.1.7)

$$J_{n,\chi}(y)=-\frac{\delta\Gamma[\chi]}{\delta\chi^n(y)}.$$

Therefore Eq. (16.4.4) may be written as

$$0=\int d^4y\,\langle F^n(y)\rangle_{J_\chi}\frac{\delta\Gamma[\chi]}{\delta\chi^n(y)}. \tag{16.4.6}$$

In other words, $\Gamma[\chi]$ is invariant under the infinitesimal transformation

$$\chi^n(y)\longrightarrow\chi^n(y)+\epsilon\langle F^n(y)\rangle_{J_\chi}. \tag{16.4.7}$$

Such symmetry conditions are known as Slavnov–Taylor identities [7].

Is this the symmetry transformation with which we started? It is for one very important class of infinitesimal symmetry transformations: those that are *linear*. For such symmetries F is

$$F^n[x;\chi] = s^n(x) + \int t^n{}_m(x,y)\chi^m(y)\,d^4y. \tag{16.4.8}$$

(In the most common case $s^n(x)$ vanishes and $t^n{}_m(x,y)$ is a constant matrix times $\delta^4(x-y)$.) For any linear F, we have

$$\langle F^n(x)\rangle_J = s^n(x) + \int t^n{}_m(x,y)\langle\chi^m(y)\rangle_J\,d^4y.$$

But for any fixed χ, J_χ is defined as the value of the current J that makes $\langle\chi^m(y)\rangle_J$ equal to $\chi^m(y)$, so

$$\langle F^n(x)\rangle_{J_\chi} = s^n(x) + \int t^n{}_m(x,y)\chi^m(y)\,d^4y = F^n[x;\chi]. \tag{16.4.9}$$

Hence Eq. (16.4.6) requires that $\Gamma[\chi]$ be invariant under all the functional linear transformations $\chi^n \to \chi^n + \epsilon F^n$ that leave $I[\chi]$ and the measure invariant.

We occasionally have to deal with symmetry transformations that are not linear. One important example is provided by the BRST transformation discussed in Section 15.7. For non-linear transformations, the symmetry transformation (16.4.7) under which the effective action is invariant is not generally the same as the assumed symmetry transformation (16.4.1) that leaves the original action invariant, because the average of a non-linear functional of fields is not generally the same as the functional of the average fields. Indeed, the form of $\langle F\rangle_{J_\chi}$ as a functional of χ depends in general on the dynamics of the system, and is usually non-local. This complication will be dealt with in the next chapter by the method of antibrackets.

* * *

Up to now, we have tacitly been assuming that the fields χ^n and the corresponding transformation functions Δ^n and currents J_n are all bosonic. We will need to take note of the sign factors that appear when some of these are fermionic, as in particular for supersymmetry or BRST transformations, where ϵ is fermionic and χ^n and Δ^n have opposite statistics. With currents inserted to the right of fields, as in Eqs. (16.1.1) and (16.4.5), Eqs. (16.1.5) and (16.1.7) hold in the form

$$\frac{\delta_R W[J]}{\delta J_m(y)} = \chi^m_J(y), \tag{16.4.10}$$

$$\frac{\delta_L \Gamma[\chi]}{\delta\chi^m(y)} = -J_{\chi,m}(y), \tag{16.4.11}$$

with the subscripts R and L indicating that the derivative is to act from the right or left. In consequence, the Slavnov–Taylor identity (16.4.6) should be written

$$0 = \int d^4y\,\langle F^n(y)\rangle_{J_\chi}\frac{\delta_L\Gamma[\chi]}{\delta\chi^n(y)}. \tag{16.4.12}$$

Problems

1. Consider a theory of real pseudoscalars $\phi(x)$ and complex Dirac fields $\psi(x)$, with masses M and m respectively, and interaction $g\bar{\psi}\gamma_5\psi\phi$. Evaluate the effective potential for $\phi =$ constant, $\psi = 0$, to one-loop order.

2. Derive general formulas for

$$\frac{\delta^3 W[J]}{\delta J_n(x)\delta J_m(y)\delta J_\ell(z)} \quad \text{and}$$
$$\frac{\delta^4 W[J]}{\delta J_n(x)\delta J_m(y)\delta J_\ell(z)\delta J_k(w)}$$

   in terms of the variational derivatives of $\Gamma[\phi]$ with respect to ϕ. Show which Feynman diagrams correspond to each term in these formulas.

3. Calculate the effective potential to one-loop order for the theory of a neutral scalar field ϕ with interaction Lagrangian density $g\phi^3/6$ in six spacetime dimensions.

4. Suppose that the action $I[\phi]$ is invariant under a *finite* matrix transformation $\phi_n(x) \longrightarrow \sum_m M_{nm}\phi_m(x)$. Under which transformation of the currents is $W[J]$ then invariant? Use this result to derive a symmetry property of $\Gamma[\phi]$.

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17 Renormalization of Gauge Theories

We now return to gauge theories, and use the external field formalism described in the previous chapter to study the renormalizability of these theories and to carry out an important calculation.

17.1 The Zinn–Justin Equation

In this section the BRST symmetry described in Section 15.7 will be used to demonstrate a fundamental property of the quantum effective action, first derived by Zinn–Justin [1]. According to the general rules outlined in Section 16.4, the BRST invariance of the action $I[\chi]$ imposes on the effective action $\Gamma[\chi]$ the condition

$$\int d^4x \, \langle \Delta^n(x) \rangle_{J_\chi} \frac{\delta_L \Gamma[\chi]}{\delta \chi^n(x)} = 0, \tag{17.1.1}$$

where the change in $\chi^n(x)$ under a BRST transformation with infinitesimal fermionic parameter θ is

$$\delta_\theta \chi^n(x) = \theta \Delta^n(x) \tag{17.1.2}$$

and $\langle \cdots \rangle$ here denotes a vacuum expectation value taken in the presence of a current J_χ that makes the vacuum expectation value of the operator fields $X^n(x)$ equal to the c-number functions $\chi^n(x)$. The implied sum over n runs over all of the fields in the BRST formalism; that is, over ω_α, ω^*_α, and h_α as well as the gauge and matter fields that in Section 15.8 we have collectively called ϕ^r. Because $\Delta^n(x)$ is quadratic in the fields when χ^n is a gauge or matter field or ω_α, Eq. (17.1.1) does not in general tell us that the effective action is invariant under the same BRST transformations as the action itself.

To handle this complication, we employ a trick that proves useful in dealing with any sort of nilpotent symmetry transformation. First, we introduce a set of c-number external fields $K_n(x)$, and define a new effective action by

$$\Gamma[\chi, K] \equiv W[J_{\chi,K}, K] - \int d^4x \, \chi^n(x) J_{\chi,K\,n}(x), \tag{17.1.3}$$

where the connected vacuum persistence amplitude W is here calculated with the gauge-fixed action[24] $I + \int d^4x \, \Delta^n K_n$:

$$e^{iW[J,K]} \equiv \int \left[\prod_{n,x} d\chi^n(x) \right] \exp\left(iI + i\int d^4x \, \Delta^n K_n + i\int d^4x \, \chi^n J_n \right), \tag{17.1.4}$$

and $J_{\chi,K}$ is the current required to give the fields the expectation values χ in the presence of the external fields K:

$$\left. \frac{\delta_R W[J,K]}{\delta J_n(x)} \right|_{J=J_{\chi,K}} = \chi^n(x). \tag{17.1.5}$$

[24] Here I is the action I_{NEW} modified as described in Section 15.7 to depend on the ghost and antighost fields ω_α and ω^*_α and on the auxiliary field h_α, with the subscript 'NEW' dropped from now on.

(The K_n must have the same fermionic or bosonic statistics as Δ^n, which is opposite to that of χ^n.) Since the BRST transformation is nilpotent, the quantities $\Delta^n(x)$ are BRST-invariant, so in the same way as in Section 16.4 we can show that the new effective action $\Gamma[\chi, K]$ satisfies a BRST-invariance condition:

$$\int d^4x\, \langle \Delta^n(x)\rangle_{J_{\chi,K}} \frac{\delta_L \Gamma[\chi, K]}{\delta \chi^n(x)} = 0, \tag{17.1.6}$$

where $\langle \cdots \rangle_{J,K}$ denotes a vacuum expectation value calculated in the presence of the current J and the external fields K:

$$\langle \mathcal{O}[\chi]\rangle_{J,K} \equiv \frac{\int \left[\prod_{n,x} d\chi^n(x)\right] \mathcal{O}[\chi] \exp\left(iI + i\int d^4x\, \Delta^n K_n + i\int d^4x\, \chi^n J_n\right)}{\int \left[\prod_{n,x} d\chi^n(x)\right] \exp\left(iI + i\int d^4x\, \Delta^n K_n + i\int d^4x\, \chi^n J_n\right)}. \tag{17.1.7}$$

It is convenient to express the expectation value of Δ^n as a variational derivative of the effective action. Taking the right variational derivative of Eq. (17.1.3) with respect to K gives

$$\begin{aligned}\frac{\delta_R \Gamma[\chi, K]}{\delta K_n(x)} = &\left.\frac{\delta_R W[J, K]}{\delta K_n(x)}\right|_{J=J_{\chi,K}} + \int d^4y\, \left.\frac{\delta_R W[J, K]}{\delta J_m(y)}\right|_{J=J_{\chi,K}} \frac{\delta_R J_{\chi,K\,m}(y)}{\delta K_n(x)} \\ &- \int d^4y\, \chi^m(y) \frac{\delta_R J_{\chi,K\,m}(y)}{\delta K_n(x)}.\end{aligned}$$

Using Eq. (17.1.5) we see that the last two terms cancel, and using the definitions (17.1.4) and (17.1.7) gives us then our desired relation

$$\frac{\delta_R \Gamma[\chi, K]}{\delta K_n(x)} = \left.\frac{\delta_R W[J, K]}{\delta K_n(x)}\right|_{J=J_{\chi,K}} = \langle \Delta^n(x)\rangle_{J_{\chi,K},K}. \tag{17.1.8}$$

The BRST symmetry condition (17.1.6) may now be written as a simple condition, the Zinn–Justin equation, involving the effective action alone:

$$\int d^4x\, \frac{\delta_R \Gamma[\chi, K]}{\delta K_n(x)} \frac{\delta_L \Gamma[\chi, K]}{\delta \chi^n(x)} = 0. \tag{17.1.9}$$

As remarked after Eq. (15.9.3), the interchange of fields and antifields (or in this case χ^n and K_n) results simply in a change of sign of the left-hand side of Eq. (17.1.9), so this can be written as

$$(\Gamma, \Gamma) = 0, \tag{17.1.10}$$

where the antibracket is calculated here with K_n in place of the antifield of χ^n:

$$\begin{aligned}(F, G) \equiv &\int d^4x\, \frac{\delta_R F[\chi, K]}{\delta \chi^n(x)} \frac{\delta_L G[\chi, K]}{\delta K_n(x)} \\ &- \int d^4x\, \frac{\delta_R F[\chi, K]}{\delta K_n(x)} \frac{\delta_L G[\chi, K]}{\delta \chi^n(x)}.\end{aligned} \tag{17.1.11}$$

This is formally the same as the Batalin–Vilkovisky 'master equation' discussed in Section 15.9, but appears here as a constraint on the quantum effective action $\Gamma[\chi, K]$ rather than on the fundamental action $S[\chi, \chi^\ddagger]$. The Zinn–Justin equation (17.1.10) will be used in the next two sections to show how to renormalize gauge theories, and in Section 22.6 to study anomalies in these theories.

17.2 Renormalization: Direct Analysis

The simplest non-Abelian gauge theories are renormalizable in the 'Dyson' sense that the operators in the Lagrangian density all have dimensionality (in powers of mass) four or less. As we saw in Chapter 12, this guarantees that the infinities in the quantum effective action only appear in terms that could be cancelled by the counterterms in interactions of dimensionality four or less. But there is more to renormalizability than this. The Lagrangian density is constrained by gauge invariance and other symmetries. For a theory to be renormalizable, it is necessary that the infinities in the quantum effective action satisfy the same constraints, up to possible renormalizations of the fields.

The effective action $\Gamma[\chi, K]$ is a complicated functional of both χ and K, about which the symmetry condition (17.1.9) says complicated things, but fortunately matters are much simpler for the infinite terms in Γ. We will write the action $S[\chi, K] = I[\chi] + \int d^4x\, \Delta^n K_n$ as the sum of a term $S_R[\chi, K]$ in which masses and coupling constants are set equal to their renormalized values, plus a correction $S_\infty[\chi, K]$, which contains the counterterms that we intend to cancel the infinities from loop graphs. Both S_R and S_∞ must be taken to have the symmetries of the original action $S[\chi, K]$, so the question is whether the infinite parts of the higher-order contributions to Γ share the same symmetries, so that they can be cancelled by the counterterms in S_∞.

We may expand Γ in a series of terms Γ_N arising from diagrams with just N loops, plus contributions from graphs with $N - M$ loops (where $1 \leq M \leq N$) involving various counterterms in $S_\infty[\chi, K]$ that will be used to cancel infinities in graphs with a total of M loops:

$$\Gamma[\chi, K] = \sum_{N=0}^{\infty} \Gamma_N[\chi, K]. \tag{17.2.1}$$

The symmetry condition (17.1.10) then reads,[25] for each N,

$$\sum_{N'=0}^{N} \left(\Gamma_{N'}, \Gamma_{N-N'}\right) = 0. \tag{17.2.2}$$

In the sum (17.2.1) the leading term is just $\Gamma_0[\chi, K] = S_R[\chi, K]$, which of course is finite. Suppose that, for all $M < N - 1$, all infinities arising from M-loop graphs have been cancelled by counterterms in S_∞. Then infinities can appear in Eq. (17.2.2) only in the $N' = 0$ and $N' = N$ terms, which are equal, and the infinite part of this condition tells us that the infinite part $\Gamma_{N,\infty}$ of Γ_N is subject to the condition that

$$\left(S_R, \Gamma_{N,\infty}\right) = 0. \tag{17.2.3}$$

[25] Such order-by-order relations may be derived formally by repeating the reasoning of Section 16.1. When the action S_R is replaced with $g^{-1}S_R$, the contribution of a connected L-loop graph with I internal lines and V vertices is multiplied by a factor $g^{I-V} = g^{L-1}$. If the counterterms in S_∞ associated with N-loop diagrams are also provided with factors g^N, then Γ_L is the value for $g = 1$ of the term in Γ of order g^{L-1}. Eq. (17.2.2) then follows by requiring Eq. (17.1.10) to hold in each order in g. In cgs units the action has the same dimensions as $\hbar$, and so appears in the path integral multiplied with a factor $1/\hbar$, so we can also use $\hbar$ as a loop-counting parameter in place of g.

This is a symmetry principle generated by S_R, just like that described by Eqs. (15.9.16) and (15.9.17). Note in particular that the transformation $X \mapsto (S_R, X)$ acts on the external fields K_n as well as the fields χ^n.

Up to this point, we have used none of the special properties of a renormalizable Yang–Mills theory. Now note that, according to the general power-counting rules of renormalization theory, with all infinities cancelled in subgraphs of Γ_N, the infinite part $\Gamma_{N,\infty}[\chi, K]$ of $\Gamma_N[\chi, K]$ can only be a sum of products of fields (including K) and their derivatives of dimensionality (in powers of mass) four or less. Finally, the arguments of Section 16.4 show that $\Gamma[\chi, K]$ and hence also $\Gamma_{N,\infty}[\chi, K]$ are invariant under all of the *linearly realized* symmetry transformations under which the action is invariant. (As described below, these are: Lorentz transformations, *global* gauge transformations, antighost translations, and the ghost phase transformations associated with ghost number conservation. Of course, the auxiliary fields K_n must be assigned suitable transformation properties under these symmetry transformations.) These conditions together with Eq. (17.2.3) will suffice to tell us all we need to know about the structure of $\Gamma_{N,\infty}[\chi, K]$.

To implement these conditions, we need to know the dimensionalities of the external fields K_n. If a field χ^n has dimensionality d_n (that is, d_n powers of mass), then Δ^n correspondingly has dimensionality $d_n + 1$ (as can be seen by inspection of the BRST transformation rules (15.7.7)–(15.7.11)), so in order for $\int d^4x\, K_n \Delta^n$ to be dimensionless, K_n must have dimensionality $3 - d_n$. The fields $A^{\alpha\mu}$, ω^α, and $\omega^{\alpha *}$ all have dimensionalities $d_n = +1$, so the corresponding K_n all have dimensionalities $+2$. (We do not introduce any external field corresponding to h^α, because this field is BRST-invariant.) Any spin 1/2 matter fields ψ_ℓ have dimensionalities 3/2, and the corresponding K_n thus also have dimensionalities 3/2. Thus a dimension four quantity like $\Gamma_{N,\infty}[\chi, K]$ is at most quadratic in any of the K_n. Furthermore terms that are of second order in the K_n cannot involve any other fields, except that a term of second order in the K_n for spin 1/2 matter fields may involve at most one additional field of dimensionality unity.

We can now use ghost number conservation to show that in fact $\Gamma_{N,\infty}[\chi, K]$ does not contain any terms of second order in the K_n. For this purpose, we also need the ghost quantum numbers of the K_n. If χ^n has ghost quantum number γ_n, then Δ^n has ghost quantum number $\gamma_n + 1$, so K_n must be assigned ghost quantum number $-\gamma_n - 1$. The ghost quantum numbers of the fields $A^{\alpha\mu}$, ψ^ℓ, ω^α, and $\omega^{\alpha *}$ are respectively 0, 0, +1, and −1, so the corresponding external fields K_n have ghost quantum numbers −1, −1, −2, and 0, respectively. This rules out any terms in $\Gamma_{N,\infty}[\chi, K]$ that are of second order in the K_n, with the one possible exception of a term of second order in the external fields K^*_α associated with $\omega^{\alpha *}$ (and involving no other fields). However, these last terms are also forbidden, for a different reason. The BRST transformation of $\omega^{\alpha *}$ is linear in the fields, with

$$\Delta^{\alpha *} = -h^\alpha, \tag{17.2.4}$$

so here

$$\frac{\delta_L \Gamma_{N,\infty}[\chi, K]}{\delta K^*_\alpha} = \langle \Delta^{\alpha *} \rangle_{J_\chi, K} = -h^\alpha$$

is independent of K^*_α. It follows that $\Gamma_{N,\infty}[\chi,K]$ is linear in $K^{\alpha,*}$, and depends on $K^{\alpha,*}$ only through a term $-\int d^4x\, K^*_\alpha h^\alpha$. (Both K^*_α and h^α are bosonic, so their order is immaterial.) In particular, for $N>0$, $\Gamma_{N,\infty}[\chi,K]$ is independent of $K^{\alpha *}$.

We have seen that $\Gamma_{N,\infty}[\chi,K]$ is at most linear in all of the K_n. We will write it as

$$\Gamma_{N,\infty}[\chi,K]=\Gamma_{N,\infty}[\chi,0]+\int d^4x\,\mathcal{D}^n_N[\chi;x]K_n(x). \tag{17.2.5}$$

We also recall that S_R has the K dependence:

$$S_R[\chi,K]=S_R[\chi]+\int d^4x\,\Delta^n[\chi;x]K_n(x).$$

The terms in (17.2.2) of zeroth and first order in K therefore give[26]

$$\int d^4x\left[\Delta^n[\chi;x]\frac{\delta_L\Gamma_{N,\infty}[\chi,0]}{\delta\chi^n(x)}+\mathcal{D}^n_N[\chi;x]\frac{\delta_L S_R[\chi]}{\delta\chi^n(x)}\right]=0 \tag{17.2.6}$$

and

$$\int d^4x\left[\Delta^n[\chi;x]\frac{\delta_L\mathcal{D}^m_N[\chi;y]}{\delta\chi^n(x)}+\mathcal{D}^n_N[\chi;x]\frac{\delta_L \Delta^m[\chi;y]}{\delta\chi^n(x)}\right]=0, \tag{17.2.7}$$

respectively. These relations may be made more perspicuous by introducing the quantities

$$\Gamma^{(\epsilon)}_N[\chi]\equiv S_R[\chi]+\epsilon\Gamma_{N,\infty}[\chi,0], \tag{17.2.8}$$

and

$$\Delta^{(\epsilon)n}_N(x)\equiv\Delta^n(x)+\epsilon\mathcal{D}^n_N(x), \tag{17.2.9}$$

with ϵ infinitesimal. Then (17.2.6) (together with the BRST invariance of S_R) just says that $\Gamma^{(\epsilon)}_N[\chi]$ is invariant under the transformation

$$\chi^n(x)\longrightarrow\chi^n(x)+\theta\Delta^{(\epsilon)n}_N(x), \tag{17.2.10}$$

while Eq. (17.2.7) (together with the nilpotence of the original BRST transformation) tells us that this transformation is *nilpotent*.

We must now consider what form this nilpotent transformation may take. As already mentioned, $\Gamma_{N,\infty}$ consists only of terms of dimensionality four or less, so $\mathcal{D}^n_N$ and hence $\Delta^{(\epsilon)n}_N(x)$ have at most the dimensionality of the original BRST transformation function $\Delta^n(x)$. Also, $\mathcal{D}^n_N$ and hence also $\Delta^{(\epsilon)n}_N(x)$ must have the same Lorentz transformation properties and ghost quantum numbers as $\Delta^n(x)$. Hence the most general form of the transformation (17.2.10) is

$$\psi\longrightarrow\psi+i\theta\omega^\alpha T_\alpha\psi,\qquad A_{\alpha\mu}\longrightarrow A_{\alpha\mu}+\theta\big[B_{\alpha\beta}\partial_\mu\omega_\beta+D_{\alpha\beta\gamma}A_{\beta\mu}\omega_\gamma\big],$$
$$\omega_\alpha\longrightarrow\omega_\alpha-\frac{1}{2}\theta E_{\alpha\beta\gamma}\omega_\beta\omega_\gamma,$$

[26] The second terms in Eqs. (17.2.6) and (17.2.7) have been put into the form shown here by recalling that, because χ^n and K_n have opposite statistics, for any bosonic functionals A and B,

$$\frac{\delta_R A}{\delta\chi^n}\frac{\delta_L B}{\delta K_n}=-\frac{\delta_L A}{\delta\chi^n}\frac{\delta_R B}{\delta K_n}=-\frac{\delta_R B}{\delta K_n}\frac{\delta_L A}{\delta\chi^n}.$$

where T_α is some matrix acting on the spinor fields, and $B_{\alpha\beta}$, $D_{\alpha\beta\gamma}$, and $E_{\alpha\beta\gamma}$ are constants, with $E_{\alpha\beta\gamma}$ antisymmetric in β and γ. Also, the transformations of ω^*_α and h_α are linear, and are therefore unchanged:

$$\omega^*_\alpha \longrightarrow \omega^*_\alpha - \theta h_\alpha, \qquad h_\alpha \longrightarrow h_\alpha.$$

Next we impose the condition of nilpotence. The most important requirement is that $E_{\alpha\beta\gamma}\omega_\beta\omega_\gamma$ should be invariant. This yields the requirement that $E_{\alpha\beta\gamma}E_{\beta\delta\epsilon}\omega_\delta\omega_\epsilon\omega_\gamma$ should vanish, so that the part of $E_{\alpha\beta\gamma}E_{\beta\delta\epsilon}$ that is totally antisymmetric in δ, ϵ, γ vanishes:

$$E_{\alpha\beta\gamma}E_{\beta\delta\epsilon} + E_{\alpha\beta\epsilon}E_{\beta\gamma\delta} + E_{\alpha\beta\delta}E_{\beta\epsilon\gamma} = 0.$$

But this just tells us that $E_{\alpha\beta\gamma}$ is the structure constant of some Lie algebra $\mathcal{E}$. Because $E_{\alpha\beta\gamma}$ goes to the structure constant $C_{\alpha\beta\gamma}$ of the original gauge Lie algebra $\mathcal{A}$ for $\epsilon \to 0$, $\mathcal{E}$ must be the same as $\mathcal{A}$, and the structure constants $E_{\alpha\beta\gamma}$ can differ from the original $C_{\alpha\beta\gamma}$ only by a multiplicative factor:[27]

$$E_{\alpha\beta\gamma} = \mathcal{Z}C_{\alpha\beta\gamma}.$$

(This is for simple gauge groups; in the general case we would have a separate factor $\mathcal{Z}$ for each simple subgroup.)

Next we turn to the condition that the transformation (17.2.10) be nilpotent when acting on the gauge fields. The requirement that $B_{\alpha\beta}\partial_\mu\omega_\beta + D_{\alpha\beta\gamma}A_{\beta\mu}\omega_\gamma$ be invariant tells us that

$$D_{\alpha\beta\gamma}D_{\beta\delta\epsilon} - D_{\alpha\beta\epsilon}D_{\beta\delta\gamma} = E_{\beta\epsilon\gamma}D_{\alpha\delta\beta} = \mathcal{Z}C_{\beta\epsilon\gamma}D_{\alpha\delta\beta}$$

and

$$B_{\alpha\beta}E_{\beta\gamma\delta} = D_{\alpha\beta\delta}B_{\beta\gamma}.$$

The first condition has the unique solution[28]

$$D_{\alpha\beta\gamma} = \mathcal{Z}C_{\alpha\beta\gamma}.$$

The second condition tells us that the matrix $B_{\alpha\beta}$ commutes with the adjoint representation of the gauge group, and hence (since we have chosen the structure constants totally antisymmetric) must be proportional to a Kronecker delta, with a coefficient we shall call $\mathcal{ZN}$:

$$B_{\alpha\beta} = \mathcal{ZN}\delta_{\alpha\beta}.$$

Finally, the condition that the transformation (17.2.10) be nilpotent when acting on the fermion fields (if any) requires that $\omega^\alpha T_\alpha\psi$ is invariant. This tells us that

$$[T_\beta, T_\gamma] = iE_{\alpha\beta\gamma}T_\alpha,$$

[27] The requirement of global gauge invariance rules out any non-trivial similarity transformation in the relation between $E_{\alpha\beta\gamma}$ and $C_{\alpha\beta\gamma}$.

[28] The matrix $(D_\gamma)_{\alpha\beta} = D_{\gamma\alpha\beta}/\mathcal{Z}$ satisfies the commutation relations of the gauge Lie algebra, $[D_\gamma, D_\epsilon] = C_{\beta\epsilon\gamma}D_\beta$. But the only representation of a simple Lie algebra with the same dimensionality and $\mathcal{A}$ transformation properties as the adjoint representation of $\mathcal{A}$ is the adjoint representation itself.

so T_α differs from the generator t_α in the original Lagrangian only by a factor $\mathcal{Z}$:

$$T_\alpha = \mathcal{Z} t_\alpha.$$

We have thus seen that, apart from the appearance of the new constants $\mathcal{Z}$ and $\mathcal{N}$, the transformation (17.2.10) is just the BRST transformation with which we started:

$$\psi \longrightarrow \psi + i\mathcal{Z}\theta\omega^\alpha t_\alpha \psi, \tag{17.2.11}$$

$$A_{\alpha\mu} \longrightarrow A_{\alpha\mu} + \mathcal{Z}\theta\big[\mathcal{N}\partial_\mu\omega_\alpha + C_{\alpha\beta\gamma}A_{\beta\mu}\omega_\gamma\big], \tag{17.2.12}$$

$$\omega_\alpha \longrightarrow \omega_\alpha - \frac{1}{2}\mathcal{Z}\theta C_{\alpha\beta\gamma}\omega_\beta\omega_\gamma, \tag{17.2.13}$$

$$\omega^*_\alpha \longrightarrow \omega^*_\alpha - \theta h_\alpha, \tag{17.2.14}$$

$$h_\alpha \longrightarrow h_\alpha. \tag{17.2.15}$$

Now we must use this symmetry to constrain the structure of the corrected action (17.2.8). Since this contains only the original renormalized action plus the infinite part of the N-loop contribution, it must be the integral of a Lagrangian density

$$\Gamma_N^{(\epsilon)} = \int d^4x\, \mathcal{L}_N^{(\epsilon)} \tag{17.2.16}$$

with $\mathcal{L}_N^{(\epsilon)}$ a local function of fields and field derivatives of dimensionality (in powers of mass) no greater than 4. Furthermore, as we found in Section 16.4, $\mathcal{L}_N^{(\epsilon)}$ must be invariant under all the symmetries of the original Lagrangian that act *linearly* on the fields. To identify these symmetries, recall that in generalized ξ-gauge, the 'new' Lagrangian density in Eq. (15.7.6) takes a form given by replacing the term $-(\partial_\mu A^\mu_\alpha)(\partial_\nu A^\nu_\alpha)/2\xi$ in Eq. (15.6.16) with the terms $h_\alpha f_\alpha + \frac{1}{2}\xi h_\alpha h_\alpha$ in Eq. (15.7.6):

$$\mathcal{L}_{\text{NEW}} = \mathcal{L}_M - \frac{1}{4}F_\alpha{}^{\mu\nu}F_{\alpha\mu\nu} - \partial_\mu\omega^*_\alpha\partial^\mu\omega_\alpha + C_{\alpha\beta\gamma}(\partial_\mu\omega^*_\alpha)A_\gamma{}^\mu\omega_\beta + h_\alpha\partial_\mu A_\alpha{}^\mu + \frac{1}{2}\xi h_\alpha h_\alpha. \tag{17.2.17}$$

Inspection of this formula reveals the following linear symmetries:

(1) **Lorentz invariance.**

(2) **Global gauge invariance**—that is, invariance under the transformations

$$\delta\psi_\ell(x) = i\epsilon^\alpha (t_\alpha)_\ell{}^m \psi_m(x), \tag{17.2.18}$$

$$\delta A^\beta{}_\mu(x) = C_{\beta\gamma\alpha}\epsilon^\alpha A^\gamma{}_\mu(x), \tag{17.2.19}$$

$$\delta\omega_\beta(x) = C_{\beta\gamma\alpha}\epsilon^\alpha\omega_\gamma(x), \tag{17.2.20}$$

$$\delta\omega^*_\beta(x) = C_{\beta\gamma\alpha}\epsilon^\alpha\omega^*_\gamma(x), \tag{17.2.21}$$

$$\delta h_\beta(x) = C_{\beta\gamma\alpha}\epsilon^\alpha h_\gamma(x), \tag{17.2.22}$$

with constant parameters ϵ^α.

(3) **Antighost translation invariance**—that is, invariance under the transformation

$$\omega^*_\alpha(x) \longrightarrow \omega^*_\alpha(x) + c_\alpha, \tag{17.2.23}$$

with arbitrary constant parameters c_α.

(4) **Ghost number conservation**—that is, the conservation of a ghost number equal to $+1$ for ω_α, -1 for ω^*_α, and 0 for all other fields.

We shall now proceed to work out the structure of the most general Lagrangian density that is renormalizable, in the sense that it consists only of terms of dimensionality $+4$ or less, that has these linearly acting symmetries, and that is invariant under the modified BRST transformations (17.2.11)–(17.2.15).

From Eq. (17.2.17) we may conclude that the fields A^μ_α, ω_α, ω^*_α, and h_α have the dimensionalities $+1$, $+1$, $+1$, and $+2$, respectively (in powers of mass). Note also that ghost number conservation requires that ω and ω^* come in pairs, while antighost translation invariance dictates that ω^* always appears as a derivative. Each pair of ω and $\partial_\mu\omega^*$ fields adds $+3$ to the dimensionality, so renormalizability rules out any term with more than one such pair. With one such pair we can have at most one more derivative or one additional gauge field, and Lorentz invariance dictates that we must have one or the other. The only renormalizable allowed interactions involving ghost fields are then linear combinations of terms of the form $\partial_\mu\omega^*_\alpha\partial^\mu\omega_\beta$ or $\partial_\mu\omega^*_\alpha A^\mu_\gamma\omega_\beta$.

Next, let us consider the terms that involve the field h_α and possibly other fields but not ω or ω^*. This field has dimensionality $+2$, so renormalizability and Lorentz invariance allows this field to appear only[29] multiplied with another h_β or $\partial_\mu A^\mu_\beta$ or $A^\mu_\beta A_{\gamma\mu}$.

Finally, the Lagrangian will contain renormalizable terms involving only the matter and gauge fields. We will call the sum of these terms $\mathcal{L}_{\psi A}$. Putting this all together and using global gauge invariance, the most general renormalizable interaction allowed by the assumed symmetries (aside from BRST invariance) takes the form:

$$\begin{aligned}\mathcal{L}^{(\epsilon)}_N = \mathcal{L}_{\psi A} + \frac{1}{2}\xi' h_\alpha h_\alpha + c h_\alpha \partial_\mu A^\mu_\alpha - e_{\alpha\beta\gamma} h_\alpha A^\mu_\beta A_{\gamma\mu} \\ - Z_\omega(\partial_\mu\omega^*_\alpha)(\partial^\mu\omega_\alpha) - d_{\alpha\beta\gamma}(\partial_\mu\omega^*_\alpha)\omega_\beta A^\mu_\gamma,\end{aligned} \tag{17.2.24}$$

where ξ', Z_ω, c, $d_{\alpha\beta\gamma}$, and $e_{\alpha\beta\gamma}$ are unknown constants, with no constraints except obvious symmetry properties such as global gauge invariance and $e_{\alpha\beta\gamma} = e_{\alpha\gamma\beta}$. (As mentioned earlier, we are assuming for simplicity that the gauge group is simple, but the extension to a direct sum of simple and $U(1)$ gauge groups would be trivial; for instance, instead of one term proportional to $h_\alpha h_\alpha$ we would have a sum of such terms, one for each simple subgroup of the gauge group.)

Now we impose BRST invariance. The cancellation of terms in $\delta\mathcal{L}^{(\epsilon)}_N$ proportional to $\theta\partial_\mu h_\alpha\partial^\mu\omega_\alpha$ tells us that

$$c = \frac{Z_\omega}{\mathcal{Z}\mathcal{N}}. \tag{17.2.25}$$

The cancellation of terms in $\delta\mathcal{L}^{(\epsilon)}_N$ proportional to $\theta\partial_\mu h_\alpha\omega_\beta A^\mu_\gamma$ (or $\theta\partial_\mu\omega^*_\alpha\omega_\beta\partial^\mu\omega_\gamma$) requires that

$$d_{\alpha\beta\gamma} = -(Z_\omega/\mathcal{N})C_{\alpha\beta\gamma}. \tag{17.2.26}$$

[29] In a theory with scalar fields, we could also have renormalizable terms with h_α multiplied with one or two scalar fields. Such terms cause no trouble, but for brevity they will not be considered here.

The terms in $\delta\mathcal{L}_N^{(\epsilon)}$ proportional to $\theta\partial_\mu\omega^*_\alpha\omega_\beta\omega_\gamma A^\mu_\delta$ automatically then cancel by virtue of the Jacobi identity for the structure constants. The cancellation of terms in $\delta\mathcal{L}_N^{(\epsilon)}$ proportional to $\theta h_\alpha\partial_\mu\omega_\beta A^\mu_\gamma$ (or $\theta h_\alpha A^\mu_\beta A_{\gamma\mu}\omega_\delta$) yields

$$e_{\alpha\beta\gamma} = 0. \tag{17.2.27}$$

Finally, the effect of the infinitesimal transformation (17.2.10) on matter and gauge fields is the same as a local gauge transformation with gauge parameters $\epsilon_\alpha = \mathcal{Z}\mathcal{N}\theta\omega_\alpha$ and gauge couplings renormalized by a factor $1/\mathcal{N}$ (that is, with t_α and $C_{\alpha\beta\gamma}$ replaced with $\widetilde{t}_\alpha \equiv t_\alpha/\mathcal{N}$ and $\widetilde{C}_{\alpha\beta\gamma} \equiv C_{\alpha\beta\gamma}/\mathcal{N}$), so the cancellation of terms in $\delta\mathcal{L}_N^{(\epsilon)}$ with only one factor of ω_α and no factors h_α or ω^*_α simply tells us that the Lagrangian $\mathcal{L}_{\psi A}$ for these fields is *gauge-invariant*, with a renormalized gauge coupling. We conclude then that the most general renormalizable Lagrangian density allowed by our assumed symmetry principles is

$$\begin{aligned}\mathcal{L}_N^{(\epsilon)} = & -\frac{1}{4}Z_A\widetilde{F}_\alpha{}^{\mu\nu}\widetilde{F}_{\alpha\mu\nu} - Z_\psi\bar{\psi}\gamma^\mu[\partial_\mu - i\widetilde{t}_\alpha A_{\alpha\mu}]\psi + \frac{1}{2}\xi' h_\alpha h_\alpha \\ & + \frac{Z_\omega}{\mathcal{N}\mathcal{Z}}h_\alpha\partial_\mu A^\mu_\alpha - Z_\omega(\partial_\mu\omega^*_\alpha)(\partial^\mu\omega_\alpha) + Z_\omega\widetilde{C}_{\alpha\beta\gamma}(\partial_\mu\omega^*_\alpha)\omega_\beta A^\mu_\gamma,\end{aligned} \tag{17.2.28}$$

where the tilde on $\widetilde{F}_\alpha{}^{\mu\nu}$ indicates that the field strength is to be calculated using the renormalized structure constant $\widetilde{C}_{\alpha\beta\gamma} \equiv C_{\alpha\beta\gamma}/\mathcal{N}$. But apart from the appearance of a number of new constant coefficients, this is the same Lagrangian with which we started. The new constants in this Lagrangian (including the gauge coupling constant) may be freely shifted by adjusting the Nth-order terms in the corresponding constants in the original unrenormalized Lagrangian. In particular, we can adjust these terms to make $\Gamma_N^{(\epsilon)} = S_R$, in which case $\Gamma_{N,\infty} = 0$, completing the proof.

* * *

In the above proof we made important use of the accidental invariance of the gauge-fixed Lagrangian (17.2.17) under the antighost translation transformation (17.2.23). This symmetry would not be present for gauge-fixing functionals other than $f_\alpha = \partial_\mu A^\mu_\alpha$, which are not spacetime derivatives. One frequently cited example which preserves Lorentz invariance and global gauge invariance is $f_\alpha = \partial_\mu A^\mu_\alpha + a_{\alpha\beta\gamma}A^\mu_\beta A_{\gamma\mu}$, where $a_{\alpha\beta\gamma}$ is a constant matrix, symmetric in β and γ, which transforms as a tensor under global gauge transformations. (Such constant tensors exist for all $SU(N)$ groups with $N \geq 3$.) Another more important example is the background gauge-fixing functional to be introduced in Section 17.4.

The absence of antighost translation invariance does not affect our argument that $\mathcal{L}_N^{(\epsilon)}$ is a Lorentz- and global gauge-invariant local function of fields and field derivatives of dimensionality no greater than 4, that is invariant under the renormalized BRST transformation (17.2.11)–(17.2.15). But without antighost translation invariance there are new terms in $\mathcal{L}_N^{(\epsilon)}$ that satisfy these conditions. Since the transformation (17.2.11)–(17.2.15) is nilpotent, we can construct such terms as $s'F$, where the transformation (17.2.11)–(17.2.15) is written as $\chi^n \to \chi^n + \theta s'\chi^n$, and F is an arbitrary Lorentz- and global gauge-invariant function of ghost number -1. One such term is

$$a_{\alpha\beta\gamma} s'(\omega^*_\alpha A_{\beta\mu} A^\mu_\gamma) = -a_{\alpha\beta\gamma}\left[h_\alpha A_{\beta\mu} A^\mu_\gamma + 2\mathcal{Z}\omega^*_\alpha(\mathcal{N}\partial_\mu\omega_\beta + C_{\beta\delta\epsilon}A_{\delta\mu}\omega_\epsilon)A^\mu_\gamma\right].$$

This causes no trouble: it is just a renormalized version of the usual ghost and gauge-fixing terms arising from terms $a_{\alpha\beta\gamma}A^\mu_\beta A_{\gamma\mu}$ in the gauge-fixing functional f_α. But there is also another possible term of the form

$$b_{\alpha\beta\gamma} s'(\omega^*_\alpha \omega^*_\beta \omega_\gamma) = -b_{\alpha\beta\gamma}\left[2h_\alpha\omega^*_\beta\omega_\gamma + \frac{1}{2}\mathcal{Z}C_{\gamma\delta\epsilon}\omega^*_\alpha\omega^*_\beta\omega_\delta\omega_\epsilon\right],$$

where $b_{\alpha\beta\gamma}$ is a constant, antisymmetric in α and β, which transforms as a tensor under global gauge transformations. (Such tensors exist for any Lie group; for instance, we could take $b_{\alpha\beta\gamma}$ to be proportional to $C_{\alpha\beta\gamma}$.) But the Faddeev–Popov–De Witt method cannot yield four-ghost interactions in the Lagrangian, so there are no counterterms available to absorb ultraviolet divergences in this term. This is not just a technical obstacle to proving renormalizability; for gauge-fixing functionals like $f_\alpha = \partial_\mu A^\mu_\alpha + a_{\alpha\beta\gamma}A^\mu_\beta A_{\gamma\mu}$, one-loop graphs actually do yield divergences in four-ghost amplitudes that cannot be cancelled by counterterms in the Faddeev–Popov–De Witt Lagrangian.

Aside from avoiding gauge-fixing functionals other than $f_\alpha = \partial_\mu A^\mu_\alpha$, the only solution to this problem seems to be the one mentioned in Section 15.7. We must give up the Faddeev–Popov–De Witt approach, and instead take the action from the beginning as the most general renormalizable function of the gauge, matter, ghost, and auxiliary fields that is invariant under BRST and the other symmetries of the theory. According to the arguments of Section 15.8, the action can be written in the form $I_0 + s\Psi$ with I_0 ghost-free, and the S-matrix is independent of Ψ, so we can justify this procedure by quantizing the gauge theory in axial gauge, where ghosts decouple, and then taking Ψ to be anything we like. In particular, we can include $s(\omega^*\omega^*\omega)$ terms in the action that can serve as counterterms to the divergences in four-ghost vertices.

17.3 Renormalization: General Gauge Theories[30]

The proof of the renormalizability of non-Abelian gauge theories in the previous section relied on a 'brute force' analysis of the possible terms in the action of dimensionality four or less. But as we saw in Chapter 12, this limitation on the dimensionality of terms in the action can be at best a good approximation. The successful renormalizable quantum field theories that are used to describe the strong, weak, and electromagnetic interactions are almost certainly effective field theories, accompanied with terms of dimensionality $d > 4$; these terms are normally not observed because they are suppressed by $4 - d$ powers of some very large mass, perhaps of order 10^{16}–10^{18} GeV. Gravitation too can be described by an effective field theory, in which the Lagrangian density contains not only the Einstein–Hilbert term $-\sqrt{g}R/16\pi G$, but also all scalars constructed from four or more derivatives

[30]This section lies somewhat out of the book's main line of development, and may be omitted in a first reading.

of the gravitational field. We need to show that gauge theories of this sort, which are not renormalizable in the power-counting sense, are nonetheless renormalizable in the modern sense that the ultraviolet divergences are governed by the gauge symmetries in such a way that there is a counterterm available to cancel every infinity [3].

For this purpose, let us return to the action $S[\chi, \chi^\ddagger]$ introduced in Section 15.9, taken as a function of independent fields χ^n (including gauge and matter fields ϕ^r and ghost fields ω^A as well as the non-minimal fields ω^{*A} and h^A), together with their antifields $\chi^\ddagger_n$. In theories like quantum gravity or Yang–Mills theories, that are based on a closed gauge algebra with structure constants $f^C{}_{AB}$, this action is constrained to be of the form

$$S = I[\phi] + \omega^A f^r_A[\phi]\phi^\ddagger_r + \frac{1}{2}\omega^A\omega^B f^C{}_{AB}[\phi]\omega^\ddagger_C - h^A\omega^{*\ddagger}_A, \tag{17.3.1}$$

where $I[\phi]$ is invariant under the infinitesimal gauge transformations $\phi^r \longrightarrow \phi^r + \epsilon^A f^r_A[\phi]$. (As in Section 15.9, the indices r, A, etc., include a spacetime coordinate, over which we integrate in sums over these indices.)

We shall not limit ourselves here to actions of this form, but we shall suppose that the local symmetries of the theory are imposed by requiring that the action must obey some 'structural constraints' on its antifield dependence, of which Eq. (17.3.1) provides just one example. The structural constraints are assumed to be linear, in the sense that if they are satisfied for $S + \mathcal{S}_1$ and for $S + \mathcal{S}_2$, then for any constants α_1, α_2 they are satisfied for $S + \alpha_1\mathcal{S}_1 + \alpha_2\mathcal{S}_2$. We also impose the quantum master equation (15.9.35)

$$(S, S) - 2i\hbar\Delta S = 0, \tag{17.3.2}$$

with ΔS defined by Eq. (15.9.34), and the factor $\hbar$ now made explicit as a 'loop-counting' parameter, in the sense described in the footnote of the previous section.

The action is taken as a power series in $\hbar$

$$S = S_R + \hbar S_1 + \hbar^2 S_2 + \cdots, \tag{17.3.3}$$

where S_R is an action of the same general form as S, but with all coupling parameters replaced with finite renormalized values, and the S_N are a set of infinite counterterms. The action S is supposed to satisfy the quantum master equation (17.3.2) for all $\hbar$, so S_R satisfies the classical master equation

$$(S_R, S_R) = 0, \tag{17.3.4}$$

while the counterterms satisfy

$$(S_R, S_N) = -\frac{1}{2}\sum_{M=1}^{N-1}(S_M, S_{N-M}) + i\Delta S_{N-1}. \tag{17.3.5}$$

The counterterms S_N are not by themselves sufficient to cancel the ultraviolet divergences in loop graphs. As a generalization of the conventional renormalization of fields, we also have to introduce a set of renormalized fields and antifields, defined in terms of the original fields and antifields by an arbitrary anticanonical transformation. An infinitesimal anticanonical transformation may be defined in terms of an infinitesimal generating functional δF by Eq. (15.9.26), so a sequence of anticanonical transformations $G(t) \longrightarrow G(t+\delta t) = G(t) + (F(t)\delta t, G(t))$ (where $G(t)$ is any functional of fields and antifields, and $F(t)$ is the generating functional) leads to a finite canonical transformation $G \longrightarrow \widetilde{G} \equiv G(1)$, with

$$\frac{d}{dt}G(t) = (F(t), G(t)), \qquad G(0) = G. \tag{17.3.6}$$

If $F(t)$ is given by a power series

$$F(t) = gtF_1 + \hbar^2 t^2 F_2 + \cdots , \tag{17.3.7}$$

then Eqs. (17.3.3), (17.3.6), and (17.3.7) yield a transformed action

$$\begin{aligned}\widetilde{S} = S_R + \hbar\big[S_1 + (F_1, S_R)\big] \\ + \hbar^2\big[S_2 + (F_1, S_1) + (F_2, S_R) + \tfrac{1}{2}(F_1, (F_1, S_R))\big] + \cdots .\end{aligned} \tag{17.3.8}$$

The question is whether we can use what freedom we have to choose the F_N and S_N so as to cancel all infinities arising from loop graphs.

As already noted, the first term S_R in Eq. (17.3.3) is automatically finite. Suppose that by cancellation of infinities with S_M and F_M for $M < N$ it has been possible to eliminate all infinities in the terms Γ_M of order $\hbar^M$ in the quantum effective action with $M < N$. As we saw in the previous section, the Zinn–Justin equation (derived here by setting $\chi_n^\ddagger = K_n + \delta\Psi/\delta\chi^n$) tells us in this case that the infinite part $\Gamma_{N,\infty}$ of the term in the quantum effective action of order $\hbar^N$ satisfies the condition

$$(S_R, \Gamma_{N,\infty}) = 0. \tag{17.3.9}$$

The field and antifield variables χ^n and $\chi_n^\ddagger$ are related to the variables χ^n and K_n by an anticanonical transformation, which preserves all antibrackets, so the antibracket in Eq. (17.3.9) may be calculated in terms of χ^n and $\chi_n^\ddagger$ instead of χ^n and K_n.

The condition (17.3.5) satisfied by the counterterm S_N is not the same as the condition (17.3.9) satisfied by $\Gamma_{N,\infty}$. However, given any S_N^0 that satisfies Eq. (17.3.5), we can find a class of other solutions

$$S_N = S_N^0 + S'_N, \tag{17.3.10}$$

where S'_N is arbitrary, except for the condition that $S_R + S'_N$, like $S_R + S_N^0$, satisfies the same symmetry conditions as S_R, and that

$$(S_R, S'_N) = 0 \tag{17.3.11}$$

so as not to invalidate Eq. (17.3.5). The infinite part of the Nth-order term in the quantum effective action may therefore be written

$$\Gamma_{N,\infty} = S'_{N,\infty} + (F_{N,\infty}, S_R) + X_{N,\infty}, \tag{17.3.12}$$

where X_N consists of terms from loop graphs, as well as from the term S_N^0 and various terms in Γ that involve S_M and F_M for $M < N$. For instance, for $N = 2$ Eq. (17.3.8) gives

$$X_2 = S_2^0 + 2(F_1, S_1) + (F_1, (F_1, S_R)) + \text{two-loop terms involving only } S_R$$
$$+ \text{one-loop terms involving } S_R,\ S_1 \text{ and } F_1.$$

For our purposes the only thing we need to know about X_N is that it does not involve S'_N or F_N, and that it is invariant under any linearly realized global symmetries of S_R.

Now, because $(S_R, S_R) = 0$, the operation $F \longmapsto (S_R, F)$ is nilpotent; for all F,

$$(S_R, (S_R, F)) = 0. \tag{17.3.13}$$

Hence it follows from Eqs. (17.3.9) and (17.3.11)–(17.3.13) that

$$(S_R, X_{N,\infty}) = 0. \tag{17.3.14}$$

Any term in $X_{N,\infty}$ of the form (S_R, Y) may be cancelled in Eq. (17.3.12) by choosing $F_{N,\infty}$ equal to Y. Thus the space of possible remaining infinite terms in $X_{N,\infty}$ that need to be cancelled by the counterterm $S'_{N,\infty}$ consist of those functionals X that satisfy $(S_R, X) = 0$, counting as equivalent functionals that differ only by terms of the form (S_R, Y). In other words, the infinities in Γ_N that need to be cancelled by the counterterms S'_N belong to the *cohomology* of the mapping $X \longmapsto (S_R, X)$.

The possible form of the counterterm S'_N is limited by the requirement that $S_R + S'_N$ must satisfy whatever structural constraints are imposed on the action S. Thus we can complete the proof of renormalizability if we can show that the cohomology of the mapping $X \longmapsto (S_R, X)$ consists only of functionals that satisfy this structural constraint.

In the case of quantum gravity coupled to the Yang–Mills fields of a semisimple gauge symmetry, the symmetries of the theory are implemented by the structural constraint (17.3.1). In this case there is a theorem [4] that states that the cohomology of the mapping $X \longmapsto (S_R, X)$ (on the space of local functionals, rather than spacetime-dependent functions, of ghost number zero) consists [31] of functionals $A[\phi]$ that are invariant under the gauge transformation $\phi^r \longrightarrow \phi^r + \epsilon^A f^r_A[\phi]$ with structure constants $f^C{}_{AB}$. Any Nth-order infinity of this sort may be cancelled with a counterterm S'_N of the same form, so although these theories are not renormalizable in the conventional power-counting sense, they are renormalizable in the sense that all infinities can be eliminated by a choice of parameters in the original bare action $I[\phi]$ and by a suitable renormalization of fields and antifields.

In other theories the cohomology of the map $X \longmapsto (S_R, X)$ contains additional terms. This does not necessarily require a weakening of the structural constraints, because the additional terms in the cohomology may not correspond to actual ultraviolet divergences. For instance, in gauge theories with $U(1)$ factors the cohomology contains terms [4] corresponding to a redefinition of the action of the $U(1)$ gauge symmetry on the various fields

[31] Strictly speaking, this is valid if one requires that the constant coefficients in S do not take special values for which S would be invariant under a larger group of local symmetries. For instance, this excludes the case $S = 0$.

of the theory, which if infinite would require us to weaken the structural constraints by leaving the normalization of the transformation functions $f^r_A[\phi]$ in Eq. (17.3.1) arbitrary. But this infinity is forbidden by the same soft-photon theorems that tell us that ratios of $U(1)$ couplings to various fields (like the ratios of the various lepton charges in quantum electrodynamics) are unaffected by radiative corrections. (See Section 10.4.) Where the extra terms in the cohomology do contain ultraviolet divergences, it is necessary to weaken the structural constraint imposed on S in order that there should be a counterterm for every possible ultraviolet divergence. It is not known whether this will always be possible; if not, some theories may have to be rejected because of their unremovable ultraviolet divergences.

17.4 Background Field Gauge

We next turn to a method of calculation that explicitly preserves a sort of gauge invariance, and that therefore proves extremely convenient, especially in one-loop calculations. We consider the effective action $\Gamma[A, \psi, \omega, \omega^*]$ as a functional[32] of classical external gauge, matter, ghost and antighost fields: $A_{\alpha\mu}(x)$, $\psi_\ell(x)$, $\omega_\alpha(x)$, $\omega^*_\alpha(x)$. Even though ghosts and antighosts never appear in initial or final states, we are considering background ghost and antighost fields as well as gauge and matter fields in order to deal with parts of diagrams that have external ghost or antighost lines.

As described in Section 16.1, $\Gamma[A, \psi, \omega, \omega^*]$ is the sum of connected one-particle-irreducible graphs for the vacuum–vacuum amplitude, calculated in a theory in which the quantum fields A', ψ', ω', ω'^* over which we integrate are replaced in the action with shifted fields $A + A'$, $\psi + \psi'$, $\omega + \omega'$, $\omega^* + \omega'^*$, the path integral being taken over primed fields with the unprimed fields held fixed. We are free to choose the gauge-fixing function $f_\alpha(x)$ pretty much any way we like; instead of our previous choice $f_\alpha = \partial_\mu A^\mu_\alpha$ (or $\partial_\mu[A'^\mu_\alpha + A^\mu_\alpha]$) we shall now take [5]

$$f_\alpha = \partial_\mu A'^\mu_\alpha + C_{\alpha\beta\gamma} A_\beta{}^\mu A'_{\gamma\mu}. \tag{17.4.1}$$

The reason for this choice is that it makes the gauge-fixing term $f_\alpha f_\alpha$ invariant under a formal transformation, in which the background field A^μ_α transforms as a gauge field, while the quantum field A'^μ_α transforms homogeneously, like an ordinary matter field that happens to belong to the adjoint representation of the gauge group

$$\delta A^\mu_\alpha = \partial^\mu \epsilon_\alpha - C_{\alpha\beta\gamma} \epsilon_\beta A^\mu_\gamma, \tag{17.4.2}$$

$$\delta A'^\mu_\alpha = -C_{\alpha\beta\gamma} \epsilon_\beta A'^\mu_\gamma. \tag{17.4.3}$$

The transformation properties of f_α can be seen most easily by writing it as a new sort of covariant derivative

$$f_\alpha \equiv \bar{D}_\mu A'^\mu_\alpha, \tag{17.4.4}$$

[32]We are now returning to the specific choice of the gauge-fixing functional $B[f]$ as the Gaussian (15.7.4), and we are integrating out the auxiliary field h_α, so that the gauge-fixing term in the modified Lagrangian is just $-f_\alpha f_\alpha/2\xi$.

where for any field ϕ_α in the adjoint representation

$$\bar{D}_\mu \phi_\alpha \equiv \partial_\mu \phi_\alpha + C_{\alpha\beta\gamma} A_{\beta\mu} \phi_\gamma . \tag{17.4.5}$$

We see that under the transformations (17.4.2), (17.4.3), the function (17.4.1) transforms just like A'_α:

$$\delta f_\alpha = -C_{\alpha\beta\gamma} \epsilon_\beta f_\gamma , \tag{17.4.6}$$

so the term $f_\alpha f_\alpha$ in the modified Lagrangian is invariant

$$\delta(f_\alpha f_\alpha) = -C_{\alpha\beta\gamma} f_\alpha \epsilon_\beta f_\gamma = 0. \tag{17.4.7}$$

Also, the original Lagrangian density $\mathcal{L}$ depends on A and A' only through the sum $A + A'$, which under the combined transformation (17.4.2), (17.4.3) undergoes an ordinary gauge transformation

$$\delta(A^\mu_\alpha + A'^\mu_\alpha) = \partial^\mu \epsilon_\alpha - C_{\alpha\beta\gamma} \epsilon_\beta (A^\mu_\gamma + A'^\mu_\gamma). \tag{17.4.8}$$

If we transform the background and quantum matter fields by

$$\delta\psi = i t_\alpha \epsilon_\alpha \psi , \tag{17.4.9}$$

$$\delta\psi' = i t_\alpha \epsilon_\alpha \psi' , \tag{17.4.10}$$

then also

$$\delta(\psi + \psi') = i t_\alpha \epsilon_\alpha (\psi + \psi'). \tag{17.4.11}$$

The original Lagrangian $\mathcal{L}$ is invariant under the original gauge transformations (17.4.8), (17.4.11), and only depends on $A + A'$ and $\psi + \psi'$, so it is also invariant under the new formal transformations (17.4.2), (17.4.3), (17.4.9), (17.4.10).

It will be useful to make this invariance property more explicit, by writing $\mathcal{L}$ in terms of the background-covariant derivative $\bar{D}_\mu$. In general, we have

$$\begin{aligned}
\mathcal{L} = & -\frac{1}{4}\Big(\partial_\mu[A_{\alpha\nu} + A'_{\alpha\nu}] - \partial_\nu[A_{\alpha\mu} + A'_{\alpha\mu}] \\
& \qquad + C_{\alpha\beta\gamma}[A_{\beta\mu} + A'_{\beta\mu}][A_{\gamma\nu} + A'_{\gamma\nu}]\Big)^2 \\
& + \mathcal{L}_M\Big(\psi + \psi',\, \partial_\mu(\psi + \psi') - i t_\alpha (A_{\alpha\mu} + A'_{\alpha\mu})(\psi + \psi')\Big) \\
= & -\frac{1}{4}\Big(F_{\alpha\mu\nu} + \bar{D}_\mu A'_{\alpha\nu} - \bar{D}_\nu A'_{\alpha\mu} + C_{\alpha\beta\gamma} A'_{\beta\mu} A'_{\gamma\nu}\Big)^2 \\
& + \mathcal{L}_M\Big(\psi + \psi',\, \bar{D}_\mu(\psi + \psi') - i t_\alpha A'_{\alpha\mu}(\psi + \psi')\Big),
\end{aligned} \tag{17.4.12}$$

where, as in Eq. (17.4.5),

$$\bar{D}_\mu A'_{\alpha\nu} \equiv \partial_\mu A'_{\alpha\nu} + C_{\alpha\beta\gamma} A_{\beta\mu} A'_{\gamma\nu} , \tag{17.4.13}$$

$$\bar{D}_\mu \psi \equiv \partial_\mu \psi - i t_\alpha A_{\alpha\mu} \psi , \tag{17.4.14}$$

and $F_{\alpha\mu\nu}$ is the background field strength

$$F_{\alpha\mu\nu} \equiv \partial_\mu A_{\alpha\nu} - \partial_\nu A_{\alpha\mu} + C_{\alpha\beta\gamma} A_{\beta\mu} A_{\gamma\nu} . \tag{17.4.15}$$

(The square in the first term of $\mathcal{L}$ is intended to imply obvious index contractions.) Clearly $\mathcal{L}$ is invariant under the new transformations (17.4.2), (17.4.3), (17.4.9), (17.4.10), because it involves $A_{\alpha\mu}$ only in the field strength $F_{\alpha\mu\nu}$ and in background covariant derivatives $\bar{D}_\mu$ of 'matter' fields $A'_{\alpha\mu}$, ψ' and ψ.

This new transformation should be carefully distinguished from a true gauge transformation. Such a transformation can have no effect on A or ψ, which are just prescribed classical background fields, and induces an ordinary gauge transformation on $A + A'$ and $\psi + \psi'$, so

$$\delta_{\rm TRUE} A^\mu_\alpha = 0, \tag{17.4.16}$$

$$\begin{aligned} \delta_{\rm TRUE} A'^\mu_\alpha &= \partial^\mu \epsilon_\alpha - C_{\alpha\beta\gamma}\epsilon_\beta (A^\mu_\gamma + A'^\mu_\gamma) \\ &= \bar{D}^\mu \epsilon_\alpha - C_{\alpha\beta\gamma}\epsilon_\beta A'^\mu_\gamma , \end{aligned} \tag{17.4.17}$$

and

$$\delta_{\rm TRUE}\psi = 0, \tag{17.4.18}$$

$$\delta_{\rm TRUE}\psi' = it_\alpha \epsilon_\alpha (\psi + \psi'). \tag{17.4.19}$$

Of course, this is the same as the formal transformations (17.4.2), (17.4.3), (17.4.9), (17.4.10) in its effect on $A + A'$ and $\psi + \psi'$, and therefore also leaves the original Lagrangian $\mathcal{L}$ invariant. However, for our new choice (17.4.1) of f_α, the term $f_\alpha f_\alpha$ does not depend only on $A + A'$, and is not invariant under (17.4.16) and (17.4.17). Instead,

$$\delta_{\rm TRUE} f_\alpha = \bar{D}_\mu \big(\bar{D}^\mu \epsilon_\alpha - C_{\alpha\beta\gamma}\epsilon_\beta A'^\mu_\gamma\big) \tag{17.4.20}$$

with $\bar{D}_\mu$ given by (17.4.5).

Finally, let us consider the ghost Lagrangian in this new gauge. The quantity (15.7.3) in the ghost action is given in general by just replacing ϵ_α with the ghost field $\omega_\alpha + \omega'_\alpha$ in $\delta_{\rm TRUE} f_\alpha$:

$$\Delta_\alpha = \bar{D}_\mu \left[\bar{D}^\mu(\omega_\alpha + \omega'_\alpha) - C_{\alpha\beta\gamma}(\omega_\beta + \omega'_\beta) A'^\mu_\gamma\right]. \tag{17.4.21}$$

The ghost Lagrangian in Eq. (15.6.2) is therefore

$$\mathcal{L}_{\rm GH} = (\omega^*_\alpha + \omega'^*_\alpha)\bar{D}_\mu \left[\bar{D}^\mu(\omega_\alpha + \omega'_\alpha) - C_{\alpha\beta\gamma}(\omega_\beta + \omega'_\beta) A'^\mu_\gamma\right] \tag{17.4.22}$$

or, integrating by parts,

$$\mathcal{L}_{\rm GH} = -\Big(\bar{D}_\mu(\omega^*_\alpha + \omega'^*_\alpha)\Big)\Big(\bar{D}^\mu(\omega_\alpha + \omega'_\alpha) - C_{\alpha\beta\gamma}(\omega_\beta + \omega'_\beta) A'^\mu_\gamma\Big). \tag{17.4.23}$$

This is manifestly invariant under the joint transformations (17.4.2), (17.4.3), supplemented now with transformations on ω and ω':

$$\delta\omega_\alpha = -C_{\alpha\beta\gamma}\epsilon_\beta \omega_\gamma, \tag{17.4.24}$$

$$\delta\omega'_\alpha = -C_{\alpha\beta\gamma}\epsilon_\beta \omega'_\gamma, \tag{17.4.25}$$

and likewise

$$\delta\omega^*_\alpha = -C_{\alpha\beta\gamma}\epsilon_\beta \omega^*_\gamma, \tag{17.4.26}$$

$$\delta\omega'^{\,*}_\alpha = -C_{\alpha\beta\gamma}\epsilon_\beta\omega'^{\,*}_\gamma. \tag{17.4.27}$$

We see that the formal combined transformations (17.4.2), (17.4.3), (17.4.9), (17.4.10), and (17.4.24)–(17.4.27) leave invariant the complete Lagrangian in the modified action (15.6.4):

$$\mathcal{L}_{\rm MOD} = \mathcal{L} - \frac{1}{2\xi}f_\alpha f_\alpha + \mathcal{L}_{\rm GH}. \tag{17.4.28}$$

We are integrating over A', ψ', ω', and ω'^* with a measure that is presumed to be invariant under the simple matrix transformations (17.4.3), (17.4.10), (17.4.25), and (17.4.27), so the effective action $\Gamma[A,\psi,\omega,\omega^*]$ is invariant under the remaining transformations (17.4.2), (17.4.9), (17.4.24), and (17.4.26). In other words, it is gauge-invariant in the same sense as the original action $I[A,\psi,\omega,\omega^*]$.

This formal gauge invariance sets powerful constraints on the infinities that can occur in the effective action. The ultraviolet divergences in Γ appear in the coefficients of terms whose dimensionalities are [mass]^d with $d \leq 4$, but here these terms are invariant under the background gauge transformations (17.4.2), (17.4.9), (17.4.24), and (17.4.26). For instance, in a gauge theory based on a simple gauge group, with spin 1/2 fermions belonging to an irreducible representation of this group, the only such terms are of the form

$$\Gamma_\infty = \int d^4x\, \mathcal{L}_\infty, \tag{17.4.29}$$

$$\begin{aligned}\mathcal{L}_\infty = &-\frac{1}{4}L_A F_{\alpha\mu\nu}F_\alpha{}^{\mu\nu} - L_\psi\bar\psi\gamma^\mu\bar D_\mu\psi \\ &- mL_m\bar\psi\psi - L_\omega(\bar D_\mu\omega^*_\alpha)(\bar D^\mu\omega_\alpha),\end{aligned} \tag{17.4.30}$$

where here $F_{\alpha\mu\nu}$, $\bar D_\mu\psi$, $\bar D_\mu\omega_\alpha$, and $\bar D_\mu\omega^*_\alpha$ are constructed entirely from background fields:[33]

$$F_{\alpha\mu\nu} \equiv \partial_\mu A_{\alpha\nu} - \partial_\nu A_{\alpha\mu} + C_{\alpha\beta\gamma}A_{\beta\mu}A_{\gamma\nu}, \tag{17.4.31}$$

$$\bar D_\mu\psi \equiv \partial_\mu\psi - it_\alpha A_{\alpha\mu}\psi, \tag{17.4.32}$$

$$\bar D_\mu\omega_\alpha \equiv \partial_\mu\omega_\alpha + C_{\alpha\beta\gamma}A_{\beta\mu}\omega_\gamma, \tag{17.4.33}$$

$$\bar D_\mu\omega^*_\alpha \equiv \partial_\mu\omega^*_\alpha + C_{\alpha\beta\gamma}A_{\beta\mu}\omega^*_\gamma. \tag{17.4.34}$$

Dimensional analysis leads us to expect that the constants L_A, L_ψ, L_m, and L_ω are logarithmically divergent.

To deal with these infinities, we note that the Lagrangian (17.4.12) contains a purely classical piece

$$\mathcal{L}_{\rm CLASS} = -\frac{1}{4}F_{\alpha\mu\nu}F_\alpha{}^{\mu\nu} - \bar\psi(\gamma^\mu\bar D_\mu + m)\psi - (\bar D_\mu\omega^*_\alpha)(\bar D^\mu\omega_\alpha) \tag{17.4.35}$$

[33]The condition of a simple gauge group insures that there is just a single kinetic term for A and for ω, proportional to $F_{\alpha\mu\nu}F_\alpha{}^{\mu\nu}$ and $(\bar D_\mu\omega^*_\alpha)(\bar D^\mu\omega_\alpha)$ respectively, while the condition that ψ transforms irreducibly insures that there is just a single kinetic term and a single mass term for ψ. It would be easy to treat more general possibilities, at the cost of a slight complication in notation. We are also implicitly using the conservation of ghost number.

obtained by dropping all terms in $\mathcal{L}_{\rm MOD}$ that involve the quantum fields A', ψ', ω', ω'^*. We define renormalized fields

$$A^R_{\alpha\mu} \equiv \sqrt{1+L_A}\, A_{\alpha\mu}, \tag{17.4.36}$$

$$\psi^R_\ell \equiv \sqrt{1+L_\psi}\, \psi_\ell, \tag{17.4.37}$$

$$\omega^R_\alpha \equiv \sqrt{1+L_\omega}\, \omega_\alpha, \tag{17.4.38}$$

$$\omega^{R*}_\alpha \equiv \sqrt{1+L_\omega}\, \omega^*_\alpha, \tag{17.4.39}$$

so that the sum of the terms (17.4.30) and (17.4.35) takes the form

$$\begin{aligned}\mathcal{L}_{\rm CLASS} + \mathcal{L}_\infty = & -\frac{1}{4}F^R_{\alpha\mu\nu}F^{R\mu\nu}_\alpha - \bar{\psi}^R\gamma^\mu D^R_\mu\psi^R \\ & - m^R\bar{\psi}^R\psi^R - (D^R_\mu\omega^{*R}_\alpha)(D^{\mu R}\omega^R_\alpha),\end{aligned} \tag{17.4.40}$$

where m^R is the renormalized mass

$$m^R \equiv m(1+L_m)/(1+L_\psi), \tag{17.4.41}$$

and

$$F^R_{\alpha\mu\nu} \equiv \partial_\mu A^R_{\alpha\nu} - \partial_\nu A^R_{\alpha\mu} + C^R_{\alpha\beta\gamma}A^R_{\beta\mu}A^R_{\gamma\nu}, \tag{17.4.42}$$

$$D^R_\mu\psi^R \equiv \partial_\mu\psi^R - it^R_\alpha A^R_{\alpha\mu}\psi^R, \tag{17.4.43}$$

$$D^R_\mu\omega^R_\alpha \equiv \partial_\mu\omega^R_\alpha + C^R_{\alpha\beta\gamma}A^R_{\beta\mu}\omega^R_\gamma, \tag{17.4.44}$$

$$D^R_\mu\omega^{*R}_\alpha \equiv \partial_\mu\omega^{*R}_\alpha + C^R_{\alpha\beta\gamma}A^R_{\beta\mu}\omega^{*R}_\gamma. \tag{17.4.45}$$

The renormalized structure constants and group generators here are just

$$C^R_{\alpha\beta\gamma} \equiv (1+L_A)^{-1/2}C_{\alpha\beta\gamma}, \tag{17.4.46}$$

$$t^R_\alpha \equiv (1+L_A)^{-1/2}t_\alpha. \tag{17.4.47}$$

Because we assumed that the Lie algebra here is simple, the structure constant $C_{\alpha\beta\gamma}$ and group generator t_α are fixed by the group structure except for a single common factor, the unrenormalized gauge coupling constant g. Eqs. (17.4.46) and (17.4.47) thus simply tell us that the gauge coupling constant g^R factor in C^R and t^R is renormalized by

$$g^R = g(1+L_A)^{-1/2}. \tag{17.4.48}$$

This result exhibits the particular virtue of the background field gauge. In a general gauge we would encounter independent renormalization factors for the gauge field and the gauge coupling constant, and we would have to calculate two separate amplitudes (say, the vacuum polarization and three-gauge-field vertex function) in order to be able to sort these out. In background field gauge, background field gauge invariance ties these two renormalizations together by requiring that the infinite terms in the effective Lagrangian involve the field strength in its original form (17.4.31), and so we can calculate the charge renormalization factor by studying just one gauge field amplitude.

17.5 A One–Loop Calculation in Background Field Gauge

As an exercise, we are now going to calculate the one-loop renormalization factor for the gauge coupling constant in a general non-Abelian gauge theory. As we will see in the next chapter, this provides an essential input in so-called 'renormalization group' calculations of physical processes at high energy; the results we obtain here will be used there to demonstrate the asymptotic freedom of non-Abelian gauge theories.

The method to be employed here is somewhat novel. Usually, one considers the effective action in a spacetime-dependent background gauge field, and calculates the terms quadratic in this field, extracting a factor $(q_\mu A_{\alpha\nu} - q_\nu A_{\alpha\mu})^2$ (where q is the gauge field four-momentum), and only then isolating the logarithmic divergence by setting $q = 0$ in the coefficient of this factor. Instead we shall follow the much simpler course of taking the gauge field to be spacetime-independent from the beginning. In this case the terms in the effective action that are quadratic or cubic in the gauge field of course vanish, but there is a non-vanishing quartic term that is ultraviolet divergent, and which can be used to calculate the coupling constant renormalization factor $(1 + L_A)^{-1/2}$. In this way our one-loop calculation becomes a matter of simple matrix algebra. Note that this procedure can only work in background field gauge; otherwise there would be independent logarithmic divergences in the parts of the effective action that are quartic and quadratic in the background gauge field.

With this motivation, we turn to the calculation of the one-loop effective action in a background field for which $A_{\alpha\mu}$ is constant and $\psi = \omega = \omega^* = 0$. For such a background field, the full modified Lagrangian is[34]

$$\mathcal{L}_{\text{MOD}} = \mathcal{L} + \mathcal{L}_f + \mathcal{L}_{GH}, \tag{17.5.1}$$

$$\begin{aligned}\mathcal{L} = &-\frac{1}{4}\left(F_{\alpha\mu\nu} + \bar{D}_\mu A'_{\alpha\nu} - \bar{D}_\nu A'_{\alpha\mu} + C_{\alpha\beta\gamma}A'_{\beta\mu}A'_{\gamma\nu}\right)^2 \\ &-\bar{\psi}'\left(\bar{\not{D}} - it_\alpha \not{A}'_\alpha + m\right)\psi',\end{aligned} \tag{17.5.2}$$

$$\mathcal{L}_f = -\frac{1}{2\xi}f_\alpha f_\alpha = -\frac{1}{2\xi}\left(\bar{D}_\mu A'^{\mu}_\alpha\right)^2, \tag{17.5.3}$$

$$\mathcal{L}_{GH} = -\left(\bar{D}_\mu \omega'^*_\alpha\right)\left(\bar{D}^\mu \omega'_\alpha - C_{\alpha\beta\gamma}\omega'_\beta A'^{\mu}_\gamma\right). \tag{17.5.4}$$

One-loop graphs for vacuum–vacuum amplitudes are calculated from the part of the action that is quadratic in the quantum fields $A', \psi', \omega', \omega'^*$ over which one integrates. Keeping only such quadratic terms, we have

$$\begin{aligned}\mathcal{L}_{\text{QUAD}} = &-\frac{1}{4}\left(\bar{D}_\mu A'_{\alpha\nu} - \bar{D}_\nu A'_{\alpha\mu}\right)^2 - \frac{1}{2}F^{\mu\nu}_\alpha C_{\alpha\beta\gamma}A'_{\beta\mu}A'_{\gamma\nu} \\ &-\bar{\psi}'(\bar{\not{D}} + m)\psi' - \frac{1}{2\xi}\left(\bar{D}_\mu A'^{\mu}_\alpha\right)^2 - \left(\bar{D}_\mu \omega'^*_\alpha\right)\left(\bar{D}^\mu \omega'_\alpha\right).\end{aligned} \tag{17.5.5}$$

[34]See Eqs. (17.4.12), (17.4.4), and (17.4.23). We are here specializing to the case of matter fields forming a multiplet of spin 1/2 fermions. The squares in $\mathcal{L}$ and $\mathcal{L}_f$ include obvious index contractions.

The corresponding action may be put in the general quadratic form:

$$
\begin{aligned}
I_{\rm QUAD} &\equiv \int d^4x\, \mathcal{L}_{\rm QUAD} \\
&= -\frac{1}{2}\int d^4x\, d^4y\, A'^{\mu}_{\alpha}(x)A'^{\nu}_{\beta}(y)\mathcal{D}^A_{x\alpha\mu,y\beta\nu} - \int d^4x\, d^4y\, \bar{\psi}'_k(x)\psi'_\ell(y)\mathcal{D}^\psi_{xk,y\ell} \\
&\quad - \int d^4x\, d^4y\, \omega'^*_\alpha(x)\omega'_\beta(y)\mathcal{D}^\omega_{x\alpha,y\beta},
\end{aligned}
\tag{17.5.6}
$$

with

$$
\begin{aligned}
\mathcal{D}^A_{x\alpha\mu,y\beta\nu} \equiv{}& \eta_{\mu\nu}\left(-\delta_{\gamma\alpha}\frac{\partial}{\partial x_\lambda} + C_{\gamma\delta\alpha}A^\lambda_\delta(x)\right)\left(-\delta_{\gamma\beta}\frac{\partial}{\partial y^\lambda} + C_{\gamma\epsilon\beta}A_{\epsilon\lambda}(y)\right)\delta^4(x-y) \\
&- \left(-\delta_{\gamma\alpha}\frac{\partial}{\partial x^\nu} + C_{\gamma\delta\alpha}A_{\delta\nu}(x)\right)\left(-\delta_{\gamma\beta}\frac{\partial}{\partial y^\mu} + C_{\gamma\epsilon\beta}A^\mu_\epsilon(y)\right)\delta^4(x-y) \\
&+ F_{\gamma\mu\nu}(x)C_{\gamma\alpha\beta}\delta^4(x-y) \\
&+ \frac{1}{\xi}\left(-\delta_{\gamma\alpha}\frac{\partial}{\partial x^\mu} + C_{\gamma\delta\alpha}A_{\delta\mu}(x)\right)\left(-\delta_{\gamma\beta}\frac{\partial}{\partial y^\nu} + C_{\gamma\epsilon\beta}A_{\epsilon\nu}(y)\right)\delta^4(x-y),
\end{aligned}
\tag{17.5.7}
$$

$$
\mathcal{D}^\psi_{xk,y\ell} \equiv \left(-\gamma^\mu\frac{\partial}{\partial y^\mu} - it_\alpha \not{A}_\alpha(y) + m\right)_{k\ell}\delta^4(x-y),
\tag{17.5.8}
$$

$$
\mathcal{D}^\omega_{x\alpha,y\beta} \equiv \left(-\delta_{\gamma\alpha}\frac{\partial}{\partial x^\lambda} + C_{\gamma\delta\alpha}A_{\delta\lambda}(x)\right)\left(-\delta_{\gamma\beta}\frac{\partial}{\partial y_\lambda} + C_{\gamma\epsilon\beta}A^\lambda_\epsilon(y)\right)\delta^4(x-y).
\tag{17.5.9}
$$

(The minus signs in front of $\partial/\partial x$ and $\partial/\partial y$ drop out when we integrate by parts.)

The one-loop contribution to the effective action is given (as in Section 16.2) by

$$
\begin{aligned}
\exp\big(i\Gamma^{(1\ \rm loop)}[A]\big) &\propto \int_{\rm 1PI}\Big(\prod dA'\Big)\Big(\prod d\psi'\Big)\Big(\prod d\bar{\psi}'\Big)\Big(\prod d\omega'\Big)\Big(\prod d\omega'^*\Big) \\
&\qquad \times \exp\big(iI_{\rm QUAD}[A',\psi',\bar{\psi}',\omega',\omega'^*;A]\big) \\
&\propto \big(\mathrm{Det}\,\mathcal{D}^A\big)^{-1/2}\big(\mathrm{Det}\,\mathcal{D}^\psi\big)^{+1}\big(\mathrm{Det}\,\mathcal{D}^\omega\big)^{+1}.
\end{aligned}
\tag{17.5.10}
$$

(The exponents $-1/2$ and $+1$ appear because A' is a real boson field, while ψ', $\bar{\psi}'$, ω', and ω'^* are distinct fermionic fields.) The calculation of such determinants is generally not an easy task. However, it becomes much simpler in the case of constant external fields, where the $\mathcal{D}$s can be diagonalized by passing to momentum space.

Let us therefore now consider the case of constant background field $A_{\alpha\mu}$. For non-Abelian gauge theories such a constant field *cannot* be removed by a gauge transformation, as shown by the non-zero values of various gauge-covariant fields

$$
F_{\alpha\mu\nu} = C_{\alpha\beta\gamma}A_{\beta\mu}A_{\gamma\nu},
\tag{17.5.11}
$$

$$
D_\lambda F_{\delta\mu\nu} = C_{\delta\epsilon\alpha}C_{\alpha\beta\gamma}A_{\epsilon\lambda}A_{\beta\mu}A_{\gamma\nu},
\tag{17.5.12}
$$

and so on. Lorentz and gauge invariance tell us how to express the part of $\Gamma[A]$ of a given dimensionality as the integral of a finite number of local functions of $F_{\alpha\mu\nu}$, $D_\lambda F_{\alpha\mu\nu}$, etc.; the coefficients of the terms in this expression can be inferred by comparing the contribution that these terms make to $\Gamma[A]$ for constant background field $A_{\alpha\mu}$ with the results of a perturbative expansion.

We transform each of the 'matrices' $\mathcal{D}^A$, $\mathcal{D}^\psi$, and $\mathcal{D}^\omega$ to a momentum basis by the usual normalized Fourier transform

$$\mathcal{D}_{q\cdots,p\cdots} = \int \frac{d^4x}{(2\pi)^2} e^{-iq\cdot x} \int \frac{d^4y}{(2\pi)^2} e^{ip\cdot y} \mathcal{D}_{x\cdots,y\cdots}. \tag{17.5.13}$$

With A constant, this gives

$$\mathcal{D}_{q\cdots,p\cdots} = \delta^4(p-q)\mathcal{M}_{\cdots,\cdots}(q), \tag{17.5.14}$$

where $\cdots$ denotes discrete indices, and the $\mathcal{M}$ are finite q-dependent matrices

$$\begin{aligned}\mathcal{M}^A_{\alpha\mu,\beta\nu}(q) = \eta_{\mu\nu}&\big(-iq^\lambda\delta_{\gamma\alpha} + A^\lambda_\delta C_{\gamma\delta\alpha}\big)\big(iq_\lambda\delta_{\gamma\beta} + A_{\epsilon\lambda}C_{\gamma\epsilon\beta}\big)\\ &- \big(-iq_\nu\delta_{\gamma\alpha} + A_{\delta\nu}C_{\gamma\delta\alpha}\big)\big(iq_\mu\delta_{\gamma\beta} + A^\mu_\epsilon C_{\gamma\epsilon\beta}\big) + F_{\gamma\mu\nu}C_{\gamma\alpha\beta}\\ &+ \frac{1}{\xi}\big(-iq_\mu\delta_{\gamma\alpha} + A_{\delta\mu}C_{\gamma\delta\alpha}\big)\big(iq_\nu\delta_{\gamma\beta} + A_{\epsilon\nu}C_{\gamma\epsilon\beta}\big) + \epsilon \text{ terms},\end{aligned} \tag{17.5.15}$$

$$\mathcal{M}^\psi_{k\ell}(q) = \big(i\not{q} - it_\alpha \not{A}_\alpha + m\big)_{k\ell} + \epsilon \text{ terms}, \tag{17.5.16}$$

$$\mathcal{M}^\omega_{\alpha\beta}(q) = \big(-iq_\lambda\delta_{\gamma\alpha} + A_{\delta\lambda}C_{\gamma\delta\alpha}\big)\big(iq^\lambda\delta_{\gamma\beta} + A^\lambda_\epsilon C_{\gamma\epsilon\beta}\big) + \epsilon \text{ terms}, \tag{17.5.17}$$

with $F_{\gamma\mu\nu}$ given by Eq. (17.5.11). From Eq. (17.5.10) we have then

$$\begin{aligned}i\Gamma^{(1\text{ loop})}[A] &= -\frac{1}{2}\ln\mathrm{Det}\,\mathcal{D}^A + \ln\mathrm{Det}\,\mathcal{D}^\psi + \ln\mathrm{Det}\,\mathcal{D}^\omega\\ &= -\frac{1}{2}\mathrm{Tr}\ln\mathcal{D}^A + \mathrm{Tr}\ln\mathcal{D}^\psi + \mathrm{Tr}\ln\mathcal{D}^\omega\\ &= \delta^4(p-p)\int d^4q\left[-\frac{1}{2}\mathrm{tr}\ln\mathcal{M}^A(q) + \mathrm{tr}\ln\mathcal{M}^\psi(q) + \mathrm{tr}\ln\mathcal{M}^\omega(q)\right].\end{aligned} \tag{17.5.18}$$

We denote traces by 'tr' instead of 'Tr' in the last line of Eq. (17.5.18) (and from now on in this section) to indicate that these are the usual traces of finite matrices rather than of integral operators.

Since we are aiming here at a calculation of the infinite factor L_A multiplying FF terms in the effective action, let's isolate the term in (17.5.18) that is of fourth order in the background field A. For this purpose, it is convenient to divide each $\mathcal{M}$ into terms $\mathcal{M}_n$ containing $n = 0, 1$, or 2 factors of A:

$$\mathcal{M} = \mathcal{M}_0 + \mathcal{M}_1 + \mathcal{M}_2. \tag{17.5.19}$$

It is then elementary algebra to show that the term in (17.5.18) of fourth order in $A_{\alpha\mu}$ is

$$[\mathrm{tr}\ln\mathcal{M}]_{A^4} = \mathrm{tr}\left\{-\frac{1}{2}\left[\mathcal{M}_0^{-1}\mathcal{M}_2\right]^2 + \left[\mathcal{M}_0^{-1}\mathcal{M}_1\right]^2\mathcal{M}_0^{-1}\mathcal{M}_2 - \frac{1}{4}\left[\mathcal{M}_0^{-1}\mathcal{M}_1\right]^4\right\}. \tag{17.5.20}$$

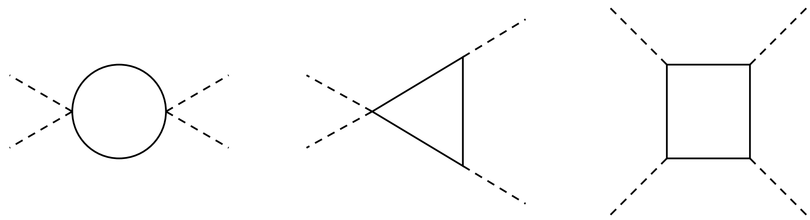

Figure 17.1. One-loop Feynman diagrams for the term in the quantum effective action that is quartic in a constant background gauge field A^μ_α. Here solid lines represent internal gauge, ghost, or matter lines; dashed lines indicate factors of A^μ_α. These three diagrams correspond to the three terms in Eq. (17.5.20).

(To see this, insert factors ϵ and ϵ^2 multiplying $\mathcal{M}_1$ and $\mathcal{M}_2$ in Eq. (17.5.19), differentiate $\operatorname{tr}\ln\mathcal{M}$ four times with respect to ϵ, divide by 4! and set $\epsilon = 0$.) The $\mathcal{M}_0^{-1}$ factors here are just the usual propagators; for $\xi = 1$, these are

$$\left[\mathcal{M}_0^A(q)\right]^{-1}_{\alpha\mu,\beta\nu} = \delta_{\alpha\beta}\eta_{\mu\nu}(q^2 - i\epsilon)^{-1}, \tag{17.5.21}$$

$$\left[\mathcal{M}_0^\psi(q)\right]^{-1}_{k\ell} = \left[i\not{q} + m\right]^{-1}_{k\ell}, \tag{17.5.22}$$

$$[\mathcal{M}_0^\omega(q)]^{-1}_{\alpha,\beta} = \delta_{\alpha\beta}(q^2 - i\epsilon)^{-1}. \tag{17.5.23}$$

Indeed, the three terms in Eq. (17.5.20) just correspond to the three Feynman diagrams shown in Figure 17.1; the present method of calculation saves us from having to think about signs and combinatoric factors.

For the A loop, Eq. (17.5.15) gives for $\xi = 1$:

$$\begin{aligned}
\left[\mathcal{M}_1^A(q)\right]_{\alpha\mu,\beta\nu} &= -2\eta_{\mu\nu}q^\lambda[\mathcal{A}_\lambda]_{\alpha\beta}, \\
\left[\mathcal{M}_2^A(q)\right]_{\alpha\mu,\beta\nu} &= \left[\eta_{\mu\nu}\mathcal{A}_\lambda\mathcal{A}^\lambda - \mathcal{A}_\nu\mathcal{A}_\mu - \mathcal{A}_\mu\mathcal{A}_\nu\right]_{\alpha\beta} + F_{\gamma\mu\nu}C_{\gamma\alpha\beta},
\end{aligned}$$

where $\mathcal{A}_\lambda$ is the matrix

$$[\mathcal{A}_\lambda]_{\alpha\beta} \equiv -iC_{\alpha\beta\gamma}A_{\gamma\lambda}$$

for which

$$\begin{aligned}
[\mathcal{A}_\lambda, \mathcal{A}_\rho]_{\alpha\beta} &= -C_{\alpha\delta\gamma}C_{\delta\beta\epsilon}(A_{\gamma\lambda}A_{\epsilon\rho} - A_{\gamma\rho}A_{\epsilon\lambda}) \\
&= -\big(C_{\alpha\delta\gamma}C_{\delta\beta\epsilon} + C_{\alpha\delta\epsilon}C_{\delta\beta\gamma}\big)A_{\gamma\lambda}A_{\epsilon\rho} \\
&= +C_{\alpha\delta\beta}C_{\delta\epsilon\gamma}A_{\gamma\lambda}A_{\epsilon\rho} = C_{\alpha\delta\beta}F_{\delta\rho\lambda}.
\end{aligned}$$

The integrals have the structure

$$\begin{aligned}
\int d^4q\, q^\mu q^\nu f(q^2) &= \frac{1}{4}\eta^{\mu\nu}\int d^4q\, q^2 f(q^2), \\
\int d^4q\, q^\mu q^\nu q^\rho q^\sigma f(q^2) &= \frac{1}{24}\left[\eta^{\mu\nu}\eta^{\rho\sigma} + \eta^{\mu\rho}\eta^{\nu\sigma} + \eta^{\mu\sigma}\eta^{\nu\rho}\right]\int d^4q\,(q^2)^2 f(q^2).
\end{aligned}$$

We then find, for $\xi = 1$:

$$\int d^4q \; \text{tr}\left\{\left[\mathcal{M}_0^A(q)^{-1}\mathcal{M}_2^A(q)\right]^2\right\}$$
$$= 4\mathcal{I}\;\text{tr}[\mathcal{A}^\lambda\mathcal{A}_\lambda\mathcal{A}^\eta\mathcal{A}_\eta] + 4\mathcal{I}\,C_{\gamma\alpha\beta}C_{\delta\alpha\beta}F_{\gamma\mu\nu}F_\delta^{\mu\nu},$$

$$\int d^4q \; \text{tr}\left\{\left[\mathcal{M}_0^A(q)^{-1}\mathcal{M}_1^A(q)\right]^2\mathcal{M}_0^A(q)^{-1}\mathcal{M}_2^A(q)\right\}$$
$$= 4\mathcal{I}\;\text{tr}[\mathcal{A}^\lambda\mathcal{A}_\lambda\mathcal{A}^\eta\mathcal{A}_\eta],$$

$$\int d^4q \; \text{tr}\left\{\left[\mathcal{M}_0^A(q)^{-1}\mathcal{M}_1^A(q)\right]^4\right\}$$
$$= \frac{8}{3}\mathcal{I}\;\text{tr}\begin{bmatrix}2\mathcal{A}^\lambda\mathcal{A}_\lambda\mathcal{A}^\eta\mathcal{A}_\eta \\ +\mathcal{A}^\lambda\mathcal{A}^\eta\mathcal{A}_\lambda\mathcal{A}_\eta\end{bmatrix},$$

where $\mathcal{I}$ is the divergent integral

$$\mathcal{I} \equiv \int d^4q\,[q^2 - i\epsilon]^{-2}, \tag{17.5.24}$$

whose significance is discussed below. Putting this together in Eq. (17.5.20), we have

$$\int d^4q\left[\text{tr}\ln\mathcal{M}^A(q)\right]_{A^4} = \frac{2}{3}\mathcal{I}\;\text{tr}[\mathcal{A}^\lambda\mathcal{A}_\lambda\mathcal{A}^\eta\mathcal{A}_\eta - \mathcal{A}^\lambda\mathcal{A}^\eta\mathcal{A}_\lambda\mathcal{A}_\eta]$$
$$- 2\mathcal{I}\,C_{\gamma\alpha\beta}C_{\delta\alpha\beta}F_{\gamma\mu\nu}F_\delta^{\mu\nu}.$$

Both terms are actually of the same form, and when combined yield

$$\int d^4q\left[\text{tr}\ln\mathcal{M}^A(q)\right]_{A^4} = -\frac{5}{3}\mathcal{I}\,C_{\gamma\alpha\beta}C_{\delta\alpha\beta}F_{\gamma\mu\nu}F_\delta^{\mu\nu}. \tag{17.5.25}$$

Now jumping ahead to the ghost loop, we see from Eq. (17.5.17) that

$$[\mathcal{M}_1^\omega(q)]_{\alpha\beta} = -2[\mathcal{A}^\lambda]_{\alpha\beta}q_\lambda, \tag{17.5.26}$$

$$[\mathcal{M}_2^\omega(q)]_{\alpha\beta} = [\mathcal{A}^\lambda\mathcal{A}_\lambda]_{\alpha\beta}. \tag{17.5.27}$$

We then find

$$\int d^4q \; \text{tr}\left\{[\mathcal{M}_0^\omega(q)^{-1}\mathcal{M}_2^\omega(q)]^2\right\}$$
$$= \mathcal{I}\;\text{tr}[\mathcal{A}^\lambda\mathcal{A}_\lambda\mathcal{A}^\eta\mathcal{A}_\eta],$$

$$\int d^4q \; \text{tr}\left\{[\mathcal{M}_0^\omega(q)^{-1}\mathcal{M}_1^\omega(q)]^2\mathcal{M}_0^\omega(q)^{-1}\mathcal{M}_2^\omega(q)\right\}$$
$$= \mathcal{I}\;\text{tr}[\mathcal{A}^\lambda\mathcal{A}_\lambda\mathcal{A}^\eta\mathcal{A}_\eta],$$

$$\int d^4q \; \text{tr}\left\{[\mathcal{M}_0^\omega(q)^{-1}\mathcal{M}_1^\omega(q)]^4\right\}$$
$$= \frac{2}{3}\mathcal{I}\;\text{tr}\begin{bmatrix}2\mathcal{A}^\lambda\mathcal{A}_\lambda\mathcal{A}^\eta\mathcal{A}_\eta \\ +\mathcal{A}^\lambda\mathcal{A}^\eta\mathcal{A}_\lambda\mathcal{A}_\eta\end{bmatrix}.$$

Thus for the ghost loop the integral of the quantity (17.5.20) is

$$\begin{aligned}\int d^4q\,[\mathrm{Tr}\ln\mathcal{M}^\omega(q)]_{A^4} &= \frac{1}{6}\mathcal{I}\ \mathrm{Tr}[\mathcal{A}^\lambda\mathcal{A}_\lambda\mathcal{A}^\eta\mathcal{A}_\eta - \mathcal{A}^\lambda\mathcal{A}^\eta\mathcal{A}_\lambda\mathcal{A}_\eta] \\ &= \frac{1}{12}\mathcal{I}\,C_{\gamma\alpha\beta}C_{\delta\alpha\beta}F_{\gamma\mu\nu}F_\delta^{\mu\nu}. \end{aligned} \tag{17.5.28}$$

Finally, the vertices in the matter loop are

$$[\mathcal{M}_1^\psi(q)]_{k\ell} = -i(t_\alpha \not{\mathcal{A}}_\alpha)_{k\ell}, \qquad [\mathcal{M}_2^\psi(q)]_{k\ell} = 0,$$

so here there is only one term in Eq. (17.5.20):

$$\int d^4q\left[\mathrm{Tr}\ln\mathcal{M}^\psi(q)\right]_{A^4} = -\frac{1}{4}\int d^4q\ \mathrm{Tr}\left\{\left[(i\not{q}+m)^{-1}t_\alpha\not{\mathcal{A}}_\alpha\right]^4\right\}.$$

We are interested in the ultraviolet-divergent part of this integral, so we may drop the mass (which is negligible for large q^ν) and write

$$\begin{aligned}\int d^4q\left[\mathrm{Tr}\ln\mathcal{M}^\psi(q)\right]_{A^4} =& -\frac{1}{4}\mathrm{Tr}\{t_\alpha t_\beta t_\gamma t_\delta\}A_{\alpha\mu}A_{\beta\nu}A_{\gamma\rho}A_{\delta\sigma} \\ &\times\int d^4q\,\frac{\mathrm{Tr}\{\not{q}\gamma^\mu\not{q}\gamma^\nu\not{q}\gamma^\rho\not{q}\gamma^\sigma\}}{(q^2-i\epsilon)^8} \\ =& -\frac{\mathcal{I}}{96}\mathrm{Tr}\{t_\alpha t_\beta t_\gamma t_\delta\}A_{\alpha\mu}A_{\beta\nu}A_{\gamma\rho}A_{\delta\sigma} \\ &\times\mathrm{Tr}\left\{\begin{matrix}2\gamma_\lambda\gamma^\mu\gamma^\lambda\gamma^\nu\gamma_\eta\gamma^\rho\gamma^\eta\gamma^\sigma \\ +\gamma_\lambda\gamma^\mu\gamma_\eta\gamma^\nu\gamma^\lambda\gamma^\rho\gamma^\eta\gamma^\sigma\end{matrix}\right\}. \end{aligned} \tag{17.5.29}$$

where $\mathcal{I}$ is the same divergent integral as in Eq. (17.5.24). To calculate the traces of Dirac matrices, we use the anticommutation relations of these matrices to write

$$\begin{aligned}\mathrm{Tr}\left\{\begin{matrix}2\gamma_\lambda\gamma^\mu\gamma^\lambda\gamma^\nu\gamma_\eta\gamma^\rho\gamma^\eta\gamma^\sigma \\ +\gamma_\lambda\gamma^\mu\gamma_\eta\gamma^\nu\gamma^\lambda\gamma^\rho\gamma^\eta\gamma^\sigma\end{matrix}\right\} &= 8\,\mathrm{Tr}\{\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma\} \\ &\quad -4\,\mathrm{Tr}\{\gamma^\nu\gamma^\mu\gamma^\rho\gamma^\sigma\} - 4\,\mathrm{Tr}\{\gamma^\mu\gamma^\rho\gamma^\nu\gamma^\sigma\} \\ &= -64\eta^{\mu\rho}\eta^{\nu\sigma} + 32\eta^{\mu\nu}\eta^{\rho\sigma} + 32\eta^{\mu\sigma}\eta^{\nu\rho}. \end{aligned}$$

Eq. (17.5.29) then gives

$$\begin{aligned}\int d^4q\left[\mathrm{Tr}\ln\mathcal{M}^\psi(q)\right]_{A^4} &= \frac{1}{3}\mathcal{I}\ \mathrm{Tr}\{[t_\alpha,t_\beta][t_\gamma,t_\delta]\}A_{\alpha\mu}A_{\beta\nu}A_\gamma^\mu A_\delta^\nu \\ &= -\frac{1}{3}\mathcal{I}\,F_{\gamma\mu\nu}F_\delta^{\mu\nu}\,\mathrm{Tr}\{t_\gamma,t_\delta\}. \end{aligned} \tag{17.5.30}$$

Using Eqs. (17.5.25), (17.5.28), and (17.5.30) in Eq. (17.5.18) gives at last

$$\Gamma_{A^4}^{(1\text{ loop})} = \frac{-i\mathcal{I}}{(2\pi)^4}\int d^4x\,F_{\gamma\mu\nu}F_\delta^{\mu\nu}\left[\left(\frac{5}{6}+\frac{1}{12}\right)C_{\gamma\alpha\beta}C_{\delta\alpha\beta} - \frac{1}{3}\mathrm{Tr}\{t_\gamma,t_\delta\}\right], \tag{17.5.31}$$

where we have expressed the momentum-space delta-function in (17.5.14) as

$$\delta^4(p-p) = (2\pi)^{-4}\int d^4x\,1. \tag{17.5.32}$$

It is important that the result turns out to depend on $A_{\alpha\mu}$ only through the field strength (17.5.11), as required by background gauge invariance.

Let us now (for the first time in this section) use the assumed simplicity of the gauge group and irreducibility of the matter field multiplet. In this case

$$C_{\gamma\alpha\beta}C_{\delta\alpha\beta} = g^2 C_1 \delta_{\gamma\delta}, \tag{17.5.33}$$

$$\mathrm{Tr}\{t_\gamma, t_\delta\} = g^2 C_2 \delta_{\gamma\delta}, \tag{17.5.34}$$

where g is the common gauge coupling constant appearing as a factor in $C_{\gamma\alpha\beta}$ and t_γ, and C_1 and C_2 are numerical constants that characterize the gauge group and the representation of this group provided by the matter multiplet. For instance, in the original Yang–Mills theory the gauge group is $SU(2)$ (or equivalently $SO(3)$) and the structure constants are

$$C_{\gamma\alpha\beta} = g\epsilon_{\gamma\alpha\beta}$$

with α, β, and γ running over the values 1, 2, 3. Comparing with Eq. (17.5.33), we see that here

$$C_1 = 2.$$

Also in this theory the matter field forms a doublet with t_α given by $g/2$ times the usual Pauli matrix σ_α, so

$$C_2 = 1/2.$$

Somewhat more generally, for the group $SU(N)$ with n_f fermions in the defining representation, with a conventional normalization of generators, we have[35]

$$C_1 = N, \qquad C_2 = n_f/2. \tag{17.5.35}$$

Returning now to the general case, Eqs. (17.5.33) and (17.5.34) give

$$\Gamma^{(1\text{ loop})}_{A^4} = \frac{-ig^2\mathcal{I}}{(2\pi)^4}\int d^4x\, F_{\gamma\mu\nu}F_\gamma^{\mu\nu}\left[\frac{11}{12}C_1 - \frac{1}{3}C_2\right]. \tag{17.5.36}$$

That is, the infinite constant L_A in Eq. (17.4.30) is

$$L_A = \frac{4ig^2\mathcal{I}}{(2\pi)^4}\left(\frac{11}{12}C_1 - \frac{1}{3}C_2\right). \tag{17.5.37}$$

It remains to say a word about the interpretation of the divergent integral $\mathcal{I}$. First, before we try to integrate over the three-momentum $\mathbf{q}$, we can rotate the contour of integration of q^0 in Eq. (17.5.24) to the imaginary axis; as usual, the $-i\epsilon$ in the denominator forces us to rotate counterclockwise, so that $q^0 = iq^4$, with q^4 running from $-\infty$ to $+\infty$. The integral is then

$$\mathcal{I} = i\int_0^\infty \frac{2\pi^2 q^3\, dq}{q^4}, \tag{17.5.38}$$

[35]For $SU(3)$ the t_α are taken as $g/2$ times the Gell-Mann matrices λ_α used in Section 19.7, so that $C_{\alpha\beta\gamma} = (g/2)f_{\alpha\beta\gamma}$.

where q is now the magnitude of the Euclidean four-vector (q^1, q^2, q^3, q^4). To go further, we evidently need some method of regulating the integral. The simplest way of dealing with the ultraviolet divergence is just to cut off the integral for q above a scale Λ. However, we also need a lower cut-off to deal with the infrared divergence. This is provided by the physics of the situation. If the momenta of the four vector particles is not zero, then a momentum flows through the internal lines of the diagrams, providing an infrared cut-off at the scale μ of these momenta. Similarly, if we evaluate the fourth variational derivative of $\Gamma[A]$ with respect to A not at $A = 0$ but at a finite A, then the propagators of the internal lines do not blow up at zero momenta, and we have an infrared cut-off at a scale $\mu \sim gA$. Either way, $\mathcal{I}$ takes the form

$$\mathcal{I} = 2\pi^2 i \int_\mu^\Lambda \frac{dq}{q} = 2\pi^2 i \ln\left(\frac{\Lambda}{\mu}\right) \tag{17.5.39}$$

and hence

$$L_A = -\frac{g^2}{2\pi^2}\left(\frac{11}{12}C_1 - \frac{1}{3}C_2\right)\ln\left(\frac{\Lambda}{\mu}\right) + O(g^4). \tag{17.5.40}$$

Eq. (17.4.48) then gives the renormalized coupling as

$$g_R = g\left[1 + \frac{g^2}{4\pi^2}\ln\left(\frac{\Lambda}{\mu}\right)\left(\frac{11}{12}C_1 - \frac{1}{3}C_2\right) + O(g^4)\right]. \tag{17.5.41}$$

We note that while in quantum electrodynamics the radiative corrections discussed in Section 11.2 decrease the physical coupling g_R relative to the bare coupling g, in non-Abelian gauge theories they *increase* the physical coupling over the bare coupling, provided that the fermion multiplet is small enough so that $C_2 < 11C_1/4$. The importance of this point will be explored in Chapter 18.

Alternatively, we can deal with the ultraviolet divergence by the methods of dimensional regularization, discussed in Section 11.2. Here, in place of Eq. (17.5.39), we write

$$\mathcal{I} = i\int_0^\infty \frac{2\pi^2 q^{d-1}\,dq}{(q^2+\mu^2)^2}, \tag{17.5.42}$$

where d is a complex dimensionality, allowed to approach 4 at the end of the calculation, and μ is an infrared cut-off, again taken of the order of the external momenta (or of the background fields times g). As long as d is complex with $\mathrm{Re}\, d < 0$ and $\mu^2 > 0$, this has the finite value

$$\mathcal{I} = -i\pi^2\left(\frac{d}{2} - 1\right)\mu^{d-4}\frac{\pi}{\sin[(d/2-2)\pi]}.$$

Analytically continuing to $d \to 4$, this is

$$\mathcal{I} \longrightarrow -2i\pi^2\left[\frac{1}{d-4} + \ln\mu + \cdots\right], \tag{17.5.43}$$

where $\cdots$ denotes finite μ-independent terms. Here we have

$$L_A = \frac{g^2}{2\pi^2}\left(\frac{11}{12}C_1 - \frac{1}{3}C_2\right)\left(\frac{1}{d-4} + \ln\mu + \cdots\right) + O(g^4)$$

and so

$$g_R = g\left[1 - \frac{g^2}{4\pi^2}\left(\frac{11}{12}C_1 - \frac{1}{3}C_2\right)\left(\frac{1}{d-4} + \ln\mu + \cdots\right) + O(g^4)\right]. \tag{17.5.44}$$

Note that the ultraviolet divergence here takes a different form, but the dependence on the infrared cut-off μ is the same. Eq. (17.5.44) will provide an important input to our discussion of asymptotic freedom in Section 18.7.

Problems

1. Carry out the proof of renormalizability given in Section 17.2, including elementary scalar fields in the Lagrangian.

2. Carry out the quantization of a non-Abelian gauge theory in background field gauge, using the BRST method of quantization discussed at the end of Section 17.2.

3. Derive the relation (17.5.44) between the renormalized and unrenormalized gauge couplings by calculating the terms in $\Gamma^{(1\ \text{loop})}$ that are quadratic in a spacetime-dependent gauge field.

4. Calculate the one-loop relation between the renormalized and unrenormalized gauge couplings in a gauge theory containing elementary scalar fields.

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18 Renormalization Group Methods

The method of the renormalization group was originally introduced by Gell-Mann and Low [1] as a means of dealing with the failure of perturbation theory at very high energies in quantum electrodynamics. An n-loop contribution to an amplitude involving momenta of order q, such as the vacuum polarization $\Pi_{\mu\nu}(q)$, is found to contain up to n factors of $\ln(q^2/m_e^2)$ as well as a factor α^n, so perturbation theory will break down when $\alpha|\ln(q^2/m_e^2)|$ is large, even though the fine structure constant α is small. Even in a massless theory like a non-Abelian gauge theory we must introduce some scale μ to specify a renormalization point at which the renormalized coupling constants are to be defined, and in this case we encounter logarithms $\ln(E/\mu)$, so that perturbation theory may break down if $E \gg \mu$ or $E \ll \mu$, even if the coupling constant is small.

Fortunately, there is a modified version of perturbation theory that can often be used in such cases. The key idea of this approach consists in the introduction of coupling constants g_μ defined at a sliding renormalization scale μ—that is, a scale that is not related to particle masses in any fixed way. By then choosing μ to be of the same order of magnitude as the energy E that is typical of the process in question, the factors $\ln(E/\mu)$ are rendered harmless. We can then do perturbation theory as long as g_μ remains small. In particular, given the coupling constants defined at scale μ, we can use perturbation theory to calculate physical amplitudes at an energy $\mu + d\mu$, and use these to calculate the coupling constants defined at a renormalization scale $\mu + d\mu$. By integrating the resulting differential equation we can then relate the coupling constants at the scale of interest to the coupling constants as conventionally defined. (The name 'renormalization group' arose originally because one is concerned here with equations that describe how the appearance of a theory changes under a redefinition of the renormalized coupling constants, but it really has nothing to do with group theory.) The method of the renormalization group can also provide qualitative guidance regarding asymptotic behavior at very high or (in massless theories) at very low energy, even where the coupling constants at the scale of interest are too large to allow the use of perturbation theory.

Although the method of the renormalization group arose originally in connection with changes in the prescription used to define renormalized coupling constants, it has come to have a wider meaning. When we replace bare couplings and fields with renormalized couplings and fields defined in terms of matrix elements evaluated at a characteristic energy scale μ, the integrals over virtual momenta will be effectively cut off at energy and momentum scales of order μ. Thus as we change μ, we are in effect changing the scope of the degrees of freedom taken into account in our calculations. The lesson of the renormalization group, that in order to avoid large logarithms we should take μ to be of the order of the energy E typical of the process being studied, is a special case of a broader principle, that in order to do calculations at a given energy we should first get rid of the degrees of freedom of much higher energy.

There are various other ways to accomplish this. As we saw in Section 12.4, in the approach to the renormalization group pioneered by Wilson [2] one introduces a *finite* explicit cut-off accompanied by a change in the parameters of the theory designed to keep physical quantities cut-off-independent. This approach requires introduction of an infinite number of interaction types, all those allowed by the symmetries of the theory, and is therefore not particularly convenient in dealing with theories that are actually renormalizable, like quantum electrodynamics (although, as discussed in Section 12.3, quantum electrodynamics is today regarded as only a very good approximation to a non-renormalizable theory in which the higher-dimensional interactions are suppressed by negative powers of some very large mass.) Where the cut-off is imposed by quantizing a gauge theory on a finite spacetime lattice, the Wilson approach has the advantage that calculations can be done while maintaining manifest gauge invariance (the volume of the gauge group equaling the volume of the global symmetry group times the number of lattice sites), but it has the disadvantage of not maintaining manifest Lorentz or rotational invariance. In any case much of the formalism of the renormalization group remains the same whatever approach is used to eliminate the high-energy degrees of freedom.

18.1 Where do the Large Logarithms Come From?

Let us first consider how large logarithms can arise at very high energies. Consider a physical amplitude or cross section or other rate parameter $\Gamma(E, x, g, m)$, that depends on an over-all energy scale E, on various angles and energy ratios collectively called x, on various dimensionless coupling constants collectively called g, and on various masses collectively called m. If Γ has dimensionality $[\text{mass}]^D$ (as, for instance, a cross section would have $D = -2$) then simple dimensional analysis tells us that

$$\Gamma(E, x, g, m) = E^D \Gamma\left(1, x, g, \frac{m}{E}\right). \tag{18.1.1}$$

We might expect that in the limit $E \to \infty$ such an amplitude would behave as a simple power

$$\Gamma(E, x, g, m) \longrightarrow E^D \Gamma(1, x, g, 0).$$

But this is not what is found. Instead, in perturbation theory calculations the factor E^D is found to be accompanied by powers of $\ln(E/m)$, which invalidate this simple power-law behavior.

Clearly, powers of $\ln(E/m)$ can enter as $E \to \infty$ with fixed m only if the amplitude Γ at *fixed energy* E becomes singular as $m \to 0$. There are two classes of such mass singularities, one which is simply eliminated by calculating the right sort of amplitude or rate constant, the other of which requires a change in our renormalization procedure.

Zero-mass singularities of the first sort arise from a confluence of poles of propagators on the mass shells of the corresponding particles. For instance, suppose that a Feynman diagram has an incoming line with total four-momentum p^μ, attached at a vertex to internal lines of mass $m_1, m_2, \ldots, m_n$. According to the arguments of Chapter 10, the corresponding Feynman diagram will have a cut running along the negative real p^2 axis, from $p^2 = -(m_1 + \cdots + m_n)^2$ to $-\infty$. This does not lead to singularities if the external line is for a stable particle, with a mass $M < m_1 + \cdots + m_n$, because then $p^2 = -M^2$ is off the cut. However, when $M, m_1, \ldots, m_n$ all go to zero, the value of $-p^2$ on the mass shell and the branch point at the tip of this cut move together to join at $p^2 = 0$, producing a singularity.

This suggests that we can avoid the infrared divergences at $m = 0$ by simply staying off the mass shell, as, for instance, by letting p^2 for all external lines go to $+\infty$ along with all energy variables. We would then have to employ dispersion relations or some other technique of analytic continuation to use the results for the behavior of Feynman amplitudes in this limit to tell us anything about S-matrix elements. Often this continuation is unnecessary because we are interested not in on-shell S-matrix elements but rather in the matrix elements of currents carrying momenta q unrelated to any masses. For instance, the vacuum polarization function $\pi(q^2)$ defined in Section 10.5 is free of zero-mass singularities of the first sort except for $q^2 < 0$.

Another approach to the elimination of mass singularities of the first type is suggested by the observation that zero-mass singularities typically occur if we try to calculate a cross section that becomes unmeasurable in the limit $m \to 0$. For instance, in quantum electrodynamics the cross section for any process involving definite numbers of electrons and photons becomes infrared-divergent in the limit $m_e \to 0$, even if we sum over unlimited numbers of soft photons, because for $m_e \to 0$ it is impossible to distinguish an electron from a jet of electrons, positrons, and photons with total charge $-e$, all moving in the same direction at the same speed. As shown in Chapter 13, such infrared divergences can be cured by considering only suitably integrated cross sections, which *would* be measurable for $m_e \to 0$. For instance, instead of trying to calculate the cross section for a specific Compton scattering process, we would calculate the cross section for the scattering of a jet with total charge $-e$ with another of total charge zero into two other such jets, plus soft photons. Such inclusive rates or cross sections, which remain finite when all masses vanish, are known as 'infrared safe.'

Our troubles are not over. Even where we avoid infrared divergences by integrating over cross sections or staying off the mass shell, the resulting integrated cross sections or off-shell amplitudes for energies E contain mass singularities of a second type, leading to factors $\ln(E/m)$ that invalidate the naive power-law behavior suggested by dimensional analysis. The reason can be traced to the fact that renormalized coupling constants are conventionally defined in terms of amplitudes that become infrared-divergent when all masses vanish. For instance, consider the theory of a real scalar field with Lagrangian density

$$\mathcal{L} = -\frac{1}{2}\partial_\lambda\phi\partial^\lambda\phi - \frac{1}{2}m^2\phi^2 - \frac{1}{24}g\phi^4. \tag{18.1.2}$$

To one-loop order, the invariant elastic scattering amplitude for a scattering process with initial four-momenta p_1, p_2, and final four-momenta p'_1, p'_2, is given by Eq. (12.2.24) as

$$A = g - \frac{g^2}{32\pi^2}\int_0^1 dx\Bigg\{\ln\left(\frac{\Lambda^2}{m^2 - sx(1-x)}\right) + \ln\left(\frac{\Lambda^2}{m^2 - tx(1-x)}\right) + \ln\left(\frac{\Lambda^2}{m^2 - ux(1-x)}\right) - 3\Bigg\} + O(g^3), \tag{18.1.3}$$

where s, t, and u are the Mandelstam variables

$$s = -(p_1 + p_2)^2, \qquad t = -(p_1 - p'_1)^2, \qquad u = -(p_1 - p'_2)^2$$

and Λ is an ultraviolet cut-off. This has no zero-mass singularities as long as we keep s, t, and u away from the positive real axis, and in particular if they all go to $-\infty$ (which violates the mass shell condition $s+t+u=4m^2$.) Of course, the amplitude depends on the cut-off Λ as well as on m, so even though there are no zero-mass singularities, we do not find the result $A \to$ constant that would be found on the basis of naive scaling arguments in the limit as s, t, and u go to $-\infty$. The dependence on the cut-off can be buried by renormalization; we replace the bare coupling g with a renormalized coupling g_R, defined as the value of A at some convenient renormalization point. For instance, we might take

$$\begin{aligned} g_R &\equiv A(s=t=u=0) \\ &= g - \frac{3g^2}{32\pi^2}\left\{\ln\frac{\Lambda^2}{m^2} - 1\right\} + O(g^3). \end{aligned} \tag{18.1.4}$$

Then (18.1.3) becomes

$$\begin{aligned} A = g_R + \frac{g_R^2}{32\pi^2}\int_0^1 dx\Bigg\{\ln\left(1-\frac{sx(1-x)}{m^2}\right) + \ln\left(1-\frac{tx(1-x)}{m^2}\right) \\ + \ln\left(1-\frac{ux(1-x)}{m^2}\right)\Bigg\} + O(g_R^3). \end{aligned} \tag{18.1.5}$$

(We can freely replace g^2 with g_R^2 in the second term, because the difference is only of order g^3.) This is free of ultraviolet divergences, but it now has a singularity at $m=0$, even where s, t, and u are all kept negative. In consequence, where s, t, and u all go to $-\infty$, we again find an asymptotic behavior in disagreement with expectations based on naive scaling

$$A \longrightarrow g_R + \frac{g_R^2}{32\pi^2}\left\{\ln\left(\frac{-s}{m^2}\right) + \ln\left(\frac{-t}{m^2}\right) + \ln\left(\frac{-u}{m^2}\right) - 6\right\}. \tag{18.1.6}$$

(Much the same happens with any other 'natural' definition of the renormalized coupling; for instance, we might define g_R as the value of A at the on-shell symmetrical point $s = t = u = 4m^2/3$, and would recover the same asymptotic behavior as in Eq. (18.1.6), except that -6 would be replaced some other numerical constant.) It is clear that the zero-mass singularity that is encountered when A is expressed in terms of g_R arises entirely from the $\ln m^2$ term in the formula (18.1.4) for the renormalized coupling g_R in terms of the bare coupling g.

There are in addition other zero-mass singularities that are encountered when we calculate matrix elements of operators (such as off-shell Feynman amplitudes) rather than integrals of cross sections. These are due to the necessity of renormalizing these operators as well as coupling constants. For instance, suppose that in the scalar field theory with Lagrangian (18.1.2), we wish to calculate some matrix element $\langle\beta\|O(p)\|\alpha\rangle$ of the operator

$$O(p) \equiv \int d^4x\, e^{-ip\cdot x}\phi^2(x). \tag{18.1.7}$$

In Feynman diagram terms, this corresponds to inserting a vertex in which two internal ϕ-lines come together, and through which flows a total four-momentum p in diagrams for the transition $\alpha \to \beta$. (See Figure 18.1.)

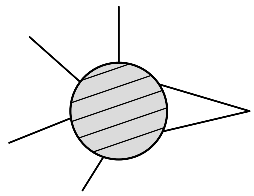

Figure 18.1. Momentum-space Feynman diagrams for the matrix element of the operator $\int d^4x \, \exp(-ip \cdot x)\, \phi^2(x)$ in the theory of an elementary scalar field $\phi(x)$. The cross-hatched disk represents the sum of diagrams with the indicated external lines. Apart from the pair of external lines that meet in a ϕ^2 vertex, the other external lines attached to the disk represent the particles in the initial and final states between which the matrix element is evaluated.

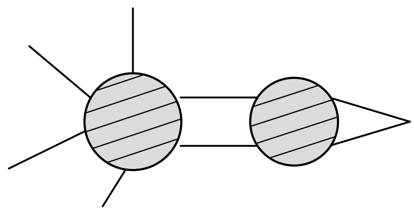

Figure 18.2. A class of Feynman diagrams for the matrix element of the operator $\int d^4x \, \exp(-ip \cdot x)\, \phi^2(x)$ that exhibit an ultraviolet divergence. The notation is the same as in Figure 18.1.

Ultraviolet divergences arise from a class of diagrams in which this new vertex is part of a subdiagram that is connected to the rest of the graph by just two ϕ-lines. (See Figure 18.2.) Dimensional analysis shows that the subgraph would be convergent if connected to the rest of the diagram by more than two ϕ-lines, because the interaction ϕ^2 has dimensionality $+2$, and hence a subdiagram in which this vertex is connected to the rest of the graph by $n > 2$ lines has dimensionality $4 - 2 - n < 0$. The divergent part of this subdiagram is just a logarithmically divergent constant, so

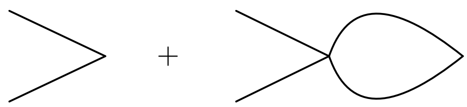

Figure 18.3. The divergent part of the diagram in Figure 18.2, to one-loop order.

the matrix elements of ϕ^2 can be made finite[36] by multiplying ϕ^2 with a suitable divergent constant Z_{ϕ^2}. To order g^2, the relevant subdiagram is given by the diagrams of Figure 18.3, and hence contributes to matrix elements of $O(p)$ a divergent factor

$$F(p) = 1 + \frac{1}{2}\big[-i(2\pi)^4 g\big] \left[\frac{-i}{(2\pi)^4}\right]^2 \int \frac{d^4k}{[k^2 + m^2 - i\epsilon][(p-k)^2 + m^2 - i\epsilon]}. \tag{18.1.8}$$

[36]In making this argument it was assumed that any divergences arising from subsubdiagrams containing the ϕ^2 vertex which are attached to the rest of the subdiagram by just two ϕ-lines are eliminated in the same way.

Combining denominators, rotating the k^0 integration contour, and imposing an ultraviolet cut-off Λ gives a result, for $\Lambda \to \infty$

$$F(p) = 1 - \frac{g}{32\pi^2}\int_0^1 dx\left[\ln\left(\frac{\Lambda^2}{m^2+p^2x(1-x)}\right) - 1\right] + O(g^2). \tag{18.1.9}$$

This has no zero-mass singularity (as long as we keep p^2 positive), but of course it does depend on the cut-off. This logarithmic divergence is eliminated by defining a renormalized ϕ^2 operator

$$(\phi^2)_R = N^{(\phi^2)}\phi^2 \tag{18.1.10}$$

with $N^{(\phi^2)}$ chosen so that $N^{(\phi^2)}F(p)$ has some definite finite value at some definite renormalization point. For instance, we could define the renormalized ϕ^2 operator so that

$$N^{(\phi^2)}F(0) = 1, \tag{18.1.11}$$

in which case

$$N^{(\phi^2)} = 1 + \frac{g}{32\pi^2}\left(\ln\frac{\Lambda^2}{m^2} - 1\right) + O(g^2). \tag{18.1.12}$$

The matrix elements of the renormalized operator $(\phi^2)_R$ then contain a factor

$$F_R(p) \equiv N^{(\phi^2)}F(p) = 1 + \frac{g}{32\pi^2}\int_0^1 dx\,\ln\left(1+\frac{p^2x(1-x)}{m^2}\right) + O(g^2). \tag{18.1.13}$$

This is finite for all $p^2 > 0$ and $m^2 > 0$, but it now contains an infrared singularity for $m \to 0$, corresponding to large logarithms in the asymptotic behavior when $p^2 \to +\infty$. Of course in order to eliminate the cut-off in higher-order calculations we would have to introduce a renormalized coupling constant as well as a renormalized ϕ^2 operator, and we would encounter logarithms arising from both sources.

Similar renormalization factors are needed for any sort of operator, not just $\phi^2(x)$. In particular, taking a matrix element of one of the elementary fields ψ of a theory introduces ultraviolet divergences that arise from radiative corrections to the corresponding propagator. As we saw in Chapter 12, these infinities can be cancelled by working with a renormalized field ψ_R:

$$\psi_R = N^{(\psi)}\psi \tag{18.1.14}$$

with $N^{(\psi)}$ chosen to make the matrix element of ψ_R between a one-particle state and the vacuum the same as for a conventionally normalized field in the absence of interactions. This is related to the usual Z-factor of renormalization theory by

$$Z^{(\psi)} = |N^{(\psi)}|^{-2}. \tag{18.1.15}$$

For instance, let's recall our earlier results for the renormalization of the photon field in spinor quantum electrodynamics.[37] In this case, the renormalized electromagnetic field A^μ_R is conventionally written in terms of the 'bare' field A^μ_B as

$$A^\mu_R = Z_3^{-1/2}A^\mu_B$$

[37]In the scalar field theory with ϕ^4 interaction used as an example above, the lowest-order terms in $N^{(\phi)}$ arise from a two-loop graph, so it will not be convenient to use this theory to illustrate the calculation of the N-factors.

with Z_3 given by Eq. (11.2.21) as

$$Z_3 \equiv 1 - \frac{e^2}{12\pi^2} \ln\left(\frac{\Lambda^2}{m^2}\right) + O(e^4). \tag{18.1.16}$$

This has a zero-mass singularity, which will affect the asymptotic behavior of matrix elements of the renormalized photon field. In particular, Eq. (11.2.22) gives the self-energy function of the renormalized electromagnetic field as

$$\pi(q^2) = \frac{e_R^2}{2\pi^2} \int_0^1 dx\, x(1-x) \ln\left[1 + \frac{q^2 x(1-x)}{m^2}\right] + O(e_R^4). \tag{18.1.17}$$

This has a singularity at $m = 0$, and so has large logarithms in its asymptotic behavior: for $q^2 \to +\infty$

$$\pi_R(q^2) \longrightarrow \frac{e^2}{2\pi^2}\left[\frac{1}{6}\ln\frac{q^2}{m^2} - \frac{5}{18}\right] + O(e^4). \tag{18.1.18}$$

Electrodynamics has the special feature that the constant Z_3 that appears in the renormalization of the electromagnetic field also appears in the renormalization of the electric charge:

$$e_R = Z_3^{1/2} e_{\text{BARE}}, \tag{18.1.19}$$

but this is not generally the case. Renormalization group techniques were first applied in quantum electrodynamics, but the scalar field theory discussed here gives a more typical illustration of these methods, with separate renormalizations for fields and couplings.

18.2 The Sliding Scale

We have seen in the previous section that the large logarithms that appear at high energy in suitably integrated cross sections or off-shell Feynman amplitudes can be traced to the prescription used to define renormalized coupling constants and operators. The central idea of the renormalization group method is to change this prescription.

Suppose we find some way of defining a new kind of renormalized coupling constant $g(\mu)$, that depends on a sliding energy scale μ, but that (at least for $\mu \gg m$) has no dependence on the scale m of the masses of the theory. Then suitably integrated cross sections or other infrared-safe rate parameters may be expressed as functions of g_μ and μ instead of g_R. By dimensional analysis such functions may be written as

$$\Gamma(E, x, g_\mu, m, \mu) = E^D \Gamma\left(1, x, g_\mu, \frac{m}{E}, \frac{\mu}{E}\right). \tag{18.2.1}$$

(Our notation here is the same as in Section 18.1; in particular, x stands for all dimensionless angles, energy ratios, etc. on which Γ may depend.) Since μ is a completely arbitrary renormalization scale, we can choose $\mu = E$, in which case Eq. (18.2.1) reads

$$\Gamma(E, x, g_\mu, m, \mu) = E^D \Gamma\left(1, x, g_E, \frac{m}{E}, 1\right). \tag{18.2.2}$$

This now has no zero-mass singularities because g_E does not depend on m for $m \ll E$, so there are no large logarithms, and we can use perturbation theory to calculate Γ in terms of g_E as long as g_E itself remains sufficiently small. In particular, in any finite order of perturbation theory Γ has the asymptotic behavior, for $E \gg m$,

$$\Gamma(E, x, g_\mu, m, \mu) \longrightarrow E^D \Gamma(1, x, g_E, 0, 1). \tag{18.2.3}$$

(Non-perturbative corrections are considered in Section 18.4.)

It remains to calculate g_E. For instance, in the scalar field theory with Lagrangian (18.1.2), we may define g_μ in terms of the value of the scattering amplitude at a renormalization point $s = t = u = -\mu^2$:

$$\begin{aligned} g_\mu &\equiv A(s = t = u = -\mu^2) \\ &= g - \frac{3g^2}{32\pi^2} \int_0^1 dx \left\{ \ln\left(\frac{\Lambda^2}{m^2 + \mu^2 x(1-x)}\right) - 1 \right\} + O(g^3). \end{aligned} \tag{18.2.4}$$

or, in terms of the conventional renormalized coupling (18.1.4),

$$g_\mu = g_R + \frac{3g_R^2}{32\pi^2} \int_0^1 dx \ln\left(1 + \frac{\mu^2 x(1-x)}{m^2}\right) + O(g_R^3). \tag{18.2.5}$$

But this formula is reliable only if the correction term is smaller than g_R; that is, only if $|g_R \ln(\mu/m)| \ll 1$. If this were the case for $\mu \simeq E$, then we would not need the methods of the renormalization group; ordinary perturbation theory would be good enough.

Instead of using formulas like (18.2.5) directly for large μ, we must instead proceed in stages: g_μ may be calculated in terms of g_R as long as μ/m is not much larger than unity; then $g_{\mu'}$ may be calculated in terms of g_μ as long as μ'/μ is not much larger than unity; and so on, up to g_E. Instead of discrete stages, this may also be done continuously. Dimensional analysis tells us that the relation between $g_{\mu'}$ and g_μ takes the form

$$g_{\mu'} = G(g_\mu, \mu'/\mu, m/\mu). \tag{18.2.6}$$

Differentiating with respect to μ' and then setting $\mu' = \mu$ yields the differential equation

$$\mu \frac{d}{d\mu} g_\mu = \beta\left(g_\mu, \frac{m}{\mu}\right), \tag{18.2.7}$$

where

$$\beta\left(g_\mu, \frac{m}{\mu}\right) \equiv \left[\frac{\partial}{\partial z} G(g_\mu, z, m/\mu)\right]_{z=1}. \tag{18.2.8}$$

There are no zero-mass singularities here, so for $\mu \gg m$ the differential equation becomes simply

$$\mu \frac{d}{d\mu} g_\mu = \beta(g_\mu, 0) \equiv \beta(g_\mu), \tag{18.2.9}$$

which is often known as the Callan–Symanzik equation [1]. We are to calculate g_E by integrating the differential equation (18.2.9), with an initial value g_M at some scale $\mu = M$, chosen in practice large enough so that for $\mu \geq M$ we can neglect the masses m compared with μ, but small enough so that large logarithms $\ln(M/m)$ do not prevent us from using perturbation theory to calculate g_M in terms of the conventional renormalized coupling constant g_R. The solution may be formally written

$$\ln(E/M) = \int_{g_M}^{g_E} \frac{dg}{\beta(g)} \tag{18.2.10}$$

as long as $\beta(g)$ does not vanish between g_M and g_E.

The results of the previous paragraph do not rely on perturbation theory, but we usually need to use perturbation theory to calculate the functions G and β. As an example, suppose we calculate $g_{\mu'}$ in the scalar field theory with interaction $g\phi^4/24$, renormalizing by expressing g in terms of g_μ rather than g_R. Following the same procedure that led to Eq. (18.2.5), this gives

$$g_{\mu'} = g_\mu - \frac{3g_\mu^2}{32\pi^2} \int_0^1 dx \ln\left(\frac{m^2 + \mu^2 x(1-x)}{m^2 + \mu'^2 x(1-x)}\right) + O(g_\mu^3).$$

Then Eq. (18.2.8) gives

$$\beta\left(g_\mu, \frac{m}{\mu}\right) = +\frac{3g_\mu^2}{16\pi^2} \int_0^1 dx \frac{\mu^2 x(1-x)}{m^2 + \mu^2 x(1-x)} + O(g_\mu^3). \tag{18.2.11}$$

For $\mu \gg m$, this is

$$\beta(g_\mu) = \frac{3g_\mu^2}{16\pi^2} + O(g_\mu^3). \tag{18.2.12}$$

In next order, the beta-function for $\mu \gg m$ is [3]

$$\beta(g_\mu) = g_\mu \left[3\left(\frac{g_\mu}{16\pi^2}\right) - \frac{17}{3}\left(\frac{g_\mu}{16\pi^2}\right)^2 + \cdots\right].$$

If we are content with the one-loop approximation the calculation of $\beta(g)$ can be done even more easily. In order to avoid large radiative corrections in matrix elements at energies of order μ, we must write the bare coupling g in terms of a finite renormalized coupling g_μ, as

$$g = g_\mu + B(g_\mu) \ln\frac{\Lambda}{\mu} + \cdots. \tag{18.2.13}$$

For instance, from Eq. (18.1.3) we could have immediately read off that the $\ln\Lambda$ terms in g have coefficient

$$B(g) = -\frac{3}{2}\left[-i(2\pi)^4 g^2\right] \left\{\left[\frac{-i}{(2\pi)^4}\right]^2 2\pi^2 i\right\} = \frac{3g^2}{16\pi^2}. \tag{18.2.14}$$

The unrenormalized coupling is of course independent of μ, so to lowest order

$$\mu\frac{d}{d\mu} g_\mu - B(g_\mu) = 0$$

and so to lowest order

$$\beta(g) = B(g). \tag{18.2.15}$$

With $B(g)$ given by Eq. (18.2.14), this agrees with our previous result (18.2.12) for $\beta(g)$ in the ϕ^4 scalar field theory.

Instead of using a simple ultraviolet cut-off, which can interfere with gauge invariance, it is often more convenient to deal with ultraviolet divergences by use of dimensional regularization. For a spacetime dimensionality $d < 4$, we find in place of $\ln(\Lambda/\mu)$ the convergent integral

$$\int_\mu^\infty k^{d-4}\frac{dk}{k} = \frac{\mu^{d-4}}{4-d} \xrightarrow[d\to 4]{} \left[\frac{1}{4-d} - \ln\mu\right].$$

Thus instead of eliminating the cut-off dependence by writing the unrenormalized coupling constant as in (18.2.13), we instead write

$$g = g_\mu + B(g_\mu)\left[\frac{1}{4-d} - \ln\mu\right] \tag{18.2.16}$$

with the same function $B(g_\mu)$ as before. Thus in order to calculate $\beta(g_\mu)$, all we need to do is to pick out the coefficient of the singular factor $1/(4-d)$ in the renormalized coupling. This argument is extended to all orders of perturbation theory in Section 18.6.

As long as g_μ is sufficiently small in the scalar field theory with Lagrangian (18.1.2), the solution of Eqs. (18.2.9) and (18.2.12) can be well approximated by

$$g_\mu = -\frac{16\pi^2}{3\ln(\mu/M)}, \tag{18.2.17}$$

where M is an integration constant. This expression illustrates a common aspect of renormalization group calculations, that dimensionless couplings like g_R become replaced with parameters like M that have the dimensionality of mass. The value of M may be related to g_R by comparing the solution (18.2.17) with the behavior of the coupling for values of μ that are large enough to allow us to use approximations based on $\mu \gg m$, but small enough so that $|g_R \ln(\mu/m)| \ll 1$, where (18.2.5) gives

$$g_\mu \simeq g_R + \frac{3g_R^2}{16\pi^2}\ln\left(\frac{\mu}{m}\right). \tag{18.2.18}$$

In this way, we find

$$M \simeq m\exp\left(\frac{16\pi^2}{3g_R}\right), \tag{18.2.19}$$

so that Eq. (18.2.17) may be put in a more conventional form

$$g_\mu = g_R\left[1 - \frac{3g_R}{16\pi^2}\ln\frac{\mu}{m}\right]^{-1}. \tag{18.2.20}$$

To repeat, this is valid provided g_μ is small, even if $|g_R \ln(\mu/m)|$ is of order unity, so it represents a significant improvement over the perturbative result (18.2.18). Of course, the condition that g_μ should be small will become violated when $g_R \ln(\mu/m)$ is sufficiently close to the critical value $16\pi^2/3$. But at least Eq. (18.2.20) makes the unequivocal prediction that g_E becomes large enough to invalidate perturbation theory at some energy E below the critical value (18.2.19).

In calculating off-shell matrix elements of operators instead of integrated cross sections, we also need to take into account the N-factors that appear in the definition of the renormalized operators whose matrix elements are finite. We saw in the previous section that if these N-factors are defined in a conventional way (say, so that the correction factors produced by the divergent subgraphs are cancelled when the operator carries zero four-momentum, or is a field on its mass shell) then the formula for the N-factor involves zero-mass singularities as in Eq. (18.1.12) or Eq. (18.1.16), resulting in large logarithms at energies $E \gg m$. The cure is to define renormalization constants $N_\mu^{(\mathcal{O})}$ at a sliding scale μ, so that in matrix elements of the renormalized operator

$$\mathcal{O}_\mu = N_\mu^{(\mathcal{O})}\mathcal{O} \tag{18.2.21}$$

the correction factor produced by divergent subgraphs containing operator $\mathcal{O}$ are cancelled at a renormalization point characterized by four-momenta of order μ. If M_R is a matrix element of operators that are conventionally renormalized, and M is one in which the operators are renormalized as in Eq. (18.2.21), then for any μ

$$M_R = \left[\prod_{\mathcal{O}} \left(N^{(\mathcal{O})}/N_\mu^{(\mathcal{O})}\right)\right] M(E, x, g_\mu, m, \mu). \tag{18.2.22}$$

We can again use dimensional analysis (assuming M has dimensionality D) and set $\mu = E$, to write this as

$$M_R = E^D \left[\prod_{\mathcal{O}} \left(N^{(\mathcal{O})}/N_E^{(\mathcal{O})}\right)\right] M\left(1, x, g_E, \frac{m}{E}, 1\right). \tag{18.2.23}$$

Thus to find the high energy behavior of an off-shell amplitude M_R, we need to know how N_μ varies with the renormalization scale μ.

For any two renormalization scales μ and μ', the renormalized operators $N_\mu^{(\mathcal{O})}\mathcal{O}$ and $N_{\mu'}^{(\mathcal{O})}\mathcal{O}$ both have finite matrix elements, so the ratio $N_{\mu'}^{(\mathcal{O})}/N_\mu^{(\mathcal{O})}$ must be cut-off independent. On dimensional grounds, this ratio must take the form

$$N_{\mu'}^{(\mathcal{O})}/N_\mu^{(\mathcal{O})} = G^{(\mathcal{O})}(g_\mu, \mu'/\mu, m/\mu). \tag{18.2.24}$$

Differentiating with respect to μ' and then setting $\mu' = \mu$ gives

$$\mu\frac{d}{d\mu}N_\mu^{(\mathcal{O})} = \gamma^{(\mathcal{O})}(g_\mu, m/\mu)N_\mu^{(\mathcal{O})}, \tag{18.2.25}$$

where

$$\gamma^{(\mathcal{O})}(g_\mu, m/\mu) \equiv \left[\frac{\partial}{\partial z}G^{(\mathcal{O})}(g_\mu, z, m/\mu)\right]_{z=1}. \tag{18.2.26}$$

The solution is

$$N_E^{(\mathcal{O})} \propto \exp\left[\int^E \gamma^{(\mathcal{O})}\left(g_\mu, \frac{m}{\mu}\right)\frac{d\mu}{\mu}\right]. \tag{18.2.27}$$

This is a useful result because the introduction of the sliding scale prevents the appearance of zero-mass singularities in $N_\mu^{(\mathcal{O})}$ and $N_{\mu'}^{(\mathcal{O})}$, and hence also in $G^{(\mathcal{O})}$ and $\gamma^{(\mathcal{O})}$. Hence as long as g_μ is small, there are no large logarithms that prevent the application of perturbation theory to calculate $\gamma^{(\mathcal{O})}$. Also, for $\mu \gg m$, $\gamma^{(\mathcal{O})}(g_\mu, m/\mu)$ has a smooth limit

$$\gamma^{(\mathcal{O})}(g_\mu) \equiv \gamma^{(\mathcal{O})}(g_\mu, 0). \tag{18.2.28}$$

As an example, consider the operator $\mathcal{O} = \phi^2$ in the scalar field theory with interaction $g\phi^4/24$. Instead of renormalizing it so that the correction factor (18.1.9) is cancelled at $p^2 = 0$, we cancel it at a sliding scale $p^2 = \mu^2$, by introducing a new renormalized ϕ^2 operator $N_\mu^{(\phi^2)}\phi^2$, with

$$N_\mu^{(\phi^2)} \equiv F^{(\phi^2)}(\mu^2)^{-1} = 1 + \frac{g}{32\pi^2}\int_0^1 dx\left[\ln\left(\frac{\Lambda^2}{m^2+\mu^2 x(1-x)}\right) - 1\right] + O(g^2).$$

Then the function (18.2.24) for this operator is

$$\begin{aligned} G^{(\phi^2)}(g_\mu, \mu'/\mu, m/\mu) &\equiv \frac{N_{\mu'}^{(\phi^2)}}{N_\mu^{(\phi^2)}} \\ &= 1 + \frac{g_\mu}{32\pi^2}\int_0^1 dx \ln\left[\frac{m^2+\mu^2 x(1-x)}{m^2+\mu'^2 x(1-x)}\right] + O(g_\mu^2). \end{aligned}$$

(We can use g_μ here instead of g or $g_{\mu'}$, because the difference only affects the terms of order g_μ^2 or higher.) From Eq. (18.2.26), we have then

$$\gamma^{(\phi^2)}\left(g_\mu, \frac{m}{\mu}\right) = -\frac{g_\mu}{16\pi^2}\int_0^1 \frac{\mu^2 x(1-x)}{m^2+\mu^2 x(1-x)}\,dx + O(g_\mu^2)$$

or for $\mu \gg m$,

$$\gamma^{(\phi^2)}(g_\mu) = -\frac{g_\mu}{16\pi^2} + O(g_\mu^2). \tag{18.2.29}$$

Another good example is provided by the N-factor associated with the renormalization of the electromagnetic field in quantum electrodynamics. Recall that the photon propagator can be made finite for all momenta by evaluating it for renormalized electromagnetic fields, or equivalently by multiplying the propagator of the unrenormalized field by Z_3^{-1}:

$$\widetilde{\Delta}_{\rho\sigma}(q) = Z_3^{-1}\Delta_{\rho\sigma}(q). \tag{18.2.30}$$

Eq. (10.5.17) shows that this renormalized propagator may be written

$$\widetilde{\Delta}_{\rho\sigma}(q) = \frac{\eta_{\rho\sigma}}{[q^2-i\epsilon][1-\pi(q^2)]} + q_\rho q_\sigma\text{-terms}. \tag{18.2.31}$$

Suppose we instead define a renormalized field $N_\mu^{(A)}A^\rho$, whose propagator has a term proportional to $\eta_{\rho\sigma}/[q^2-i\epsilon]$ with a coefficient that is equal to unity at a sliding renormalization scale $q^2 = \mu^2$. For this purpose, we must clearly take

$$N_\mu^{(A)} = Z_3^{-1/2}[1-\pi(\mu^2)]^{1/2}. \tag{18.2.32}$$

Using Eq. (11.2.22), the function (18.2.24) is then

$$G^{(A)}(g_\mu, \mu'/\mu, m/\mu) = \left[\frac{1-\pi(\mu'^2)}{1-\pi(\mu^2)}\right]^{1/2}$$
$$= 1 - \frac{e_\mu^2}{4\pi^2}\int_0^1 dx\, x(1-x)\ln\left[\frac{m^2+\mu'^2 x(1-x)}{m^2+\mu^2 x(1-x)}\right] + O(e_\mu^4) \quad (18.2.33)$$

and so Eq. (18.2.26) gives

$$\gamma^{(A)}(e_\mu, m/\mu) = -\frac{e_\mu^2}{2\pi^2}\int_0^1 dx\, \frac{x^2(1-x)^2\mu^2}{m^2+\mu^2 x(1-x)} + O(e_\mu^4). \quad (18.2.34)$$

As promised, this has a smooth limit for $\mu \gg m$

$$\gamma^{(A)}(e_\mu) \equiv \gamma^{(A)}(e_\mu, 0) = -\frac{e_\mu^2}{12\pi^2} + O(e_\mu^4). \quad (18.2.35)$$

As already mentioned, electrodynamics is a special case, because the renormalization constant $Z_3^{-1/2}$ in the definition of the renormalized electromagnetic field is just the reciprocal of the constant used to define the renormalized electric charge of the electron: $e_R = Z_3^{1/2}e$. The natural definition of the renormalized electric charge at a sliding scale μ is then

$$e_\mu = N_\mu^{(A)-1} e = Z_3^{-1/2} N_\mu^{(A)-1} e_R, \quad (18.2.36)$$

so that e_μ times the field $N_\mu^{(A)} A^\rho$ renormalized at scale μ is independent of μ. From Eq. (18.2.25) we see that the function $\beta(e)$ which gives the μ dependence of e_μ according to Eq. (18.2.9) for $\mu \gg m$ has the value

$$\beta(e) = -e\gamma^{(A)}(e) = \frac{e^3}{12\pi^2} + O(e^5). \quad (18.2.37)$$

Using an earlier calculation [3a] of the fourth-order term in the vacuum polarization function $\pi(q^2)$, Gell-Mann and Low were able also to give the term in $\beta(e)$ of next order in e:

$$\beta(e) = \frac{e^3}{12\pi^2} + \frac{e^5}{64\pi^4} + O(e^7). \quad (18.2.38)$$

In other words, the electric charge at a sliding scale μ satisfies the renormalization group equation

$$\mu\frac{d}{d\mu}e_\mu = \frac{e_\mu^3}{12\pi^2} + \frac{e_\mu^5}{64\pi^4} + O(e_\mu^7). \quad (18.2.39)$$

This shows, for small e_μ, that e_μ *increases* with increasing μ.

We also need an initial condition. This is provided by the known value of the conventionally renormalized charge $e_R = Z_3^{1/2}e$, for which $\alpha \equiv e_R^2/4\pi = 1/137.036\ldots$. Eqs. (18.2.32) and (18.2.36) yield

$$e_R/e_\mu = Z_3^{1/2} N_\mu^{(A)} = \sqrt{1-\pi(\mu^2)}$$
$$= 1 - \frac{e_R^2}{4\pi^2}\int_0^1 dx\, x(1-x)\ln\left[1+\frac{\mu^2 x(1-x)}{m_e^2}\right] + O(e_R^4). \quad (18.2.40)$$

We need to match this with the solution of Eq. (18.2.39) at a value of μ which is large enough to justify the approximation $\mu \gg m_e$ in (18.2.39) but small enough so that the logarithm in Eq. (18.2.40) is still small enough compared with $4\pi^2/e_R^2$ to justify the use of perturbation theory. (For instance, we might take μ to be of order 100 MeV.) For such values of μ, Eq. (18.2.40) gives

$$e_\mu \simeq e_R + \frac{e_R^3}{12\pi^2}\left[\ln\frac{\mu}{m_e} - \frac{5}{6}\right]. \tag{18.2.41}$$

On the other hand, the solution of Eq. (18.2.39) for e_μ small (keeping only the leading term on the right-hand side) is

$$e_\mu = \left[\text{constant} - \frac{\ln\mu}{6\pi^2}\right]^{-1/2}. \tag{18.2.42}$$

Comparing Eqs. (18.2.41) and (18.2.42) gives the solution

$$e_\mu = e_R\left[1 - \frac{e_R^2}{6\pi^2}\left(\ln\left(\frac{\mu}{m_e}\right) - \frac{5}{6}\right)\right]^{-1/2}. \tag{18.2.43}$$

Unlike Eq. (18.2.41), Eq. (18.2.43) is valid as long as $e_\mu^2/6\pi^2$ is small, whether or not $(e_R^2/6\pi^2)\ln(\mu/m_e)$ is small.

For instance, we have already seen in Section 11.3 that the leading fourth-order radiative correction to the magnetic moment of the muon can be obtained by multiplying the second-order (Schwinger) term (11.3.16) by the vacuum polarization function $\pi_e(k^2)$ at $k^2 \simeq m_\mu^2$, which according to Eq. (18.2.40) is the same (to this order) as using $e_{m_\mu}^2/4\pi$ in place of α in the Schwinger term.

Another example: Experiments at high energy electron–positron colliders such as LEP at CERN or SLC at SLAC now study physical processes at energies of the order of the mass of the Z^0 particle, or 91 GeV. Eq. (18.2.43) shows that at these energies, radiative corrections in pure quantum electrodynamics should be calculated using a value for the fine structure constant which is not $\alpha = 1/137.036$ but rather

$$\frac{e^2(91\ \text{GeV})}{4\pi} = \frac{\alpha}{1 - 2(11.25)\alpha/3\pi} = \frac{1}{134.6}. \tag{18.2.44}$$

This is for a theory in which electrons are the only charged particles with masses below m_Z. In the real world there are many such particle types, and the effective fine structure constant [4] at m_Z is $(128.87 \pm 0.12)^{-1}$.

* * *

The sliding scale at which the coupling parameters of a theory are calculated may be the value of an external *field* rather than the momentum of an external line. In one of the early applications of the renormalization group method, Coleman and E. Weinberg [4a] considered the effective potential $V(\phi)$ for a spacetime-independent external scalar field ϕ. In the simple case where the field interacts with itself alone, their one-loop result is given by Eq. (16.2.15). It is particularly interesting to consider this potential in the case where the renormalized mass m_R vanishes, where to one-loop order we may set $\mu^2(\phi)$ in the last term equal to $g_R\phi^2/2$, so that Eq. (16.2.15) here reads (with g a slightly redefined coupling):

$$V(\phi) = \lambda_R + \frac{g}{24}\phi^4 + \frac{g^2\phi^4 \ln \phi^2}{256\pi^2}. \tag{18.2.45}$$

This looks like at first sight as if for $g > 0$ the potential becomes less than λ_R for very small ϕ, so that the point $\phi = 0$ is a local maximum instead of a minimum, but for such small values of ϕ the third term is larger than the second, and the perturbation theory is obviously untrustworthy. Also, we would like to be able to argue that g must be positive in order for the potential to be bounded below at large fields, but Eq. (18.2.45) shows that however small g is, there is some sufficiently large ϕ where perturbation theory breaks down, and hence Eq. (18.2.45) cannot be trusted to tell us whether the potential is bounded below at large fields.

We can do much better by using a coupling constant defined at a sliding scale μ of field strength. Suppose we define a coupling g_μ by the condition that

$$V(\mu) = \lambda_R + \frac{g_\mu}{24}\mu^4. \tag{18.2.46}$$

If we had used g_μ as the coupling parameter from the beginning, then in place of Eq. (18.2.45) we would have obtained

$$V(\phi) = \lambda_R + \frac{g_\mu}{24}\phi^4 + \frac{g_\mu^2\phi^4}{256\pi^2}\ln\left(\frac{\phi^2}{\mu^2}\right), \tag{18.2.47}$$

which obviously satisfies Eq. (18.2.46). [38] The renormalization group equation for g_μ can be obtained from the condition that this effective potential is independent of μ, [39]

$$\mu\frac{dg_\mu}{d\mu} = \frac{3g_\mu^2}{16\pi^2}. \tag{18.2.48}$$

(Terms involving the derivative of g_μ^2 are dropped here, because they are of higher order in g_μ, and are hence negligible as long as g_μ is sufficiently small.) It is not a coincidence that this takes the same form as the renormalization group equation (18.2.9), (18.2.12), where μ was a renormalization momentum, because as we shall see in the next section the first two terms in the renormalization group equation are always independent of the way that we define the sliding scale. The solution of this equation is given by Eq. (18.2.17), in general

[38] Eq. (18.2.47) differs from what we would get from Eq. (18.2.45) by simply using Eq. (18.2.46) to express g in terms of g_μ, in that the last term is proportional to g_μ^2 rather than g^2. The difference is of higher order in g, but can become significant if $V(\phi)$ is evaluated at ϕ very different from μ, where large logarithms can compensate for powers of coupling. If we use g_μ as the coupling parameter from the beginning and take μ to be of order ϕ then no such large logarithms occur, and the approximation Eq. (18.2.47) is valid as long as g_μ remains small.

[39] To eliminate all cut-off dependence, ϕ should be written in terms of a renormalized field $\phi_\mu = N_\mu^\phi \phi$, which gives $V(\phi_\mu)$ a μ dependence arising from the μ dependence of the renormalization constant N_μ^ϕ. This point is ignored here, because in the scalar field theory with interaction $\propto \phi^4$ the lowest-order graphs contributing to the μ dependence of N_μ^ϕ have two loops, so that to the order of the calculations presented here we can take $N_\mu^\phi = 1$.

with a different integration constant M. Hence by taking $\mu = \phi$ in Eq. (18.2.17), we now have

$$V(\phi) = \lambda_R - \frac{32\pi^2\phi^4}{3\ln(\phi^2/M^2)}. \tag{18.2.49}$$

$$g_\phi = -\frac{32\pi^2}{3\ln(\phi^2/M^2)}. \tag{18.2.50}$$

This result should be used with some care, because it is only valid where the coupling constant g_ϕ is small. The problem is not just that Eq. (18.2.49) loses its validity for ϕ near M; we also cannot integrate the renormalization group equation through the singularity at $\phi = M$, so a knowledge of g_ϕ on one side of this singularity tells us nothing about its behavior on the other side.

If g_{ϕ_0} is found to have a small *positive* value for some ϕ_0 then from Eq. (18.2.50) we know that $M > |\phi_0|$, so g_ϕ remains small and Eq. (18.2.49) is valid for $|\phi| < |\phi_0|$. This shows that the point $\phi = 0$ is a local *minimum* of $V(\phi)$, contrary to what Eq. (18.2.45) might have led us to suppose. This means that the vacuum which is invariant under the symmetry transformation $\phi \to -\phi$ is stable in this model, aside from the possibility of quantum mechanical barrier penetration. On the other hand, we do not know that Eq. (18.2.49) becomes valid for $|\phi|$ sufficiently large compared with M. The coupling might remain too large to use perturbation theory for all $|\phi| > M$, and even if it becomes small for some $|\phi| > M$, the potential might be given for such ϕ by Eq. (18.2.49) with a renormalization scale $M' > \phi$, producing a second singularity. Thus in this case we cannot conclude that $V(\phi) \to -\infty$ for $|\phi| \to \infty$.

Similarly, if g_ϕ is found to have a small *negative* value for some ϕ_0 then we know that $M < |\phi_0|$, so g_ϕ remains small and Eq. (18.2.49) is valid for $|\phi| > |\phi_0|$. In this case we cannot conclude anything about the behavior of the potential for $|\phi| < M$, but here we *can* use Eq. (18.2.49) to see that $V(\phi) \to -\infty$ for $|\phi| \to \infty$, ruling out the possibility of any stable vacuum. Because we are here considering the limit $|\phi| \to \infty$, this conclusion holds also for scalar fields of mass $m > 0$, provided $m \ll |\phi_0|$. This is why it is necessary to assume that the ϕ^4 coupling (renormalized at any scale much larger than the scalar mass) is positive.

18.3 Varieties of Asymptotic Behavior

The renormalization group method provides useful insight into the types of possible asymptotic behavior that are encountered in quantum field theories, even in cases where the running coupling g_μ does not remain small enough to allow the use of perturbation theory. We will distinguish four different ways that g_μ may behave for $\mu \to \infty$, that correspond to four different shapes of the function $\beta(g)$ in theories with a single coupling constant. In the next section we will take up the case of theories with several independent couplings.

Let's first recall the results for $\beta(g)$ obtained for the two examples considered in the previous section. One of these is the scalar field theory with interaction $g\phi^4/24$, for which the beta-function for small g is

$$\beta(g) = \frac{3g^2}{16\pi^2} - \frac{18}{3}\frac{g^3}{(16\pi^2)^2} + O(g^4). \tag{18.3.1}$$

The other is quantum electrodynamics. Instead of writing the beta-function here in the form (18.2.38), we will emphasize the similarities between this theory and the scalar field theory by writing

$$g \equiv e^2, \tag{18.3.2}$$

with $\beta(g_\mu)$ now understood to be $\mu\, dg_\mu/d\mu$, so that for small g

$$\begin{aligned}\beta(g) &= 2e\left[\frac{e^3}{12\pi^2} + \frac{e^5}{64\pi^4} + O(e^7)\right] \\ &= \frac{g^2}{6\pi^2} + \frac{g^3}{32\pi^4} + O(g^4).\end{aligned} \tag{18.3.3}$$

Note that in both cases the physically allowed coupling constants fall in the range $g \geq 0$, where $\beta(g) \geq 0$ for small g. In electrodynamics this is simply

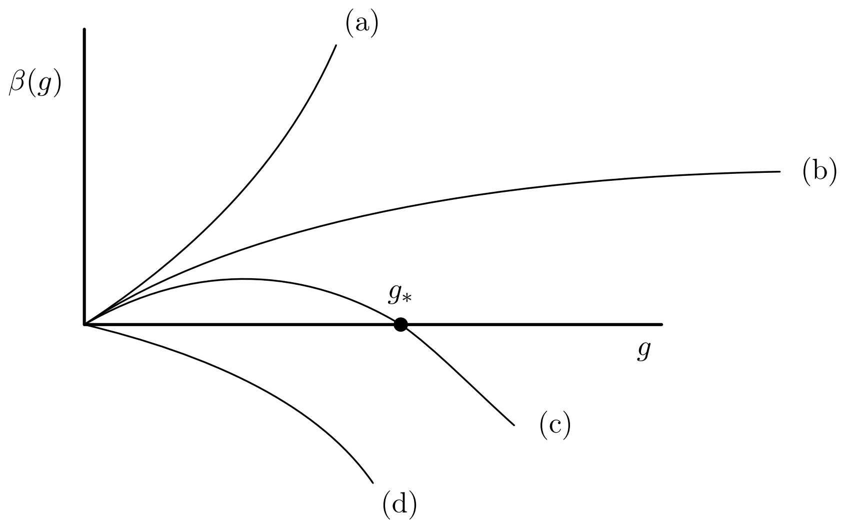

Figure 18.4. Schematic representation of four possible forms of the function $\beta(g)$. For such forms of $\beta(g)$, the running coupling g_μ would: (a) approach infinity at a finite value of μ; (b) continue to grow as μ increases; (c) approach a finite limit g_* for $\mu \to \infty$; (d) approach zero for $\mu \to \infty$.

because the reality of the Lagrangian requires e to be real. In the scalar field theory, as we saw at the end of the previous section, it is necessary to have $g > 0$ in order to have any stable vacuum state. However, there are other examples that have $\beta(g) \leq 0$ for $g \geq 0$. For instance, we can consider a scalar field theory with interaction Hamiltonian $-g\phi^4/24$, with g taken positive. This may be unphysical, but stability problems will not bother us as long as we stick to perturbation theory. Eq. (18.2.9) shows that if we redefine $g \to -g$ the beta-function undergoes the change $\beta(g) \to -\beta(-g)$, so our previous result that $\beta(g) = 3g^2/16\pi^2 + O(g^3)$ for an interaction $g\phi^4/24$ now gives

$$\beta(g) = -3g^2/16\pi^2 + O(g^3) \tag{18.3.4}$$

for an interaction Hamiltonian density $-g\phi^4/24$. Of greater relevance to physics, we shall see in Section 18.7 that non-Abelian gauge theories with not too many spinor fields have

$\beta \leq 0$ for small positive gauge coupling constants. In what follows we will always define the coupling g so that $g \geq 0$, but we shall consider the cases of $\beta(g)$ either positive or negative for small g.

Now let us turn to our list of possibilities. (See Figure 18.4.)

(a) Singularity at Finite Energy

Suppose that $\beta(g) > 0$ for small positive g (as is the case for Eqs. (18.3.1) and (18.3.3)), and that $\beta(g)$ remains positive and continues to rise sufficiently rapidly with increasing g, so that the integral $\int^\infty dg/\beta(g)$ converges:

$$\int^\infty \frac{dg}{\beta(g)} < \infty. \tag{18.3.5}$$

Then g_μ will move steadily away from $g_\mu = 0$, and Eq. (18.2.10) shows that g_E must become infinite at a *finite* value of E:

$$E_\infty = \mu \exp\left(\int_{g_\mu}^\infty \frac{dg}{\beta(g)}\right), \tag{18.3.6}$$

where μ is any renormalization scale with $\mu \gg m$. We saw an example of this phenomenon in the previous section: if the lowest-order formula $\beta(g) = 3g^2/16\pi^2$ for the beta-function in the scalar field theory were taken as exact for all values of g, then the running coupling (18.2.17) would become infinite at the energy (18.2.19). Similarly, if the lowest- order formula $\beta(g) = g^2/6\pi^2$ (with $g \equiv e^2$) for the beta-function in spinor quantum electrodynamics were taken as exact for all g, then the energy (18.3.6) at which g_E and e_E become infinite would be

$$E_\infty = \mu \exp(6\pi^2/g_\mu). \tag{18.3.7}$$

Using Eq. (18.2.43), we may express this in terms of the conventional renormalized charge:

$$E_\infty \simeq m_e \exp\left(\frac{6\pi^2}{e_R^2} + \frac{5}{6} + O(e_R^2)\right) = e^{646.6\ldots} m_e. \tag{18.3.8}$$

Of course, the approximation that $\beta(g) = g^2/6\pi^2$ will break down before this energy is reached, so all we can say with confidence is that e_E will become large enough to invalidate perturbation theory at some energy E below E_∞.

(b) Continued Growth

Now suppose that in some theory $\beta(g)$ remains positive definite for $g \to \infty$, but rises slowly enough (or decreases) so that $\int^\infty dg/\beta(g)$ is divergent. The coupling constant g_E then continues to increase as $E \to \infty$, but becomes infinite only for $E = \infty$. Furthermore, the leading term in the asymptotic behavior of g_E for $E \to \infty$ is independent of the conventionally renormalized coupling. For instance, if $\beta(g)$ behaves for large g like bg^k with $b > 0$ and $k < 1$, then the solution of Eq. (18.2.9) is

$$g_E = \left[1 + (1-k) b\, g_\mu^{k-1} \ln \frac{E}{\mu}\right]^{1/(1-k)} g_\mu. \tag{18.3.9}$$

If g_μ is small for some μ (say, of order m) then the growth of g_E is seen only at energies which are exponentially large compared with this μ. However, in the extreme high energy limit the coupling grows according to

$$g_E \to [(1-k)b\ln E]^{1/(1-k)}, \tag{18.3.10}$$

a limiting behavior independent of g_μ!

(c) Fixed Point at Finite Coupling

Suppose next that $\beta(g)$ remains positive-definite for $0 < g < g_*$, but drops to zero at $g = g_*$ and is negative thereafter. Then Eq. (18.2.9) dictates that as μ increases g_μ will increase for $g_\mu < g_*$ and decrease for $g_\mu > g_*$, in either case approaching the fixed point g_* for $\mu \to \infty$. If the zero of $\beta(g)$ at g_* is simple, then in the neighborhood of this point we have

$$\beta(g) \to a(g_* - g) \quad \text{for} \quad g \to g_* \tag{18.3.11}$$

with $a > 0$. The solution of Eq. (18.2.9) is then

$$g_* - g_\mu \propto \mu^{-a}. \tag{18.3.12}$$

(The behavior of type (b) described above may be regarded as the special case where the fixed point g_* is at infinity.) Also, $\gamma(g)$ for a general operator $\mathcal{O}$ may be expected to behave smoothly near g_*:

$$\gamma(g) = \gamma(g_*) + c(g_* - g) + O\big((g_* - g)^2\big). \tag{18.3.13}$$

(We are here dropping the label $\mathcal{O}$ on γ and c.) Hence in matrix elements of this (and perhaps other) operators, we encounter a factor (see Eq. (18.2.27))

$$N_E^{-1} \propto \exp\left[-\int^E \gamma(g_\mu)\frac{d\mu}{\mu}\right] \propto E^{-\gamma(g_*)}\left[1 + O(E^{-a})\right]. \tag{18.3.14}$$

The product of the factors $E^{-\gamma(g_*)}$ can be lumped together with the factor E^D in Eq. (18.2.23), with the result that the whole matrix element goes as

$$M_R \propto E^{D_*}, \tag{18.3.15}$$

where the dimensionality D_* is calculated adding an 'anomalous dimension' $-\gamma(g_*)$ to the actual dimensionality of each operator appearing in the matrix element.

(d) Asymptotic Freedom

In the examples that have been discussed so far, $\beta(g)$ was positive for small positive g, so that g_μ is driven away from $g = 0$ as μ increases. Suppose that for some other theory $\beta(g)$ is negative for small positive g. Then

$$\beta(g) \to -bg^n, \tag{18.3.16}$$

where $b > 0$. Here n is the order of the lowest-order diagrams that contribute to $\beta(g)$, and hence is always an integer greater than unity. (In the theories used as examples here, $n = 2$.) The solution of Eq. (18.2.9) here is

$$g_E = g_\mu\left[1 + b(n-1)g_\mu^{n-1}\ln\frac{E}{\mu}\right]^{-1/(n-1)}. \tag{18.3.17}$$

For $E \to \infty$, this has a limit independent of g_μ:

$$g_E \to [b(n-1)\ln E]^{-1/(n-1)}. \tag{18.3.18}$$

Since this gives a vanishing g_E for $E \to \infty$, we can trust perturbation theory in this limit, provided only that g_E for some finite E is within the region around $g = 0$ where g and $\beta(g)$ have opposite signs. The anomalous dimensions $\gamma^{(\mathcal{O})}$ of various operators $\mathcal{O}$ are expected to have the weak-coupling behavior (dropping the label $\mathcal{O}$)

$$\gamma(g) \to c g^m, \tag{18.3.19}$$

where m is the order of the lowest-order diagrams that contribute to the renormalization of the operator, and c is a real constant that can be positive or negative. The asymptotic behavior at high energies of the factor introduced into matrix elements by renormalization of this operator is then

$$\begin{aligned} N_E^{-1} &\propto \exp\left[-\int^E \gamma(g_\mu)\frac{d\mu}{\mu}\right] \\ &\to \exp\left[-c\int^E [b(n-1)\ln\mu]^{-m/(n-1)}\frac{d\mu}{\mu}\right] \\ &\propto \exp\left[-\frac{c[b(n-1)]^{-m/(n-1)}}{1-m/(n-1)}(\ln E)^{1-m/(n-1)}\right] \end{aligned} \tag{18.3.20}$$

except that for $m = n-1$

$$N_E^{-1} \propto (\ln E)^{-c/b(n-1)}. \tag{18.3.21}$$

We see that in the case of asymptotic freedom there are no corrections to those effective dimensionalities that determine the powers of energy appearing in the asymptotic behavior of the matrix element, but this asymptotic behavior is instead modified by powers of $\ln E$.

For a toy model that exhibits asymptotic freedom, we can use the scalar field theory with interaction Hamiltonian density $-g\phi^4/24$, with g taken positive. Eq. (18.3.4) here gives the parameters of Eq. (18.3.16) as:

$$b = 3/16\pi^2, \qquad n = 2, \tag{18.3.22}$$

so Eq. (18.3.17) gives, for $E \to \infty$,

$$g_E \to \left[\frac{3\ln E}{16\pi^2}\right]^{-1}. \tag{18.3.23}$$

Also, the operator ϕ^2 in this theory has an anomalous dimension given by Eq. (18.2.29) as

$$\gamma(g) = -\frac{g}{16\pi^2} + O(g^2). \tag{18.3.24}$$

Hence Eq. (18.3.19) applies here, with

$$c = -1/16\pi^2, \qquad m = 1. \tag{18.3.25}$$

Each ϕ^2 operator in a matrix element therefore contributes a factor given by Eq. (18.3.21) as

$$N_E^{-1} \propto (\ln E)^{1/3}. \tag{18.3.26}$$

The scalar field ϕ itself in this theory has $\gamma(g) \propto g^2$, and hence $m = 2$, so each ϕ operator in a matrix element therefore contributes a factor given by Eq. (18.3.20) as

$$N_E^{-1} \propto 1 + O\left(\frac{1}{\ln E}\right). \tag{18.3.27}$$

We will see another, more physical, example of asymptotic freedom when we take up quantum chromodynamics in Section 18.7.

In all cases where g_E can be extended to infinite energy, its behavior in this limit turns out to be independent of the renormalized coupling g_R. However, this does not necessarily mean that the theory involves no arbitrary dimensionless parameters. In all cases, in order to describe how g_E approaches its limit for $E \to \infty$, we need to specify a free parameter λ with the dimensions of energy. For case (b), Eq. (18.3.10) may be written

$$g_E \to [(1-k)b\ln(E/\lambda)]^{1/(1-k)}.$$

For case (c), Eq. (18.3.12) may be written

$$g_E \to g_* \left[1 - \left(\frac{\lambda}{E}\right)^b\right].$$

Finally, for case (d), Eq. (18.3.18) may be written

$$g_E \to [b(n-1)\ln(E/\lambda)]^{-1/(n-1)}.$$

Such theories in general do have a free dimensionless parameter: the ratio of λ to the mass m. Coupling constants like e_R that are renormalized at scales tied to m may be expressed as functions of m/λ. It is only when all masses in a theory vanish that we can say that the theory has no free dimensionless parameters.

Of the four types of asymptotic behavior described here, types (a) and (b) lead to the apparently unphysical behavior that the running coupling g_E becomes infinite, either at a finite energy (case (a)) or for $E \to \infty$ (case (b)). This does not in itself mean disaster: we have to look at how the coupling is defined. For instance, if g_μ drops smoothly from a finite value g_m at $\mu = m$ to zero for $\mu \to \infty$, and we define a new coupling $\widetilde{g}_\mu \equiv g_\mu/[1 - g_\mu/g_{2m}]$, then $\widetilde{g}_\mu$ becomes infinite at $\mu = 2m$, but this is just an artifact of this particular choice of coupling parameter. However, the conventional renormalized couplings g_μ in both the ϕ^4 scalar field theory and quantum electrodynamics have been defined here in terms of the values of matrix elements at energies of order μ. Specifically, g_μ in ϕ^4 scalar field theory is defined as the invariant Feynman amplitude A for scalar–scalar scattering at $s = t = u = \mu^2$, where A is supposed to be analytic. Also, $g_\mu = e_\mu^2$ in spinor electrodynamics is given by

$$e_\mu^2/e_R^2 = Z_3/N_\mu^{(A)2} = \left[1 - \pi(\mu^2)\right]^{-1}. \tag{18.3.28}$$

An infinity in e_μ^2 at a point μ_∞ would therefore produce a pole or other singularity in the renormalized photon propagator at a positive value of p^2, that is, at $p^2 = \mu_\infty^2$, where the propagator is supposed to be analytic. Thus, with g_μ as defined here, the type of asymptotic behavior described in case (a) is ruled out physically.

How then do our various quantum field theories behave? Years ago, Landau [4b] argued that in quantum electrodynamics the increasing powers of $\ln(E/M)$ encountered at each order of perturbation theory would add up to give singularities (so-called 'Landau ghosts') at finite values of E. In modern terms, Landau could be said to have discovered possibility (a) above, but he did not give any argument against cases (b) or (c).

Nevertheless, there is today a widespread view that interacting quantum field theories that are not asymptotically free, like quantum electrodynamics or the scalar field theory with ϕ^4 interaction, are not mathematically consistent. In quantum electrodynamics there is some evidence against case (c), the existence of a finite fixed point e_*. Such a fixed point would only be possible [4c] if non-perturbative effects changed the qualitative nature of the operator product expansion, the subject of Chapter 20, or if there were a non-perturbative renormalization of the triangle anomaly discussed in Chapter 22. But even if case (c) is indeed ruled out in quantum electrodynamics, there is still the possibility of case (b), a fixed point at infinite coupling.

Most of the evidence against the consistency of interacting non- asymptotically free quantum field theories comes from the study of the scalar field theory in four spacetime dimensions with ϕ^4 interaction, quantized on a finite spacetime lattice. There are rigorous theorems [4d] to the effect that this theory (with arbitrary dependence of the parameters of the theory on lattice spacing) does not have an interacting continuum theory as its limit for zero lattice spacing unless the theory is asymptotically free, which of course is contrary to what is found for this theory in perturbation theory. This argument also seems inconclusive. It is true that if there were a consistent continuum scalar field theory that is not asymptotically free, then it would be possible to construct a lattice theory by integrating out the values of the scalar field at all points except on a spacetime lattice. *But this would not be the lattice theory that is considered in these theorems.* It would be a lattice theory with every possible coupling allowed by symmetry principles—not just a term proportional to ϕ^4, but also terms proportional to ϕ^6, $\phi^3\Box\phi$, etc., with coefficients having a dependence on the cut-off (the inverse lattice spacing) governed by the Wilson renormalization group equations discussed in Section 12.4. No one has proved anything about the continuum limit of such a theory.

If it is really true that there is no interacting continuum scalar field theory in the limit of zero lattice spacing, then we must encounter some obstruction when we try to solve the Wilson renormalization group equations, which for weak renormalized couplings would have to be at very small lattice spacings. Such a theory would appear like an interacting continuum field theory unless examined at very short distances. The renormalized coupling constant in this approximate continuum theory presumably has a singularity at finite energy, as in case (a) above, so that it too breaks down at short distances. (But for strong couplings there is no direct connection between the forms of the Wilson and Gell-Mann–Low renormalization group equations, so the existence of a singularity in the bare couplings at finite lattice spacing does not necessarily imply a singularity in the renormalized coupling constants at a finite renormalization scale.)

Theories of this sort are sometimes called *trivial*, either because, under various assumptions about the bare couplings of the theory quantized on a lattice, the continuum limit turns out to be a free field theory, or because the only way to make a continuum theory of type (a) physically satisfactory at all energies is to adopt the solution $g_\mu = 0$ of the renormalization group equation (18.2.9). Even if a field theory is trivial in either sense, there is no reason not to include it as part of a realistic theory of physical phenomena. The existence of an obstruction to the solution of the Wilson renormalization group equations for a field theory at very small lattice spacings is not important if in the real world there are other fields that must also be taken into account at such short distances. Similarly, the fact that a given quantum field theory has unphysical singularities at some large energy E_∞ is not a physical problem if the theory in question is only a low-energy approximation to a larger theory, an approximation valid only at energies far below E_∞. In particular, long before we reach the energies near (18.3.8) where quantum electrodynamics could be expected to become singular, it becomes necessary to take even gravitation into account, and no one knows how to calculate the effects of strong gravitational forces at such energies.

Despite these reassuring remarks, it is possible that in order to avoid unphysical singularities, all our separate quantum field theories like spinor quantum electrodynamics will eventually have to be integrated into an asymptotically free theory. Fortunately, the question of whether a theory is asymptotically free for *some* finite range of coupling constants can be settled by perturbative calculations: if $\beta(g)$ is negative as $g \to 0+$, then the theory is asymptotically free for all renormalized couplings g_μ lying between zero and the first zero of $\beta(g)$.

* * *

In this connection, it is worth noting that although the detailed form of $\beta(g)$ depends on the gauge and on precisely how the running coupling is defined, the first two terms in the power series for $\beta(g)$ do not. Suppose we have two definitions g_μ and $\widetilde{g}_\mu$ of the running coupling, perhaps employing different definitions of the renormalization scale μ or different gauges. Since both g_μ and $\widetilde{g}_\mu$ are dimensionless and cut-off independent, there is no way that $\widetilde{g}_\mu$ for $\mu \gg m$ can depend on anything but g_μ:

$$\widetilde{g}_\mu = \widetilde{g}(g_\mu).$$

We then have

$$\widetilde{\beta}(\widetilde{g}_\mu) \equiv \mu \frac{d}{d\mu} \widetilde{g}_\mu = \frac{d\widetilde{g}(g_\mu)}{dg_\mu} \beta(g_\mu)$$

and so

$$\widetilde{\beta}(\widetilde{g}) = \frac{d\widetilde{g}(g)}{dg} \beta(g). \tag{18.3.29}$$

As long as we are sticking to the same definition of the unrenormalized coupling, all renormalized couplings are equal in lowest order, so the power series for $\widetilde{g}$ in terms of g may be written

$$\widetilde{g}(g) = g + ag^2 + O(g^3)$$

or, equivalently,

$$g = \widetilde{g} - a\widetilde{g}^2 + O(\widetilde{g}^3).$$

The derivative is

$$\frac{d\widetilde{g}}{dg} = 1 + 2ag + O(g^2) = 1 + 2a\widetilde{g} + O(\widetilde{g}^2).$$

Also, for the couplings we have been considering here (including $g = e^2$) the power series for $\beta(g)$ takes the form

$$\beta(g) = bg^2 + b'g^3 + O(g^4)$$

or in terms of $\widetilde{g}$

$$\beta(\widetilde{g}) = b\widetilde{g}^2 + (b' - 2ab)\widetilde{g}^3 + O(\widetilde{g}^4).$$

From (18.3.29), we have then

$$\begin{aligned}\widetilde{\beta}(\widetilde{g}) &= \left[1 + 2a\widetilde{g} + O(\widetilde{g}^2)\right]\left[b\widetilde{g}^2 + (b' - 2ab)\widetilde{g}^3 + O(\widetilde{g}^4)\right] \\ &= b\widetilde{g}^2 + b'\widetilde{g}^3 + O(\widetilde{g}^4).\end{aligned} \tag{18.3.30}$$

We see that the first two terms in the power series for $\widetilde{\beta}$ in terms of $\widetilde{g}$ have the same coefficients as in the power series for β in terms of g. However, this is definitely not the case for the higher-order terms. In fact, it is always possible to choose the function $\widetilde{g}(g)$ so that all terms in $\widetilde{\beta}(\widetilde{g})$ of higher than third order in $\widetilde{g}$ vanish, so we can describe the asymptotic behavior of $\widetilde{g}_E$ for $E \to \infty$ by inspection of the first two terms in the perturbation series for $\beta(g)$. But this is of little value, since we would need to carry our calculations to all orders to determine how g depends on $\widetilde{g}$, and without this we cannot use our knowledge of the asymptotic behavior of $\widetilde{g}$ to say anything about the asymptotic behavior of g, or of physical quantities.

The same argument that led to Eq. (18.3.30) shows that, at *small couplings*, the Wilson renormalization group equation for the bare coupling constant as a function of lattice spacing for inverse lattice spacings greater than particle masses is the same as the Gell-Mann–Low renormalization group equation for the renormalized coupling constant as a function of renormalization scale for scales greater than particle masses. Hence if a continuum theory is asymptotically free, then there will be no obstruction in passing to the continuum limit of the theory quantized on a lattice.

18.4 Multiple Couplings and Mass Effects

Up to now, we have considered theories with only one dimensionless coupling g. It is easy to extend the formalism to incorporate several such couplings g^ℓ: for each g^ℓ, we have a renormalization group equation that for $\mu \gg m$ takes the form

$$\mu \frac{d}{d\mu} g^\ell(\mu) = \beta^\ell\big(\mathbf{g}(\mu)\big), \tag{18.4.1}$$

with each β^ℓ depending in general on all the gs. There are now many more possibilities for the asymptotic behavior of the $g^\ell(\mu)$ as $\mu \to \infty$; in a given theory we may have some trajectories in g-space that go off to infinity, at finite or infinite values of μ, other trajectories that approach fixed points, and yet other trajectories that approach closed curves known as 'limit cycles'. To get a taste of some of the various possibilities, let's consider the behavior of $g^\ell(\mu)$ near a fixed point.

Eq. (18.4.1) has a fixed-point solution $g^\ell(\mu) = g^\ell_*$ if

$$\beta^\ell(\mathbf{g}_*) = 0. \tag{18.4.2}$$

In the neighborhood of this point, Eq. (18.4.1) becomes

$$\mu \frac{d}{d\mu}\big[g^\ell(\mu) - g^\ell_*\big] = \sum_k M^\ell{}_k \big[g^k(\mu) - g^k_*\big], \tag{18.4.3}$$

where M is the matrix

$$M^\ell{}_k \equiv \left[\frac{\partial \beta^\ell(\mathbf{g})}{\partial g^k}\right]_{\mathbf{g}=\mathbf{g}_*}. \tag{18.4.4}$$

The solution can be expanded in eigenvectors of this matrix

$$g^\ell(\mu) = g^\ell_* + \sum_m c_m V^\ell_m \mu^{\lambda_m}, \tag{18.4.5}$$

where V_m is an eigenvector of M with eigenvalue λ_m (normalized in any convenient way):

$$\sum_k M^\ell{}_k V^k_m = \lambda_m V^\ell_m, \tag{18.4.6}$$

and the c_m are a set of expansion coefficients.[40] (The summation convention is suspended in this section.)

Eq. (18.4.5) shows that the coupling constants approach the fixed point as $\mu \to \infty$ if and only if $c_m = 0$ for all eigenvectors with $\lambda_m > 0$. (For simplicity we are assuming here that none of the eigenvalues vanish.) Thus in general the trajectories that are attracted to the fixed point lie on an N_--dimensional surface, where N_- is the number of negative eigenvalues of M; the tangents to this surface at $\mathbf{g}_*$ are the eigenvectors with negative eigenvalues. Trajectories that are not on this surface may approach close to the fixed point, but are eventually repelled, predominantly in the direction of eigenvectors with the largest positive eigenvalues. Of course, if all eigenvalues are negative then there is a finite region around the fixed point within which all trajectories converge on this point.

Since the eigenvalues λ_m are evidently important in learning the asymptotic behavior of trajectories that approach a fixed point, it is useful to note that these eigenvalues are

[40] We are assuming here that the eigenvectors V_m form a complete set. This is not always so, but it is the generic case; the eigenvectors of a finite matrix M will form a complete set if all of the roots of the secular equation $\mathrm{Det}(M - \lambda\mathbf{1}) = 0$ are different. A matrix whose eigenvectors do not form a complete set can be regarded as a limiting case of a matrix with a complete set of eigenvectors when some of its eigenvalues become degenerate.

independent of the definition of the couplings. Suppose we introduce a new set of couplings $\widetilde{g}^\ell$, defined as functions of the gs. These satisfy renormalization group equations

$$\mu\frac{d}{d\mu}\widetilde{g}^\ell(\mu) = \sum_m \frac{\partial\widetilde{g}^\ell(\mathbf{g})}{\partial g^m}\bigg|_{\mathbf{g}=\mathbf{g}(\mu)} \beta^m\big(\mathbf{g}(\mu)\big) \equiv \widetilde{\beta}^\ell\big(\widetilde{\mathbf{g}}(\mu)\big),$$

so

$$\widetilde{\beta}^\ell(\widetilde{\mathbf{g}}) = \sum_m \frac{\partial\widetilde{g}^\ell(\mathbf{g})}{\partial g^m}\beta^m(\mathbf{g}). \tag{18.4.7}$$

(That is, β transforms as a contravariant vector in coupling-constant space.) Differentiating, we have

$$\sum_m \frac{\partial\widetilde{\beta}^\ell(\widetilde{\mathbf{g}})}{\partial\widetilde{g}^m}\frac{\partial\widetilde{g}^m}{\partial g^k} = \sum_m \frac{\partial^2\widetilde{g}^\ell}{\partial g^m\partial g^k}\beta^m(\mathbf{g}) + \sum_m \frac{\partial\widetilde{g}^\ell}{\partial g^m}\frac{\partial\beta^m(\mathbf{g})}{\partial g^k}.$$

At a fixed point $\mathbf{g}_*$ the first term on the right vanishes, so this gives the matrix equation

$$\widetilde{M}S = SM, \tag{18.4.8}$$

where

$$\widetilde{M}^\ell{}_k \equiv \left[\frac{\partial\widetilde{\beta}^\ell}{\partial\widetilde{g}^k}\right]_{\widetilde{\mathbf{g}}=\widetilde{\mathbf{g}}(\mathbf{g}_*)}, \tag{18.4.9}$$

$$S^\ell{}_k \equiv \left[\frac{\partial\widetilde{g}^\ell}{\partial g^k}\right]_{\mathbf{g}=\mathbf{g}_*}. \tag{18.4.10}$$

As long as the transformation $\mathbf{g} \to \widetilde{\mathbf{g}}$ is non-singular, Eq. (18.4.8) is a similarity transformation, and hence M and $\widetilde{M}$ have the same eigenvalues λ_m.

The renormalization group formalism may be extended to non-renormalizable as well as renormalizable theories. As explained in Section 12.3, the infinities in non-renormalizable theories are eliminated by a suitable renormalization of coupling constants and masses, just as in renormalizable theories; the only difference is that in non-renormalizable theories the Lagrangian must be supposed to contain all possible interactions allowed by the symmetries of the theory. If g_B^ℓ is the unrenormalized coupling constant multiplying an operator of dimensionality D_ℓ in the Lagrangian (that is, a product of fields and spacetime derivatives of fields whose dimensionality in powers of mass or energy is D_ℓ), then g_B^ℓ will have a dimensionality $\Delta_\ell = 4 - D_\ell$. We may then re-express the bare couplings in terms of a set of *dimensionless* renormalized couplings $g^\ell(\mu)$ and a cut-off Λ, through relations of the general form

$$g_B^\ell \equiv \mu^{\Delta_\ell}\left[g^\ell(\mu) + \sum_{k,m} b^\ell{}_{km}g^k(\mu)g^m(\mu)\ln\left(\frac{\Lambda}{\mu}\right) + O\big(g(\mu)^3\big)\right], \tag{18.4.11}$$

with the dimensionless numerical coefficients $b^\ell{}_{km}$ and similar coefficients in higher-order terms chosen to cancel the cut-off dependence of physical quantities. (In some theories the leading term might be trilinear or even higher order in couplings; the modifications that would be needed here are obvious.) From the requirement that g^ℓ_B be cut-off-independent, we obtain the renormalization group equation (18.4.1), with

$$\beta^\ell(\mathbf{g}) = -\Delta_\ell g^\ell - \sum_{k,m} b^\ell{}_{km} g^k g^m + O(g^3). \tag{18.4.12}$$

Non-renormalizable interactions are those with $D_\ell > 4$, or $\Delta_\ell < 0$, so as long as the $g^\ell(\mu)$ all remain sufficiently small we expect the non-renormalizable renormalized couplings to have positive β^ℓ and hence to grow with μ, but no one knows what happens when the couplings become large enough to invalidate perturbation theory.

However, as explained in the next section, even theories with infinite number of independent parameters commonly have fixed points $\mathbf{g}_*$ at which the number N_- of negative eigenvalues of the matrix (18.4.6) is *finite*, just as it is at zero coupling. (In particular, often $N_- = 1$.) Where $N_- \neq 0$, the fixed point lies on an N_--dimensional *critical surface*, consisting of trajectories that are attracted into the fixed point as $\mu \to \infty$. A non-renormalizable theory with coupling parameters on such a critical surface, although not of course asymptotically free, is said to be asymptotically *safe* [5], because the renormalized couplings remain finite for large values of μ. The condition of asymptotic safety in such a theory would play the role that used to be associated with the principle of renormalizability, of eliminating all but a finite number of free parameters, the coordinates of the critical surface.

In a renormalizable theory, all physical quantities are made cut-off-independent by adjusting the cut-off dependence of a *finite* number of bare couplings. These bare couplings may be expressed in terms of an equal number of μ-dependent renormalized couplings, and the condition that the bare couplings are μ-independent yields renormalization group equations relating only these renormalized couplings. From the broader point of view which allows non-renormalizable as well as renormalizable couplings, a renormalizable theory just corresponds to a finite-dimensional *invariant surface* in the infinite-dimensional space of all renormalizable and non-renormalizable theories; that is, it is a surface for which $\beta^\ell(\mathbf{g})$ at any point $\mathbf{g}$ on the surface is tangent to the surface at that point.

So far in this section we have tacitly assumed that $\mu \gg m$, so that we could neglect the dependence of β^ℓ on m/μ. However, this is not necessary; we can if we like treat a mass as just another coupling parameter [6]. That is, all renormalized couplings can be defined as before in terms of various Green's functions at off-mass-shell momenta of order μ, but now evaluated with all bare masses zero. The dimensionless renormalized mass parameters for Dirac fields ψ or scalar fields ϕ may be defined as

$$m_\psi(\mu) \equiv N^{(\bar\psi\psi)}(\Lambda/\mu)^{-1} \frac{m_{\psi,\text{BARE}}(\Lambda)}{\mu}, \tag{18.4.13}$$

$$m^2_\phi(\mu) \equiv N^{(\phi^2)}(\Lambda/\mu)^{-1} \frac{m^2_{\phi,\text{BARE}}(\Lambda)}{\mu^2}, \tag{18.4.14}$$

where $N^{(\mathcal{O})}(\Lambda/\mu)$ are the dimensionless constants which, when multiplied into corresponding operators $\mathcal{O}$, cancel the infinities in the matrix elements of these operators, also evaluated with all bare masses zero. (See Section 18.1.) These new renormalized masses and couplings have no direct physical significance, but the true physical masses and all physical matrix elements can be expressed in terms of them. These matrix elements take the form of sums of matrix elements for zero bare mass, with any number of insertions of the renormalized mass operators $N^{(\phi^2)}\phi^2$ and $N^{(\bar\psi\psi)}\bar\psi\psi$, times the corresponding renormalized mass parameters.

In this renormalization scheme the beta-functions for the various couplings are obviously mass-independent, and the beta-functions for the mass parameters are proportional to these parameters, with coefficients that depend on all the various couplings; using Eq. (18.2.25), we have

$$\mu\frac{d}{d\mu}m_\psi(\mu) = \left[-1-\gamma_{\bar\psi\psi}(\mathbf{g}_\mu)\right]m_\psi(\mu), \tag{18.4.15}$$

$$\mu\frac{d}{d\mu}m_\phi^2(\mu) = \left[-2-\gamma_{\phi^2}(\mathbf{g}_\mu)\right]m_\phi^2(\mu). \tag{18.4.16}$$

For instance, we noted in Section 18.2 that in the scalar field theory with Lagrangian (18.1.2), the mass operator ϕ^2 has anomalous dimension (18.2.29) for $m=0$, so here

$$\mu\frac{d}{d\mu}m_\phi^2(\mu) = \left[-2+\frac{g_\mu}{16\pi^2}+O(g_\mu^2)\right]m_\phi^2(\mu). \tag{18.4.17}$$

Also, Eq. (11.4.3) shows that the effect of higher-order corrections to the electron propagator is to replace the electron mass by $m_e-\Sigma^*(p,m_e)$, so the effect of these corrections on matrix elements of the operator $\bar\psi_e\psi_e$ between one-electron states of four-momentum p^μ is to multiply them by a factor

$$F(p) = 1-\left(\frac{\partial\Sigma^*(p,m_e)}{\partial m_e}\right)_{m_e=0}.$$

The renormalization constant $N^{(\bar\psi\psi)}$ for the operator $\bar\psi_e\psi_e$ is therefore equal to $F^{-1}(p)$, evaluated with p^2 equal to some renormalization scale, say $+\mu^2$. According to Eq. (11.4.8), to one-loop order this is

$$\begin{aligned}
N^{(\bar\psi\psi)} &= 1+\left(\frac{\partial\Sigma^*_{1\text{ loop}}(p,m_e)}{\partial m_e}\right)_{m_e=0,\,p^2=\mu^2} \\
&= 1-\frac{4\pi^2}{(2\pi)^4}\int_0^1 dx\,\ln\left[1+\left(\frac{\Lambda^2}{\mu^2}\right)(1-x)\right] \\
&\longrightarrow 1-\frac{e^2}{4\pi^2}\left[\ln\left(\frac{\Lambda^2}{\mu^2}\right)-1\right],
\end{aligned} \tag{18.4.18}$$

where Λ is an ultraviolet cut-off,[41] and we take the limit $\Lambda \gg \mu$. The anomalous dimension of the operator $\bar\psi_e\psi_e$ is therefore given by (18.2.25) as

$$\gamma^{(\bar\psi\psi)} = \mu\frac{d}{d\mu}\ln N^{(\bar\psi\psi)} = \frac{e_\mu^2}{2\pi^2}+O(e_\mu^4), \tag{18.4.19}$$

[41] This notation is different from that of Eq. (11.4.8), where the ultraviolet cut-off was called μ.

so Eq. (18.4.15) here reads

$$\mu\frac{d}{d\mu}m_e(\mu) = \left[-1 - \frac{e_\mu^2}{2\pi^2} + O(e_\mu^4)\right] m_e(\mu). \tag{18.4.20}$$

The same formula holds in general gauge theories, with e^2 replaced with the value of $\sum_\alpha (t_\alpha)^2$ for the particular species of fermion in question.

The important difference between the $m(\mu)$ and the other renormalized parameters of the theory is of course that bare masses have positive dimensionality, so as long as the couplings remain small the $m(\mu)$ all decrease in magnitude. Our previous assumption that masses may be neglected as $\mu \to \infty$ is justified if in fact $m(\mu)$ does vanish for $\mu \to \infty$. However this is only known to be the case in asymptotically free theories, where the couplings all do remain small for $\mu \to \infty$; in all other cases this assumption is just an educated guess.

18.5 Critical Phenomena[42]

For some purposes we may be interested in the limit of very low rather than high energies or wave numbers. The arguments of Section 18.2 can be repeated to study this limit, except that here we must examine the case $\mu \to 0$ rather than $\mu \to \infty$. This limit is of course simplest if there are no masses in the theory, as, for instance, in quantum electrodynamics with a symmetry under the chiral transformation $\psi \to \gamma_5\psi$ which forbids an electron mass. In this particular case the only renormalizable coupling $eA^\mu\bar{\psi}\gamma_\mu\psi$ as well as all non-renormalizable couplings have $\beta^\ell > 0$ for sufficiently small couplings, so all trajectories in at least a finite region around the origin are attracted into the point $g^\ell = 0$ as $\mu \to 0$.

The same considerations may be applied even to theories with very small but non-zero masses if we include these masses among the coupling parameters of the theory, as described in the previous section. The coefficient Δ in Eq. (18.4.12) is positive for a mass parameter, so in this case the trajectories can never reach the point $g = 0$, but they may come close if the masses are small.

Of course, even if we can regard some degree of freedom like the electron field as having zero or very small mass, in the real world there are many other degrees of freedom whose masses are not small. The renormalization group should properly be applied not to the true theory that encompasses all these heavy degrees of freedom, but to an 'effective' field theory, in which only massless or nearly massless degrees of freedom appear explicitly, with interactions that include the effects of internal heavy particle lines. (We shall have more to say about effective field theories in Chapter 19.)

The low wave number limit is of particular interest in the study of critical phenomena, such as long-range correlations at or near a second-order phase transition (a smooth phase transition, with no latent heat) in condensed matter. Because we are interested in the limit $\mu \to 0$, the important eigenvectors of the matrix (18.4.4) are those with eigenvalues $\lambda < 0$,

[42]This section lies somewhat out of the book's main line of development, and may be omitted in a first reading.

which are called *relevant*. The eigenvectors with $\lambda = 0$ and $\lambda > 0$ are called *marginal* and *irrelevant*, respectively.

Suppose that there is a non-trivial fixed point g_* with just one negative eigenvalue λ_0, perhaps corresponding approximately to a mass operator. The set of trajectories of $g^\ell(\mu)$ that are attracted into this fixed point for $\mu \to 0$ therefore forms a critical surface of codimension one; that is, a surface defined by a single condition on the couplings, the condition that for $g \to g_*$, the tangents $g^\ell - g_*^\ell$ have no components in the direction of the eigenvector with negative eigenvalue. There is a phase transition as the physical value of the couplings at any fixed characteristic scale approaches this surface. Because the critical surface has codimension one, the phase transition can be reached by adjusting any one parameter on which the couplings depend, such as the pressure or temperature. The fact that a wide variety of substances do exhibit phase transitions of this sort shows that it is common to encounter fixed points for which the matrix (18.4.4) has a single negative eigenvalue, as already mentioned in the previous section.

To be specific, as the temperature T approaches its critical value T_c we expect the coefficient c_0 of the growing term in Eq. (18.4.5) to become proportional to $T - T_c$, because there is no reason why it should be singular or why it should vanish faster than this. Hence for $\mu \to 0$ and then $T \to T_c$, the couplings go as

$$g^\ell(\mu) \to (T - T_c) V_0^\ell \mu^{\lambda_0}, \tag{18.5.1}$$

where λ_0 is the only negative eigenvalue at g_*, and V_0^ℓ is the corresponding eigenvector.[43] Applying our renormalization group arguments to wave numbers instead of energies, the N-point function (the Nth partial derivative of the effective action with respect to a field ϕ of dimensionality [wave number]$^{D_\phi}$) at a small characteristic wave number scale κ has the form[44]

$$\Gamma_N(\kappa) \to \kappa^{d - N[D_\phi + \gamma_\phi(g_*)]} F_N\big((T - T_c)\kappa^{\lambda_0}\big), \tag{18.5.2}$$

where $\gamma_\phi(g)$ is the anomalous dimension associated with the field ϕ, and d is the spacetime dimensionality, or in classical statistical mechanics the spatial dimensionality. It is convenient to rewrite this in the equivalent form

$$\Gamma_N(\kappa) \to (T - T_c)^{-[d - N(D_\phi + \gamma_\phi(g_*))]/\lambda_0} G_N\big(\kappa (T - T_0)^{1/\lambda_0}\big). \tag{18.5.3}$$

This shows for one thing that the correlation length ξ (the characteristic length that determines the scale over which the Fourier transform of Γ_N varies) increases as T approaches T_c like

$$\xi \propto (T - T_c)^{-\nu}, \tag{18.5.4}$$

where ν is a conventionally defined positive 'critical exponent', given by Eq. (18.5.3) as

$$\nu = -1/\lambda_0. \tag{18.5.5}$$

[43] Other contributions to the couplings will go as $(T - T_0)^0 \mu^{\lambda_1}$ with $\lambda_1 > 0$. Thus Eq. (18.5.1) is valid here provided $T - T_0$ does not go to zero as fast as $\mu^{\lambda_1 - \lambda_0}$.

[44] The function F_N also depends on dimensionless angles and ratios of wave numbers. Note that Γ_N has 'naive' dimensionality $d - ND_\phi$ because $\delta^d(\sum \kappa)\Gamma$ must be dimensionless.

Also, the zero-field effective action Γ_0 (or in statistical physics, the free energy) must be κ-independent because it corresponds to graphs with no external lines. It follows that Eq. (18.5.3) here becomes for $T \to T_c$:

$$\Gamma_0 - F_0 \propto (T - T_c)^{\nu d}, \tag{18.5.6}$$

where the constant F_0 is the effective action or free energy due to the heavy degrees of freedom which have been integrated out. Thus the exponent ν also governs the behavior of the part of the free energy which is not analytic in temperature for T near T_c.

In 1972 Wilson and Fisher [7] used an expansion in powers of $d-4$ both to show that the theory of a scalar field actually fits the above description, and also to carry out an approximate calculation of critical exponents such as ν. Consider a theory with a single 'light' degree of freedom, a scalar field ϕ, such as the magnetization in a ferromagnet, with a symmetry under $\phi \to -\phi$ that rules out interactions odd in ϕ. In addition to the 'mass' term $-g_2\phi^2/2$, the Lagrangian density of the effective field theory will contain interactions $-g_4\phi^4/4!$, $-g_6\phi^6/6!$, etc. The dimensionality of the field ϕ in powers of wave number is $(d-2)/2$ (so that $\int d^dx\,(\nabla\phi)^2$ should be dimensionless) so the dimensionalities of the couplings g_2, g_4, g_6, etc. in d dimensions are $+2$, $4-d$, $6-2d$, etc. For the fixed point at zero coupling in three dimensions, there are two relevant couplings, g_2 and g_4, but this conclusion is changed by interactions at non-trivial fixed points. Let's examine the surface in coupling constant space in which only g_2 and g_4 are non-zero, and take g_4 to be small.[45] Eq. (18.2.12) gives $\beta(g_4) = 3g_4^2/16\pi^2 + O(g_4^3)$ for $d=4$, and Eq. (18.4.12) tells us that for $d = 4-\epsilon$ dimensions we must add to this a term $-\epsilon g_4$, so

$$\mu\frac{d}{d\mu}g_4(\mu) = -\epsilon g_4(\mu) + \frac{3g_4^2(\mu)}{16\pi^2} + O\big(g_4^3(\mu)\big). \tag{18.5.7}$$

Also, Eq. (18.4.17) gives

$$\mu\frac{d}{d\mu}g_2(\mu) = \left[-2 + \frac{g_4(\mu)}{16\pi^2} + O\big(g_4(\mu)\big)\right]g_2(\mu). \tag{18.5.8}$$

Therefore for small ϵ there is a non-trivial fixed point at

$$g_{4*} = \frac{16\pi^2\epsilon}{3}, \qquad g_{2*} = 0. \tag{18.5.9}$$

The matrix (18.4.4) at this fixed point is diagonal, with eigenvalues

$$\lambda_4 = M^4{}_4 = -\epsilon + \frac{3g_{4*}}{8\pi^2} + O(g_{4*}^2) = +\epsilon + O(\epsilon^2), \tag{18.5.10}$$

$$\lambda_2 = M^2{}_2 = -2 + \frac{g_{4*}}{16\pi^2} + O(g_{4*}^2) = -2 + \frac{\epsilon}{3} + O(\epsilon^2). \tag{18.5.11}$$

[45]These are the only renormalizable couplings for $3 \le d \le 4$, so for such d this is an invariant surface. Note that we do not include the coefficient of $(\nabla\phi)^2$ among the couplings here, because this is a redundant coupling, in the sense described in Section 7.7.

From Eq. (18.5.10) we see that the coupling g_4 is actually irrelevant, so that there is just one relevant coupling here, signalling the presence of a second-order phase transition. From Eq. (18.5.11) we see that the anomalous exponent (18.5.5) is

$$\nu = -\frac{1}{\lambda_2} = \frac{1}{2} + \frac{\epsilon}{12} + O(\epsilon^2). \tag{18.5.12}$$

For the physical value $\epsilon = 1$ the first two terms give $\nu \simeq 0.58$. Three-loop calculations [8a] give this critical exponent to order ϵ^3 as

$$\nu = \frac{1}{2} + \frac{\epsilon}{12} + \frac{7\epsilon^2}{162} - 0.01904\epsilon^3, \tag{18.5.13}$$

which for $\epsilon = 1$ gives $\nu = 0.61$.

In the calculation presented here nothing was assumed about the system under study except that there is a second-order phase transition, near which the only long-wavelength degree of freedom is a single scalar field. There are a number of different physical systems that fit this description, such as the spontaneous appearance of magnetization (represented here by ϕ) in ferromagnetic and antiferromagnetic materials, and also second-order phase transitions between liquids and gases and in binary fluids. All of these systems are therefore expected to have the same value of ν. This is confirmed by experiment, which gives a value [8] $\nu = 0.63 \pm 0.04$, in good agreement with the three-loop result (18.5.13), and in fair agreement even with the one-loop result (18.5.12). It is fortunate though still somewhat mysterious that an expansion in powers of 1 should work so well.

More generally, all the systems that are described by the same set of long-wavelength degrees of freedom near their second-order phase transitions are said to belong to the same universality class. All the critical exponents are the same for all systems in a given universality class.

18.6 Minimal Subtraction

We saw in Section 11.2 that dimensional regularization provides a particularly convenient method for calculating radiative corrections in quantum electrodynamics, because it preserves the conservation laws associated with gauge invariance. For the same reason, dimensional regularization also turns out to provide a very convenient alternative definition for the sliding scale of the renormalization group in general gauge theories [9].

In calculations using dimensional regularization, ultraviolet divergences arise as poles in physical amplitudes when the spacetime dimensionality d approaches the physical value $d = 4$. (For an example, see Eq. (11.2.13).) To cancel these poles, the bare coupling constants $g_B^\ell(d)$ (including masses) must themselves have such poles, with residues fixed by the condition that physical amplitudes be regular as $d \to 4$. These bare couplings in general have non-zero dimensionalities $\Delta_\ell(d)$ that depend on the spacetime dimensionality d, so it is convenient to consider the dimensionless quantity $g_B^\ell(d)\mu^{-\Delta_\ell(d)}$, where μ is a sliding scale with the dimensions of energy or mass. This rescaled bare coupling may be expressed as a sum of terms proportional to positive-definite powers ν of $1/(d-4)$, with coefficients b_ν^ℓ fixed by the requirement of cancellation of singularities as $d \to 4$ in physical amplitudes, plus a remainder that is analytic in d at $d = 4$. This remainder is identified as the dimensionless renormalized coupling constant $g^\ell(\mu, d)$, so

$$g_B^\ell(d)\mu^{-\Delta_\ell(d)} = g^\ell(\mu, d) + \sum_{\nu=1}^{\infty}(d-4)^{-\nu} b_\nu^\ell\big(\mathbf{g}(\mu, d)\big). \tag{18.6.1}$$

We are free to give the bare couplings any d dependence we like, as long as the singularities at $d = 4$ in physical amplitudes are cancelled; we shall remove this ambiguity by requiring that $g^\ell(\mu, d)$ be analytic in d not only at $d = 4$, but for all d.

To calculate the renormalization group equation satisfied by $g^\ell(\mu, d)$, first differentiate Eq. (18.6.1) with respect to μ:

$$\begin{aligned} -\Delta_\ell(d)\left[g^\ell + \sum_{\nu=1}^{\infty}(d-4)^{-\nu} b_\nu^\ell(\mathbf{g})\right] &= \beta^\ell(\mathbf{g}, d) \\ &\quad + \sum_{\nu=1}^{\infty}\sum_m b_{\nu m}^\ell(\mathbf{g})\beta^m(\mathbf{g}, d)(d-4)^{-\nu}, \end{aligned} \tag{18.6.2}$$

where

$$b_{\nu m}^\ell(\mathbf{g}) \equiv \frac{\partial}{\partial g^m} b_\nu^\ell(\mathbf{g}) \tag{18.6.3}$$

and as before

$$\mu\frac{d}{d\mu} g^\ell(\mu, d) = \beta^\ell\big(\mathbf{g}(\mu, d), d\big). \tag{18.6.4}$$

Note that β^ℓ is a function of all of the $g^m(\mu, d)$ and also of d, but it cannot depend separately on μ because, with rescaled masses included among the dimensionless coupling parameters, there are no other dimensionful parameters besides μ.

As we have seen, the dimensionalities $\Delta_\ell(d)$ are always linear functions of d, which we shall now write as

$$\Delta_\ell(d) = \Delta_\ell + \rho_\ell(d-4). \tag{18.6.5}$$

We rewrite the left-hand side of Eq. (18.6.2) as

$$-\rho_\ell g^\ell(d-4) - \left[\Delta_\ell g^\ell + b_1^\ell(\mathbf{g})\rho_\ell\right] - \sum_{\nu=1}^{\infty}(d-4)^{-\nu}\left[\rho_\ell b_{\nu+1}^\ell(\mathbf{g}) + \Delta_\ell b_\nu^\ell(\mathbf{g})\right].$$

The highest power of d in the analytic part here is of first order, so the same must be true on the right-hand side of Eq. (18.6.2), and therefore $\beta(\mathbf{g}, d)$ must be linear in d:

$$\beta^\ell(\mathbf{g}, d) = \beta^\ell(\mathbf{g}) + (d-4)\alpha^\ell(\mathbf{g}). \tag{18.6.6}$$

Equating terms of first and zeroth order in (18.6.2) gives then

$$\alpha^\ell(\mathbf{g}) = -\rho_\ell g^\ell \tag{18.6.7}$$

and, more importantly,

$$\beta^\ell(\mathbf{g}) = -\Delta_\ell g^\ell - b_1^\ell(\mathbf{g})\rho_\ell + \sum_{\nu=1}^{\infty}\sum_m b_{1m}^\ell(\mathbf{g})\rho_m g^m. \tag{18.6.8}$$

It is noteworthy that the beta-function depends only on the coefficients of the simple pole in the bare couplings. In fact, these coefficients also determine the coefficients of all the higher poles; equating the pole terms on the right and left of Eq. (18.6.2) yields the recursion relation

$$\rho_\ell b^\ell_{\nu+1}(\mathbf{g}) - \sum_m \rho_m g^m b^\ell_{\nu+1\,m}(\mathbf{g}) = -\Delta_\ell b^\ell_\nu(\mathbf{g}) - \sum_m b^\ell_{\nu m}(\mathbf{g})\beta^m(\mathbf{g}). \tag{18.6.9}$$

For instance, in order for $\int d^dx\, F_{\mu\nu}F^{\mu\nu}$ to be dimensionless, any gauge field A^μ must have dimensions (in powers of mass) $(d-2)/2$, and since $g_B A^\mu$ must have the same dimensions as $\partial/\partial x^\mu$, g_B must have dimensions $(4-d)/2$, so that for gauge couplings $\Delta = 0$ and $\rho = -1/2$. Eq. (18.6.8) gives then for a gauge theory with a single coupling constant:

$$\beta(g) = \frac{1}{2}\big[b_1(g) - g b_1'(g)\big]. \tag{18.6.10}$$

In particular, Eq. (11.2.20) shows that in quantum electrodynamics in one-loop order the bare electric charge has a pole at $d \to 4$ with

$$e_B = Z_3^{-1/2} e \to e - \frac{e^3}{12\pi^2}\frac{1}{d-4}. \tag{18.6.11}$$

Setting $b_1(e) = -e^3/12\pi^2$ in Eq. (18.6.10) gives then

$$\beta(e) = \frac{e^3}{12\pi^2}, \tag{18.6.12}$$

in agreement with the previous result (18.2.37).

The coupling constants $g^\ell(\mu)$ introduced in this section are said to be defined by *minimal subtraction*. There is a slightly different scheme that is somewhat more convenient. The simple poles $(d-4)^{-1}$ typically arise from functions $(4\pi)^{d/2-2}\Gamma(2-d/2)$ (as in Eq. (11.2.13)) which for $d \to 4$ have the limit

$$(4\pi)^{2-d/2}\Gamma\left(2-\frac{d}{2}\right) \to \frac{1}{2-d/2} - \gamma + \ln 4\pi, \tag{18.6.13}$$

where γ is the Euler constant, $\gamma = 0.5772157$. It is therefore convenient to make the replacement everywhere in Eq. (18.6.1):

$$\frac{1}{d-4} \to \frac{1}{d-4} + \frac{\gamma}{2} - \frac{1}{2}\ln 4\pi. \tag{18.6.14}$$

With this prescription, the coupling constants are said to be defined by *modified minimal subtraction*.

One of the distinguishing characteristics of the definition of couplings by minimal subtraction (or modified minimal subtraction) is that, since no factors of an ultraviolet cut-off ever appear in any calculation, loop diagrams have poles at $d = 4$ that correspond only to logarithmic ultraviolet divergences, not divergences that are linear, quadratic, etc. Hence a residue function $b^\ell_1(\mathbf{g})$ can contain a term of order $g^a g^b g^c \cdots$ only if at $d = 4$ the dimensionality of g^ℓ_B equals the total dimensionality of the couplings g^a_B, g^b_B, g^c_B, etc.:

$$\Delta_\ell = \Delta_a + \Delta_b + \Delta_c + \cdots, \tag{18.6.15}$$

and it follows then from (18.6.8) that the same is true of $\beta^\ell(\mathbf{g})$. In particular, in a theory with no superrenormalizable couplings (like a gauge theory with massless spinors and no scalars, such as quantum electrodynamics with vanishing electron mass) all couplings have $\Delta_\ell \leq 0$, so *the renormalization group equations for the renormalizable couplings (with $\Delta_\ell = 0$) are unaffected by the presence of any non-renormalizable interactions* [5]. Also, in such a theory the beta-functions for the non-renormalizable couplings are polynomials of finite order in the non-renormalizable couplings, with each coefficient in each polynomial given by an infinite series of powers of the renormalizable couplings. For instance, in the theory of photons and massless electrons (assuming invariance under $\psi \to \gamma_5\psi$ and P), there are no nonrenormalizable interactions of dimensionality $+5$, and several interactions of dimensionality $+6$ (four-fermion interactions as well as the purely photonic interaction $F_{\mu\nu}\Box F^{\mu\nu}$) with couplings f_i of dimensionality -2. The beta-function for f_i is of the form $\sum_j b_{ij}(e) f_j$, with the coefficients $b_{ij}(e)$ given by a power series in e.

18.7 Quantum Chromodynamics

Quantum chromodynamics is the modern theory of strong interactions. It is a non-Abelian gauge theory, based on the gauge group $SU(3)$. In addition to the gauge fields, quantum chromodynamics involves fields of spin $\frac{1}{2}$ particles known as quarks. There are quarks of six types, or 'flavors', the u, c, and t quarks having charge $2e/3$, and the d, s, and b quarks having charge $-e/3$. Quarks of each flavor come in three 'colors' which furnish the defining representation **3** of the $SU(3)$ gauge group.[46] Baryons like the protons and neutron may be approximately regarded as color-neutral bound states of three quarks, totally antisymmetric in quark colors, while mesons like the rho meson behave approximately like color-neutral bound states of quarks and antiquarks [11].

In the approximation where the quark masses may be regarded as negligible compared with the energies of interest, the inversion of Eq. (17.5.44) shows that the bare coupling constant in a general gauge theory has a pole at spacetime dimensionality $d \to 4$ with residue given by

$$g_B \longrightarrow \left[\frac{g^3}{4\pi^2}\left(\frac{11}{12}C_1 - \frac{1}{3}C_2\right) + O(g^5)\right]\frac{1}{d-4},$$

where C_1 and C_2 are defined by Eqs. (17.5.33) and (17.5.34). That is, in the notation of the previous section,

$$b_1(g) = \frac{g^3}{4\pi^2}\left(\frac{11}{12}C_1 - \frac{1}{3}C_2\right) + O(g^5). \tag{18.7.1}$$

Using this in Eq. (18.6.10) gives

$$\beta(g) = -\frac{g^3}{4\pi^2}\left(\frac{11}{12}C_1 - \frac{1}{3}C_2\right) + O(g^5). \tag{18.7.2}$$

[46]Before the final formulation of quantum chromodynamics several authors had speculated that there might be three varieties of quarks of each flavor [10], both in order to account for the rate of decays like $\pi^0 \to \gamma + \gamma$ (see Sections 22.1 and 22.2) and to introduce an additional degree of freedom that would explain how the wave function of fermionic quarks in a baryon could be symmetric in spin, space, and flavor coordinates [11].

For an $SU(3)$ theory with n_f massless quarks in the defining representation **3** of $SU(3)$, Eq. (17.5.35) gives

$$C_1 = 3, \qquad C_2 = n_f/2. \tag{18.7.3}$$

Because we are taking the quarks as massless here, this formula may be applied only in the effective field theory obtained by integrating out all quarks heavier than the typical energy E under consideration, so that n_f is the number of quark flavors with masses much less than E. With this understanding, Eqs. (18.7.2) and (18.7.3) yield

$$\beta(g) = -\frac{g^3}{4\pi^2}\left(\frac{11}{4} - \frac{1}{6}n_f\right) + O(g^5). \tag{18.7.4}$$

We see that the theory is asymptotically free as long as there are no more than 16 quark flavors with masses below the energy scale of interest. Since in fact there seem to be only six quark flavors of any mass, the theory of strong interactions based on the gauge group $SU(3)$ is asymptotically free.

It was the 1973 discovery of asymptotic freedom in non-Abelian gauge theories of this sort by Gross and Wilczek [12] and Politzer [13] that convinced theoretical physicists that this is the correct theory of strong interactions. Their calculation immediately explained the puzzling result of a famous 1968 experiment [14] at SLAC on deep-inelastic electron–nucleon scattering, that strong interactions seem to get weaker at high energies.[47] (This experiment will be discussed further in Section 20.6.) But the historical importance of the discovery of asymptotic freedom in Yang–Mills theories is not just that it explained an old experimental result; it for the first time opened up the prospect of doing reliable perturbative calculations of strong interaction processes, at least at high energy.

Asymptotic freedom was soon found to have another important implication. At first after the discovery of asymptotic freedom it was widely assumed that the gauge bosons in a realistic Yang–Mills theory of strong interactions would have to be quite heavy, to explain why these strongly-interacting bosons had not been discovered long before. Following the precedent of the theory of weak and electromagnetic interactions (discussed in Chapter 21), it was supposed that the masses of the gauge bosons arose from a spontaneous breakdown of the color $SU(3)$ gauge group, triggered by the vacuum expectation values of scalar fields in a non-trivial representation of this group. But these strongly interacting scalars would contribute positive terms to $\beta(g)$, which could destroy asymptotic freedom. Even worse, in a theory with strongly interacting scalar fields, radiative corrections involving weak interactions would introduce large violations of various symmetries like charge conjugation invariance and flavor conservation which, as we shall see, would not be violated without the scalars [17]. Then it was suggested to drop the strongly-interacting scalars, and accept the

[47] Zee [15] and perhaps other theorists had already understood that this experimental result could be understood in a theory with a beta-function that becomes negative for small positive coupling, but calculations of $\beta(g)$ in all renormalizable field theories except non-Abelian gauge theories gave $\beta(g) > 0$. On the other hand, by 1972 't Hooft had developed techniques for calculating $\beta(g)$ in Yang–Mills theories, and in June 1972 he announced at a conference on gauge theory at Marseilles [16] that $\beta(g) < 0$, but he waited to publish this result and work out its implications while he was doing other things, so his result did not attract much attention.

consequence that the gluons, the $SU(3)$ gauge bosons, have zero mass [18]. The decrease of the strong coupling constant at high energy or short distance of course implies an increase at low energy or large distance, and it was suggested that this might explain why massless gluons and quarks had not been detected. According to this hypothesis, only color-neutral particles like baryons or mesons will ever appear in isolation [19]. This is unfortunately still a hypothesis rather than a theorem, but after two decades there seems to be little doubt that it is correct.

Even though quarks cannot materialize as free particles, they are in a sense observed as jets produced in high energy collision processes. For instance, in many events in electron–positron annihilation, the final state consists of two narrowly collimated hadron jets, with a distribution in the angle θ between the colliding lepton momenta and the jet directions (in the center-of-mass system) given by $1 + \sin^2\theta$, just as expected from the tree graph for electron–positron annihilation into quark–antiquark final states [20]. This can be understood [21] in terms of the general analysis of infrared divergences in Section 13.4. At extremely high energies we would expect the rate for a physical process to be given by lowest-order perturbation theory, provided it is 'infrared-safe,' in the sense of not becoming infrared divergent when all masses are taken to zero. The total rate for electron–positron annihilation into hadrons is infrared-safe, since we sum over all hadronic final states. (We are ignoring higher-order electromagnetic effects here.) Therefore we can rely on perturbation theory, which immediately tells us that the ratio R of this rate to the rate for $e^+ + e^- \to \mu^+ + \mu^-$ is $R = 3\sum_q Q_q^2$, where the sum runs over all quark flavors, Q_q is their charge in units of e, and the factor 3 is the number of colors. (For instance, in the wide energy range between $m_b \simeq 4.5$ GeV and $m_t \simeq 180$ GeV, $R \simeq 3\big(2(2/3)^2 + 3(-1/3)^2\big) = 11/3$.) On the other hand, the rate for electron–positron annihilation into some definite state of quarks and gluons is not infrared-safe, and its rate therefore cannot be calculated in perturbation theory at all; in fact, it is zero. In between these two extremes is the rate for electron–positron annihilation into a definite number of jets, each jet carrying a definite total momentum and charge, together with a set of unobserved hadrons with limited total energy outside the jets. As discussed in Section 13.4, this rate is infrared-safe. It can therefore be calculated at high energy in the tree approximation of perturbation theory, identifying jets (in this order of perturbation theory) with the outgoing quarks, antiquarks, and gluons. We can even calculate the rate for three-jet events, arising from tree diagrams in which a gluon is emitted from the outgoing quark or antiquark, and use the comparison of the results with experiment to measure the value of $\alpha_s(\mu)$ [22]. But we cannot use perturbation theory to predict the distribution of momenta within a jet, because such a differential rate is not infrared-safe. Similar remarks apply to the production of jets in deep inelastic lepton–hadron collisions, to be discussed in Section 20.6, but the presence of hadrons in the initial state makes the analysis more complicated.

Following the same reasoning as in Section 12.5, with no scalar fields the most general renormalizable Lagrangian for quantum chromodynamics can be put in the form

$$\mathcal{L} = -\frac{1}{4}F_\alpha^{\mu\nu}F_{\alpha\mu\nu} - \sum_n \bar{\psi}_n\big[\partial\!\!\!/ - ig A\!\!\!/_\alpha t_\alpha + m_n\big]\psi_n, \tag{18.7.5}$$

where A^μ_α is the color gauge vector potential; $F^{\mu\nu}_\alpha$ is the color gauge-covariant field strength tensor; g is the strong coupling constant; t_α are a complete set of generators of color $SU(3)$ in the **3** representation (that is, Hermitian traceless $3 \cdot 3$ matrices with rows and columns labelled by the three quark colors), normalized so that $\text{Tr}(t_\alpha t_\beta) = \frac{1}{2}\delta_{\alpha\beta}$; and the subscript n labels quark flavors, with quark color indices suppressed. Just as we found for electrodynamics in Section 12.5, this Lagrangian has important accidental symmetries: it conserves space parity,[48] charge conjugation parity, and the numbers of quarks of each flavor (minus the number of the corresponding antiquarks), including the long-established 'strangeness' quantum number, which counts the numbers of 's' quarks. Thus quantum chromodynamics immediately explained the mysterious fact that the strong interactions respect various symmetries that are not symmetries of all interactions. This argument also makes it clear why, as mentioned earlier, in this theory the weak interactions do not introduce large violations of parity, charge conjugation, strangeness, etc. Since all renormalizable interactions among quarks and gluons conserve these symmetries, at energies E much less than the masses m_W of the particles that carry the weak interactions these symmetries could be violated only by non-renormalizable terms in the effective field theory, such as $\bar\psi\psi\bar\psi\psi$ interactions, which as discussed in Section 12.3 would be suppressed by negative powers of m_W as well as by the coupling constants of the weak interactions.

Of course, it is possible that the quarks and gluons exhibit some new kind of strong interaction at an energy scale Λ' much larger than the scale Λ characteristic of quantum chromodynamics. For instance, as discussed in Section 22.5, the quarks might be bound states of more fundamental fermions, which interact with gauge fields whose asymptotically free couplings become strong at energies of order Λ', trapping them into the quarks. In that case the effective Lagrangian density for quarks at energies $E \ll \Lambda'$ would contain non-renormalizable interactions such as $\bar\psi\psi\bar\psi\psi$, which are suppressed only by powers of E/Λ'. These interactions could show up, not only in small violations of symmetries like parity and quark flavor conservation at ordinary energies, but also in departures [23] from the quantitative predictions of quantum chromodynamics at energies approaching Λ'.

Now let us consider the behaviour of the coupling constant of quantum chromodynamics in greater detail. In lowest order, the renormalization group equation is given by Eq. (18.7.4) as

$$\mu \frac{d}{d\mu} g(\mu) = -\frac{g^3(\mu)}{4\pi^2}\left(\frac{11}{4} - \frac{1}{6} n_f\right). \tag{18.7.6}$$

The solution is

$$\alpha_s(\mu) \equiv \frac{g^2(\mu)}{4\pi} = \frac{12\pi}{(33 - 2n_f)\ln(\mu^2/\Lambda^2)}, \tag{18.7.7}$$

where Λ is an integration constant. This formula exhibits a characteristic property of theories of massless (or, for the quarks, approximately massless) particles: in such theories one of the dimensionless couplings in the Lagrangian is exchanged for a free dimensionful

[48] Although it was not known in 1973, we shall see in Section 23.6 that non-perturbative effects can violate parity in quantum chromodynamics. Various ways of avoiding strong parity violation have been suggested, but it is not yet clear which is correct.

parameter. Eq. (18.7.7) involves no free dimensionless parameters, but it does involve one free parameter with the dimensions of mass, the integration constant Λ.

These calculations have been carried to three-loop order. The renormalization group equations to this order are [24]

$$\mu\frac{d}{d\mu}g(\mu) = -\beta_0\frac{g^3(\mu)}{16\pi^2} - \beta_1\frac{g^5(\mu)}{128\pi^4} - \beta_2\frac{g^7(\mu)}{8192\pi^6}, \tag{18.7.8}$$

where β_n are the numerical coefficients:

$$\beta_0 = 11 - \frac{2}{3}n_f, \tag{18.7.9}$$

$$\beta_1 = 51 - \frac{19}{3}n_f, \tag{18.7.10}$$

$$\beta_2 = 2857 - \frac{5033}{9}n_f - \frac{325}{27}n_f^2. \tag{18.7.11}$$

The solution is

$$\begin{aligned}\alpha_s(\mu) &\equiv \frac{g^2(\mu)}{4\pi} \\ &= \frac{4\pi}{\beta_0\ln(\mu^2/\Lambda^2)}\left[1 - \frac{2\beta_1}{\beta_0^2}\frac{\ln\left[\ln(\mu^2/\Lambda^2)\right]}{\ln(\mu^2/\Lambda^2)}\right. \\ &\left.+\frac{4\beta_1^2}{\beta_0^4\ln^2(\mu^2/\Lambda^2)}\left\{\left(\ln\left[\ln(\mu^2/\Lambda^2)\right] - \frac{1}{2}\right)^2 + \frac{8\beta_2\beta_0}{\beta_1^2} - \frac{5}{4}\right\}\right].\end{aligned} \tag{18.7.12}$$

It should be recalled that n_f in the above results is the number of quark flavors with masses below the energies of interest. In each energy range between any two successive quark masses we have a different value of n_f, and also a different Λ, chosen to make $g(\mu)$ continuous at each quark mass. In particular, experiments on the deep inelastic scattering of electrons typically involve energies above only the first four quark flavors (u, d, s, and c), so here we must take $n_f = 4$. On the other hand, experiments at electron–positron colliders like PEP, PETRA, TRISTAN, and LEP are at energies well above the fifth (b) quark mass, so in these experiments we must take $n_f = 5$. But these results may be expressed in terms of those for $n_f = 4$ by matching the solutions of the renormalization group equations at the b quark mass. In this way it is found [25] (using the modified minimum subtraction prescription in calculating β_2) that the strong coupling extrapolated to $m_Z = 91.2$ GeV is $\alpha_s(m_Z) \equiv g_s^2(m_Z)/4\pi = 0.118\pm0.006$, corresponding to $\Lambda \simeq 250$ MeV for energies μ with $m_b \ll \mu \ll m_t$, where $n_f = 5$. A more recent study [26] of hadronic production in e^+–e^- annihilation at the Z resonance has given a directly measured value $\alpha_s(m_Z) = 0.1200 \pm 0.0025$, with a theoretical uncertainty of ±0.0078, corresponding to $\Lambda = 253^{+130}_{-96}$ MeV.

18.8 Improved Perturbation Theory[49]

[49]This section lies somewhat out of the book's main line of development, and may be omitted in a first reading.

The ground-breaking paper [1] of Gell-Mann and Low was in large part directed to the problem of 'improving' perturbation theory—that is, of using the ideas of the renormalization group and the results of perturbation theory to a given order to say something about the next order of perturbation theory. To illustrate this, let's return to the specific case studied by Gell-Mann and Low: vacuum polarization in quantum electrodynamics.

Recall that the renormalized electric charge e_μ at a sliding scale μ is given by Eq. (18.2.36) in terms of the bare charge e_B as

$$e_\mu = N_\mu^{(A)-1} e_B, \tag{18.8.1}$$

where $N_\mu^{(A)}$ is the constant which, when multiplied into the unrenormalized electromagnetic field, gives a field renormalized at scale μ. (See Eq. (18.2.21).) Thus we can define a renormalized (and hence cut-off-independent) complete photon propagator $\Delta'_{\rho\sigma}(q,\mu,e_\mu)$ in terms of the complete propagator $\Delta'_{B\rho\sigma}(q,e_B)$ of the unrenormalized field as

$$\Delta'_{\rho\sigma}(q,\mu,e_\mu) = N_\mu^{(A)2} \Delta'_{B\rho\sigma}(q,e_B) \tag{18.8.2}$$

in such a way that the function $e_\mu^2 \Delta'_{\rho\sigma}(q,\mu,e_\mu)$ is both independent of μ, because it equals $e_B^2 \Delta'_{B\rho\sigma}(q,e_B)$, and independent of the cut-off, because both e_μ and $\Delta'_{\rho\sigma}(q,\mu,e_\mu)$ are renormalized quantities. (We are not explicitly displaying cut-off dependence here.) But Lorentz invariance and dimensional analysis tell us that this function must take the form:

$$e_\mu^2 \Delta'_{\rho\sigma}(q,\mu,e_\mu) = \frac{\eta_{\rho\sigma} d(q^2/\mu^2, e_\mu)}{q^2} + q_\rho q_\sigma \text{ terms}. \tag{18.8.3}$$

Since Eq. (18.8.3) is μ-independent, we can set $\mu = \sqrt{q^2} \equiv q$ here, so that

$$d(q^2/\mu^2, e_\mu) = d(1, e_q). \tag{18.8.4}$$

Now let us see what this tells us about the structure of the perturbation series for $d(q^2/\mu^2, e_\mu)$. The beta-function for e has the expansion:

$$\beta(e) = b_1 e^3 + b_2 e^5 + b_3 e^7 + \cdots . \tag{18.8.5}$$

The renormalization group equation for e_μ then has a power series solution

$$\begin{aligned} e_q^2 = e_\mu^2 &- b_1 e_\mu^4 \ln\frac{q^2}{\mu^2} - b_2 e_\mu^6 \ln\frac{q^2}{\mu^2} \\ &- \left(\frac{b_1 b_2}{2} \ln^2\frac{q^2}{\mu^2} + b_3 \ln\frac{q^2}{\mu^2}\right) e_\mu^8 + \cdots . \end{aligned} \tag{18.8.6}$$

If we also expand d:

$$d(1,e) = e^2 + d_1 e^4 + d_2 e^6 + d_3 e^8 + \cdots \tag{18.8.7}$$

then

$$\begin{aligned} d(q^2/\mu^2, e_\mu) = d(1, e_q) = e_\mu^2 &- \left(b_1 \ln\frac{q^2}{\mu^2} - d_1\right) e_\mu^4 - \left(b_2 \ln\frac{q^2}{\mu^2} - d_2\right) e_\mu^6 \\ &- \left(\frac{1}{2} b_1 b_2 \ln^2\frac{q^2}{\mu^2} + (b_3 - b_1 d_2) \ln\frac{q^2}{\mu^2} - d_3\right) e_\mu^8 + \cdots . \end{aligned} \tag{18.8.8}$$

Note that the leading powers of $\ln(q^2/\mu^2)$ in each order of $d(q^2/\mu^2, e_\mu)$ are respectively $0, 1, 1, 2, 3, \ldots$. Also, if we calculate $d(q^2/\mu^2, e_\mu)$ to order e_μ^6 and thus determine b_1 and b_2, we can immediately write down the coefficient of the leading logarithm in order e_μ^8, as $-\frac{1}{2}b_1b_2$. None of this would be easy to infer without using the method of the renormalization group.

Problems

1. Consider an $SU(N)$ gauge theory with a scalar field in the defining representation of $SU(N)$. Calculate the beta-function for the gauge coupling to one-loop order, including the contribution of a scalar loop. (Recommendation: Use the background field gauge, with a constant background field.)

2. Suppose that the beta-function $\beta(g)$ for a theory with positive coupling constant g has a simple zero at $g = g_*$, where $\beta(g) \to a(g_* - g)$ with $a > 0$. What is the asymptotic behavior of the correction to the leading term $\propto E^{-\gamma^{\mathcal{O}}(g_*)}$ in the factor $N_E^{\mathcal{O}-1}$ associated with the inclusion of an operator $\mathcal{O}$ in a vacuum expectation value.

3. Show that in a theory with $\beta(g) = bg^2 + b'g^3 + b''g^4 + \cdots$, it is possible by a redefinition of the coupling constant to make the coefficient b'' anything we want.

4. Calculate the effective electric charge that should be used in studying processes at energy 100 GeV, taking account of all known charged quarks and leptons with masses below 100 GeV.

5. Calculate the asymptotic behavior for large four-momentum of the electron propagator in quantum electrodynamics. (You may use the one-loop value of Z_2 calculated elsewhere, for instance in Section 11.4.)

6. Calculate the anomalous exponent ν to first order in the expansion in $\epsilon = 4 - d$ for an $O(N)$-invariant theory of scalar fields $\phi_n(x)$ with $n = 1, \ldots, N$ belonging to the vector representation of $O(N)$, and an interaction $\frac{1}{4}g(\sum_n \phi_n^2)^2$.

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19 Spontaneously Broken Global Symmetries

19.1 Degenerate Vacua

19.2 Goldstone Bosons

19.3 Spontaneously Broken Approximate Symmetries

19.4 Pions as Goldstone Bosons

19.5 Effective Field Theories: Pions and Nucleons

19.6 Effective Field Theories: General Broken Symmetries

19.7 Effective Field Theories: $SU(3) \cdot SU(3)$

19.8 Anomalous Terms in Effective Field Theories*

19.9 Unbroken Symmetries

19.10 The $U(1)$ Problem

Problems

References

20 Operator Product Expansions

21 Spontaneously Broken Gauge Symmetries

22 Anomalies

22.1 The π^0 Decay Problem

22.2 Transformation of the Measure: The Abelian Anomaly

22.3 Direct Calculation of Anomalies: The General Case

22.4 Anomaly–Free Gauge Theories

22.5 Massless Bound States*

22.6 Consistency Conditions

22.7 Anomalies and Goldstone Bosons

Problems

References

23 Extended Field Configurations

23.1 The Uses of Topology

23.2 Homotopy Groups

23.3 Monopoles

23.4 The Cartan–Maurer Integral Invariant

23.5 Instantons

23.6 The Theta Angle

23.7 Quantum Fluctuations around Extended Field Configurations

23.8 Vacuum Decay

Appendix A Euclidean Path Integrals

Appendix B A List of Homotopy Groups

Problems

References

Author Index

Subject Index